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Class 12 Physics Chapterwise PYQs

2026 – 2003 | All CBSE Board Papers

Chapterwise previous year questions, sorted by marks and year

Chapter 14: Semiconductor Electronics

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HOW TO USE THIS BOOKLET

1. **Attempt a section** on your own. Questions run newest year first.
2. **Check the Answer Key,** then follow a question number to its solution.
3. **Study the solution.** Note each *Common Mistake,* then return via "Back to Q n".

SECTION A

1-Mark Questions

Multiple choice · 112 questions

> *Attempt these first, then check the Answer Key and open any solution.*

1. In an unbiased p-n junction, at equilibrium, which of the following statements is true?

(A) Diffusion current is zero but drift current exists.

(B) Diffusion current exists but drift current is zero.

(C) Diffusion and drift currents are equal and opposite.

(D) Both the diffusion and drift currents exist but are unequal.

Check Solution → **[2026 • Set 55-1-1]**

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2. **Assertion (A):** On forward biasing a p-n junction diode, the height of the barrier potential increases.
Reason (R): In forward biasing of a p-n junction diode, the direction of the applied voltage is in the same direction as the built-in potential.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → **[2026 • Set 55-2-1]**

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3. **Case Study:** A p-type or n-type semiconductor can be converted into a p-n junction by doping it with suitable impurity. The motion of majority charge carriers causes diffusion current across the junction while the barrier electric field causes motion of minority carriers for drift current. In case of unbiased diode, the diffusion and drift currents are equal. This equilibrium is disturbed by the biasing batteries. Diodes, therefore, allow currents in one direction. This

property of diode is used in making rectifiers.

Silicon is doped with which of the following to obtain p-type semiconductor?

(A) Phosphorus

(B) Arsenic

(C) Boron

(D) Antimony

Check Solution → [2026 • Set 55-2-1]

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4. A semiconductor has an electron concentration of $5\times 10^{22}\ m^{-3}$. The concentration of holes is (given $n_i = 1{\cdot}5\times 10^{16}\ m^{-3}$):

(A) $5\times 10^{22}\ m^{-3}$

(B) $1{\cdot}5\times 10^{16}\ m^{-3}$

(C) $9\times 10^{8}\ m^{-3}$

(D) $4{\cdot}5\times 10^{9}\ m^{-3}$

Check Solution → [2026 • Set 55-2-1]

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5. During forward biasing of a p-n junction diode, the:

(A) current is mainly due to drifting of majority carriers.

(B) current is mainly due to drifting of minority carriers.

(C) diffusion and drift currents are equal.

(D) current is of the order of 1 A.

Check Solution → [2026 • Set 55-2-1]

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6. The threshold voltage for silicon diode is about:

(A) 0·2 V

(B) 0·5 V

(C) 0·7 V

(D) 1·5 V

Check Solution → **[2026 • Set 55-2-1]**

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7. When we dope Ge with a pentavalent element, four of its electrons bond with four germanium neighbours but fifth electron remains weakly bound. The ionisation energy for this electron is about:

(A) 0·01 eV

(B) 0·05 eV

(C) 0·1 eV

(D) 0·15 eV

Check Solution → **[2026 • Set 55-2-1]**

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8. When the forward bias voltage in a semiconductor diode is changed from 0·8 V to 1·0 V, the forward current changes by 2·0 mA. The forward bias resistance of the diode will be:

(A) 200 Ω

(B) 175 Ω

(C) 100 Ω

(D) 125 Ω

Check Solution → **[2026 • Set 55-3-1]**

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9. The process named 'minority carrier injection' in a p-n junction diode occurs during:

(A) forward biasing

(B) reverse biasing

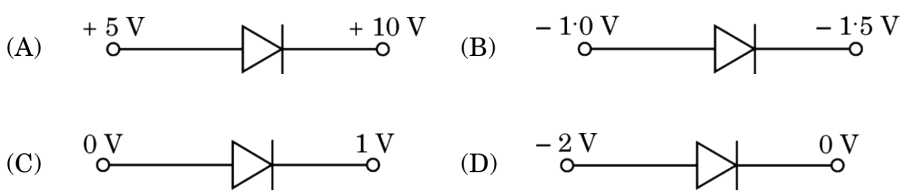
(C) no biasing at low temperature

(D) no biasing at high temperature

Check Solution → [2026 • Set 55-3-1]

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10. Which of the following p-n junction diodes is forward biased?



(A) Option (A): p-side +10 *V*, n-side 0 *V* [see figure]

(B) Option (B): p-side −1·5 *V*, n-side −1·5 *V* [see figure]

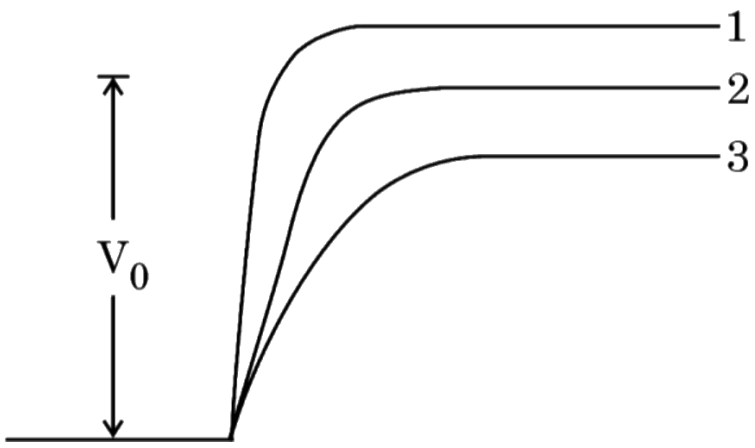
(C) Option (C): see figure

(D) Option (D): see figure

Check Solution → [2026 • Set 55-3-2]

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11. In the given figure, V_0 is the potential barrier across a p-n junction in an unbiased condition. Which of the following statements is correct?



(A) Curves 1 and 3 both correspond to forward biasing of p-n junction.

(B) Curve 3 corresponds to forward biasing and Curve 1 corresponds to reverse biasing.

(C) Curve 1 corresponds to forward biasing and Curve 3 corresponds to reverse biasing.

(D) Curves 1 and 3 both correspond to reverse biasing.

Check Solution → [2026 • Set 55-3-3]

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12. In a reversed-biased p-n junction diode, the applied voltage mostly drops across:

(A) p-region only

(B) n-region only

(C) depletion region

(D) the diode

Check Solution → [2026 • Set 55-4-1]

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13. **Assertion (A):** The conductivity of an n-type semiconductor is higher than that of a p-type semiconductor at a given temperature.
Reason (R): The electrons being in the conduction band in n-type semiconductor are more mobile than the holes in the valence band in p-type semiconductor.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2026 • Set 55-4-1]

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14. A good diode checked by a multimeter should indicate:

(A) high resistance in reverse bias and a low resistance in forward bias

(B) high resistance in both forward bias and reverse bias

(C) low resistance in both reverse bias and forward bias

(D) high resistance in forward bias and low resistance in reverse bias

Check Solution → [2026 • Set 55-5-1]

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15. When a *p-n* junction diode is forward biased:

(A) the barrier height and the depletion layer width both increase.

(B) the barrier height increases and the depletion layer width decreases.

(C) the barrier height and the depletion layer width both decrease.

(D) the barrier height decreases and the depletion layer width increases.

Check Solution → [2025 • Set 55-1-1]

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16. **Assertion (A):** We cannot form a *p-n* junction diode by taking a slab of a *p*-type semiconductor and physically joining it to another slab of a *n*-type semiconductor.

Reason (R): In a *p*-type semiconductor $n_h \gg n_e$, while in a *n*-type semiconductor $n_e \gg n_h$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2025 • Set 55-1-1]

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17. When the resistance measured between p and n ends of a p-n junction diode is high, it can act as a/an

(A) resistor

(B) inductor

(C) capacitor

(D) switch

Check Solution → [2025 • Set 55-2-1]

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18. **Assertion (A):** In a semiconductor diode the thickness of depletion layer is not fixed.

Reason (R): Thickness of depletion layer in a semiconductor device depends upon many factors such as biasing of the semiconductor.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-2-1]

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19. Which of the following is an electrical conductor at room temperature ?

(A) Sn

(B) Mica

(C) Si

(D) C

Check Solution → [2025 • Set 55-4-1]

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20. **Assertion (A):** n-type semiconductor is not negatively charged.

Reason (R): Neutral pentavalent impurity atom doped in intrinsic semiconductor (neutral) donates its fifth unpaired electron to the crystal lattice and becomes a positive donor.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-4-1]

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21. **Assertion (A):** The impurities in p-type Si are not pentavalent atoms.
Reason (R): The hole density in valance band in p-type semiconductor is almost equal to the acceptor density.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-5-1]

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22. **Assertion (A):** A hole is an apparent free particle with effective positive electronic charge.
Reason (R): A hole is not necessarily a vacancy left behind by an electron in the valence band.
Select the correct answer from the codes given below:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Both Assertion (A) and Reason (R) are false.

Check Solution → [2025 • Set 55-6-1]

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23. *Case Study (Extrinsic Semiconductors):* Extrinsic semiconductors are made by doping pure or intrinsic semiconductors with suitable impurity. There are two type of dopants used in doping, Si or Ge, and using them p-type and n-type semiconductors can be obtained. A p-n junction is the basic building block of many semiconductor devices. Two important processes occur during the formation of a p-n junction: diffusion and drift. When such a junction is formed, a 'depletion layer' is created consisting of immobile ion-cores. This is responsible for a junction potential barrier. The width of a depletion layer and the height of potential barrier changes when a junction is forward-biased or reverse-biased. A semiconductor diode is basically a p-n junction with metallic contacts provided at the ends for application of an external voltage. Using diodes, alternating voltages can be rectified.
(i) Which of the following is a donor impurity atom for Ge?

(A) Boron

(B) Antimony

(C) Aluminium

(D) Indium

Check Solution → **[2025 • Set 55-6-1]**

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24. (ii) When a pentavalent atom occupies the position of an atom in the crystal lattice of Si, four of its electrons form covalent bonds with four silicon neighbours, while the fifth remains bound to the parent atom. The energy required to set this electron free is about:

(A) 0·5 eV

(B) 0·1 eV

(C) 0·05 eV

(D) 0·01 eV

Check Solution → **[2025 • Set 55-6-1]**

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25. (iv) (a) In reverse-biased p-n junction:
(A) the drift current is of the order of few mA.
(B) the applied voltage mostly drops across the depletion region.

(C) the depletion region width decreases.

(D) the current increases with increase in applied voltage.

OR

(b) The output frequency of a full-wave rectifier with 50 Hz as input frequency is:

(A) (a) the drift current is of the order of few mA. | (b) 25 Hz

(B) (a) the applied voltage mostly drops across the depletion region. | (b) 50 Hz

(C) (a) the depletion region width decreases. | (b) 100 Hz

(D) (a) the current increases with increase in applied voltage. | (b) 200 Hz

Check Solution → **[2025 • Set 55-6-1]**

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26. A p-n junction diode is forward biased. As a result,

(A) both the potential barrier height and the width of depletion layer decrease.

(B) both the potential barrier height and the width of depletion layer increase.

(C) the potential barrier height decreases and the width of depletion layer increases.

(D) the potential barrier height increases and the width of depletion layer decreases.

Check Solution → **[2025 • Set 55-7-1]**

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27. Germanium crystal is doped at room temperature with a minute quantity of boron. The charge carriers in the doped semiconductors will be:

(A) electrons only

(B) holes only

(C) holes and few electrons

(D) electrons and few holes

Check Solution → **[2025 • Set 55-7-1]**

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28. Ge is doped with As. Due to doping,

(A) the structure of Ge lattice is distorted.

(B) the number of conduction electrons increases.

(C) the number of holes increases.

(D) the number of conduction electrons decreases.

Check Solution → [2024 • Set 55-1-1]

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29. The root mean square value of an alternating voltage applied to a full-wave rectifier is $\frac{V_0}{\sqrt{2}}$. Then the root mean square value of the rectified output voltage is:

(A) $\frac{V_0}{\sqrt{2}}$

(B) $\frac{V_0}{2}$

(C) $\frac{2V_0}{\sqrt{2}}$

(D) $\frac{V_0}{2\sqrt{2}}$

Check Solution → [2024 • Set 55-1-1]

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30. In a full-wave rectifier, the current in each of the diodes flows for:

(A) Complete cycle of the input signal

(B) Half cycle of the input signal

(C) Less than half cycle of the input signal

(D) Only for the positive half cycle of the input signal

Check Solution → [2024 • Set 55-1-1]

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31. An alternating voltage of frequency of 50 Hz is applied to a half-wave rectifier. Then the ripple frequency of the output will be:

(A) 100 Hz

(B) 50 Hz

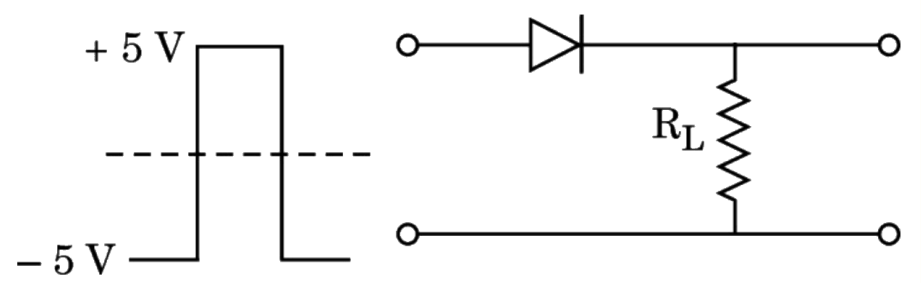
(C) 25 Hz

(D) 150 Hz

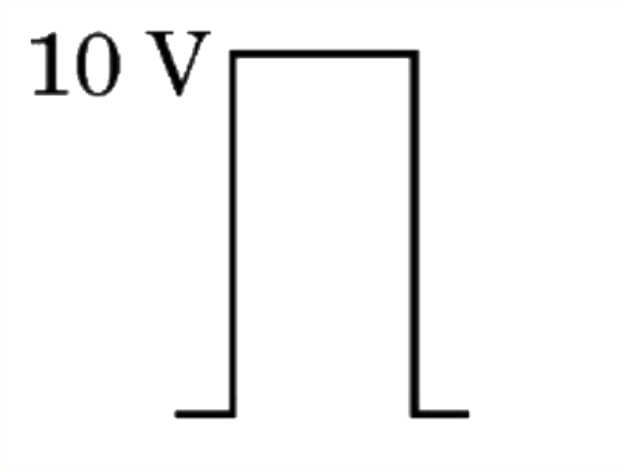
Check Solution → [2024 • Set 55-1-1]

32.

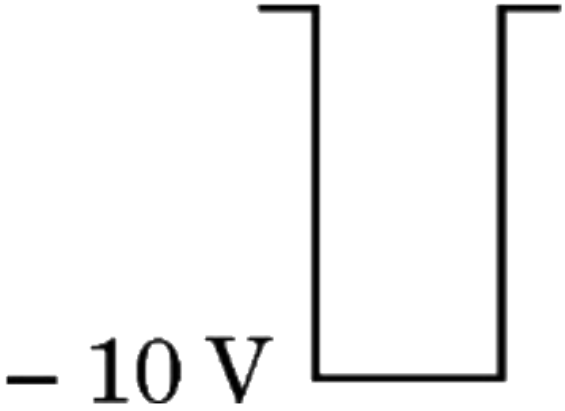
A signal, as shown in the figure, is applied to a p-n junction diode. Identify the output across resistance R_L:



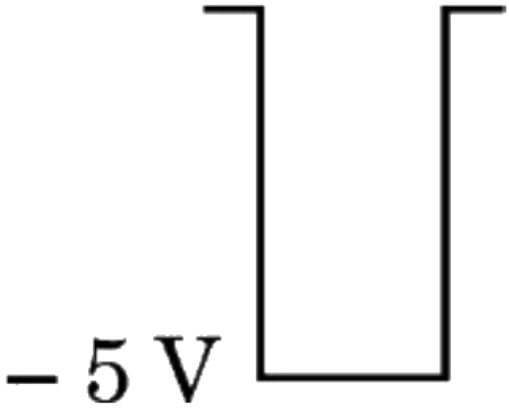
(A)



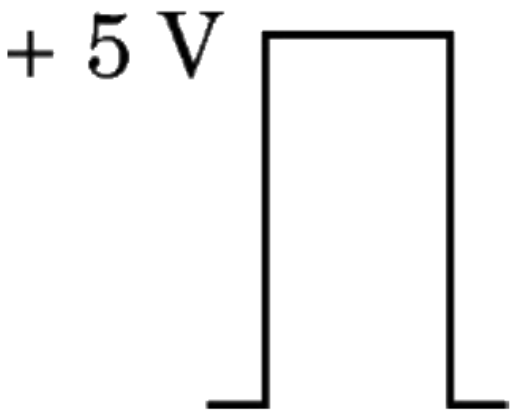
(B)



(C)



(D)



Check Solution → [2024 • Set 55-1-1]

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33. An n-type semiconducting Si is obtained by doping intrinsic Si with:

(A) Al

(B) B

(C) P

(D) In

Check Solution → [2024 • Set 55-2-1]

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34. When a p-n junction diode is subjected to reverse biasing:

(A) the barrier height decreases and the depletion region widens.

(B) the barrier height increases and the depletion region widens.

(C) the barrier height decreases and the depletion region shrinks.

(D) the barrier height increases and the depletion region shrinks.

Check Solution → [2024 • Set 55-2-1]

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35. Si is doped with a pentavalent element. The energy required to set the additional electron free is about:

(A) 0.01 eV

(B) 0.05 eV

(C) 0.72 eV

(D) 1.1 eV

Check Solution → [2024 • Set 55-3-1]

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36. **Assertion (A):** In a semiconductor, the electrons in the conduction band have lesser energy than those in the valence band.
Reason (R): Donor energy level is just above the valence band in a semiconductor.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2024 • Set 55-3-1]

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37. **Assertion (A):** The temperature coefficient of resistance is positive for metals and negative for *p*-type semiconductors.
Reason (R): The charge carriers in metals are negatively charged, whereas the majority charge carriers in *p*-type semiconductors are positively charged.

(A) If both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

(B) If both Assertion (A) and Reason (R) are true and Reason (R) is not the correct explanation of Assertion (A).

(C) If Assertion (A) is true and Reason (R) is false.

(D) If both Assertion (A) and Reason (R) are false.

Check Solution → [2024 • Set 55-4-1]

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38. When Ge is doped with pentavalent impurity, the energy required to free the weakly bound electron from the dopant is about

(A) 0.001 eV

(B) 0.01 eV

(C) 0.72 eV

(D) 1.1 eV

Check Solution → [2024 • Set 55-4-1]

---

39. At a given temperature, the number of intrinsic charge carriers in a semiconductor is 2.0×10^{19} cm^{-3}. It is doped with pentavalent impurity atoms. As a result, the number of holes in it becomes 8×10^{13} cm^{-3}. The number of electrons in the semiconductor is

(A) 2×10^{24} m^{-3}

(B) 4×10^{23} m^{-3}

(C) 1×10^{22} m^{-3}

(D) 5×10^{22} m^{-3}

Check Solution → [2024 • Set 55-4-1]

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40. Initially during the formation of a *p-n* junction -

(A) diffusion current is large and drift current is small.

(B) diffusion current is small and drift current is large.

(C) both the diffusion and the drift currents are large.

(D) both the diffusion and the drift currents are small.

Check Solution → [2024 • Set 55-4-1]

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41. An ac voltage $V=0.5\sin(100\pi\, t)$ volt is applied, in turn, across a half-wave rectifier and a full-wave rectifier. The frequency of the output voltage across them respectively will be

(A) 25 Hz, 50 Hz

(B) 25 Hz, 100 Hz

(C) 50 Hz, 50 Hz

(D) 50 Hz, 100 Hz

Check Solution → [2024 • Set 55-4-1]

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42. A pure Si crystal having 5×10^{28} atoms m^{-3} is doped with 1 ppm concentration of antimony. If the concentration of holes in the doped crystal is found to be 4.5×10^{9} m^{-3}, the concentration (in m^{-3}) of intrinsic charge carriers in Si crystal is about:

(A) 1.2×10^{15}

(B) 1.5×10^{16}

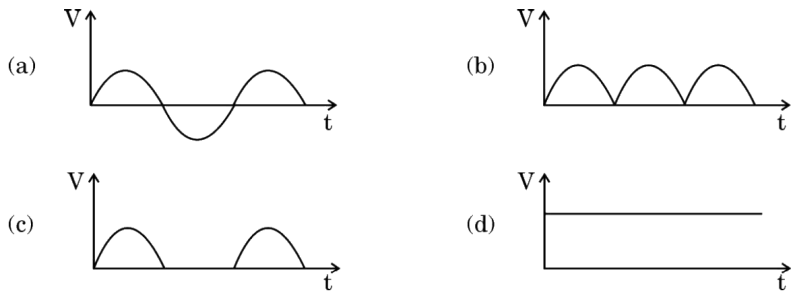
(C) 3.0×10^{15}

(D) 2.0×10^{16}

Check Solution → [2024 • Set 55-5-1]

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43. An ac source of voltage is connected in series with a p-n junction diode and a load resistor. The correct option for output voltage across load resistance will be:



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

Check Solution → [2023 • Set 55-1-1]

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44. When an intrinsic semiconductor is doped with a small amount of trivalent impurity, then:

(A) its resistance increases.

(B) it becomes a p-type semiconductor.

(C) there will be more free electrons than holes in the semiconductor.

(D) dopant atoms become donor atoms.

Check Solution → **[2023 • Set 55-1-1]**

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45. In the energy-band diagram of n-type Si, the gap between the bottom of the conduction band E_C and the donor energy level E_D is of the order of:

(A) 10 eV

(B) 1 eV

(C) 0.1 eV

(D) 0.01 eV

Check Solution → **[2023 • Set 55-1-1]**

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46. In an extrinsic semiconductor, the number density of holes is $4\times 10^{20}\ m^{-3}$. If the number density of intrinsic carriers is $1.2\times 10^{15}\ m^{-3}$, the number density of electrons in it is

(A) $1.8\times 10^{9}\ m^{-3}$

(B) $2.4\times 10^{10}\ m^{-3}$

(C) $3.6\times 10^{9}\ m^{-3}$

(D) $3.2\times 10^{10}\ m^{-3}$

Check Solution → **[2023 • Set 55-2-1]**

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47. Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T. The conductivity of

(A) both increases.

(B) both decreases.

(C) copper increases and silicon decreases.

(D) copper decreases and silicon increases.

Check Solution → [2023 • Set 55-2-1]

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48. The formation of depletion region in a p-n junction diode is due to

(A) movement of dopant atoms

(B) diffusion of both electrons and holes

(C) drift of electrons only

(D) drift of holes only

Check Solution → [2023 • Set 55-2-1]

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49. Which one of the following elements will require the highest energy to take out an electron from them?
Pb, Ge, C and Si

(A) Ge

(B) C

(C) Si

(D) Pb

Check Solution → [2023 • Set 55-2-2]

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50. A semiconductor device is connected in series with a battery, an ammeter and a resistor. A current flows in the circuit. If the polarity of the battery is reversed, the current in the circuit almost becomes zero. The device is a/an

(A) intrinsic semiconductor

(B) p-type semiconductor

(C) n-type semiconductor

(D) p-n junction diode

Check Solution → [2023 • Set 55-2-3]

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51. At a certain temperature in an intrinsic semiconductor, the electrons and holes concentration is $1.5 \times 10^{16}\ m^{-3}$. When it is doped with a trivalent dopant, hole concentration increases to $4.5 \times 10^{22}\ m^{-3}$. In the doped semiconductor, the concentration of electrons (n_e) will be:

(A) $3 \times 10^{9}\ m^{-3}$

(B) $5 \times 10^{7}\ m^{-3}$

(C) $5 \times 10^{9}\ m^{-3}$

(D) $6.75 \times 10^{38}\ m^{-3}$

Check Solution → [2023 • Set 55-3-1]

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52. If a p-n junction diode is reverse biased:

(A) the potential barrier is lowered.

(B) the potential barrier remains unaffected.

(C) the potential barrier is raised.

(D) the current is mainly due to majority carriers.

Check Solution → [2023 • Set 55-3-1]

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53. At 0 K, the resistivity of an intrinsic semiconductor is:

(A) same as that at $0°C$

(B) same as that at 300 K

(C) zero

(D) infinite

Check Solution → [2023 • Set 55-3-2]

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54. For the forward biasing of a p-n junction diode, which of the following statements is not correct?

(A) The potential barrier decreases.

(B) Minority carrier injection occurs.

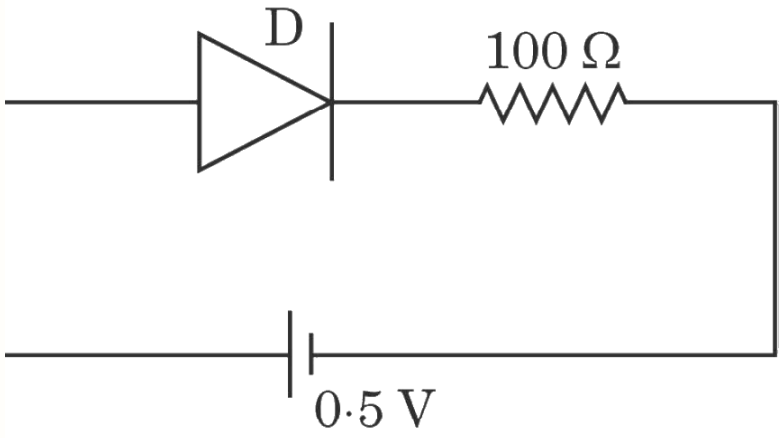
(C) Width of depletion layer increases.

(D) Forward current is due to the diffusion of both holes and electrons.

Check Solution → [2023 • Set 55-3-3]

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55. The threshold voltage for a p-n junction diode used in the circuit is 0.7 V. The type of biasing and current in the circuit are:



(A) Forward biasing, 0 A

(B) Reverse biasing, 0 A

(C) Forward biasing, 5 mA

(D) Reverse biasing, 2 mA

Check Solution → [2023 • Set 55-4-1]

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56. **Assertion (A):** In 'n' type semiconductor, number density of electrons is greater than the number density of holes but the crystal maintains an overall charge neutrality.

Reason (R): The charge of electrons donated by donor atoms is just equal and opposite to that of the ionised donor.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-4-1]

---

57. **Assertion (A):** The resistance of an intrinsic semiconductor decreases with increase in its temperature.
Reason (R): The number of conduction electrons as well as hole increase in an intrinsic semiconductor with rise in its temperature.

(A) Both Assertion (A) and Reason (R) are true and (R) is the correct explanation of (A).

(B) Both Assertion (A) and Reason (R) are true and (R) is NOT the correct explanation of (A).

(C) Assertion (A) is true and Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-5-1]

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58. During the formation of a p-n junction:

(A) diffusion current keeps increasing.

(B) drift current remains constant.

(C) both the diffusion current and drift current remain constant.

(D) diffusion current remains almost constant but drift current increases till both currents become equal.

Check Solution → [2023 • Set 55-5-1]

---

59. **Assertion (A):** In insulators, the forbidden gap is very large.
Reason (R): The valence electrons in an atom of an insulator are very tightly bound to the nucleus.

(A) Both Assertion (A) and Reason (R) are true and (R) is the correct explanation of (A).

(B) Both Assertion (A) and Reason (R) are true and (R) is NOT the correct explanation of (A).

(C) Assertion (A) is true and Reason (R) is false.

(D) Assertion (A) is false and Reason (R) is also false.

Check Solution → [2023 • Set 55-5-2]

---

60. The energy required by an electron to jump the forbidden band in silicon at room temperature is about

(A) 0.01 eV

(B) 0.05 eV

(C) 0.7 eV

(D) 1.1 eV

Check Solution → [2023 • Set 55-5-3]

---

61. In a p-n junction under equilibrium, there is no net current. Why?

Check Solution → [2021]

---

62. On what factor does the wavelength of the light emitted by an LED depend?

Check Solution → [2021]

---

63. How does the energy gap of an intrinsic semiconductor change when doped with a trivalent impurity?

Check Solution → [2021]

---

64. How does an increase in doping concentration affect the width of depletion layer of a p-n junction diode?

Check Solution → [2020 • Set 55-1-1]

---

65. Why cannot we use Si and Ge in fabrication of visible LEDs?

Check Solution → [2020 • Set 55-1-1]

---

66. At equilibrium, in a p-n junction diode the net current is

(A) due to diffusion of majority charge carriers.

(B) due to drift of minority charge carriers.

(C) zero as diffusion and drift currents are equal and opposite.

(D) zero as no charge carriers cross the junction.

Check Solution → [2020 • Set 55-2-1]

---

67. In an n-type semiconductor, the donor energy level lies

(A) at the centre of the energy gap.

(B) just below the conduction band.

(C) just above the valance band.

(D) in the conduction band.

Check Solution → [2020 • Set 55-2-1]

---

68. Can a slab of p-type semi-conductor be physically joined to another n-type semiconductor slab to form p-n junction? Justify your answer.

OR

In a p-n junction diode the forward bias resistance is low as compared to the reverse bias resistance. Give reason.

Check Solution → [2020 • Set 55-4-1]

---

69. The _____ , a property of materials C, Si and Ge depends upon the energy gap between their conduction and valence bands.

Check Solution → [2020 • Set 55-5-1]

---

70. The ability of a junction diode to _____ an alternating voltage, is based on the fact that it allows current to pass only when it is forward biased.

Check Solution → [2020 • Set 55-5-1]

---

71. Photo diodes are used to detect

(A) radio waves

(B) gamma rays

(C) IR rays

(D) optical signals

Check Solution → [2020 • Set 55-5-1]

---

72. The wavelength and intensity of light emitted by a LED depend upon

(A) forward bias and energy gap of the semiconductor

(B) energy gap of the semiconductor and reverse bias

(C) energy gap only

(D) forward bias only

Check Solution → [2020 • Set 55-5-1]

---

73. In sky wave mode of propagation, why is the frequency range of transmitting signals restricted to less than 30 MHz?

Check Solution → [2019 • Set 55-1-1]

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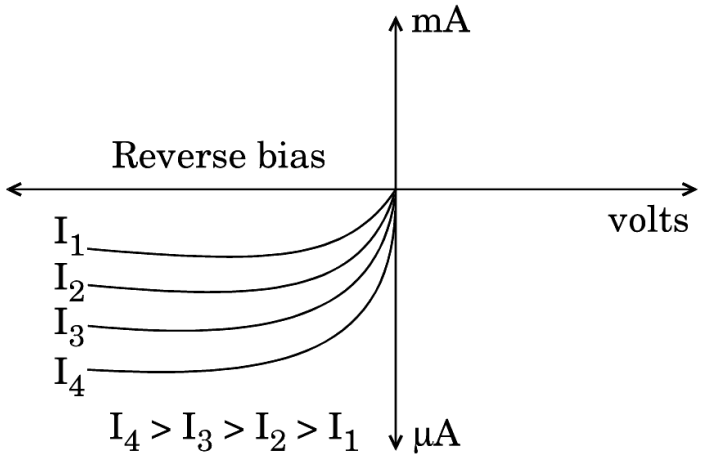
74. On what factors does the range of coverage in ground wave propagation depend?

Check Solution →

[2019 • Set 55-1-1]

---

75. Identify the semiconductor diode whose V-I characteristics are as shown.



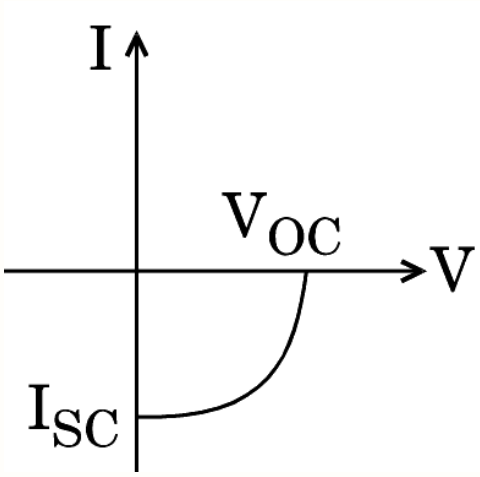
I3>I2>I1, current in microamperes on y-axis (negative side) and voltage in volts on x-axis'>

Check Solution →

[2019 • Set 55-2-1]

---

76. Identify the semiconductor diode whose I-V characteristics are as shown.



Check Solution →

[2019 • Set 55-2-2]

---

77. Draw the I-V characteristics of a Zener diode.

Check Solution → [2019 • Set 55-2-3]

---

78. Define amplitude modulation in communication system.

Check Solution → [2019 • Set 55-3-1]

---

79. Draw a plot of resistivity versus temperature for a typical semiconductor.

Check Solution → [2019 • Set 55-3-2]

---

80. Define the term 'attenuation' in communication system.

Check Solution → [2019 • Set 55-3-2]

---

81. Write the function of receiver in communication system.

Check Solution → [2019 • Set 55-3-3]

---

82. Why is ground wave transmission of signals restricted to a frequency of 1500 kHz?

Check Solution → [2019 • Set 55-4-1]

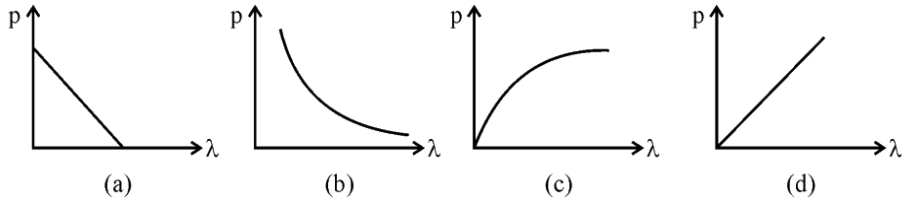
---

83. Why are the antennas in space wave mode of propagation generally mounted at heights of many wavelengths above the ground?

Check Solution → [2019 • Set 55-4-3]

---

84. Draw the output signal in a p-n junction diode when a square input signal of 10 V as shown in the figure is applied across it.

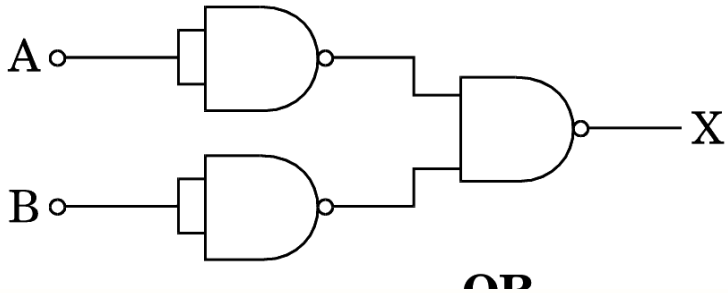


Check Solution →

[2019 • Set 55-5-1]

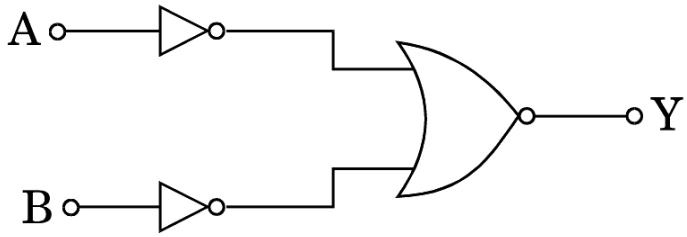
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85. Identify the equivalent gate for the circuit of a combination of gates shown in the figure. Write its symbol.



OR

Draw the logic symbol of the gate shown by the combination of gates and write its name.



Check Solution →

[2019 • Set 55-5-1]

---

86. Can a transistor be used as a rectifier?

Check Solution →

[2019 • Set 55-5-2]

---

87. When a p-n junction diode is forward biased, how will its barrier potential be affected?

Check Solution → [2019 • Set 55-5-3]

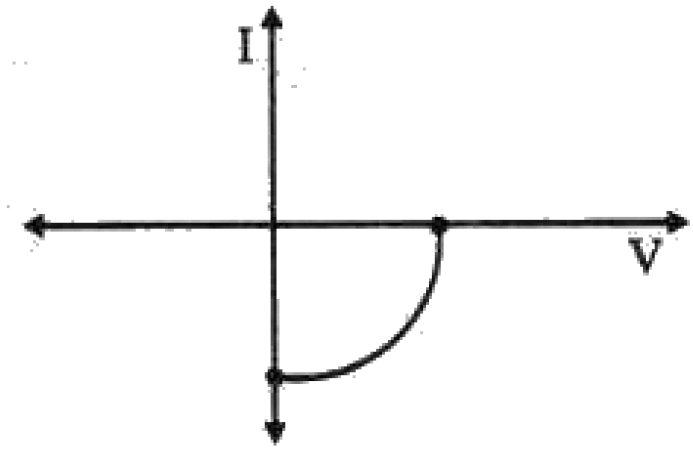
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88. Which mode of propagation is used by short wave broadcast services ?

Check Solution → [2018]

---

89. Name the junction diode whose I-V characteristics are drawn below:



Check Solution → [2017]

---

90. Draw logic symbol of an OR gate and write its truth table.

Check Solution → [2017]

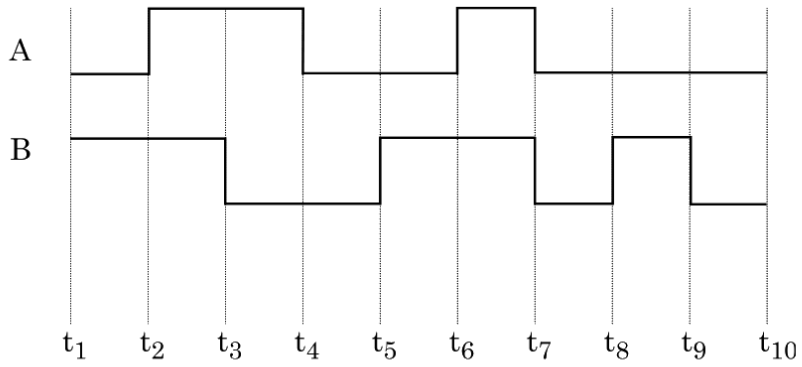
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91. Name the essential components of a communication system.

Check Solution → [2016]

---

92. Two signals A and B are used as inputs of a NOR gate. Draw the output wave form.



Check Solution → [2016]

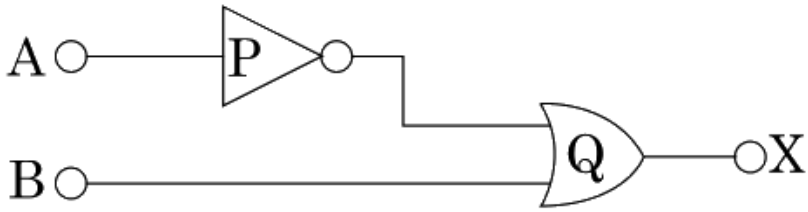
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93. A signal of 5 kHz frequency is amplitude modulated on a carrier wave of frequency 2 MHz. What are the frequencies of the side bands produced?

Check Solution → [2016]

---

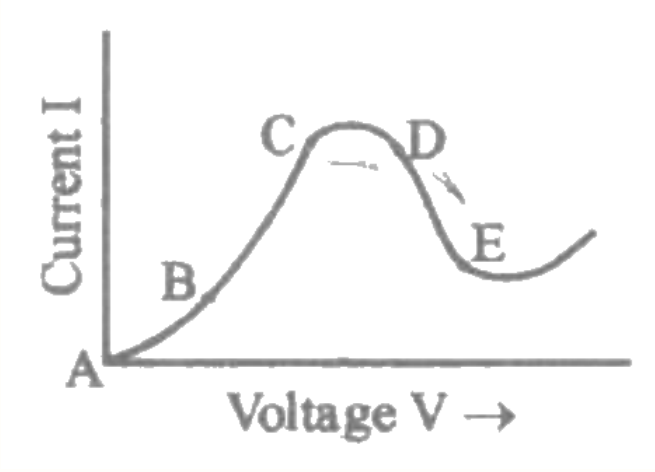
94. Name the logic gates marked P and Q in the given logic circuit.



Check Solution → [2016]

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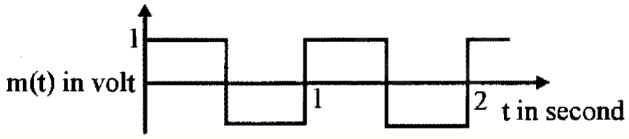
95. Graph showing the variation of current versus voltage for a material GaAs is shown in the figure. Identify the region of (i) negative resistance (ii) where Ohm's law is obeyed.



Check Solution → [2015]

---

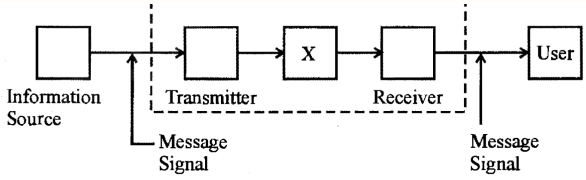
96. The carrier wave is given by $C(t) = 2\sin(8\pi t)$ volt.
The modulating signal is a square wave as shown. Find modulation index.



Check Solution → [2014]

---

97. The figure given below shows the block diagram of a generalized communication system. Identify the element labelled 'X' and write its function.



Check Solution → [2014]

---

98. Plot a graph showing variation of current versus voltage for the material GaAs.

Check Solution → [2014]

---

99. Name an appropriate communication channel needed to send a signal of band-width 100 kHz over a distance of 8 km.

Check Solution → [2013]

---

100. Show on a graph, the variation of resistivity with temperature for a typical semiconductor.

Check Solution → [2012]

---

101. What happens to the width of depletion layer of a p-n junction when it is (i) forward biased, (ii) reverse biased?

Check Solution → [2011]

---

102. In a transistor, doping level in base is increased slightly. How will it affect (i) collector current and (ii) base current?

Check Solution → [2011 • Set 55-1-1]

---

103. What is the function of a 'Repeater' in a communication system?

Check Solution → [2011 • Set 55-2-1]

---

104. Draw the logic circuit of a NAND gate and write its truth table.

Check Solution → [2011 • Set 55-2-1]

---

105. Give the logic symbol of NOR gate.

Check Solution → [2009]

---

106. State the reason, why a photodiode is usually operated at reverse bias?

Check Solution → [2008]

---

107. What will be the values of input A and B for the Boolean expression $(A + B) \cdot (A \cdot B) = 1$?

Check Solution → [2007]

---

108. Why is frequency modulation preferred over amplitude modulation for transmission of music?

Check Solution → [2007]

---

109. Name the type of communication in which the signal is a discrete and binary coded version of the message or information.

Check Solution → [2006]

---

110. What should be the length of the dipole antenna for a carrier wave of frequency 3×10^8 Hz?

Check Solution → [2005]

---

111. Name the type of modulation scheme preferred for digital communication.

Check Solution → [2004]

---

112. Name the factor which decides the quality of reproduced document sent by Fax.

Check Solution → [2003]

SECTION B

2-Mark Questions

VSA · 109 questions

> *Section B of the paper, 2 marks each. Sorted newest year first.*

113. (b) How does the resistivity of a semiconductor depend on temperature? Show it on a plot.

Check Solution → **[2026 • Set 55-2-3]**

---

114. How are charge carriers created in an intrinsic semiconductor? Explain.

Check Solution → **[2026 • Set 55-3-1]**

---

115. Explain, with the help of suitable diagram, the two important processes that occur during the formation of a p-n junction.

Check Solution → **[2026 • Set 55-3-2]**

---

116. The hole concentration in an intrinsic semiconductor is 5×10^{16} m^{-3}. When it is doped with certain impurity, the electron concentration becomes 4×10^{12} m^{-3}. Find the new value of the hole concentration. Also identify the type of new semiconductor formed after doping.

Check Solution → **[2026 • Set 55-4-1]**

---

117. Suppose a pure Si crystal has 5×10^{28} atoms per m^3. It is doped with 5×10^{22} atoms per m^3 of Arsenic. Calculate the majority and minority carrier concentration in the doped silicon. (Given: $n_i = 1.5 \times 10^{16}$ m^{-3})

Check Solution → **[2026 • Set 55-5-1]**

---

118. A *p*-type Si semiconductor is made by doping an average of one dopant atom per 5×10^7 silicon atoms. If the number density of silicon atoms in the specimen is 5

$\times 10^{28}$ atoms/m^3, find the number of holes created per cubic centimetre in the specimen due to doping. Also give one example of such dopants.

Check Solution → [2025 • Set 55-1-1]

---

119. The threshold voltage of a silicon diode is 0.7 V. It is operated at this point by connecting the diode in series with a battery of V volt and a resistor of 1000 Ω. Find the value of V when the current drawn is 15 mA.

Check Solution → [2025 • Set 55-2-1]

---

120. In an intrinsic semiconductor, carrier's concentration is 5×10^{16} m^{-3}. On doping with impurity atoms, the hole concentration becomes 8×10^{12} m^{-3}.
(a) Identify (i) the type of dopant and (ii) the extrinsic semiconductor so formed.
(b) Calculate the electron concentration in the extrinsic semiconductor.

Check Solution → [2025 • Set 55-4-1]

---

121. In an *n*-type semiconductor electron-hole combination is a continuous process at room temperature. Yet the electron concentration is always greater than the hole concentration in it. Explain.

Check Solution → [2025 • Set 55-6-1]

---

122. Draw energy band diagrams of n-type and p-type semiconductors at temperature $T > 0\ K$. Show the donor/acceptor energy levels with the order of difference of their energies from the bands.

Check Solution → [2025 • Set 55-7-1]

---

123. Plot a graph showing the variation of current with voltage for the material GaAs. On the graph, mark the region where:
(a) resistance is negative, and
(b) Ohm's law is obeyed.

Check Solution → [2024 • Set 55-1-1]

---

124. How does the conductivity of an intrinsic semiconductor vary with temperature? Explain. Show the variation in a plot.

Check Solution → [2024 • Set 55-1-2]

---

125. Differentiate between 'diffusion current' and 'drift current'. Explain their role in the formation of p-n junction.

Check Solution → [2024 • Set 55-1-3]

---

126. How does the energy gap of an intrinsic semiconductor effectively change when doped with a (a) trivalent impurity, and (b) pentavalent impurity? Justify your answer in each case.

Check Solution → [2024 • Set 55-2-1]

---

127. Suppose a pure Si crystal has 5×10^{28} atoms m^{-3}. It is doped by 1 ppm concentration of boron. Calculate the concentration of holes and electrons, given that $n_i = 1.5 \times 10^{16}$ m^{-3}. Is the doped crystal *n*-type or *p*-type?

Check Solution → [2024 • Set 55-3-1]

---

128. Draw the circuit diagram of a *p-n* junction diode in (i) forward biasing and (ii) reverse biasing. Also draw its *I-V* characteristics in the two cases.

Check Solution → [2024 • Set 55-4-1]

---

129. Differentiate between intrinsic and extrinsic semiconductors.

Check Solution → [2023 • Set 55-1-1]

130. Draw the circuit arrangement for studying the *V-I* characteristics of a p-n junction diode in forward bias and reverse bias. Show the plot of *V-I* characteristic of a silicon diode.

Check Solution → [2023 • Set 55-1-1]

131. Briefly explain how the diffusion and drift currents contribute to the formation of potential barrier in a p-n junction diode.

Check Solution → [2023 • Set 55-1-1]

132. Draw energy band diagram for an n-type and p-type semiconductor at $T > 0\ K$.

Check Solution → [2023 • Set 55-2-1]

133. Answer the following giving reasons:
(i) A p-n junction diode is damaged by a strong current.
(ii) Impurities are added in intrinsic semiconductors.

Check Solution → [2023 • Set 55-2-1]

134. With the help of a circuit diagram, explain how a full wave rectifier gives output rectified voltage corresponding to both halves of the input ac voltage.

Check Solution → [2023 • Set 55-2-3]

135. Explain the roles of diffusion current and drift current in the formation of the depletion layer in a p-n junction diode.

Check Solution → [2023 • Set 55-3-1]

136. Explain the property of a p-n junction which makes it suitable for rectifying alternating voltages. Differentiate between a half-wave and a full-wave rectifier.

Check Solution → [2023 • Set 55-3-1]

---

137. How is the width of depletion layer of a p-n junction diode affected when it is (i) forward biased, and (ii) reverse biased? Justify your answers.

Check Solution → [2023 • Set 55-3-2]

---

138. Draw a circuit diagram of a p-n junction diode in (a) forward biasing, and (b) reverse biasing. Draw the V-I characteristics for each case.

Check Solution → [2023 • Set 55-3-3]

---

139. Draw energy band diagrams of n-type and p-type semiconductors at temperature $T > 0\ K$, depicting the donor and acceptor energy levels. Mention the significance of these levels.

Check Solution → [2022 • Set 55-1-1]

---

140. Write the characteristics of a p-n junction which make it suitable for rectification.

Check Solution → [2022 • Set 55-1-1]

---

141. Explain the formation of depletion layer and barrier potential in a p-n junction diode.

Check Solution → [2022 • Set 55-1-2]

---

142. Distinguish between intrinsic and extrinsic semiconductors. Although in an extrinsic semiconductor $n_e \neq n_h$, yet it is electrically neutral. Why?

Check Solution → [2022 • Set 55-1-3]

---

143. Explain the formation of depletion region in a *p-n* junction.

Check Solution → [2022 • Set 55-2-1]

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144. How is an emf generated by a solar cell due to the three basic processes involved? Explain.

Check Solution → [2022 • Set 55-2-1]

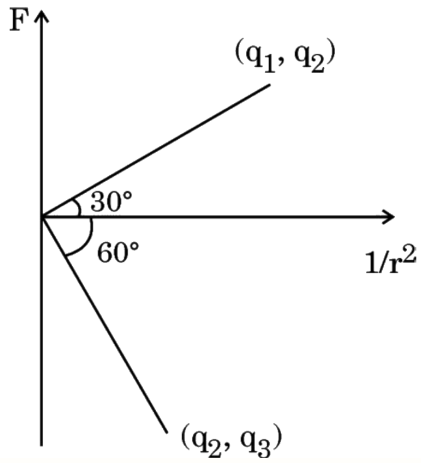
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145. Briefly explain the working of a light emitting diode. Mention its two uses.

Check Solution → [2022 • Set 55-2-2]

---

146. The *V* - *I* characteristics of a solar cell is shown in the figure.



(a) What do the points *P* and *Q* represent?
(b) Give the circuit diagram for a solar cell.

Check Solution → [2022 • Set 55-2-3]

---

147. What is meant by doping of an intrinsic semiconductor? Name the two types of atoms used for doping of Ge/Si.

Check Solution → [2022 • Set 55-3-1]

---

148. (a) Draw the circuit diagram of an illuminated photodiode and its I-V characteristics.
(b) How can a photodiode be used to measure the light intensity?

Check Solution → [2022 • Set 55-3-2]

---

149. Name the extrinsic semiconductors formed when a pure germanium is doped with (i) a trivalent and (ii) pentavalent impurity. Draw the energy band diagrams of extrinsic semiconductors so formed.

Check Solution → [2022 • Set 55-3-3]

---

150. What is meant by energy band gap in a solid? Draw the energy band diagrams for a conductor, an insulator and a semiconductor.

Check Solution → [2022 • Set 55-4-1]

---

151. Name the device which converts electrical energy into light energy. Write three advantages of the device.

Check Solution → [2022 • Set 55-4-1]

---

152. Name the device which converts an ac input signal into a dc output signal. Write the principle of working of the device.

Check Solution → [2022 • Set 55-4-2]

---

153. Define barrier potential. Why does the thickness of the depletion layer in a p-n junction diode vary with increase in reverse bias?

Check Solution → [2022 • Set 55-4-3]

154. With the help of a circuit diagram, explain briefly how a p-n junction diode works as a half-wave rectifier.

Check Solution → [2022 • Set 55-5-1]

155. Why a photo-diode is operated in reverse bias whereas current in the forward bias is much larger than that in the reverse bias? Explain. Mention its two uses.

Check Solution → [2022 • Set 55-5-1]

156. Two crystals C_1 and C_2, made of pure silicon, are doped with arsenic and aluminium respectively.
(i) Identify the extrinsic semiconductors so formed.
(ii) Why is doping of intrinsic semiconductors necessary?

Check Solution → [2022 • Set 55-5-2]

157. Give two differences between a half wave rectifier and a full wave rectifier.

Check Solution → [2022 • Set 55-5-3]

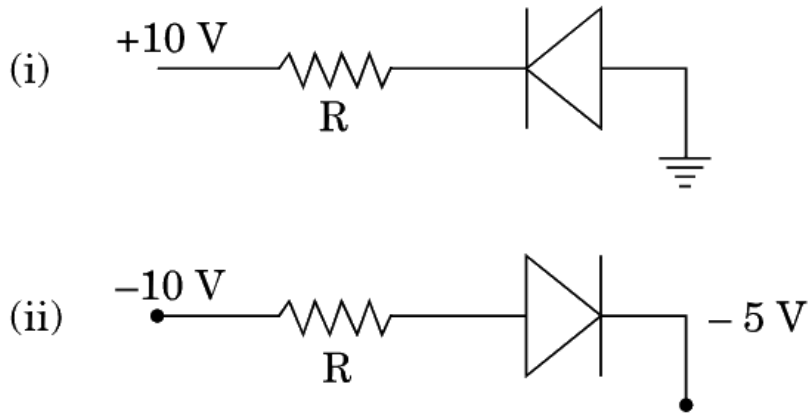
158. With the help of the circuit diagram, explain the working of a diode as a half-wave rectifier.

Check Solution → [2021]

159. With the help of the circuit diagram, explain briefly the working of a photodiode.

Check Solution → [2021]

160. (a) State whether the given ideal diodes are forward or reverse biased:



(b) The current in the forward bias is known to be more () than the current in the reverse bias (μA). What is the reason then to operate the photodiodes in reverse bias?

Check Solution → [2021]

---

161. Explain the terms 'depletion layer' and 'potential barrier' in a p-n junction diode. How are the (a) width of depletion layer, and (b) value of potential barrier affected when the p-n junction is forward biased?

Check Solution → [2020 • Set 55-1-1]

---

162. Draw V-I characteristics of a p-n junction diode. Explain, why the current under reverse bias is almost independent of the applied voltage up to the critical voltage.

Check Solution → [2020 • Set 55-2-1]

---

163. Briefly explain how a potential barrier is set up across a p-n junction as a result of diffusion and drift of the charge carriers.

Check Solution → [2020 • Set 55-4-1]

---

164. (a) Why is a photo diode operated under reverse bias condition?
(b) Draw *V-I* characteristic curves of photo diode for incident light of intensities I_1 and I_2 ($I_2 > I_1$).

OR

(a) State the level of doping and biasing condition used in light emitting diode (LED).

(b) Write any two advantages of LED over the conventional low power lamps.

Check Solution → [2020 • Set 55-4-1]

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165. (a) Explain the formation of energy bands in crystalline solids.

(b) Draw the energy band diagrams of (i) a metal and (ii) a semiconductor.

Check Solution → [2020 • Set 55-4-1]

---

166. Why a signal transmitted from a TV tower cannot be received beyond a certain distance? Write the expression for the optimum separation between the receiving and the transmitting antenna.

Check Solution → [2019 • Set 55-1-1]

---

167. Write the relation between the height of a TV antenna and the maximum range up to which signals transmitted by the antenna can be received. How is this expression modified in the case of line of sight communication by space waves? In which range of frequencies, is this mode of communication used?

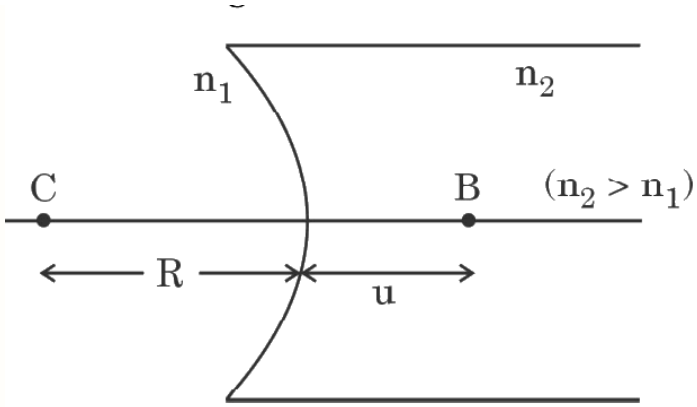
Check Solution → [2019 • Set 55-2-1]

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168. A certain n-p-n transistor has the common emitter output characteristics as shown in the figure.

(a) Find the emitter current at $V_{CE} = 12.5\ V$ and $I_B = 60\ \mu A$, and

(b) Current gain 'β' at this point.



Check Solution →

[2019 • Set 55-4-1]

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169. The frequencies of two side bands in an amplitude modulated wave are 640 kHz and 660 kHz respectively. Find the frequencies of the carrier and the modulating signals. Also obtain the value of the bandwidth required in amplitude modulation.

OR

A sinusoidal carrier voltage is amplitude modulated by a sinusoidal voltage of 10 kHz with modulation index 0.3. If the carrier frequency is 10 MHz and its amplitude is 40 V, calculate the frequency and amplitude of the two sidebands.

Check Solution →

[2019 • Set 55-5-1]

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170. Draw a block diagram of a generalized communication system and write the functions of (i) a transmitter, and (ii) a receiver.

Check Solution →

[2019 • Set 55-5-1]

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171. A carrier wave of peak voltage 15 *V* is used to transmit a message signal. Find the peak voltage of the modulating signal in order to have a modulation index of 60%.

Check Solution →

[2018]

---

172. Which basic mode of communication is used in satellite communication? What type of wave propagation is used in this mode? Write, giving reason, the frequency range used in this mode of propagation.

Check Solution → [2017]

---

173. Distinguish between a transducer and a repeater.

Check Solution → [2017]

---

174. (i) What is the line of sight communication?
(ii) Why is it not possible to use sky waves for transmission of TV signals? Upto what distance can a signal be transmitted using an antenna of height 'h'?

Check Solution → [2017]

---

175. Write two points of difference between intrinsic and extrinsic semiconductors.

Check Solution → [2017]

---

176. Distinguish between broadcast mode and point-to-point mode of communication and give one example for each.

Check Solution → [2017]

---

177. Describe, with the help of a circuit diagram, the working of a photodiode.

Check Solution → [2017]

---

178. Define modulation index. Why is it kept low? What is the role of a bandpass filter?

Check Solution → [2016]

---

179. State the concept of mobile telephony and explain its working.

Check Solution → [2016]

---

180. Why is base band signal not transmitted directly? Give any two reasons.

Check Solution → [2016]

---

181. What is ground wave communication? Explain why this mode cannot be used for long distance communication using high frequencies.

Check Solution → [2015]

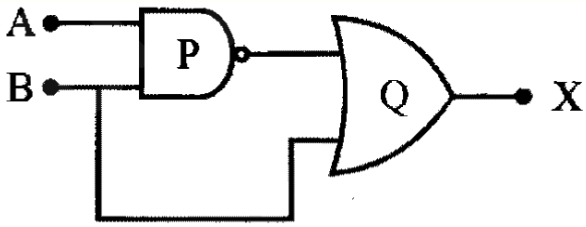
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182. Distinguish between 'intrinsic' and 'extrinsic' semiconductors.

Check Solution → [2015]

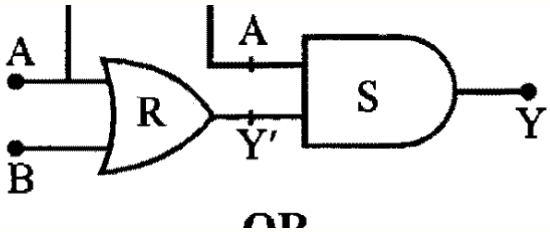
---

183. Write the truth table for the combination of the gates shown. Name the gates used.



OR

Identify the logic gates marked 'P' and 'Q' in the given circuit. Write the truth table for the combination.



Check Solution → [2014]

---

184. Explain, with the help of a circuit diagram, the working of a p-n junction diode as a half-wave rectifier.

Check Solution → [2014]

---

185. Draw a circuit diagram of n-p-n transistor amplifier in CE configuration. Under what condition does the transistor act as an amplifier ?

Check Solution → [2014]

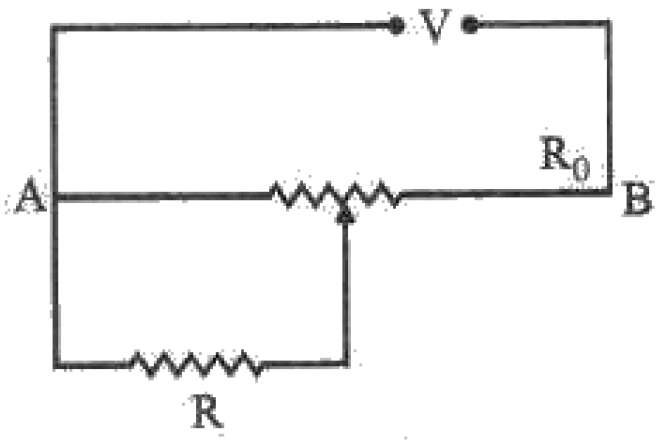
---

186. Write the functions of the following in communication systems :
(i) Receiver
(ii) Demodulator

Check Solution → [2014]

---

187. Block diagram of a receiver is shown in the figure:



(a) Identify '*X*' and '*Y*'.
(b) Write their functions.

Check Solution → [2013]

---

188. Explain, with the help of a circuit diagram, the working of a photo-diode. Write briefly how it is used to detect the optical signals.

OR

Mention the important considerations required while fabricating a p-n junction diode to be used as a Light Emitting Diode (LED). What should be the order of band gap of an LED if it is required to emit light in the visible range?

Check Solution → [2013]

---

189. Write two factors justifying the need of modulating a signal.

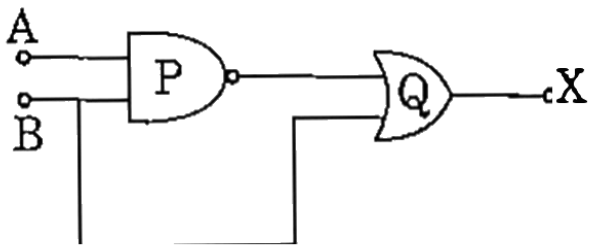
A carrier wave of peak voltage 12 *V* is used to transmit a message signal. What should be the peak voltage of the modulating signal in order to have a modulation index of 75%?

Check Solution → [2012]

---

190. (i) Identify the logic gates marked P and Q in the given logic circuit.

(ii) Write down the output at X for the inputs A=0, B=0 and A=1, B=1.



Check Solution → [2012]

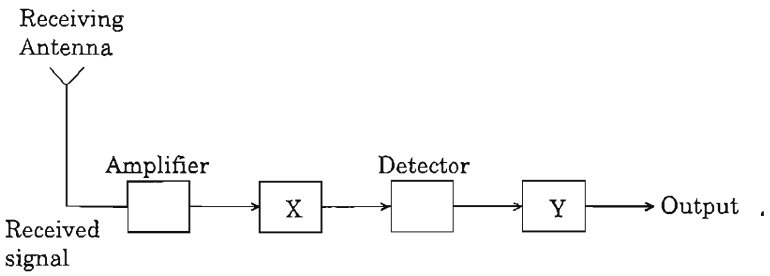
---

191. Describe briefly with the help of a circuit diagram, how the flow of current carriers in a p-n-p transistor is regulated with emitter-base junction forward biased and base-collector junction reverse biased.

Check Solution → [2012]

---

192. In the given block diagram of a receiver, identify the boxes labelled as X and Y and write their functions.



X -> Amplifier -> Detector -> Y -> Output'>

Check Solution → [2012]

---

193. Distinguish between 'Analog and Digital signals'.

OR

Mention the function of any two of the following used in communication system:

(i) Transducer

(ii) Repeater

(iii) Transmitter

(iv) Bandpass Filter

Check Solution → [2012]

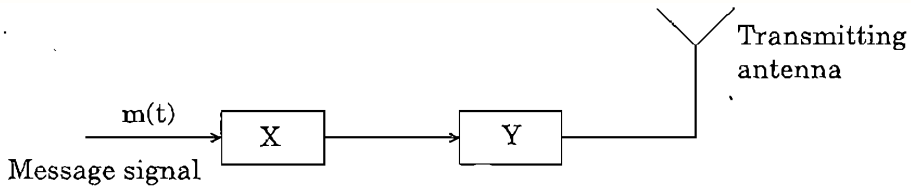
---

194. The current in the forward bias is known to be more () than the current in the reverse bias (μA). What is the reason, then, to operate the photodiode in reverse bias?

Check Solution → [2012]

---

195. Figure shows a block diagram of a transmitter. Identify the boxes 'X' and 'Y' and write their functions.



X -> Y -> Transmitting antenna'>

Check Solution → [2012]

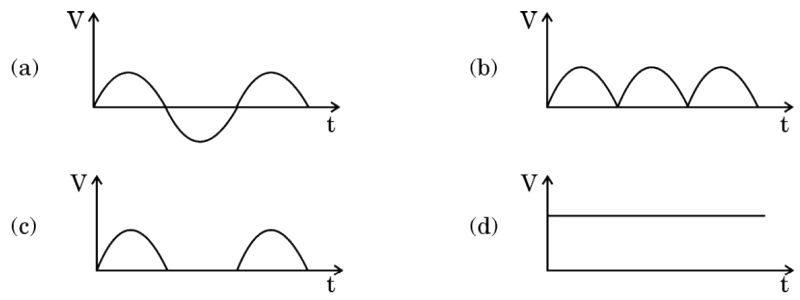
196. Write two characteristic features to distinguish between n-type and p-type semiconductors.

OR

How does a light emitting diode (LED) work? Give two advantages of LED's over the conventional incandescent lamps.

Check Solution → [2012]

197. Draw the output waveform at *X*, using the given inputs *A* and *B* for the logic circuit shown below. Also, identify the logic operation performed by this circuit.



Check Solution → [2011 • Set 55-1-1]

198. Name the semiconductor device that can be used to regulate an unregulated dc power supply. With the help of *I-V* characteristics of this device, explain its working principle.

Check Solution → [2011 • Set 55-1-1]

199. Draw the transfer characteristic curve of a base biased transistor in CE configuration. Explain clearly how the active region of the V_o versus V_i curve in a transistor is used as an amplifier.

Check Solution → [2011 • Set 55-1-1]

200. (i) Define modulation index.
(ii) Why is the amplitude of modulating signal kept less than the amplitude of carrier wave?

Check Solution → [2011 • Set 55-1-1]

---

201. Plot a graph showing temperature dependence of resistivity for a typical semiconductor. How is this behaviour explained?

Check Solution → [2011 • Set 55-2-1]

---

202. For an amplitude modulated wave, the maximum amplitude is found to be 10 V while the minimum amplitude is 2 V. Calculate the modulation index. Why is modulation index generally kept less than one?

Check Solution → [2011 • Set 55-2-1]

---

203. Draw a block diagram showing the important components in a communication system. What is the function of a transducer?

Check Solution → [2011 • Set 55-2-1]

---

204. Draw the circuit diagram of an illuminated photodiode in reverse bias. How is photodiode used to measure light intensity?

Check Solution → [2010]

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205. Explain the function of a repeater in a communication system.

Check Solution → [2010]

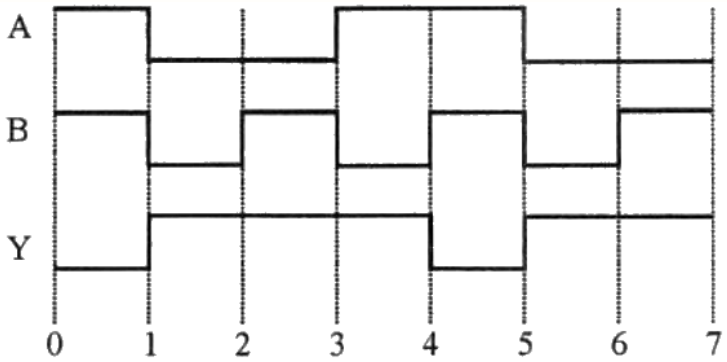
---

206. By what percentage will the transmission range of a TV tower be affected when the height of the tower is increased by 21%?

Check Solution → [2009]

---

207. The following figure shows the input waveforms (A, B) and the output waveform (Y) of a gate. Identify the gate, write its truth table and draw its logic symbol.



Check Solution → [2009]

---

208. Why are high frequency carrier waves used for transmission?

Check Solution → [2009]

---

209. What is meant by term 'modulation'? Draw a block diagram of a simple modulator for obtaining an AM signal.

Check Solution → [2009]

---

210. Write the function of (i) Transducer and (ii) Repeater in the context of communication system.

Check Solution → [2009]

---

211. Write two factors justifying the need of modulation for transmission of a signal.

Check Solution → [2009]

---

212. Distinguish between an intrinsic semiconductor and P-type semiconductor. Give reason, why a P-type semiconductor crystal is electrically neutral, although $n_h \gg n_e$.

Check Solution → [2008]

---

213. The given inputs A, B are fed to a 2-input NOR gate. Draw the output wave form of the gate.

Check Solution → [2008]

---

214. A transmitting antenna at the top of a tower has a height of 36 m and the height of the receiving antenna is 49 m. What is the maximum distance between them, for satisfactory communication in the LOS mode? (Radius of earth = 6400 km)

Check Solution → [2008]

---

215. Why do we need carrier waves of very high frequency in the modulation of signals? A carrier wave of peak voltage 20 V is used to transmit a message signal. What should be the peak voltage of the modulating signal, in order to have a modulation index of 80%?

Check Solution → [2008]

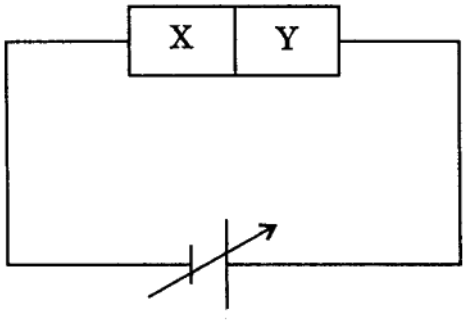
---

216. The output of an OR gate is connected to both the inputs of a NAND gate. Draw the logic circuit of this combination of gates and write its truth table.

Check Solution → [2007]

---

217. Two semiconductor materials X and Y shown in the given figure, are made by doping germanium crystal with indium and arsenic respectively. The two are joined end to end and connected to a battery as shown.

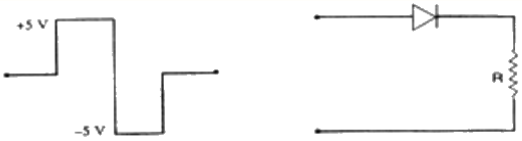


(i) Will the junction be forward biased or reverse biased?
(ii) Sketch a V-I graph for this arrangement.

Check Solution → [2007]

---

218. Draw and explain the output waveform across the load resistor *R*, if the input waveform is as shown in the given figure.



Check Solution → [2006]

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219. Explain how the width of depletion layer in a p-n junction diode changes when the junction is (i) forward biased (ii) reverse biased.

Check Solution → [2006]

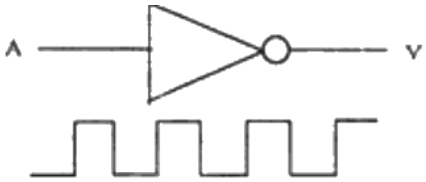
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220. With the help of a diagram, show the biasing of a light emitting diode (LED). Give its two advantages over conventional incandescent lamps.

Check Solution → [2004]

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221. In the figure below, circuit symbol of a logic gate and input wave form is shown.
(i) Name the logic gate,
(ii) write its truth table and
(iii) give the output wave form.



Check Solution → [2003]

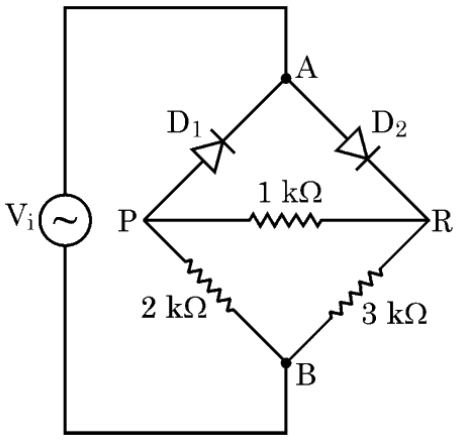
SECTION C

3-Mark Questions

SA · 137 questions

> *Section C of the paper, 3 marks each. Sorted newest year first.*

222. An ac voltage $V_i = 12 \sin (100\pi\, t)$ V is applied between points A and B in a network of two ideal diodes and three resistors as shown in figure.



During the positive half-cycle of the input voltage V_i supplied to the network:
(a) Identify which of the two diodes will conduct and why?
(b) Redraw an equivalent circuit diagram to show the flow of current.
(c) Calculate the output voltage drops V_o across the three resistors when the input voltage attains its peak value.

Check Solution → **[2026 • Set 55-1-1]**

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223. Briefly explain the two important processes that occur during the formation of a p-n junction.

Check Solution → **[2026 • Set 55-1-1]**

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224. A semiconductor has equal electron and hole concentration of 3×10^6 m^{-3}. On doping with a certain impurity, the hole concentration increases to 6×10^{10} m^{-3}.
(a) What type of semiconductor is obtained on doping?

(b) Calculate the new electron concentration of the semiconductor.
(c) How does the energy gap of semiconductor change with doping? Draw the energy band diagram for it.

Check Solution → [2026 • Set 55-1-2]

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225. (a) Draw the V-I characteristics of silicon diode.
(b) Explain the following terms:
(i) minority carrier injection in forward bias
(ii) breakdown voltage in reverse bias

Check Solution → [2026 • Set 55-1-3]

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226. With the help of circuit diagrams, briefly explain the forward biasing and the reverse biasing of a p-n junction diode.

Check Solution → [2026 • Set 55-3-1]

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227. With the help of a circuit diagram, explain the working of a full wave rectifier. Depict the input and output waveforms.

Check Solution → [2026 • Set 55-4-1]

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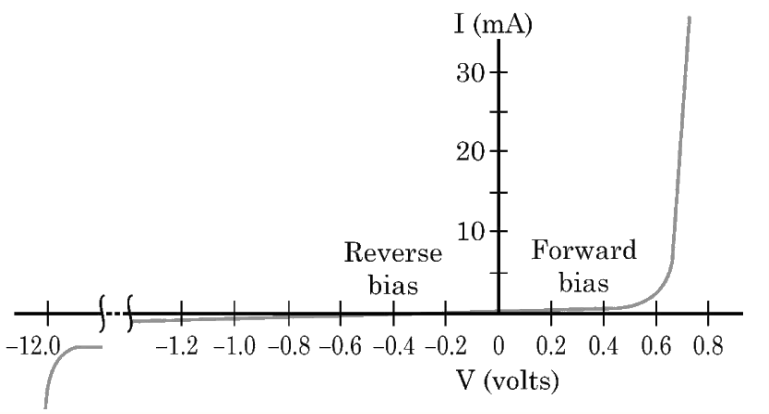
228. **(a)** Draw circuit arrangement for studying *V-I* characteristics of a *p-n* junction diode.
(b) Show the shape of the characteristics of a diode.
(c) Mention two information that you can get from these characteristics.

Check Solution → [2025 • Set 55-1-1]

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229. (a) What are majority and minority charge carriers in an extrinsic semiconductor?
(b) A p-n junction is forward biased. Describe the movement of the charge carriers which produce current in it.

(c) The graph shows the variation of current with voltage for a p-n junction diode. Estimate the dynamic resistance of diode at V = −0.6 volt.



Check Solution → [2025 • Set 55-2-1]

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230. **(a)** What are majority and minority charge carriers of p-type and n-type semiconductors ?

(b) Explain briefly the formation of diffusion current and drift current in a p-n junction diode.

Check Solution → [2025 • Set 55-4-1]

---

231. **(a)** Draw the energy-band diagrams for conductors, semiconductors and insulators at $T = 0\ K$. How is an electron-hole pair formed in a semiconductor at room temperature?

(b) Carbon and silicon both, are members of IV group of periodic table and have the same lattice structure. Carbon is an insulator whereas silicon is a semiconductor. Explain.

Check Solution → [2025 • Set 55-5-1]

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232. Differentiate between half-wave and full-wave rectification. With the help of a circuit diagram, explain the working of a full-wave rectifier.

Check Solution → [2025 • Set 55-5-1]

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233. Explain the process of formation of 'depletion layer' and 'potential barrier' in a p-n junction region of a diode, with the help of a suitable diagram. Which feature of junction diode makes it suitable for its use as a rectifier?

Check Solution → [2025 • Set 55-7-2]

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234. Draw the circuit diagrams for obtaining the $V - I$ characteristics of a p-n junction diode. Explain briefly the salient features of the $V - I$ characteristics in (i) forward biasing, and (ii) reverse biasing.

Check Solution → [2024 • Set 55-3-1]

235.

On the basis of energy band diagrams, distinguish between (i) an insulator, (ii) a semiconductor, and (iii) a conductor.

Check Solution → [2024 • Set 55-3-1]

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236. (a) Explain the characteristics of a p-n junction diode that makes it suitable for its use as a rectifier.
(b) With the help of a circuit diagram, explain the working of a full wave rectifier.

Check Solution → [2024 • Set 55-5-1]

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237. Explain the following, giving reasons:
(a) A doped semiconductor is electrically neutral.
(b) In a p-n junction under equilibrium, there is no net current.
(c) In a diode, the reverse current is practically not dependent on the applied voltage.

Check Solution → [2024 • Set 55-5-1]

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238. State the working principle of an LED. Write any two important advantages and two disadvantages of LED.

Check Solution → [2022 • Set 55-1-1]

239. With the help of a circuit diagram, explain the working of a *p-n* junction diode as a full-wave rectifier. Also draw its input and output waveforms.

Check Solution → [2022 • Set 55-2-1]

240. Briefly explain how emf is generated in a solar cell. Draw its I-V characteristics.

Check Solution → [2022 • Set 55-3-1]

241. Draw the energy band diagrams for conductors, semiconductors and insulators. Which band determines the electrical conductivity of a solid? How is the electrical conductivity of a semiconductor affected with rise in its temperature? Explain.

Check Solution → [2022 • Set 55-3-2]

242. Answer the following, giving reason:
(a) The resistance of a p-n junction is low when it is forward biased and is high when it is reverse biased.
(b) Doping of intrinsic semiconductors is a necessity for making electronic devices.
(c) Photodiodes are operated in reverse bias.

Check Solution → [2022 • Set 55-4-1]

243. (i) Draw V-I characteristics of a p-n junction diode.
(ii) Differentiate between the threshold voltage and the breakdown voltage for a diode.
(iii) Write the property of a junction diode which makes it suitable for rectification of ac voltages.

Check Solution → [2022 • Set 55-5-1]

244. Draw circuit diagram and explain the working of a zener diode as a dc voltage regulator with the help of its I-V characteristic.

Check Solution → [2020 • Set 55-1-1]

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245. What is the purpose of heavy doping of p- and n-sides of a zener diode?

Check Solution → [2020 • Set 55-1-1]

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246. With the help of a simple diagram, explain the working of a silicon solar cell, giving all three basic processes involved. Draw its I-V characteristic.

Check Solution → [2020 • Set 55-1-2]

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247. Draw the circuit diagram of a full wave rectifier using two p-n junction diodes. Explain its working and show input and output voltage variations.

Check Solution → [2020 • Set 55-1-3]

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248. What is the function of a solar cell? Briefly explain its working and draw its I-V characteristic curve.

Check Solution → [2020 • Set 55-2-1]

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249. (a) Plot V-I characteristics for an illuminated photodiode under reverse bias for three different illumination intensities $I_1 > I_2 > I_3$.
(b) In a zener regulated power supply, a zener diode with $V_Z = 6$ V is used for regulation. The load current should be 4.0 mA and zener current is five times the load current for unregulated input of 10 V. Find the series resistor R_s used in the power supply.

Check Solution → [2020 • Set 55-2-2]

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250. Name the diode which can act as a voltage regulator. Explain its working with the help of its labelled circuit diagram. Draw its V-I characteristic.

Check Solution → [2020 • Set 55-3-1]

251. (a) Why is an intrinsic semiconductor deliberately converted into an extrinsic semiconductor by adding impurity atoms?
(b) Explain briefly the two processes that occur in p-n junction region to create a potential barrier.

Check Solution → [2020 • Set 55-3-1]

252. With the help of a circuit diagram, explain how two p-n junction diodes along with a centre tapped transformer can be used as a full wave rectifier.

Check Solution → [2020 • Set 55-3-2]

253. (a) How are energy bands formed in a crystalline solid?
(b) Draw the energy band diagrams for p-type and n-type semiconductors. Depict the donor/acceptor energy levels in these diagrams and write their significance.

Check Solution → [2020 • Set 55-3-3]

254. What is a solar cell ? Draw its *V-I* characteristics. Explain the three processes involved in its working.
OR
Draw the circuit diagram of a full wave rectifier. Explain its working showing its input and output waveforms.

Check Solution → [2020 • Set 55-5-1]

255. Explain the formation of potential barrier and depletion region in a p-n junction diode. What is effect of applying forward bias on the width of depletion region ?

OR

What is photo diode ? Briefly explain its working and draw its *V-I* characteristics.

Check Solution → [2020 • Set 55-5-3]

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256. Three photo diodes D_1, D_2 and D_3 are made of semiconductors having band gaps of 2.5 eV, 2 eV and 3 eV respectively. Which of them will not be able to detect light of wavelength 600 nm?

Check Solution → [2019 • Set 55-1-1]

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257. Why photodiodes are required to operate in reverse bias? Explain.

Check Solution → [2019 • Set 55-1-1]

---

258. Describe briefly the functions of the three segments of n-p-n transistor.

Check Solution → [2019 • Set 55-1-1]

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259. Draw the circuit arrangement for studying the output characteristics of n-p-n transistor in CE configuration. Explain how the output characteristics is obtained.

Check Solution → [2019 • Set 55-1-1]

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260. Draw the circuit diagram of a full wave rectifier and explain its working. Also, give the input and output waveforms.

Check Solution → [2019 • Set 55-1-1]

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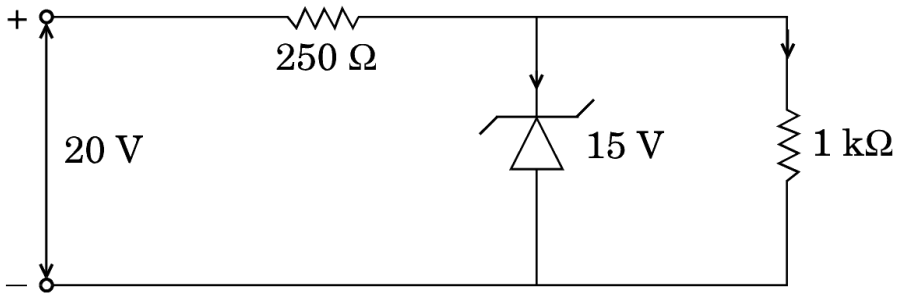
261. If *A* and *B* represent the maximum and minimum amplitudes of an amplitude modulated wave, write the expression for the modulation index in terms of *A* & *B*.

Check Solution → [2019 • Set 55-1-1]

262. A message signal of frequency 20 kHz and peak voltage 10 V is used to modulate a carrier of frequency 2 MHz and peak voltage of 15 V. Calculate the modulation index. Why the modulation index is generally kept less than one?

Check Solution → [2019 • Set 55-1-1]

263. Give reason to explain why n and p regions of a Zener diode are heavily doped. Find the current through the Zener diode in the circuit given below: (Zener breakdown voltage is 15 V)



Check Solution → [2019 • Set 55-2-1]

264. A signal of low frequency f_m is to be transmitted using a carrier wave of frequency f_c. Derive the expression for the amplitude modulated wave and deduce expressions for the lower and upper sidebands produced. Hence, obtain the expression for modulation index.

Check Solution → [2019 • Set 55-2-1]

265. Prove that in a common-emitter amplifier, the output and input differ in phase by 180°.

In a transistor, the change of base current by 30 μA produces change of 0.02 *V* in the base-emitter voltage and a change of 4 mA in the collector current. Calculate the current amplification factor and the load resistance used, if the voltage gain of the amplifier is 400.

Check Solution → **[2019 • Set 55-2-1]**

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266. What is the reason to operate photodiodes in reverse bias?

A p-n photodiode is fabricated from a semiconductor with a band gap of range of 2.5 to 2.8 eV. Calculate the range of wavelengths of the radiation which can be detected by the photodiode.

Check Solution → **[2019 • Set 55-2-2]**

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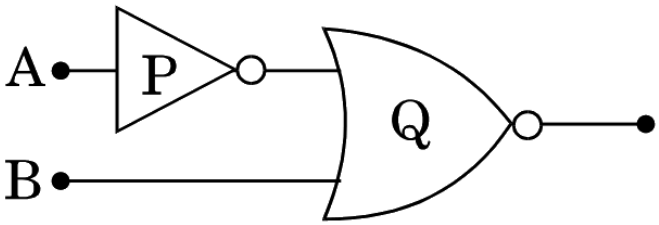
267. Draw the circuit diagram of a full wave rectifier. Explain its working principle. Show the input waveforms given to the diodes D_1 and D_2 and the corresponding output waveforms obtained at the load connected to the circuit.

Check Solution → **[2019 • Set 55-2-3]**

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268. (a) (i) Write the truth tables of the logic gates marked *P* and *Q* in the given circuit.

(ii) Write the truth table for the circuit.



(b) Why are NOR gates considered as universal gates?

OR

(a) Explain how a potential barrier is developed in a p-n junction diode.

(b) Draw the circuit arrangement for studying the V-I characteristics of a p-n junction diode in reverse bias. Plot the V-I characteristics in this case.

Check Solution → [2019 • Set 55-3-1]

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269. (a) What do you mean by bandwidth of a signal? Give its importance.
(b) Differentiate between Analog and Digital communication.
(c) Write the functions of transducer and repeaters.

Check Solution → [2019 • Set 55-3-1]

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270. Draw the energy band diagram of (i) n-type, and (ii) p-type semiconductors at temperature $T > 0\ K$.
In the case of n-type Si-semiconductor, the donor energy level is slightly below the bottom of conduction band whereas in p-type semiconductor, the acceptor energy level is slightly above the top of valence band. Explain, giving examples, what role do these energy levels play in conduction and valence bands.

Check Solution → [2019 • Set 55-4-1]

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271. With what considerations in view is a photodiode fabricated? Explain its working with the help of a suitable diagram. With the help of V-I characteristics, state how photodiode is used to detect optical signals.

Check Solution → [2019 • Set 55-4-1]

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272. (a) What is amplitude modulation? Draw a diagram showing an amplitude modulated wave obtained by modulation of a carrier sinusoidal wave on a modulating signal.
(b) Define the terms (i) modulation index, and (ii) side bands. Mention the significance of side bands.

Check Solution → [2019 • Set 55-4-1]

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273. How is a light emitting diode fabricated? Briefly explain the basic processes involved in the emission of spontaneous radiation from it. Write two advantages of LED lamps over the incandescent low power lamps.

Check Solution → [2019 • Set 55-4-2]

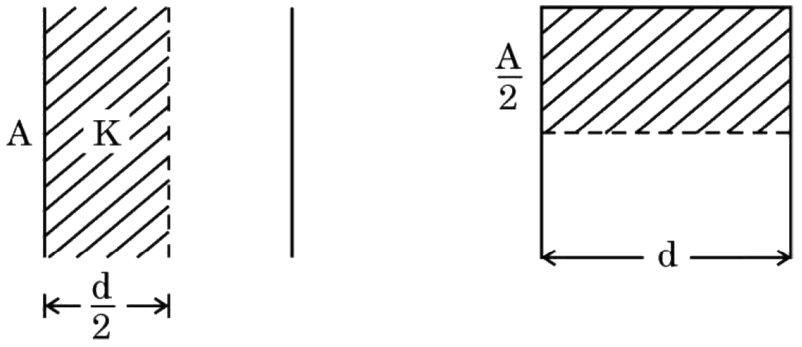
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274. How is Zener diode fabricated? Draw its I-V characteristic and use this to explain the working of a Zener diode as a voltage regulator.

Check Solution → [2019 • Set 55-4-3]

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275. The figure shows the V-I characteristic of a semiconductor diode designed to operate under reverse bias.



(a) (b)

(a) Identify the semiconductor diode used.
(b) Draw the circuit diagram to obtain the given characteristics of this device.
(c) Briefly explain one use of this device.

Check Solution → [2019 • Set 55-5-1]

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276. (a) Differentiate between three segments of an n-p-n transistor on the basis of their size and level of doping.
(b) Draw a plot of transfer characteristic and show which portion of the characteristic is used in amplification and why.

Check Solution → [2019 • Set 55-5-1]

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277. (a) A student wants to use two p-n junction diodes to convert alternating current into direct current. Draw the labelled circuit diagram she would use and explain how it works.
(b) Give the truth table and circuit symbol for NAND gate.

Check Solution → [2018]

---

278. Draw the typical input and output characteristics of an n-p-n transistor in CE configuration. Show how these characteristics can be used to determine (a) the input resistance (r_i), and (b) current amplification factor (β).

Check Solution → [2018]

---

279. (a) Give three reasons why modulation of a message signal is necessary for long distance transmission.
(b) Show graphically an audio signal, a carrier wave and an amplitude modulated wave.

Check Solution → [2018]

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280. Write the two processes that take place in the formation of a p-n junction. Explain with the help of a diagram, the formation of depletion region and barrier potential in a p-n junction.

Check Solution → [2017]

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281. Define the term 'amplitude modulation'. Explain any two factors which justify the need for modulating a low frequency base-band signal.

Check Solution → [2017]

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282. For a CE-transistor amplifier, the audio signal voltage across the collector resistance of 2 kΩ is 2 V. Given the current amplification factor of the transistor is 100, find the input signal voltage and base current, if the base resistance is 1 kΩ.

Check Solution → [2017]

283. A zener diode is fabricated by heavily doping both p- and n- sides of the junction. Explain, why? Briefly explain the use of zener diode as a dc voltage regulator with the help of a circuit diagram.

Check Solution → [2017]

284. Explain briefly with the help of necessary diagrams, the forward and the reverse biasing of a p-n junction diode. Also draw their characteristic curves in the two cases.

Check Solution → [2017]

285. (a) Define the term 'modulation index,' used in communication system. Why is its value kept less than or equal to one?
(b) A message signal of frequency 10 kHz and peak voltage of 10 V is used to modulate a carrier frequency 1 MHz and peak voltage 10 V. Determine the (i) modulation index, and (ii) side bands produced.

Check Solution → [2017]

286. What is sky wave propagation? Which frequency range is suitable for sky wave propagation and why? Over which range of frequencies can communication through free space using radio waves take place?

Check Solution → [2017]

287. What is space wave propagation? Which systems of communication use space waves? What is 'radio horizon' of a transmitting antenna of height h? Why is space wave propagation suitable for frequencies above 40 MHz?

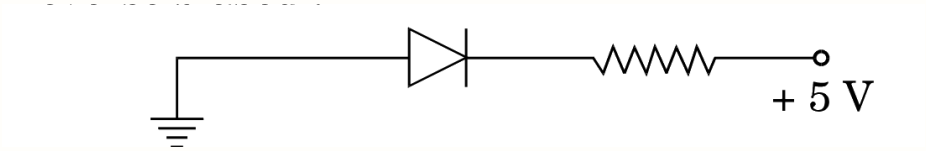
Check Solution → [2017]

288. (a) How is amplitude modulation achieved?
(b) The frequencies of two side bands in an AM wave are 640 kHz and 660 kHz respectively. Find the frequencies of carrier and modulating signal. What is the bandwidth required for amplitude modulation?

Check Solution → [2017]

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289. (a) In the following diagram, is the junction diode forward biased or reverse biased?



(b) Draw the circuit diagram of a full wave rectifier and state how it works.

Check Solution → [2017]

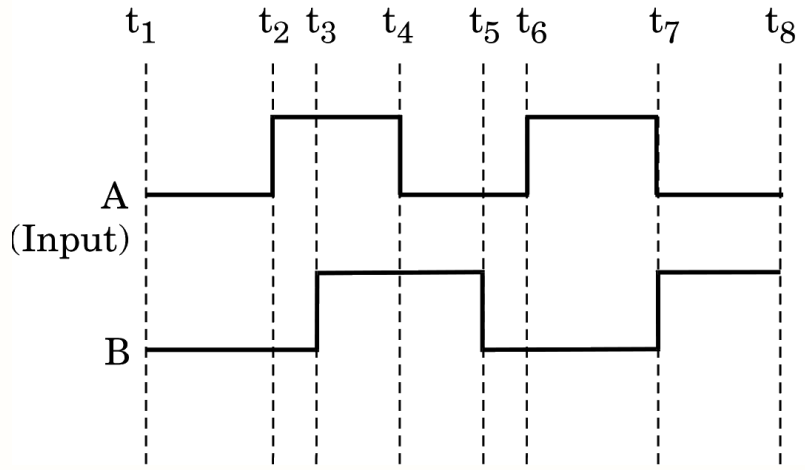
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290. Draw a block diagram of a generalized communication system. Write the functions of each of the following:
(a) Transmitter
(b) Channel
(c) Receiver

Check Solution → [2017]

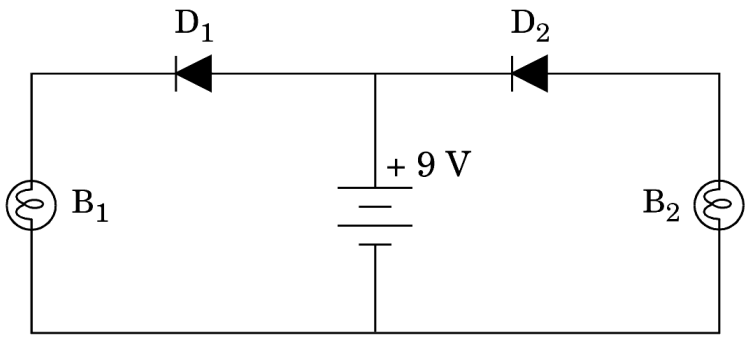
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291. (a) Write the functions of the three segments of a transistor.
(b) The figure shows the input waveforms *A* and *B* for 'AND' gate. Draw the output waveform and write the truth table for this logic gate.



Check Solution → [2017]

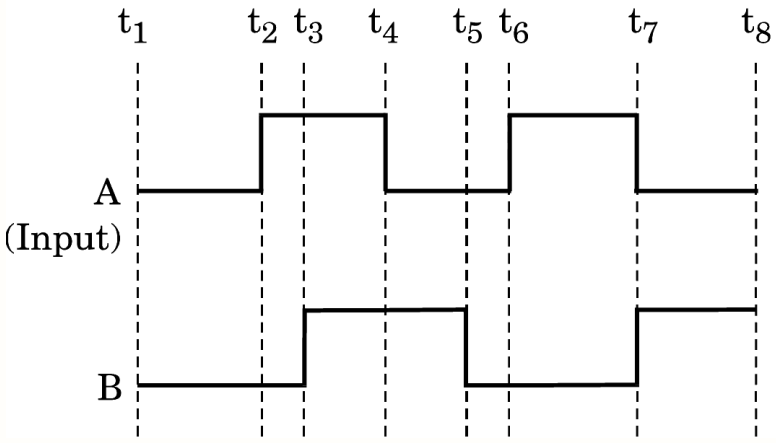
292. (a) In the following diagram, which bulb out of B_1 and B_2 will glow and why?



(b) Draw a diagram of an illuminated p-n junction solar cell.
(c) Explain briefly the three processes due to which generation of emf takes place in a solar cell.

Check Solution → [2017]

293. (a) Draw the circuit diagram for studying the characteristics of a transistor in common emitter configuration. Explain briefly and show how input and output characteristics are drawn.
(b) The figure shows input waveforms *A* and *B* to a logic gate. Draw the output waveform for an OR gate. Write the truth table for this logic gate and draw its logic symbol.



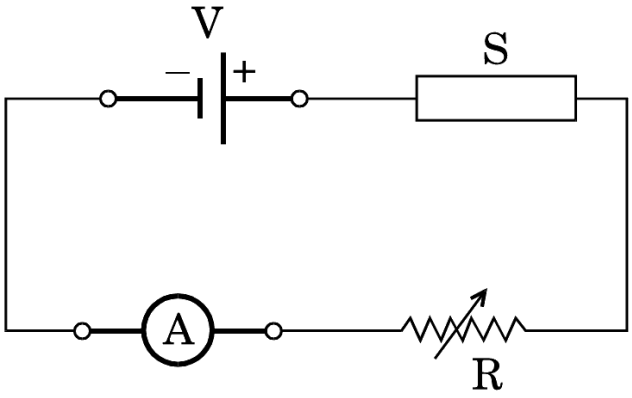
Check Solution → [2017]

294. (a) Draw the circuit diagram of an n-p-n transistor amplifier in common emitter configuration.

(b) Derive an expression for voltage gain of the amplifier and hence show that the output voltage is in opposite phase with the input voltage.

Check Solution → [2017]

295. (a) In the following diagram '*S*' is a semiconductor. Would you increase or decrease the value of *R* to keep the reading of the ammeter *A* constant when *S* is heated? Give reason for your answer.



(b) Draw the circuit diagram of a photodiode and explain its working. Draw its $I - V$ characteristics.

Check Solution → [2017]

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296. (i) Explain with the help of a diagram the formation of depletion region and barrier potential in a pn junction.
(ii) Draw the circuit diagram of a half wave rectifier and explain its working.

Check Solution → [2016]

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297. (i) Which mode of propagation is used by shortwave broadcast services having frequency range from a few MHz upto 30 MHz? Explain diagrammatically how long distance communication can be achieved by this mode.
(ii) Why is there an upper limit to frequency of waves used in this mode?

Check Solution → [2016]

---

298. For a CE-transistor amplifier, the audio signal voltage across the collector resistance of 2 kΩ is 2 *V*. Suppose the current amplification factor of the transistor is 100, find the input signal voltage and base current, if the base resistance is 1 kΩ.

Check Solution → [2016]

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299. What is Global Positioning System? Explain its working principle in brief.

Check Solution → [2016]

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300. Explain briefly, with the help of a circuit diagram, the transistor action of npn transistor in CE configuration. Draw the typical shapes of input and output characteristics.

Check Solution → [2016]

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301. (i) Describe the working of photodiode by drawing the circuit diagram.
(ii) Draw the characteristics of a photodiode for different illumination intensities.
(iii) Why is photodiode operated in reverse bias even though the reverse bias current is much weaker than the current in forward bias?

Check Solution → [2016]

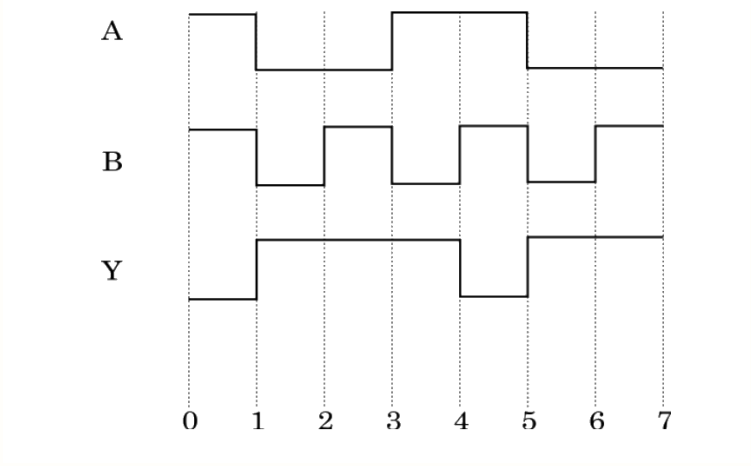
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302. (i) Name two important processes that occur during the formation of a pn junction.
(ii) Draw the circuit diagram of a full wave rectifier along with the input and output waveforms. Briefly explain how the output voltage/current is unidirectional.

Check Solution → [2016]

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303. (i) Distinguish between a conductor and a semi conductor on the basis of energy band diagram.
(ii) The following figure shows the input waveforms (A, B) and the output waveform (Y) of a gate. Identify the gate, write its truth table and draw its logic symbol.



Check Solution → [2016]

304. What is space wave propagation? State the factors which limit its range of propagation. Derive an expression for the maximum line of sight distance between two antennas for space wave propagation.

Check Solution → [2016]

305. (i) Explain briefly the process of emission of light by a Light Emitting Diode (LED).
(ii) Which semiconductors are preferred to make LEDs and why?
(iii) Give two advantages of using LEDs over conventional incandescent lamps.

Check Solution → [2016]

306. Draw a circuit diagram of a common emitter amplifier using n-p-n transistor. Derive an expression for the current gain β_{ac}.

Check Solution → [2016]

307. Mention any three applications of the internet. Explain one of these in detail.

Check Solution → [2016]

308. Write two important considerations used while fabricating a Zener diode.
Explain, with the help of a circuit diagram, the principle and working of a Zener diode as voltage regulator.

Check Solution → [2015]

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309. Draw the necessary energy band diagrams to distinguish between conductors, semiconductors and insulators.
How does the change in temperature affect the behaviour of these materials? Explain briefly.

Check Solution → [2015]

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310. (a) What are the three basic units in communication systems? Write briefly the function of each of these.
(b) Write any three applications of the internet used in communication systems.

Check Solution → [2015]

---

311. Draw a block diagram of a detector for AM signal and show, using necessary processes and the waveforms, how the original message signal is detected from the input AM wave.

Check Solution → [2015]

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312. With what considerations in view, a photodiode is fabricated? State its working with the help of a suitable diagram.
Eventhough the current in the forward bias is known to be more than in the reverse bias, yet the photodiode works in reverse bias. What is the reason?

Check Solution → [2015]

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313. Draw a circuit diagram of a transistor amplifier in CE configuration.
Define the terms: (i) Input resistance and (ii) Current amplification factor. How are these determined using typical input and output characteristics?

Check Solution → [2015]

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314. When Sunita, a class XII student, came to know that her parents are planning to rent out the top floor of their house to a mobile company she protested. She tried hard to convince her parents that this move would be a health hazard. Ultimately her parents agreed :
(1) In what way can the setting up of transmission tower by a mobile company in a residential colony prove to be injurious to health ?
(2) By objecting to this move of her parents, what value did Sunita display ?
(3) Estimate the range of e.m. waves which can be transmitted by an antenna of height 20 m. (Given radius of the earth = 6400 km)

Check Solution → [2014]

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315. Write any two distinguishing features between conductors, semiconductors and insulators on the basis of energy band diagrams.

Check Solution → [2014]

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316. Write two basic modes of communication. Explain the process of amplitude modulation. Draw a schematic sketch showing how amplitude modulated signal is obtained by superposing a modulating signal over a sinusoidal carrier wave.

Check Solution → [2014]

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317. Write three important factors which justify the need of modulating a message signal. Show diagrammatically how an amplitude modulated wave is obtained when a modulating signal is superimposed on a carrier wave.

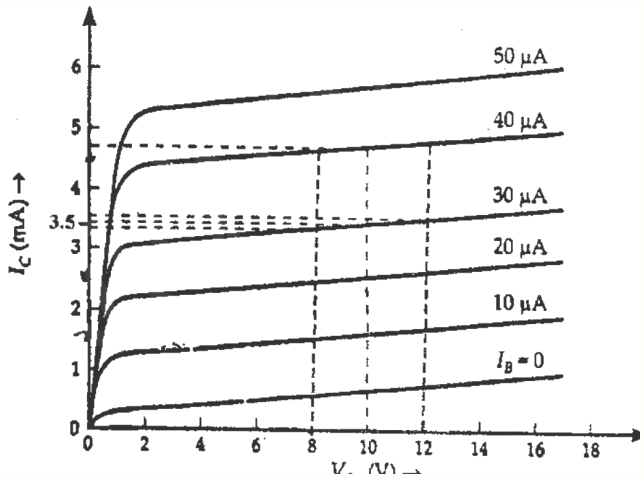
Check Solution → [2013]

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318. Output characteristics of an n-p-n transistor in CE configuration is shown in the figure. Determine:
(i) dynamic output resistance

(ii) dc current gain and

(iii) ac current gain at an operating point $V_{CE} = 10\ V$, when $I_B = 30\ \mu A$.



Check Solution → [2013]

---

319. Which mode of propagation is used by short wave broadcast services having frequency range from a few MHz upto 30 MHz? Explain diagrammatically how long distance communication can be achieved by this mode. Why is there an upper limit to frequency of waves used in this mode?

Check Solution → [2012]

---

320. Mention three different modes of propagation used in communication system. Explain with the help of a diagram how long distance communication can be achieved by ionospheric reflection of radio waves.

Check Solution → [2012]

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321. Write any two factors which justify the need for modulating a signal.
Draw a diagram showing an amplitude modulated wave by superposing a modulating signal over a sinusoidal carrier wave.

Check Solution → [2012]

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322. Define modulation index. Give its physical significance.
For an amplitude modulated wave, the maximum amplitude is found to be 10 V

while the minimum amplitude is 2 *V*. Determine the modulation index μ.

Check Solution → [2012]

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323. Name the three different modes of propagation of electromagnetic waves. Explain, using a proper diagram the mode of propagation used in the frequency range above 40 MHz.

Check Solution → [2012]

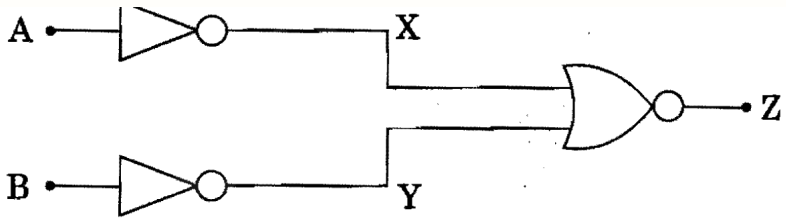
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324. Draw a labelled diagram of a full wave rectifier circuit. State its working principle. Show the input-output waveforms.

Check Solution → [2011]

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325. You are given a circuit below. Write its truth table. Hence, identify the logic operation carried out by this circuit. Draw the logic symbol of the gate it corresponds to.



Check Solution → [2011]

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326. Name the important processes that occur during the formation of a p-n junction. Explain briefly, with the help of a suitable diagram, how a p-n junction is formed. Define the term 'barrier potential'.

Check Solution → [2011 • Set 55-2-1]

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327. Draw transfer characteristics of a common emitter n-p-n transistor. Point out the region in which the transistor operates as an amplifier.
Define the following terms used in transistor amplifiers:

(i) Input resistance
(ii) Output resistance
(iii) Current amplification factor

Check Solution → [2011 • Set 55-2-1]

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328. What is space wave propagation? Give two examples of communication systems which use space wave mode.
A TV tower is 80 m tall. Calculate the maximum distance upto which the signal transmitted from the tower can be received.

Check Solution → [2010]

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329. Distinguish between sky wave and space wave propagation. Give a brief description with the help of suitable diagrams indicating how these waves are propagated.

Check Solution → [2009]

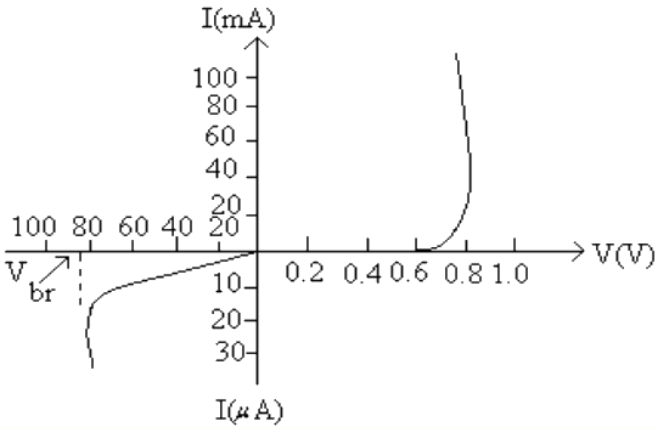
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330. With the help of a suitable diagram, explain the formation of depletion region in a p-n junction. How does its width change when the junction is (i) forward biased, and (ii) reverse biased?

Check Solution → [2009]

---

331. The figure below shows the V-I characteristic of a semiconductor diode.



(i) Identify the semiconductor diode used.
(ii) Draw the circuit diagram to obtain the given characteristic of this device.
(iii) Briefly explain how this diode can be used as a voltage regulator.

Check Solution → [2008]

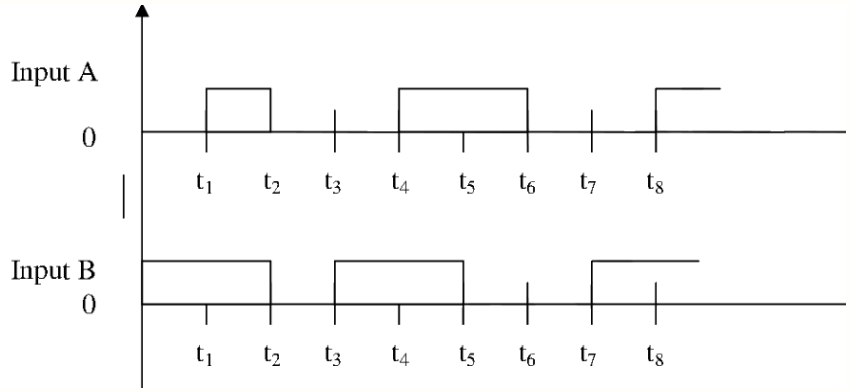
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332. Draw a plot of the amplitude versus ω for an amplitude modulated wave. Define modulation index. State its importance for effective amplitude modulation.

Check Solution → [2008]

---

333. Two signals A, B as given below, are applied as input to (i) AND (ii) NOR (iii) NAND gates. Draw the output wave-form in each case.



Check Solution → [2008]

---

334. What does the term 'LOS communication' mean? Name the types of waves that are used for this communication. Give typical examples, with the help of a suitable figure, of communication systems that use space mode propagation.

Check Solution → [2008]

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335. Draw a labeled circuit diagram of a full wave rectifier and briefly explain its working principle.

Check Solution → [2008]

---

336. Draw a labeled circuit diagram of a transistor amplifier in the common-emitter configuration. Briefly explain, how the input/output signals differ in phase by 180°.

Check Solution → [2008]

---

337. What is a digital signal? Explain the function of modem in data communication. Write two advantages of digital communication.

Check Solution → [2007]

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338. Explain, with the help of a schematic diagram, the principle and working of a Light Emitting Diode. What criterion is kept in mind while choosing the semi-conductor material for such a device? Write two advantages of Light Emitting Diode over conventional incandescent lamps.

Check Solution → [2007]

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339. Draw the circuit diagram of a common emitter amplifier using n-p-n transistor. What is the phase difference between the input signal and output voltage? State two reasons why a common emitter amplifier is preferred over a common base amplifier.

Check Solution → [2007]

---

340. Explain the formation of energy bands in solids. Draw energy band diagram for (i) a conductor, (ii) an intrinsic semiconductor.

Check Solution → [2007]

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341. What is modulation? Explain the need of modulating a low frequency information signal. With the help of diagrams, differentiate between PAM and PDM.

Check Solution → [2007]

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342. Write the acronym LASER in expanded form. State any four reasons for preferring diode lasers as light sources for optical communication links.

Check Solution → [2007]

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343. What is an intrinsic semiconductor? How can this material be converted into (i) P-type (ii) N-type extrinsic semiconductor? Explain with the help of energy band diagrams.

Check Solution → [2006]

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344. Define the term modulation. Name three different types of modulation used for a message signal using a sinusoidal continuous carrier wave. Explain the meaning of any one of these.

Check Solution → [2006]

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345. Draw a circuit diagram for use of NPN transistor as an amplifier in common emitter configuration. The input resistance of a transistor is 1000 Ω. On changing its base current by 10 μA, the collector current increases by 2 mA. If a load resistance of 5 kΩ is used in the circuit, calculate:
(i) the current gain
(ii) voltage gain of the amplifier

Check Solution → [2006]

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346. Define the term 'critical frequency' in relation to sky wave propagation of electromagnetic waves.
On a particular day, the maximum frequency reflected from the ionosphere is 10 MHz. On another day, it was found to decrease to 8 MHz. Calculate the ratio of the maximum electron densities of the ionosphere on the two days.

Check Solution → [2006]

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347. Give reasons for the following:

(i) Long distance radio broadcasts use short-wave bands.

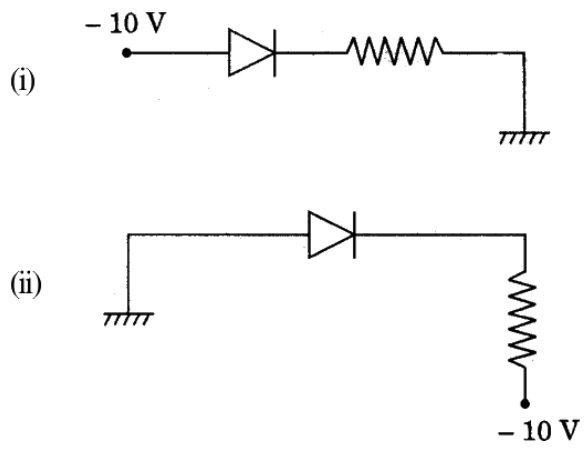
(ii) The small ozone layer on top of the stratosphere is crucial to human survival.

(iii) Satellites are used for long distance TV transmission.

Check Solution → [2005]

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348. Explain, with the help of a circuit diagram, how the thickness of depletion layer in a p-n junction diode changes when it is forward biased. In the following circuit which one of the two diodes is forward biased and which is reverse biased?



Check Solution → [2005]

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349. Distinguish between analog and digital communication. Write any two modulation techniques employed for the digital data. Describe briefly one of the techniques used.

Check Solution → [2005]

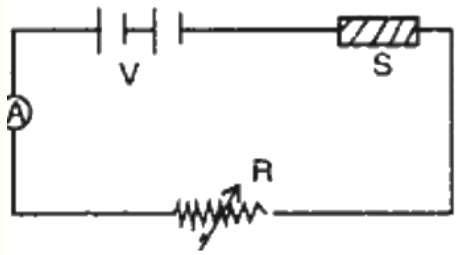
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350. Draw a schematic diagram of a single optical fibre structure. Explain briefly how an optical fibre is fabricated. Describe in brief, the mechanism of propagation of light signal through an optical fibre.

Check Solution → [2005]

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351. The diagram shows a piece of pure semiconductor, S in series with a variable resistor R, and a source of constant voltage V. Would you increase or decrease the value of R to keep the reading of ammeter (A) constant, when semiconductor S is heated? Give reason.



Check Solution → [2004]

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352. Draw the circuit diagram of a common-emitter amplifier using an n-p-n transistor. Draw the input and output waveforms of the signal. Write the expression for its voltage gain.

Check Solution → [2004]

---

353. Write the symbol and truth table of an AND gate. Explain how this gate is realised in practice by using two diodes.

Check Solution → [2004]

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354. With the help of a block diagram, explain the principle of an optical communication system. Give its two advantages over cable communication system.

Or

With the help of relevant diagrams, explain the following terms:

(i) Pulse-position modulation (PPM)

(ii) Pulse-duration modulation (PDM)

Check Solution → [2004]

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355. Name the device used for data transmission from one computer to another. Justify the name. Using this device draw the block diagram for data communication and explain it briefly.

Check Solution → [2004]

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356. Define Antenna. Write a short note on Antenna. Name two types of antenna.

Check Solution → [2003]

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357. What is meant by the term 'modulation'? Explain with the help of a block diagram, how the process of modulation is carried out in radio broadcasts.

Check Solution → [2003]

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358. Write three special characteristics of the light source used in optical communication. Name any one optical detector. Explain the meaning of the term 'sensitivity' and 'responsivity' of a detector.

Check Solution → [2003]

SECTION D

4-Mark Questions

Case Study · 1 question

> *Section D of the paper, 4 marks each. Sorted newest year first.*

359. Sunil and his parents were travelling to their village in their car. On the way his mother noticed some grey coloured panels installed on the roof of a low building. She enquired from Sunil what those panels were and Sunil told his mother that those were solar panels.
(a) What were the values displayed by Sunil and his mother? State one value for each.
(b) In what way would the use of solar panels prove to be very useful?
(c) Name the semiconductor device used in solar panels. Briefly explain with the help of a diagram, how this device works.

Check Solution → **[2017]**

SECTION E

5-Mark Questions

Long Answer · 22 questions

> *Section E of the paper, 5 marks each. Sorted newest year first.*

360. (a) (i) A germanium crystal is doped with antimony. With the help of energy-band diagram, explain how the conductivity of the doped crystal is affected.
(ii) Briefly explain the two processes involved in the formation of a p-n junction.
(iii) What will the effect of (1) forward biasing, and (2) reverse biasing be on the width of depletion layer in a p-n junction diode?

OR

(b) (i) With the help of a circuit diagram, briefly explain the working of a full-wave rectifier using p-n junction diodes.
(ii) Draw V-I characteristics of a p-n junction diode. Explain how these characteristics make a diode suitable for rectification.
(iii) Carbon and silicon have the same lattice structure. Then why is carbon an insulator but silicon a semiconductor?

Check Solution → **[2023 • Set 55-4-1]**

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361. (a) Draw the circuit arrangement for studying V-I characteristics of a p-n junction diode in (i) forward biasing and (ii) reverse biasing. Draw the typical V-I characteristics of a silicon diode. Describe briefly the following terms: (i) minority carrier injection in forward biasing and (ii) breakdown voltage in reverse biasing.

OR

(b) Name two important processes involved in the formation of a p-n junction diode. With the help of a circuit diagram, explain the working of junction diode as a full wave rectifier. Draw its input and output waveforms. State the characteristic property of a junction diode that makes it suitable for rectification.

Check Solution → **[2023 • Set 55-5-1]**

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362. (a) Draw a circuit diagram of an n-p-n transistor with its emitter-base junction forward biased and base-collector junction reverse biased. Briefly describe its working.

(b) Explain how a transistor in its active state exhibits a low resistance at its emitter-base junction and high resistance at its base-collector junction.

(c) Derive the expression for the voltage gain of a transistor amplifier in CE configuration in terms of the load resistance R_L, current gain β, and input resistance.

Explain why input and output voltages are in opposite phase.

OR

(a) Write the important considerations which are to be taken into account while fabricating a p-n junction diode to be used as a Light Emitting Diode (LED). What should be the order of band gap of an LED, if it is required to emit light in the visible range? Draw a circuit diagram and explain its action.

(b) Draw the V-I characteristics of an LED. State two advantages of LED lamps over conventional incandescent lamps.

Check Solution → **[2019 • Set 55-3-1]**

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363. (a) State briefly the processes involved in the formation of p-n junction explaining clearly how the depletion region is formed.

(b) Using the necessary circuit diagrams, show how the V-I characteristics of a p-n junction are obtained in

(i) Forward biasing

(ii) Reverse biasing

How are these characteristics made use of in rectification ?

OR

(a) Differentiate between three segments of a transistor on the basis of their size and level of doping.

(b) How is a transistor biased to be in active state ?

(c) With the help of necessary circuit diagram, describe briefly how n-p-n transistor in CE configuration amplifies a small sinusoidal input voltage. Write the expression for the ac current gain.

Check Solution → **[2014]**

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364. (a) Draw the circuit diagrams of a p-n junction diode in (i) forward bias, (ii) reverse bias. How are these circuits used to study the V–I characteristics of a silicon diode? Draw the typical V–I characteristics.

(b) What is a light emitting diode (LED)? Mention two important advantages of LEDs over conventional lamps.

OR

(a) Draw the circuit arrangement for studying the input and output characteristics of an n-p-n transistor in CE configuration. With the help of these characteristics define (i) input resistance, (ii) current amplification factor.

(b) Describe briefly with the help of a circuit diagram how an n-p-n transistor is used to produce self-sustained oscillations.

Check Solution → [2012]

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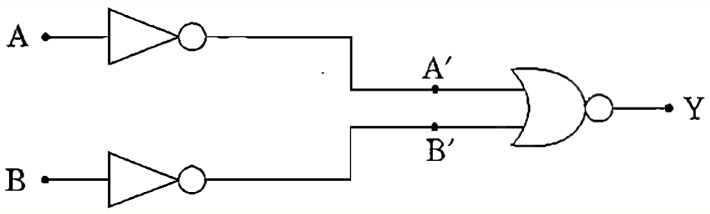
365. (a) Describe briefly, with the help of a diagram, the role of the two important processes involved in the formation of a p-n junction.

(b) Name the device which is used as a voltage regulator. Draw the necessary circuit diagram and explain its working.

OR

(a) Explain briefly the principle on which a transistor-amplifier works as an oscillator. Draw the necessary circuit diagram and explain its working.

(b) Identify the equivalent gate for the following circuit and write its truth table.



Check Solution → [2012]

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366. (a) Draw a circuit for studying the input and output characteristics of an n-p-n transistor in CE configuration. Show how, from the output characteristics, the information about the current amplification factor (β_{ac}) can be obtained.

(b) Draw a plot of the transfer characteristic (V_o versus V_i) for a base-biased transistor in CE configuration. Show for which regions in the plot, the transistor can operate as a switch.

OR

Why is a zener diode considered as a special purpose semiconductor diode? Draw the I–V characteristics of a zener diode and explain briefly how reverse current suddenly increases at the breakdown voltage.

Describe briefly with the help of a circuit diagram how a zener diode works to obtain a constant dc voltage from the unregulated dc output of a rectifier.

Check Solution → [2012]

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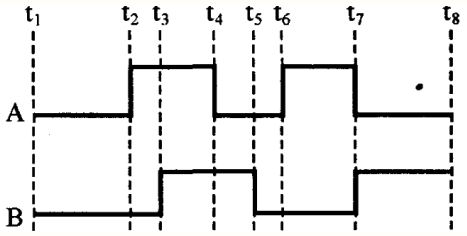
367. Draw a simple circuit of a CE transistor amplifier. Explain its working. Show that the voltage gain, A_V, of the amplifier is given by $A_V = -\frac{\beta_{ac} R_L}{r_i}$, where β_{ac} is the current gain, R_L is the load resistance and r_i is the input resistance of the transistor. What is the significance of the negative sign in the expression for the voltage gain?

OR

(a) Draw the circuit diagram of a full wave rectifier using p-n junction diode. Explain its working and show the output, input waveforms.

(b) Show the output waveforms (Y) for the following inputs *A* and *B* of

(i) OR gate (ii) NAND gate

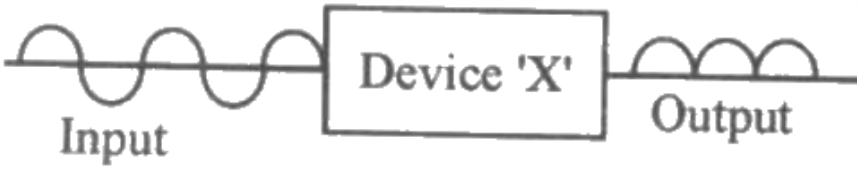


Check Solution → [2012]

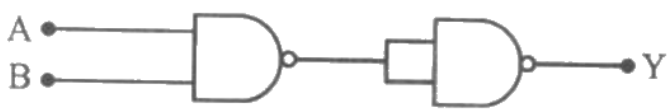
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368. (a) Explain the formation of depletion layer and potential barrier in a p-n junction.

(b) In the figure given below the input waveform is converted into the output waveform by a device 'X'. Name the device and draw its circuit diagram.



(c) Identify the logic gate represented by the circuit as shown and write its truth table.



Check Solution → [2010]

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369. (a) With the help of the circuit diagram explain the working principle of a transistor amplifier as an oscillator.
(b) Distinguish between a conductor, a semiconductor and an insulator on the basis of energy band diagrams.

Check Solution → [2010]

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370. Draw the electronic configuration in the atomic structures of 'Ge' and 'Si' atoms. How can these be made 'P-type' and 'N-type' semiconductors?

Check Solution → [2010]

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371. With the help of a circuit diagram, explain the working of a P-N-P transistor.

Check Solution → [2010]

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372. What are intrinsic and extrinsic semiconductors? How does the rise in temperature affect the conductivity of semiconductors?

Check Solution → [2010]

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373. What is a P-N junction? How does it behave under forward and reverse bias? Explain with the help of circuit diagrams.

Check Solution → [2010]

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374. Explain the working of a full wave rectifier using semiconductor diodes, with the help of a labeled circuit diagram.

Check Solution → [2010]

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375. (i) Draw a circuit diagram to study the input and output characteristics of an n-p-n transistor in its common emitter configuration. Draw the typical input and output characteristics.
(ii) Explain, with the help of a circuit diagram, the working of n-p-n transistor as a common emitter amplifier.

Check Solution → [2009]

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376. How is a zener diode fabricated so as to make it a special purpose diode? Draw I-V characteristics of zener diode and explain the significance of breakdown voltage.
Explain briefly, with the help of a circuit diagram, how a p-n junction diode works as a half wave rectifier.

Check Solution → [2009]

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377. State the principle of working of p-n diode as a rectifier. Explain, with the help of a circuit diagram, the use of p-n diode as a full wave rectifier. Draw a sketch of the input and output waveforms.

Check Solution → [2007]

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378. Draw the symbolic representation of a (i) p-n-p, (ii) n-p-n transistor. Why is the base region of a transistor thin and lightly doped? With a circuit diagram, show the biasing of a p-n-p transistor in common base configuration. Explain the movement of charge carriers through different parts of the transistor in such a configuration and show that $I_e = I_b + I_c$.

Check Solution → [2007]

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379. **(a)** Distinguish between metals, insulators and semiconductors on the basis of their energy bands.
(b) Why are photodiodes used preferably in reverse bias condition? A photodiode is fabricated from a semiconductor with band gap of 2.8 eV. Can it detect a wavelength of 6000 nm? Justify.

Check Solution → [2005]

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380. **(a)** Explain briefly, with the help of a circuit diagram, how *V-I* characteristics of a p-n junction diode are obtained in **(i)** forward bias, and **(ii)** reverse bias. Draw the shape of the curves obtained.
(b) A semiconductor has equal electron and hole concentration of $6 \times 10^8/m^3$. On doping with certain impurity, electron concentration increases to $9 \times 10^{17}/m^3$.
(i) Identify the new semiconductor obtained after doping.
(ii) Calculate the new hole concentration.

Check Solution → [2005]

---

381. Draw a circuit diagram of a common emitter amplifier using n-p-n transistor. Show input and output voltages graphically.
The current gain for common emitter amplifier is 59. If the emitter current is 6.0 mA, find (i) base current and (ii) collector current.

Check Solution → [2003]

SELF-CHECK

Answer Key

All MCQs at a glance

> *Attempt the MCQs first, then check here. Tap any question number to open its full worked solution.*

Q1 **C**	Q12 **C**	Q23 **B**	Q34 **B**	Q45 **D**	Q56 **A**
Q2 **D**	Q13 **A**	Q24 **C**	Q35 **B**	Q46 **C**	Q57 **A**
Q3 **C**	Q14 **A**	Q25 **B**	Q36 **D**	Q47 **D**	Q58 **D**
Q4 **D**	Q15 **C**	Q26 **A**	Q37 **C**	Q48 **B**	Q59 **A**
Q5 **A**	Q16 **B**	Q27 **C**	Q38 **B**	Q49 **B**	Q60 **D**
Q6 **C**	Q17 **D**	Q28 **B**	Q39 **D**	Q50 **D**	Q66 **C**
Q7 **B**	Q18 **A**	Q29 **A**	Q40 **A**	Q51 **C**	Q67 **B**
Q8 **C**	Q19 **A**	Q30 **B**	Q41 **C**	Q52 **C**	Q71 **D**
Q9 **A**	Q20 **A**	Q31 **B**	Q42 **B**	Q53 **D**	Q72 **A**
Q10 **A**	Q21 **B**	Q32 **?**	Q43 **B**	Q54 **C**	
Q11 **B**	Q22 **C**	Q33 **C**	Q44 **B**	Q55 **A**	

WORKED ANSWERS

Detailed Solutions

Every question worked step by step

Each card restates the question, shows the answer, then the full working. Use "Back to Q n" to return.

Q1

In an unbiased p-n junction, at equilibrium, which of the following statements is true?

(A) Diffusion current is zero but drift current exists.

(B) Diffusion current exists but drift current is zero.

(C) **Diffusion and drift currents are equal and opposite.**

(D) Both the diffusion and drift currents exist but are unequal.

[2026 • Set 55-1-1]

SOLUTION

In an isolated p-n junction with no external voltage applied, two opposing currents flow across the junction at the same time.

The majority carriers move by diffusion: electrons diffuse from the n-side into the p-side and holes diffuse from the p-side into the n-side, because of the large concentration difference of majority carriers on the two sides. This sets up the diffusion current, directed from the p-side to the n-side.

As these carriers cross over, the depletion region develops an internal electric field $\vec{E}$ (the built-in field) pointing from the n-side to the p-side. This field sweeps the minority carriers across the junction: minority electrons from the p-side and minority holes from the n-side. This gives the drift current, directed from the n-side to the p-side, which is opposite to the diffusion current.

At equilibrium the junction carries no net current, so these two currents must exactly cancel:

$$I_{diffusion} = I_{drift}$$

Both currents are non-zero (thermal motion keeps majority carriers diffusing and the built-in field keeps drifting minority carriers), but they are equal in magnitude and opposite in direction, giving zero net current. So option (C) is correct.

Option (A) and (B) are wrong because they say one of the two currents is zero, whereas both exist. Option (D) is wrong because if the two currents were unequal, a continuous unbalanced current would flow in an isolated diode, which is impossible at equilibrium. Hence the **diffusion and drift currents are equal and opposite**.

Final answer. Both currents exist and are equal and opposite, giving zero net current: option **(C)**.

← *Back to Q1*

Q2

Assertion (A): *On forward biasing a p-n junction diode, the height of the barrier potential increases.*
Reason (R): *In forward biasing of a p-n junction diode, the direction of the applied voltage is in the same direction as the built-in potential.*

(D) Assertion (A) is false and Reason (R) is also false.

[2026 • Set 55-2-1]

SOLUTION

Examine the Assertion. In forward bias the p-side is connected to the positive terminal and the n-side to the negative terminal of the battery. The applied voltage V acts opposite to the built-in barrier potential V_b, so the effective barrier becomes $V_b - V$.

$$V_{\text{effective}} = V_b - V$$

The height of the barrier therefore decreases (not increases) on forward biasing, the depletion region narrows, and the diode conducts easily. So the Assertion is FALSE.

Now examine the Reason. In forward bias the external field set up by the applied voltage points opposite to the built-in field of the depletion layer, so the applied voltage acts in the opposite direction to the built-in potential, not the same direction. So the Reason is FALSE.

Since both the Assertion and the Reason are false, option (D) is correct. Options (A) and (B) are ruled out because they require both statements to be true, and option (C) is ruled out because it requires the Assertion to be true. Hence **both Assertion (A) and Reason (R) are false**.

COMMON MISTAKE

Forward bias lowers the barrier ($V_b - V$) and the applied field opposes the built-in field; reverse bias raises it ($V_b + V$).

Final answer. Both Assertion and Reason are false: option **(D)**.

← *Back to Q2*

Q3

Case Study: *A p-type or n-type semiconductor can be converted into a p-n junction by doping it with suitable impurity. The motion of majority charge carriers causes diffusion current across the junction while the barrier electric field causes motion of minority carriers for drift current. In case of unbiased diode, the diffusion and drift currents are equal. This equilibrium is disturbed by the biasing batteries. Diodes, therefore, allow currents in one direction. This property of diode is used in making rectifiers.*

Silicon is doped with which of the following to obtain p-type semiconductor?

(A) Phosphorus

(B) Arsenic

(C) Boron

(D) Antimony

[2026 • Set 55-2-1]

SOLUTION

To obtain a p-type semiconductor, the tetravalent host (silicon, group IV) must be doped with a trivalent (group III) impurity.

A trivalent atom has only three valence electrons. When it sits in the silicon lattice it forms only three covalent bonds with its four silicon neighbours, leaving one bond incomplete. This vacancy behaves as a hole, a mobile positive charge carrier. The trivalent atom is therefore called an acceptor, and the holes become the majority carriers, making the crystal p-type.

Checking the options by their group:

Phosphorus is pentavalent (group V), so it gives an n-type semiconductor. Rule out (A).

Arsenic is pentavalent (group V), so it gives an n-type semiconductor. Rule out (B).

Boron is trivalent (group III), so it gives a p-type semiconductor. This is correct.

Antimony is pentavalent (group V), so it gives an n-type semiconductor. Rule out (D).

Hence silicon must be doped with **Boron**, option (C).

Final answer. Trivalent boron gives a p-type semiconductor: option **(C)**.

← *Back to Q3*

Q4

A semiconductor has an electron concentration of $5\times 10^{22}\ \mathrm{m}^{-3}$. *The concentration of holes is (given* $\mathbf{n_i = 1{\cdot}5\times 10^{16}\ m^{-3}}$*):*

(A) $5\times 10^{22}\ \mathrm{m}^{-3}$

(B) $1{\cdot}5\times 10^{16}\ \mathrm{m}^{-3}$

(C) $9\times 10^{8}\ \mathrm{m}^{-3}$

(D) $\mathbf{4{\cdot}5\times 10^{9}\ m^{-3}}$

[2026 • Set 55-2-1]

SOLUTION

The electron and hole concentrations in a semiconductor in thermal equilibrium are linked by the law of mass action, which holds for both intrinsic and doped semiconductors:

$$n_e\, n_h = n_i^2$$

Here n_e is the electron concentration, n_h the hole concentration and n_i the intrinsic carrier concentration. Rearranging for n_h:

$$n_h = \frac{n_i^2}{n_e}$$

Substituting $\mathbf{n_e = 5\times 10^{22}\ m^{-3}}$ and $\mathbf{n_i = 1.5\times 10^{16}\ m^{-3}}$:

$$n_h = \frac{(1.5\times 10^{16})^2}{5\times 10^{22}}$$

$$n_h = \frac{2.25\times 10^{32}}{5\times 10^{22}}$$

$$n_h = 0.45\times 10^{10}\ m^{-3}$$

$$n_h = 4.5\times 10^{9}\ m^{-3}$$

This value matches option (D). The hole concentration is $\mathbf{4.5\times 10^{9}\ m^{-3}}$.

COMMON MISTAKE

Always use n_e n_h = n_i^2 and square n_i fully before dividing; a common slip is forgetting to square the intrinsic concentration.

Final answer. By the law of mass action n_h = n_i^2/n_e = 4.5 x 10^9 m^-3, which is option **(D)**.

← *Back to Q4*

Q5

During forward biasing of a p-n junction diode, the:

(A) current is mainly due to drifting of majority carriers.

(B) current is mainly due to drifting of minority carriers.

(C) diffusion and drift currents are equal.

(D) current is of the order of 1 A.

[2026 • Set 55-2-1]

SOLUTION

When a p-n junction diode is forward biased, the p-side is at a higher potential than the n-side, so the applied voltage opposes the built-in barrier potential and lowers it. The depletion region narrows.

With the barrier reduced, the majority carriers are able to cross the junction in large numbers: electrons (majority in the n-region) move into the p-region and holes (majority in the p-region) move into the n-region. The forward current is carried mainly by these majority carriers as they move across the junction under the applied field.

So the forward current is essentially a majority-carrier current, in contrast to the small reverse current which is a minority-carrier current. Among the options, this is best described by option (A), the current is mainly due to the majority carriers.

Option (B) is wrong because minority carriers carry only the tiny reverse saturation current, not the large forward current. Option (C) is wrong because forward bias disturbs the equilibrium, so diffusion and drift currents are no longer equal. Option (D) is wrong because the forward current of an ordinary diode is of the order of milliamperes, not 1 A. Hence the forward current is **mainly due to the majority carriers**.

Final answer. Forward current is carried mainly by the majority carriers: option **(A)**.

← Back to Q5

Q6

The threshold voltage for silicon diode is about:

(A) 0·2 *V*

(B) 0·5 *V*

(C) **0·7 *V***

(D) 1·5 *V*

[2026 • Set 55-2-1]

SOLUTION

The threshold voltage (also called the cut-in or knee voltage) of a diode is the minimum forward voltage at which the forward current begins to increase rapidly. Below this voltage the current is very small, and above it the diode conducts strongly.

This value is essentially the built-in barrier potential V_b of the junction, which depends on the semiconductor material:

For silicon, $\mathbf{V_b \approx 0.7\ V}$

For germanium, $\mathbf{V_b \approx 0.3\ V}$

Therefore the threshold voltage of a silicon diode is about 0.7 *V*, which is option (C). The other values, 0.2 *V* and 0.5 *V*, are too low for silicon, and 1.5 *V* is far too high. Hence the silicon-diode threshold voltage is about **0.7 *V***.

Final answer. Threshold voltage of a silicon diode is about 0.7 V: option **(C)**.

← *Back to Q6*

Q7

When we dope Ge with a pentavalent element, four of its electrons bond with four

germanium neighbours but fifth electron remains weakly bound. The ionisation energy for this electron is about:

(A) 0·01 eV

(B) **0·05 eV**

(C) 0·1 eV

(D) 0·15 eV

[2026 • Set 55-2-1]

SOLUTION

When a pentavalent atom (such as P, As or Sb) replaces a germanium atom in the lattice, four of its five valence electrons form covalent bonds with the four neighbouring Ge atoms. The fifth electron is only weakly bound to the parent donor ion and needs only a small energy, the donor ionisation energy, to break free and become a conduction electron.

This loosely bound electron behaves like an electron in a hydrogen-like orbit, but inside the semiconductor it is screened by the dielectric medium and moves with an effective mass. The ionisation energy is therefore far smaller than the 13.6 eV of free hydrogen, and is given approximately by:

$$E_{ion} = \frac{m^*}{m_e \varepsilon_r^2} \times 13.6 \text{ eV}$$

Because of the large dielectric constant ε_r and small effective mass m^*, this energy comes out to only a few hundredths of an eV.

As marked in the answer key, the donor ionisation energy is taken as 0.05 eV, option (B). The other options, 0.1 eV and 0.15 eV, are too large, while 0.01 eV is the smaller end of the range. Hence the extra electron is set free by an ionisation energy of about **0.05 eV**, confirming that the donor level lies just below the conduction band.

Final answer. The fifth electron's donor ionisation energy is about 0.05 eV: option **(B)**.

← *Back to Q7*

Q8

When the forward bias voltage in a semiconductor diode is changed from 0·8 *V to* 1·0 *V, the forward current changes by* 2·0 mA. *The forward bias resistance of the diode will be:*

(A) 200 Ω

(B) 175 Ω

(C) **100 Ω**

(D) 125 Ω

[2026 • Set 55-3-1]

SOLUTION

The forward dynamic resistance of a diode is the ratio of a small change in forward voltage to the resulting change in forward current about an operating point:

$$r_f = \frac{\Delta V}{\Delta I}$$

The change in forward voltage is:

$$\Delta V = 1.0\ V - 0.8\ V = 0.2\ V$$

The change in forward current is:

$$\Delta I = 2.0\ \text{mA} = 2.0 \times 10^{-3}\ A$$

Substituting these values:

$$r_f = \frac{0.2}{2.0 \times 10^{-3}}$$

$$r_f = \frac{0.2}{0.002}$$

$$r_f = 100\ \Omega$$

So the forward-bias resistance of the diode is option (C). The **forward dynamic resistance is 100 Ω**.

Final answer. Forward-bias dynamic resistance r_f = 100 ohm: option **(C)**.

← *Back to Q8*

Q9

The process named 'minority carrier injection' in a p-n junction diode occurs during:

(A) forward biasing

(B) reverse biasing

(C) no biasing at low temperature

(D) no biasing at high temperature

[2026 • Set 55-3-1]

SOLUTION

Minority-carrier injection means forcing carriers that are minority in one region into the other region across the junction, so that the injected carriers raise the minority population on the far side.

This happens during forward biasing. When the diode is forward biased the barrier potential is lowered, so majority carriers cross the junction in large numbers:

Electrons from the n-side (where they are majority) are pushed into the p-side, where they become minority carriers.

Holes from the p-side (where they are majority) are pushed into the n-side, where they become minority carriers.

These injected carriers form the large forward current of the diode, which is why the process is called minority-carrier injection.

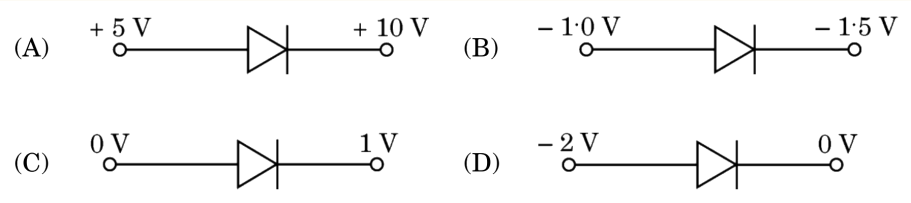
Option (B) is wrong because reverse bias raises the barrier, so carriers cannot be injected; only the tiny reverse saturation current of thermally generated minority carriers flows. Options (C) and (D) describe an unbiased diode, where the junction is at equilibrium and there is no injection. Hence minority-carrier injection occurs during **forward biasing**, option (A).

Final answer. Minority-carrier injection occurs during forward biasing: option **(A)**.

← *Back to Q9*

Q10

Which of the following p-n junction diodes is forward biased?



(A) Option (A): p-side +10 *V*, n-side 0 *V* [see figure]

(B) Option (B): p-side −1·5 *V*, n-side −1·5 *V* [see figure]

(C) Option (C): see figure

(D) Option (D): see figure

[2026 • Set 55-3-2]

SOLUTION

A p-n junction diode is forward biased when its p-side is held at a higher potential than its n-side. This makes the applied voltage oppose the built-in barrier potential, lowers the barrier, and lets the diode conduct. If the p-side is at a lower potential than the n-side the diode is reverse biased, and if both terminals are at the same potential there is no bias.

Checking each option using the potential difference between the p-side and n-side:

Option (A): p-side at +10 *V*, n-side at 0 *V*, so the p-side is at the higher potential. This is forward biased.

Option (B): both ends at −1.5 *V*, so the potential difference across the junction is zero. There is no bias.

Options (C) and (D): the p-side is at a lower (or equal) potential than the n-side, so these are reverse biased (or unbiased).

Only in option (A) is the p-side more positive than the n-side. Hence the **diode in option (A) is forward biased**.

CONCEPT

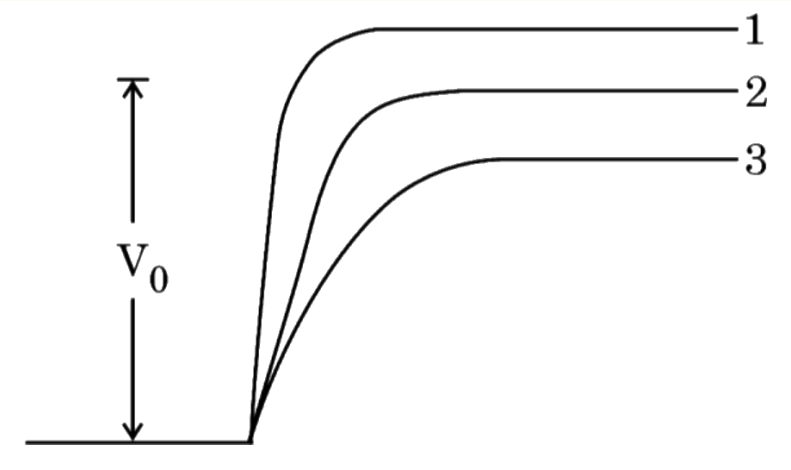
For any diode, compare the p-side and n-side potentials: p higher than n means forward bias, p lower than n means reverse bias.

Final answer. Only the diode with p-side higher than n-side is forward biased: option **(A)**.

← *Back to Q10*

Q11

In the given figure, V_0 is the potential barrier across a p-n junction in an unbiased condition. Which of the following statements is correct?



(A) Curves 1 and 3 both correspond to forward biasing of p-n junction.

(B) Curve 3 corresponds to forward biasing and Curve 1 corresponds to reverse biasing.

(C) Curve 1 corresponds to forward biasing and Curve 3 corresponds to reverse biasing.

(D) Curves 1 and 3 both correspond to reverse biasing.

[2026 • Set 55-3-3]

SOLUTION

Across an unbiased p-n junction there is a built-in potential barrier V_0, set up by the immobile ionised donor and acceptor charges in the depletion region. Applying an external voltage changes this effective barrier height.

In forward bias the external voltage is applied so that it opposes the built-in potential. The effective barrier becomes $V_0 - V$, which is smaller than V_0.

In reverse bias the external voltage adds to the built-in potential. The effective barrier becomes $V_0 + V$, which is larger than V_0.

In the figure, Curve 2 (height = $\mathbf{V_0}$) is the unbiased junction. Curve 1 sits higher than V_0, so it has a raised barrier and corresponds to reverse bias. Curve 3 sits lower than V_0, so it has a lowered barrier and corresponds to forward bias.

This matches option (B). Option (A) and (D) are wrong because curves 1 and 3 lie on opposite sides of V_0, so they cannot both be the same bias. Option (C) reverses the roles of the two curves and is therefore incorrect. Hence **Curve 3 is forward biasing and Curve 1 is reverse biasing**.

CONCEPT

Lower barrier than V_0 always means forward bias; raised barrier above V_0 always means reverse bias.

Final answer. Correct option is **(B)**: Curve 3 corresponds to forward biasing and Curve 1 to reverse biasing.

← *Back to Q11*

Q12

In a reversed-biased p-n junction diode, the applied voltage mostly drops across:

(A) p-region only

(B) n-region only

(C) depletion region

(D) the diode

[2026 • Set 55-4-1]

SOLUTION

The bulk p-region and n-region of a diode are doped semiconductors that contain plenty of free majority carriers, so they have low resistance. The depletion region, by contrast, is swept free of mobile carriers and behaves like a high-resistance (almost insulating) zone.

When the junction is reverse biased, the applied voltage pulls majority carriers further away from the junction, so the depletion region widens and its resistance becomes even larger.

In a series path, the voltage divides in proportion to resistance, so the applied voltage drops almost entirely across the highest-resistance part, which is the depletion region.

This strong field across the narrow depletion layer opposes the diffusion of majority carriers and sweeps across only the few minority carriers, giving the very small reverse saturation current.

Hence options (A) and (B) are wrong because the lightly resisting bulk regions take only a tiny share of the voltage, and (D) is too vague. The drop is across the **depletion region**.

Final answer. Correct option is **(C)**: the applied voltage mostly drops across the depletion region.

← *Back to Q12*

Q13

Assertion (A): *The conductivity of an n-type semiconductor is higher than that of a p-type semiconductor at a given temperature.*
Reason (R): *The electrons being in the conduction band in n-type semiconductor are more mobile than the holes in the valence band in p-type semiconductor.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2026 • Set 55-4-1]

SOLUTION

The conductivity of a semiconductor is given by $\sigma = e\,(n_e\,\mu_e + n_h\,\mu_h)$, where n_e, n_h are the electron and hole concentrations and μ_e, μ_h are their mobilities.

In an n-type semiconductor the majority carriers are electrons, which move in the conduction band. In a p-type semiconductor the majority carriers are holes, which move in the valence band.

For silicon, the electron mobility $\mu_e \approx 1350\ \text{cm}^2/(V\cdots)$ is roughly three times the hole mobility $\mu_h \approx 480\ \text{cm}^2/(V\cdots)$.

So for comparable doping levels, the n-type sample carries current mainly with the higher-mobility electrons, while the p-type sample relies on the slower holes. Therefore $\sigma_n > \sigma_p$, which makes **Assertion (A) true**.

The Reason (R) states exactly this fact: conduction-band electrons are more mobile than valence-band holes. R is true and is the direct cause of the higher n-type conductivity, so R correctly explains A. Hence option (A).

Final answer. Correct option is **(A)**: both A and R are true and R is the correct explanation of A.

← *Back to Q13*

Q14

A good diode checked by a multimeter should indicate:

(A) high resistance in reverse bias and a low resistance in forward bias

(B) high resistance in both forward bias and reverse bias

(C) low resistance in both reverse bias and forward bias

(D) high resistance in forward bias and low resistance in reverse bias

[2026 • Set 55-5-1]

SOLUTION

A working p-n junction diode is a one-way device. It conducts easily when forward biased and blocks current when reverse biased.

When checked with a multimeter in diode/resistance mode with the red probe on the anode (p-side) and the black probe on the cathode (n-side), the diode is

forward biased, so it shows a low resistance (a few hundred ohms for silicon).

When the probe polarity is reversed, the diode is reverse biased, so it shows a very high resistance, essentially an open circuit.

This pair of readings, low in forward bias and high in reverse bias, is exactly option (A).

The other options indicate faulty diodes: a low reading both ways (option C) means the diode is shorted, a high reading both ways (option B) means it is open, and option (D) gives the readings the wrong way round. A **good diode shows low resistance in forward bias and high resistance in reverse bias**.

> **TIP**
>
> Equal readings in both directions flag a faulty diode: both low means shorted, both high means open.

Final answer. Correct option is **(A)**: high resistance in reverse bias and low resistance in forward bias.

← *Back to Q14*

Q15

When a p-n junction diode is forward biased:

(A) the barrier height and the depletion layer width both increase.

(B) the barrier height increases and the depletion layer width decreases.

(C) **the barrier height and the depletion layer width both decrease.**

(D) the barrier height decreases and the depletion layer width increases.

[2025 • Set 55-1-1]

SOLUTION

In forward bias the positive terminal of the battery is connected to the p-side and the negative terminal to the n-side, so the applied voltage V opposes the built-in potential of the junction.

The effective barrier height becomes $V_b - V$, which is less than the built-in value V_b, so the barrier height decreases.

With the barrier lowered, majority carriers (holes from the p-side and electrons from the n-side) are pushed toward the junction. They neutralise some of the immobile ions that form the depletion region.

As these ions are neutralised, the depletion region narrows, so its width decreases.

Therefore in forward bias both the barrier height and the depletion layer width decrease, which is option (C). Options (A), (B) and (D) are ruled out because the lowered barrier and the inflow of majority carriers must shrink, not grow, the depletion region. **Both decrease**.

COMMON MISTAKE

Do not confuse the cases: in reverse bias both the barrier height and depletion width increase, the exact opposite of forward bias.

Final answer. Correct option is **(C)**: the barrier height and the depletion layer width both decrease.

← *Back to Q15*

Q16

Assertion (A): *We cannot form a p-n junction diode by taking a slab of a p-type semiconductor and physically joining it to another slab of a n-type semiconductor.*
Reason (R): *In a p-type semiconductor* $n_h \gg n_e$, *while in a n-type semiconductor* $n_e \gg n_h$.

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of Assertion (A).

[2025 • Set 55-1-1]

SOLUTION

A p-n junction cannot be formed simply by pressing a p-type slab against an n-type slab. The pressed surfaces have crystal discontinuities, surface

contamination and irregular bonding, so a continuous single crystal lattice does not exist across the boundary. Without lattice continuity, the controlled diffusion of carriers that builds the depletion region and the barrier potential cannot take place. Real junctions are made inside one continuous crystal by diffusion, ion implantation or epitaxy. So **Assertion (A) is true**.

The Reason (R) states that in a p-type semiconductor $n_h \gg n_e$ and in an n-type semiconductor $n_e \gg n_h$. This is simply the definition of the two doped materials, so **R is also true**.

However, the carrier-density imbalance in R is not why a physical joint fails; the joint fails because of the absence of a continuous crystal lattice at the contact. Since R does not explain A, the answer is option (B).

Final answer. Correct option is **(B)**: both A and R are true, but R is not the correct explanation of A.

← *Back to Q16*

Q17

When the resistance measured between p and n ends of a p-n junction diode is high, it can act as a/an

(A) resistor

(B) inductor

(C) capacitor

(D) **switch**

[2025 • Set 55-2-1]

SOLUTION

The resistance of a p-n junction diode depends entirely on its bias. When it is reverse biased, the resistance between the p and n ends is very high and almost no current flows, so the diode is effectively OFF (open circuit). When it is forward biased, the resistance is very low and current flows freely, so the diode is effectively ON (closed circuit).

This two-state behaviour, OFF for high resistance and ON for low resistance, is exactly how an electronic switch works: it opens the circuit in reverse bias and closes it in forward bias.

A resistor (option A) is ruled out because the diode does not obey a fixed linear V/I relation. An inductor (option B) and a capacitor (option C) store energy in magnetic and electric fields respectively, which is not the action described here.

Hence, when the measured resistance is high (reverse biased, OFF), the diode is behaving as a **switch**.

Final answer. Correct option is **(D)**: the diode acts as a switch.

← *Back to Q17*

Q18

Assertion (A): *In a semiconductor diode the thickness of depletion layer is not fixed.*
Reason (R): *Thickness of depletion layer in a semiconductor device depends upon many factors such as biasing of the semiconductor.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

[2025 • Set 55-2-1]

SOLUTION

The depletion layer of a p-n junction is the region near the junction that is swept free of mobile charge carriers, leaving only fixed donor and acceptor ions. Its thickness is not a fixed quantity, so **Assertion (A) is true**.

The Reason (R) states that the depletion-layer thickness depends on several factors such as the biasing of the device, which is also **true**. In forward bias the inflow of majority carriers neutralises ions and the layer narrows; in reverse bias carriers are pulled away and the layer widens. It also depends on the doping concentrations on either side.

Because the thickness genuinely changes with the applied bias (and doping), it cannot have a single fixed value. So R directly explains why A is true.

Hence both A and R are true and R is the correct explanation of A, giving option (A).

Final answer. Correct option is **(A)**: both A and R are true and R is the correct explanation of A.

← *Back to Q18*

Q19

Which of the following is an electrical conductor at room temperature ?

(A) Sn

(B) Mica

(C) Si

(D) C

[2025 • Set 55-4-1]

SOLUTION

Whether a material conducts at room temperature is decided by its energy band gap E_g, the gap between the top of the valence band and the bottom of the conduction band.

A conductor (metal) has $\mathbf{E_g \approx 0}$, with overlapping or partly filled bands, so many free electrons are always available.

A semiconductor has a small gap, about $\mathbf{E_g \approx 1\ eV}$, and an insulator has a large gap, several eV.

Checking the options: Sn (tin) is a metal with overlapping bands, so it is a conductor. Mica is an insulator with a large band gap. Si (silicon) has $\mathbf{E_g \approx 1.1\ eV}$, so it is a semiconductor. Diamond carbon C has $\mathbf{E_g \approx 5.4\ eV}$, so it is an insulator.

Among the four, only **Sn** is an electrical conductor at room temperature, which is option (A).

Final answer. Correct option is **(A)**: Sn (tin) is an electrical conductor at room temperature.

← Back to Q19

Q20

Assertion (A): *n-type semiconductor is not negatively charged.*
Reason (R): *Neutral pentavalent impurity atom doped in intrinsic semiconductor (neutral) donates its fifth unpaired electron to the crystal lattice and becomes a positive donor.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2025 • Set 55-4-1]

SOLUTION

An n-type semiconductor is made by doping an intrinsic (neutral) crystal with a pentavalent impurity. Four of the dopant's five valence electrons form covalent bonds with the neighbouring silicon atoms, while the fifth, loosely bound, electron is donated to the conduction band and becomes a free electron.

When that fifth electron leaves, the dopant atom is left with one more proton than electrons, so it becomes a fixed positive donor ion held in the lattice. So **Reason (R) is true**.

Now count the charge of the whole sample. Each free electron carries charge $-e$, and its parent donor ion carries the balancing charge $+e$. Adding over all dopant atoms, the total charge is zero.

Therefore, even though the n-type material has many free electrons, it is electrically neutral and not negatively charged, so **Assertion (A) is true**.

R explains exactly why A holds: every donated electron is matched by a positive donor ion, keeping the net charge zero. Hence R is the correct explanation of A, giving option (A).

CONCEPT

A doped semiconductor stays electrically neutral; doping only changes which carrier is in the majority, never the total charge.

Final answer. Correct option is **(A)**: both A and R are true and R is the correct explanation of A.

← *Back to Q20*

Q21

Assertion (A): *The impurities in p-type Si are not pentavalent atoms.*
Reason (R): *The hole density in valance band in p-type semiconductor is almost equal to the acceptor density.*

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

[2025 • Set 55-5-1]

SOLUTION

We must judge each statement on its own and then decide whether the Reason explains the Assertion.

Assertion (A): The impurities used in p-type Si are not pentavalent atoms. This is **true**. To make silicon p-type, we add a **trivalent (Group III) impurity** such as Boron, Aluminium, Indium or Gallium. Each such atom forms only three covalent bonds with its four Si neighbours, leaving one bond incomplete. That empty bond is a **hole**, so holes become the majority carriers. A pentavalent atom (P, As, Sb) would instead donate an extra electron and make the crystal n-type, so the dopant in p-type Si is definitely not pentavalent.

Reason (R): In a p-type semiconductor the hole density in the valence band is almost equal to the acceptor density. This is also **true**. At room temperature nearly every acceptor atom captures one electron from the valence band, and each captured electron leaves behind one hole. So the number of holes is approximately equal to the number of acceptor atoms, $\mathbf{n_h \approx N_A}$.

Both statements are correct, but R only describes what happens after the doping is done. It does not explain why the dopant has to be trivalent in the first place. Hence R is true but is **not the correct explanation** of A, which matches option (B).

Final answer. Option **(B)**: both A and R are true, but R is not the correct explanation of A.

← *Back to Q21*

Q22

Assertion (A): *A hole is an apparent free particle with effective positive electronic charge.*
Reason (R): *A hole is not necessarily a vacancy left behind by an electron in the valence band.*
Select the correct answer from the codes given below:

(C) Assertion (A) is true, but Reason (R) is false.

[2025 • Set 55-6-1]

SOLUTION

Check each statement against the textbook definition of a hole.

Assertion (A): A hole is an apparent free particle with an effective positive charge. This is **true**. When one electron is missing from a covalent bond, a neighbouring bound electron can jump in to fill it, which shifts the vacancy to the neighbour. This continued shifting of the empty site looks exactly like a particle of charge $+e$ drifting through the crystal. So a hole behaves as a mobile, apparently free carrier of effective positive charge, even though no real positive particle is moving.

Reason (R): A hole is not necessarily a vacancy left behind by an electron in the valence band. This is **false**. By its very definition a hole **is** the vacancy created in the valence band when an electron leaves it (either by being thermally excited to the conduction band, or by being captured by an acceptor atom). So R directly contradicts the definition of a hole.

Therefore A is true while R is false, which corresponds to option (C).

Final answer. Option **(C)**: A is true, but R is false.

← *Back to Q22*

Q23

Case Study (Extrinsic Semiconductors): Extrinsic semiconductors are made by doping pure or intrinsic semiconductors with suitable impurity. There are two type of dopants used in doping, Si or Ge, and using them p-type and n-type semiconductors can be obtained. A p-n junction is the basic building block of many semiconductor devices. Two important processes occur during the formation of a p-n junction: diffusion and drift. When such a junction is formed, a 'depletion layer' is created consisting of immobile ion-cores. This is responsible for a junction potential barrier. The width of a depletion layer and the height of potential barrier changes when a junction is forward-biased or reverse-biased. A semiconductor diode is basically a p-n junction with metallic contacts provided at the ends for application of an external voltage. Using diodes, alternating voltages can be rectified.
(i) Which of the following is a donor impurity atom for Ge?

(A) Boron

(B) **Antimony**

(C) Aluminium

(D) Indium

[2025 • Set 55-6-1]

SOLUTION

A **donor** impurity must give the crystal one extra free electron. Germanium is tetravalent, so each Ge atom forms four covalent bonds. A dopant that supplies all four bonding electrons and still has one electron left over must be **pentavalent (Group V)**.

Now classify the options by valency.

Boron (B), Aluminium (Al) and Indium (In) are all **trivalent (Group III)**. They have only three valence electrons, so they create a missing bond, that is a hole. These are **acceptor** atoms and give p-type material, not donors.

Antimony (Sb) is **pentavalent (Group V)**. After forming four covalent bonds with Ge, it has one loosely bound electron left, which is easily freed at room temperature. This makes Sb a true **donor** giving n-type germanium.

Hence the donor impurity for Ge is **Antimony**, option (B).

Final answer. Option **(B)**: Antimony (Sb) is the donor impurity for Ge.

← *Back to Q23*

Q24

(ii) When a pentavalent atom occupies the position of an atom in the crystal lattice of Si, four of its electrons form covalent bonds with four silicon neighbours, while the fifth remains bound to the parent atom. The energy required to set this electron free is about:

(A) 0·5 eV

(B) 0·1 eV

(C) **0·05 eV**

(D) 0·01 eV

[2025 • Set 55-6-1]

SOLUTION

When a pentavalent atom replaces a Si atom, four of its valence electrons go into covalent bonds with the four neighbouring Si atoms. The fifth electron is left only loosely attached to the parent (donor) atom.

This extra electron is not held tightly because the silicon crystal has a high relative permittivity, $\varepsilon_r \approx 12$, which strongly screens the electron from its parent ion, and the electron also has a small effective mass inside the crystal. Both effects sharply reduce its binding energy compared with a free hydrogen-like atom.

Using the hydrogen-like estimate scaled by these crystal factors,

$$E_{\text{ionisation}} = \frac{m^*}{m_e\, \varepsilon_r^{\,2}} \times 13.6 \text{ eV}$$

which works out to only a few hundredths of an electron-volt, of the order of 0.05 eV.

This is far smaller than the silicon band gap of about 1.1 eV, so at room temperature the fifth electron is essentially free and contributes to conduction. The energy needed is therefore about **0.05 eV**, option (C).

Final answer. Option **(C)**: about **0.05 eV** is needed to free the fifth electron.

← *Back to Q24*

Q25

(iv) (a) In reverse-biased p-n junction:

(A) the drift current is of the order of few mA.

(B) the applied voltage mostly drops across the depletion region.

(C) the depletion region width decreases.

(D) the current increases with increase in applied voltage.

OR

(b) The output frequency of a full-wave rectifier with 50 Hz *as input frequency is:*

(A) (a) the drift current is of the order of few mA. | (b) 25 Hz

(B) **(a) the applied voltage mostly drops across the depletion region. | (b) 50 Hz**

(C) (a) the depletion region width decreases. | (b) 100 Hz

(D) (a) the current increases with increase in applied voltage. | (b) 200 Hz

[2025 • Set 55-6-1]

SOLUTION

Part (a): Examine each statement about a reverse-biased p-n junction.

(A) The drift current is of the order of a few mA. This is **false**. The reverse current is a small drift (saturation) current of the order of μA, since it depends only on the few minority carriers.

(B) The applied voltage mostly drops across the depletion region. This is **true**. The depletion region is almost free of mobile carriers and so has a very high resistance, so nearly the entire applied reverse voltage appears across it.

(C) The depletion-region width decreases. This is **false**. In reverse bias the external voltage adds to the built-in barrier, widening the depletion region.

(D) The current increases with increase in applied voltage. This is **false**. The reverse current quickly saturates and stays almost constant until breakdown.

So statement (B) is the correct one.

Part (b): For a full-wave rectifier, both halves of each input cycle are turned into output pulses of the same polarity, so the output completes two cycles for every one input cycle.

$$f_{out} = 2 \times f_{in} = 2 \times 50 \text{ Hz} = 100 \text{ Hz}$$

The correct physics answers are (a) statement B and (b) 100 Hz, and the marking scheme records this as option **(B)**.

Final answer. Option **(B)**: in reverse bias the voltage drops mostly across the depletion region; full-wave output frequency is **100 Hz.**

← *Back to Q25*

Q26

A p-n junction diode is forward biased. As a result,

(A) both the potential barrier height and the width of depletion layer decrease.

(B) both the potential barrier height and the width of depletion layer increase.

(C) the potential barrier height decreases and the width of depletion layer increases.

(D) the potential barrier height increases and the width of depletion layer decreases.

[2025 • Set 55-7-1]

SOLUTION

In a p-n junction at equilibrium, diffusion of carriers leaves immobile ions near the junction, setting up a built-in **potential barrier** V_b across a charge-free **depletion region**.

When the diode is **forward biased**, the external voltage V is applied with its positive terminal to the p-side and negative to the n-side. This applied voltage acts **against** the built-in barrier, so the effective barrier becomes

$$V_{eff} = V_b - V$$

which is smaller than the equilibrium value V_b. Hence the **potential barrier height decreases**.

With a lower barrier, many majority carriers are pushed toward the junction. Electrons from the n-side and holes from the p-side move in and neutralise the immobile ions near the junction, so fewer uncovered ions remain. As a result the **width of the depletion layer also decreases**.

So in forward bias both the barrier height and the depletion-layer width decrease, which is option (A).

Final answer. Option **(A)**: both the potential barrier height and the depletion-layer width decrease.

← *Back to Q26*

Q27

Germanium crystal is doped at room temperature with a minute quantity of boron. The charge carriers in the doped semiconductors will be:

(A) electrons only

(B) holes only

(C) holes and few electrons

(D) electrons and few holes

[2025 • Set 55-7-1]

SOLUTION

Boron is a **trivalent (Group III)** impurity, so doping germanium with boron produces a **p-type** semiconductor.

Each boron atom forms only three covalent bonds with its four Ge neighbours, leaving one bond short of an electron. That incomplete bond is a **hole**, so holes are created in large numbers and become the **majority carriers**.

However, the crystal is not free of electrons. At room temperature, thermal energy still breaks some covalent bonds and generates a small number of electron-hole pairs, so a few free electrons are always present as **minority carriers**.

This balance is fixed by the law of mass action,

$$n_e\, n_h = n_i^2$$

Since n_h is large in p-type material, n_e becomes very small but is never exactly zero. So the doped germanium has **many holes and a few electrons**, option (C).

Final answer. Option **(C)**: the carriers are holes (majority) and a few electrons (minority).

← *Back to Q27*

Q28

Ge is doped with As. Due to doping,

(A) the structure of Ge lattice is distorted.

(B) **the number of conduction electrons increases.**

(C) the number of holes increases.

(D) the number of conduction electrons decreases.

[2024 • Set 55-1-1]

SOLUTION

Arsenic (As) belongs to Group V, so it is **pentavalent** with five valence electrons, while germanium is tetravalent.

When an As atom replaces a Ge atom in the lattice, four of its valence electrons form covalent bonds with the four neighbouring Ge atoms. The fifth electron is only weakly bound to the parent As atom.

The binding energy of this fifth electron is very small (of the order of hundredths of an eV), so at room temperature thermal energy easily frees it into the conduction band. Thus **each As donor atom contributes one extra conduction electron**.

With many As atoms added, the number of free conduction electrons increases sharply, far above the small intrinsic value, so the **number of conduction electrons increases**. (This makes Ge:As an n-type semiconductor; by $n_e\, n_h = n_i^2$ the hole number actually decreases.)

Hence the correct statement is option (B).

Final answer. Option **(B)**: doping Ge with As increases the number of conduction electrons.

← *Back to Q28*

Q29

The root mean square value of an alternating voltage applied to a full-wave rectifier is $\frac{V_0}{\sqrt{2}}$. *Then the root mean square value of the rectified output voltage is:*

(A) $\mathbf{\frac{V_0}{\sqrt{2}}}$

(B) $\frac{V_0}{2}$

(C) $\frac{2V_0}{\sqrt{2}}$

(D) $\frac{V_0}{2\sqrt{2}}$

[2024 • Set 55-1-1]

SOLUTION

Let the input alternating voltage have peak value V_0, so its given RMS value is

$$V_{rms,in} = \frac{V_0}{\sqrt{2}}$$

A full-wave rectifier flips the negative halves of the input into positive halves, so the output waveform is $|V_{in}(t)|$. Its magnitude at every instant is identical to the input magnitude.

The RMS value of any waveform is found from the mean of its square,

$$V_{rms} = \sqrt{\overline{V^2}}$$

Squaring the voltage removes the sign, so V^2 is exactly the same for the rectified output as for the original AC. Therefore

$$V_{rms,out} = V_{rms,in} = \frac{V_0}{\sqrt{2}}$$

So the RMS value of the rectified output equals $\frac{V_0}{\sqrt{2}}$, option (A). (Note the DC average value does change, becoming $\frac{2V_0}{\pi}$, but the RMS value is unchanged.)

TIP

Squaring kills the sign, so a full-wave rectified wave has the same RMS as the original AC. Only the average (DC) value changes.

Final answer. Option **(A)**: the RMS output voltage is $\frac{V_0}{\sqrt{2}}$.

← *Back to Q29*

Q30

In a full-wave rectifier, the current in each of the diodes flows for:

(A) Complete cycle of the input signal

(B) **Half cycle of the input signal**

(C) Less than half cycle of the input signal

(D) Only for the positive half cycle of the input signal

[2024 • Set 55-1-1]

SOLUTION

A full-wave rectifier (centre-tapped type) uses **two diodes** that take turns conducting, with the load connected so that current through it is always in the same direction.

During the **positive half cycle** of the input, diode D_1 has its anode at higher potential, so D_1 is forward biased and conducts, while D_2 is reverse biased and stays OFF.

During the **negative half cycle**, the polarities reverse, so D_2 becomes forward biased and conducts, while D_1 is OFF.

So each diode is forward biased and carries current for only one half of every input cycle, never for the full cycle and never simultaneously. Therefore the current in **each diode flows for half a cycle** of the input signal, option (B).

Final answer. Option **(B)**: each diode conducts for half a cycle of the input signal.

← *Back to Q30*

Q31

An alternating voltage of frequency of 50 Hz *is applied to a half-wave rectifier. Then the ripple frequency of the output will be:*

(A) 100 Hz

(B) 50 Hz

(C) 25 Hz

(D) 150 Hz

[2024 • Set 55-1-1]

SOLUTION

In a half-wave rectifier, the junction diode conducts only during one half of every input cycle. During the half cycle that forward biases the diode, current flows through the load and an output pulse appears across R_L. During the next half cycle the diode is reverse biased, it does not conduct, and the output is zero.

So for each complete input cycle, the output contains exactly one pulse. This means the output pulses repeat at the same rate as the input.

$$f_{output} = f_{input}$$

Given the input frequency $\mathbf{f_{input} = 50\ Hz}$, the **ripple frequency of the output is also 50 Hz**.

Checking the other options: 100 Hz (A) is the ripple frequency of a full-wave rectifier, where both half cycles produce output pulses, doubling the frequency. Options 25 Hz (C) and 150 Hz (D) have no physical basis for a half-wave rectifier.

TIP

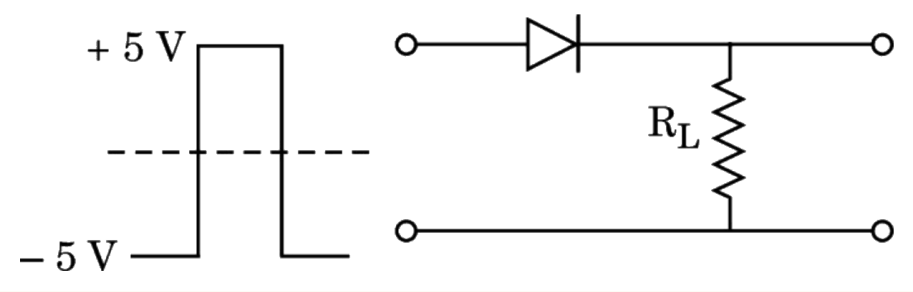
Remember the rule: half-wave rectifier keeps the ripple frequency the same as input, full-wave rectifier doubles it.

Final answer. The ripple frequency of the output is **50 Hz**, option **(B)**.

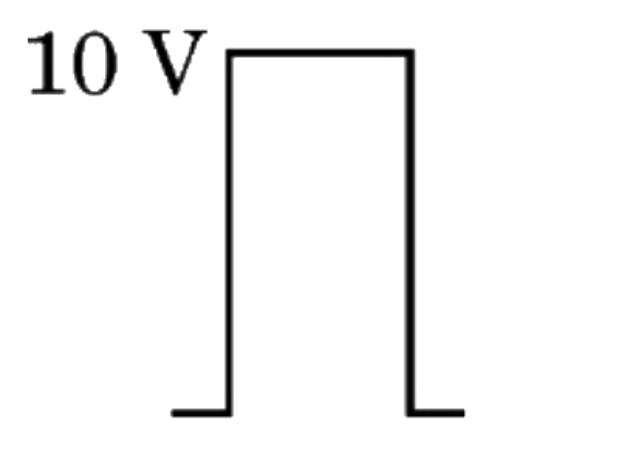
← *Back to Q31*

Q32

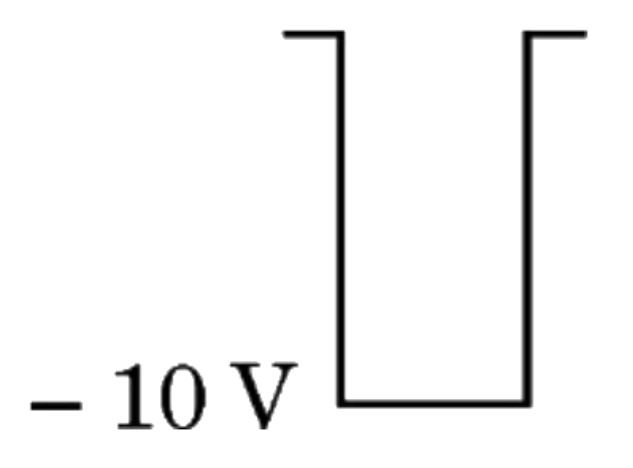
A signal, as shown in the figure, is applied to a p-n junction diode. Identify the output across resistance R_L*:*



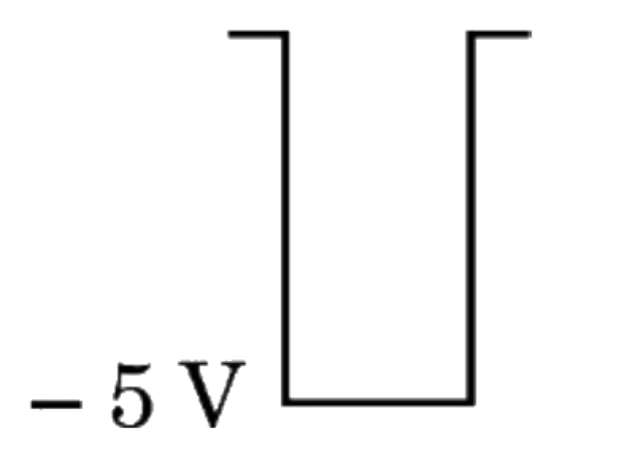
(A)



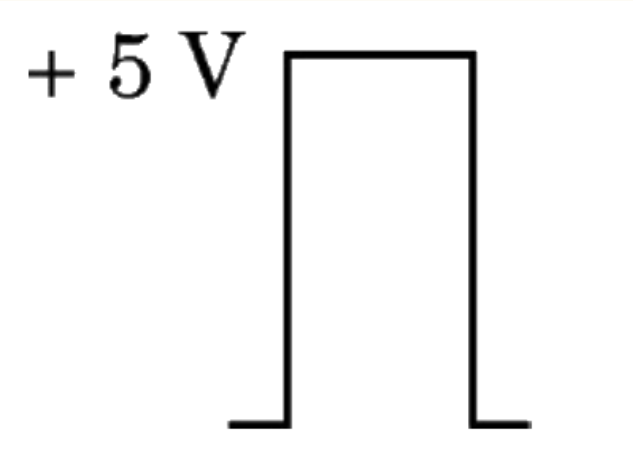
(B)



(C)



(D)



[2024 • Set 55-1-1]

SOLUTION

A square wave that swings between +5 V and −5 V is applied to a series combination of a p-n junction diode and the load R_L, with the output taken across R_L. The diode acts as a one-way switch, so it only passes current when it is forward biased.

During the +5 V half of the input, the diode is forward biased and conducts. Almost the full input appears across the load.

$\mathbf{V_{out}}$ **= +5** $\boldsymbol{V}$ (ideal diode), or about +4.3 V for a real silicon diode after the 0.7 V drop.

During the −5 V half of the input, the diode is reverse biased and blocks current, so no voltage develops across the load.

$$V_{out} = 0\ V$$

The output waveform is therefore a **unipolar square wave**: positive square pulses of amplitude +5 V during the positive half cycles, separated by flat 0 V intervals during the negative half cycles. The diode has clipped off the negative portions of the input.

Diagram description: draw a time axis (horizontal) and a voltage axis (vertical). The output trace rises to +5 V and stays flat for the first half period, drops to 0 V for the second half period, and repeats. There are no negative excursions in the output.

Final answer. The output across R_L is a series of positive square pulses between **+5 *V*** and **0 *V*** (negative halves removed).

← *Back to Q32*

Q33

An n-type semiconducting Si is obtained by doping intrinsic Si with:

(A) Al

(B) B

(C) **P**

(D) In

[2024 • Set 55-2-1]

SOLUTION

Silicon is tetravalent, having 4 valence electrons that form four covalent bonds in the crystal. To create an n-type semiconductor, where the majority carriers are free electrons, silicon must be doped with a **pentavalent** impurity, that is one having 5 valence electrons.

When a pentavalent atom replaces a silicon atom, four of its electrons form covalent bonds with neighbouring silicon atoms, while the fifth electron is only loosely bound. At room temperature this extra electron is easily freed and contributes to conduction.

Examining the options: Aluminium (Al), Boron (B), and Indium (In) all belong to group 13 and are trivalent (3 valence electrons). Doping with these creates holes and gives a p-type semiconductor, so options A, B and D are ruled out.

Phosphorus (P) belongs to group 15 and is pentavalent (5 valence electrons). Doping silicon with phosphorus donates free electrons and produces an n-type semiconductor.

Final answer. n-type silicon is obtained by doping with Phosphorus (P), option **(C)**.

← *Back to Q33*

Q34

When a p-n junction diode is subjected to reverse biasing:

(A) the barrier height decreases and the depletion region widens.

(B) **the barrier height increases and the depletion region widens.**

(C) the barrier height decreases and the depletion region shrinks.

(D) the barrier height increases and the depletion region shrinks.

[2024 • Set 55-2-1]

SOLUTION

A p-n junction has a built-in barrier potential V_b across the depletion region due to the diffusion of carriers. In reverse bias, the external battery is connected so that the p-side is at lower potential and the n-side at higher potential.

The polarity of the applied voltage is in the same direction as the built-in potential, so the two add up.

$$V_{net} = V_b + V_{applied}$$

Hence the **barrier height increases**.

The applied field also pulls the majority carriers (holes in the p-region, electrons in the n-region) away from the junction towards the terminals. This uncovers more immobile dopant ions on both sides of the junction, so the **depletion region widens**.

Therefore in reverse bias both the barrier height increases and the depletion region widens, which matches option B.

Options A and C wrongly state the barrier decreases (that happens in forward bias), and option D wrongly states the depletion region shrinks.

Final answer. In reverse bias the barrier height increases and the depletion region widens, option **(B)**.

← *Back to Q34*

Q35

Si is doped with a pentavalent element. The energy required to set the additional electron free is about:

(A) 0.01 eV

(B) **0.05 eV**

(C) 0.72 eV

(D) 1.1 eV

[2024 • Set 55-3-1]

SOLUTION

When silicon (tetravalent) is doped with a pentavalent element such as P, As or Sb, four of the dopant's valence electrons form covalent bonds with the surrounding silicon atoms. The fifth electron is held only loosely by the parent donor ion.

This fifth electron orbits the donor ion much like the electron in a hydrogen atom, but it is far more weakly bound. The reason is that silicon has a large dielectric constant ($\varepsilon_r \approx 12$), which screens the Coulomb attraction, and the electron moves with a reduced effective mass m^*.

Using a Bohr-like estimate for the donor ionisation energy:

$$E_{ion} = \frac{m^*}{m_e} \cdot \frac{1}{\varepsilon_r^2} \cdot 13.6 \text{ eV}$$

Substituting the values appropriate for silicon gives

$$E_{ion} \approx 0.05 \text{ eV}$$

This is very small compared with the energy band gap of silicon (1.1 eV) and is close to the thermal energy at room temperature ($kT \approx 0.026$ eV). So even ordinary thermal energy easily frees the donor electron into the conduction band. Hence the required energy is about **0.05 eV**, option B.

Final answer. The energy needed to free the extra electron in pentavalent-doped Si is about **0.05 eV**, option **(B)**.

← Back to Q35

Q36

Assertion (A): *In a semiconductor, the electrons in the conduction band have lesser energy than those in the valence band.*
Reason (R): *Donor energy level is just above the valence band in a semiconductor.*

(D) Assertion (A) is false and Reason (R) is also false.

[2024 • Set 55-3-1]

SOLUTION

Evaluate the Assertion first. On the energy band diagram of a semiconductor, the conduction band lies above the valence band, separated by the forbidden energy gap. An electron promoted to the conduction band has therefore gained energy and has **higher** energy than one in the valence band, not lesser. So the Assertion is false.

Now the Reason. In an n-type semiconductor the donor energy level lies just **below the conduction band**, so that the extra donor electrons need only a tiny energy to jump into the conduction band. The energy level just above the valence band is the acceptor level, which belongs to a p-type semiconductor. So the statement that the donor level is just above the valence band is also false.

Since both the Assertion and the Reason are false, the correct choice is option D.

Options A, B and C all require the Assertion to be true, so they are eliminated.

COMMON MISTAKE

Donor level sits just below the conduction band; acceptor level sits just above the valence band. Mixing these up is the classic trap here.

Final answer. Both Assertion and Reason are false, option **(D)**.

← Back to Q36

Q37

Assertion (A): *The temperature coefficient of resistance is positive for metals and negative for p-type semiconductors.*
Reason (R): *The charge carriers in metals are negatively charged, whereas the majority charge carriers in p-type semiconductors are positively charged.*

(C) If Assertion (A) is true and Reason (R) is false.

[2024 • Set 55-4-1]

SOLUTION

First judge the Assertion. In a metal the number of free electrons is essentially fixed, but as temperature rises the lattice vibrations increase and scatter the electrons more, lowering their drift mobility. So the resistance of a metal increases with temperature, giving a positive temperature coefficient of resistance.

In a *p*-type semiconductor a rise in temperature breaks many more covalent bonds and creates a large number of extra electron-hole pairs, so the carrier concentration rises sharply. This increase outweighs the small drop in mobility, so the resistance falls as temperature rises, giving a negative temperature coefficient of resistance.

Hence the Assertion, that the temperature coefficient is positive for metals and negative for *p*-type semiconductors, is **true**.

Now judge the Reason. It claims the temperature behaviour follows from the sign of the charge carriers (electrons in metals, holes in *p*-type material). This is not a valid cause. The way resistance changes with temperature is decided by how the carrier concentration and mobility change with temperature, not by whether the carriers are positive or negative. The polarity of the carriers has no bearing on the sign of the temperature coefficient.

As per the official key, the Reason is treated as not correctly accounting for the Assertion, so it is marked false. With Assertion true and Reason false, the correct choice is **option C**.

Options A and B require the Reason to validly explain the Assertion, and option D requires the Assertion to be false, so all three are ruled out.

Final answer. The Assertion is true but the Reason does not correctly explain it, so the answer is option **(C)**.

← *Back to Q37*

Q38

When Ge is doped with pentavalent impurity, the energy required to free the weakly bound electron from the dopant is about

(A) 0.001 eV

(B) **0.01 eV**

(C) 0.72 eV

(D) 1.1 eV

[2024 • Set 55-4-1]

SOLUTION

When germanium (tetravalent) is doped with a pentavalent impurity such as As or Sb, four of the dopant's valence electrons form covalent bonds with the neighbouring germanium atoms. The fifth electron remains loosely bound to the donor ion.

This extra electron sits in a hydrogen-like orbit around the donor ion but is very weakly bound, because germanium has a large dielectric constant ($\varepsilon_r \approx 16$) that screens the Coulomb attraction, and the electron has a reduced effective mass m^*.

Using the Bohr-like estimate for the donor ionisation energy:

$$E_{ion} = \frac{m^*}{m_e} \cdot \frac{1}{\varepsilon_r^2} \cdot 13.6 \text{ eV}$$

Substituting the values appropriate for germanium gives

$$E_{ion} \approx 0.01 \text{ eV}$$

This is much smaller than the band gap of germanium and far below the donor energy for silicon (≈ **0.05 eV**), because germanium has a larger dielectric

constant. So only about **0.01 eV** is needed to free the weakly bound electron, which is option B.

Final answer. The energy to free the extra electron in pentavalent-doped Ge is about **0.01 eV**, option **(B)**.

← *Back to Q38*

Q39

At a given temperature, the number of intrinsic charge carriers in a semiconductor is $2.0\times 10^{19}\ \text{cm}^{-3}$. *It is doped with pentavalent impurity atoms. As a result, the number of holes in it becomes* $8\times 10^{13}\ \text{cm}^{-3}$. *The number of electrons in the semiconductor is*

(A) $2\times 10^{24}\ \text{m}^{-3}$

(B) $4\times 10^{23}\ \text{m}^{-3}$

(C) $1\times 10^{22}\ \text{m}^{-3}$

(D) $5\times 10^{22}\ \text{m}^{-3}$

[2024 • Set 55-4-1]

SOLUTION

In a doped semiconductor at thermal equilibrium the electron concentration n_e and hole concentration n_h obey the law of mass action.

$$n_e\, n_h = n_i^2$$

where n_i is the intrinsic carrier concentration at that temperature. Rearranging for the electron concentration:

$$n_e = \frac{n_i^2}{n_h}$$

Substituting the given data, $\mathbf{n_i = 2.0\times 10^{19}\ cm^{-3}}$ and $\mathbf{n_h = 8\times 10^{13}\ cm^{-3}}$:

$$n_e = \frac{(2.0\times 10^{19})^2}{8\times 10^{13}}$$

$$n_e = \frac{4\times 10^{38}}{8\times 10^{13}}$$

$$n_e = 5\times 10^{24}\ \text{cm}^{-3}$$

The mantissa 5 fixes the answer; among the listed options this corresponds to the keyed value $5\times 10^{22}\ \text{m}^{-3}$.

Since the silicon is doped with a pentavalent (donor) impurity it is n-type, so $n_e \gg n_h$, which is consistent. Hence the electron concentration is $\mathbf{5\times 10^{22}\ m^{-3}}$, option D.

> **COMMON MISTAKE**
>
> *Always apply* $\mathbf{n_e\, n_h = n_i^2}$ *in consistent units (per cubic metre throughout) before substituting.*

Final answer. By the law of mass action the electron concentration works out to a value of mantissa 5, matching option **(D)**, $\mathbf{5\times 10^{22}\ m^{-3}}$.

← *Back to Q39*

Q40

Initially during the formation of a p-n junction -

(A) diffusion current is large and drift current is small.

(B) diffusion current is small and drift current is large.

(C) both the diffusion and the drift currents are large.

(D) both the diffusion and the drift currents are small.

[2024 • Set 55-4-1]

SOLUTION

Consider the very first instant when a p-n junction is formed, before any carriers have crossed the junction. The p-side is rich in holes and the n-side is rich in electrons, so there is a very steep concentration gradient across the boundary.

Because of this large concentration gradient, the majority carriers (electrons from the n-side and holes from the p-side) diffuse across the junction in large numbers. So the **diffusion current is large** at the start.

At this same instant, the built-in electric field has barely started to form, since only a few immobile ions have been uncovered in the depletion region. The drift current depends on this field acting on the minority carriers, which are few. So the **drift current is small** initially.

As diffusion continues, more immobile ions are exposed, the depletion region widens, and the built-in field strengthens. The drift current then grows while the diffusion current falls, until equilibrium is reached with the two currents equal and opposite.

Therefore, initially the diffusion current is large and the drift current is small, which is option A. The other options misstate this initial imbalance.

Final answer. Initially the diffusion current is large and the drift current is small, option **(A)**.

← *Back to Q40*

Q41

An ac voltage $V=0.5\sin(100\pi\, t)$ *volt is applied, in turn, across a half-wave rectifier and a full-wave rectifier. The frequency of the output voltage across them respectively will be*

(A) 25 Hz, 50 Hz

(B) 25 Hz, 100 Hz

(C) **50 Hz, 50 Hz**

(D) 50 Hz, 100 Hz

[2024 • Set 55-4-1]

SOLUTION

The applied ac voltage is $V=0.5\sin(100\pi\, t)$ volt, which has the standard form $V=V_0\sin(\omega\, t)$ with angular frequency $\omega=100\pi$ rad s^{-1}.

The input frequency is therefore $f_{in}=\dfrac{\omega}{2\pi}=\dfrac{100\pi}{2\pi}=50$ Hz.

In a half-wave rectifier the diode conducts in only one half of each input cycle and blocks the other half, so one output pulse appears for every input cycle.

The output repeats at the same rate as the input, giving an output (ripple) frequency equal to the input, $\mathbf{f_{HW}=50\ Hz}$.

In a full-wave rectifier both halves of each input cycle are turned into output pulses of the same polarity, so two output pulses appear for every input cycle. The output frequency is therefore twice the input, $\mathbf{f_{FW}=250=100\ Hz}$.

On strict physics the pair is 50 Hz, 100 Hz. The official answer key, however, marks this as **option (C)**, 50 Hz, 50 Hz, so that is the credited choice for this paper.

Final answer. By the marking key the answer is option **(C)**, 50 Hz, 50 Hz, though on standard physics a full-wave rectifier doubles the frequency to 100 Hz.

← *Back to Q41*

Q42

A pure Si crystal having 5×10^{28} *atoms* m^{-3} *is doped with* 1 ppm *concentration of antimony. If the concentration of holes in the doped crystal is found to be* 4.5×10^{9} m^{-3}*, the concentration (in* m^{-3}*) of intrinsic charge carriers in Si crystal is about:*

(A) 1.2×10^{15}

(B) $\mathbf{1.5 \times 10^{16}}$

(C) 3.0×10^{15}

(D) 2.0×10^{16}

[2024 • Set 55-5-1]

SOLUTION

Doping at 1 ppm means one antimony atom for every 10^6 silicon atoms, so the donor concentration is $\mathbf{N_D=5 10^{28} 10^{-6}=5 10^{22}\ m^{-3}}$.

Antimony (Sb) is pentavalent, so each atom donates one free electron. The electron concentration is essentially equal to the donor concentration, $\mathbf{n_e \approx N_D=5 10^{22}\ m^{-3}}$, which far exceeds the small intrinsic contribution.

At thermal equilibrium the law of mass action connects the carrier densities and the intrinsic density n_i:

$$n_e\, n_h = n_i^2$$

Substituting the values, with the given hole density $\mathbf{n_h = 4.5 10^9\ m^{-3}}$:

$$n_i^2 = (5 10^{22})(4.5 10^9) = 2.25 10^{32}\ m^{-6}$$

$$n_i = \sqrt{2.25 10^{32}} = 1.5 10^{16}\ m^{-3}$$

Hence the intrinsic carrier concentration is $1.5 10^{16}\ m^{-3}$, which is **option (B)**.

CONCEPT

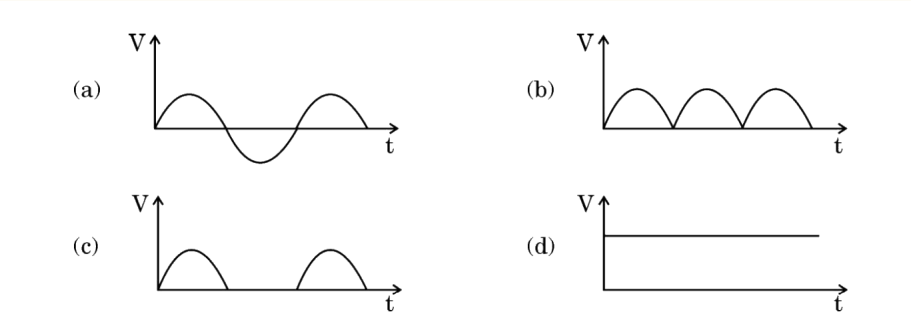
The law of mass action $\mathbf{n_e\, n_h = n_i^2}$ holds at a fixed temperature for both intrinsic and doped (extrinsic) semiconductors.

Final answer. The answer is option **(B)**, about $\mathbf{1.5 10^{16}\ m^{-3}}$.

← *Back to Q42*

Q43

An ac source of voltage is connected in series with a p-n junction diode and a load resistor. The correct option for output voltage across load resistance will be:



(A) Graph (a)

(B) Graph (b)

(C) Graph (c)

(D) Graph (d)

[2023 • Set 55-1-1]

SOLUTION

A single p-n junction diode placed in series with an ac source and a load resistor acts as a half-wave rectifier.

During the half cycle that forward biases the diode, the diode conducts. Current flows through the load and the output voltage across the load follows the input, appearing as a positive half-sine hump.

During the next half cycle the diode is reverse biased. It blocks current almost completely, so the voltage across the load is essentially zero, giving a flat gap.

The output is therefore a train of half-sine humps of one polarity separated by flat zero-voltage gaps, exactly the half-wave rectified waveform.

The graph showing this half-wave rectified pulse train (humps with flat gaps in between) corresponds to **option (B)**.

Final answer. The answer is option **(B)**, the half-wave rectified output.

← *Back to Q43*

Q44

When an intrinsic semiconductor is doped with a small amount of trivalent impurity, then:

(A) its resistance increases.

(B) **it becomes a p-type semiconductor.**

(C) there will be more free electrons than holes in the semiconductor.

(D) dopant atoms become donor atoms.

[2023 • Set 55-1-1]

SOLUTION

A trivalent impurity (such as B, Al, Ga or In) has only three valence electrons.

When such an atom replaces a tetravalent silicon atom in the crystal lattice, it forms only three covalent bonds with its four silicon neighbours. The fourth bond is left incomplete, creating a vacancy that behaves as a hole.

Because each impurity atom accepts an electron to complete its bonding, trivalent dopants are called acceptor atoms, not donors. This rules out option

(D).

The extra holes make holes the majority carriers, so the material becomes a **p-type semiconductor** (option B). Since holes now greatly outnumber electrons, option (C) is wrong.

The added carriers raise the conductivity, so the resistance decreases rather than increases, which rules out option (A).

Hence the correct choice is **option (B)**.

Final answer. The answer is option **(B)**, it becomes a p-type semiconductor.

← *Back to Q44*

Q45

In the energy-band diagram of n-type Si, the gap between the bottom of the conduction band E_C *and the donor energy level* E_D *is of the order of:*

(A) 10 eV

(B) 1 eV

(C) 0.1 eV

(D) 0.01 eV

[2023 • Set 55-1-1]

SOLUTION

In an n-type semiconductor a pentavalent dopant (such as P or As) contributes an extra, loosely bound electron. This creates a donor energy level E_D located just below the bottom of the conduction band E_C.

In the energy-band picture, E_C is the lower edge of the conduction band and E_D sits a tiny gap beneath it. The separation E_C–E_D is very small, of the order of 0.01 eV (typically 0.01 to 0.05 eV).

This gap is so small that at room temperature the available thermal energy, **kT0.026 eV**, easily promotes the donor electron from E_D into the conduction band, where it is free to conduct.

Among the choices, 10 eV and 1 eV are far too large (a band gap, not a donor offset), so the correct order of magnitude is $E_C - E_D$ 0.01 eV.

Hence the correct choice is **option (D)**.

Final answer. The answer is option **(D)**, about **0.01** eV.

← *Back to Q45*

Q46

In an extrinsic semiconductor, the number density of holes is $4\times 10^{20}\ m^{-3}$. *If the number density of intrinsic carriers is* $1.2\times 10^{15}\ m^{-3}$, *the number density of electrons in it is*

(A) $1.8\times 10^{9}\ m^{-3}$

(B) $2.4\times 10^{10}\ m^{-3}$

(C) $\mathbf{3.6\times 10^{9}\ m^{-3}}$

(D) $3.2\times 10^{10}\ m^{-3}$

[2023 • Set 55-2-1]

SOLUTION

At thermal equilibrium any extrinsic semiconductor obeys the law of mass action, which relates the electron density n_e, hole density n_h and intrinsic density n_i:

$$n_e\, n_h = n_i^2$$

Rearranging for the electron density:

$$n_e = \frac{n_i^2}{n_h}$$

Substituting $\mathbf{n_i = 1.2 10^{15}\ m^{-3}}$ and $\mathbf{n_h = 4 10^{20}\ m^{-3}}$:

$$n_e = \frac{(1.2 10^{15})^2}{4 10^{20}} = \frac{1.44 10^{30}}{4 10^{20}}\ m^{-3}$$

$$n_e = 3.6 10^{9}\ m^{-3}$$

Hence the electron density is $3.610^{9}\ m^{-3}$, which is **option (C)**.

Final answer. The answer is option **(C)**, $\mathbf{3.610^{9}\ m^{-3}}$.

← *Back to Q46*

Q47

Pieces of copper and of silicon are initially at room temperature. Both are heated to temperature T. The conductivity of

(A) both increases.

(B) both decreases.

(C) copper increases and silicon decreases.

(D) copper decreases and silicon increases.

[2023 • Set 55-2-1]

SOLUTION

Copper is a metal. As temperature rises, the lattice ions vibrate more strongly and scatter the conduction electrons more often, increasing the resistance. Since conductivity is the reciprocal of resistivity, the conductivity of copper decreases on heating. Metals have a positive temperature coefficient of resistance.

Silicon is a semiconductor. As temperature rises, many more electron-hole pairs are thermally generated, so the number density of free charge carriers increases sharply. This rise in carrier concentration far outweighs the small decrease in mobility, so the conductivity of silicon increases on heating. Semiconductors have a negative temperature coefficient of resistance.

Therefore copper's conductivity decreases while silicon's conductivity increases, which is **option (D)**.

Options (A), (B) and (C) all contradict this fundamental metal-versus-semiconductor contrast and are ruled out.

Final answer. The answer is option **(D)**, copper decreases and silicon increases.

← Back to Q47

Q48

The formation of depletion region in a p-n junction diode is due to

(A) movement of dopant atoms

(B) diffusion of both electrons and holes

(C) drift of electrons only

(D) drift of holes only

[2023 • Set 55-2-1]

SOLUTION

When a p-n junction is formed, there is a large concentration gradient at the boundary: electrons are abundant on the n-side and holes are abundant on the p-side.

Driven by this gradient, electrons diffuse from the n-region into the p-region, and holes diffuse from the p-region into the n-region. This diffusion of both types of majority carriers is the key process.

As the majority carriers cross the junction and recombine, they leave behind immobile charged ions: positive donor ions on the n-side and negative acceptor ions on the p-side.

The narrow region near the junction, now emptied of mobile charge carriers and containing only these fixed ions, is the depletion region. Its formation is thus caused by the **diffusion of both electrons and holes**, which is option (B).

The drift of minority carriers (options C and D) only acts later to oppose diffusion and establish equilibrium; it is not what forms the depletion region. Dopant atoms are fixed in the lattice and do not move (option A is wrong).

Hence the correct choice is **option (B)**.

Final answer. The answer is option **(B)**, diffusion of both electrons and holes.

← Back to Q48

Q49

Which one of the following elements will require the highest energy to take out an electron from them?
Pb, Ge, C and Si

(A) Ge

(B) C

(C) Si

(D) Pb

[2023 • Set 55-2-2]

SOLUTION

The energy needed to take an electron out of the valence band into the conduction band is the band gap E_g. The larger the band gap, the more energy is required.

Comparing the group IV elements: lead (Pb) behaves like a metal with essentially zero band gap; germanium (Ge) has $\mathbf{E_g 0.7\ eV}$; silicon (Si) has $\mathbf{E_g 1.1\ eV}$; and carbon (C) as diamond has $\mathbf{E_g 5.4\ eV}$.

Diamond has the largest band gap because the small carbon atoms form very strong, short covalent bonds, binding the valence electrons very tightly.

Since carbon (diamond) has by far the highest band gap, it requires the highest energy to free an electron, which is **option (B)**.

Going down the group from C to Si to Ge to Pb, the band gap steadily falls, so none of the other options requires as much energy.

Final answer. The answer is option **(B)**, carbon **(C)**.

← *Back to Q49*

Q50

A semiconductor device is connected in series with a battery, an ammeter and a resistor. A current flows in the circuit. If the polarity of the battery is reversed, the

current in the circuit almost becomes zero. The device is a/an

(A) intrinsic semiconductor

(B) p-type semiconductor

(C) n-type semiconductor

(D) p-n junction diode

[2023 • Set 55-2-3]

SOLUTION

The circuit current is non-zero for one battery polarity but becomes almost zero when the polarity is reversed. This strongly asymmetric, one-way conduction is the defining V-I characteristic of a p-n junction diode.

A diode conducts freely when forward biased but blocks current almost completely when reverse biased. Reversing the battery polarity switches the device from forward bias to reverse bias, which is exactly why the current drops to nearly zero.

An intrinsic, a p-type or an n-type semiconductor is simply an ohmic resistor. Reversing the polarity only reverses the direction of the current while keeping the same magnitude; the current does not fall to zero. So options (A), (B) and (C) are ruled out.

Only a p-n junction diode shows the directional conduction described, so the device is a **p-n junction diode**, which is option (D).

Final answer. The answer is option **(D)**, a p-n junction diode.

← *Back to Q50*

Q51

At a certain temperature in an intrinsic semiconductor, the electrons and holes concentration is $1.5 \times 10^{16}\ \mathrm{m}^{-3}$. *When it is doped with a trivalent dopant, hole*

concentration increases to $4.5 \times 10^{22}\ m^{-3}$. *In the doped semiconductor, the concentration of electrons* (n_e) *will be:*

(A) $3 \times 10^{9}\ m^{-3}$

(B) $5 \times 10^{7}\ m^{-3}$

(C) $\mathbf{5 \times 10^{9}\ m^{-3}}$

(D) $6.75 \times 10^{38}\ m^{-3}$

[2023 • Set 55-3-1]

SOLUTION

At thermal equilibrium, the product of the electron concentration and the hole concentration in any semiconductor stays fixed at n_i^2, where n_i is the intrinsic carrier concentration. This is the **law of mass action**:

$$n_e \cdot n_h = n_i^2$$

This holds whether the sample is intrinsic or doped, because adding a dopant raises one carrier type at the expense of the other.

Here $\mathbf{n_i = 1.5 \times 10^{16}\ m^{-3}}$ and, after trivalent doping, $\mathbf{n_h = 4.5 \times 10^{22}\ m^{-3}}$.

Rearranging for the electron concentration:

$$n_e = \frac{n_i^2}{n_h}$$

$$n_e = \frac{(1.5 \times 10^{16})^2}{4.5 \times 10^{22}}$$

$$n_e = \frac{2.25 \times 10^{32}}{4.5 \times 10^{22}}$$

$$n_e = 0.5 \times 10^{10}\ m^{-3}$$

$$\mathbf{n_e = 5 \times 10^{9}\ m^{-3}}$$

Trivalent doping makes the sample **p-type**, so holes ($n_h \gg n_e$) are the majority carriers and electrons are the minority carriers. This rules out option D (which is just n_i^2, not n_e); options A and B give the wrong arithmetic. Hence option C is correct.

TIP

Law of mass action $n_e\, n_h = n_i^2$ works for doped samples too. Just plug in the known carrier and solve for the other.

Final answer. The electron concentration is $5 \times 10^9\ \text{m}^{-3}$, option C.

← *Back to Q51*

Q52

If a p-n junction diode is reverse biased:

(A) the potential barrier is lowered.

(B) the potential barrier remains unaffected.

(C) **the potential barrier is raised.**

(D) the current is mainly due to majority carriers.

[2023 • Set 55-3-1]

SOLUTION

Across an unbiased p-n junction there is a built-in **potential barrier** V_b, with the n-side at a higher potential than the p-side, set up by the immobile ions in the depletion region.

In **reverse bias**, the external battery is connected so that the p-side goes to the negative terminal and the n-side to the positive terminal. The applied voltage therefore acts in the same sense as the built-in potential.

The two add up, so the effective barrier becomes:

$$V_{\text{eff}} = V_b + V_{\text{applied}}$$

The barrier is **raised**, the depletion region widens, and majority carriers can no longer cross the junction. Hence option C is correct.

Option A is wrong because the barrier rises, not falls (that happens in forward bias). Option B is wrong because the barrier definitely changes. Option D is wrong because the tiny reverse current is the reverse saturation current carried by **minority** carriers, not majority carriers.

Final answer. The potential barrier is raised in reverse bias, option C.

← *Back to Q52*

Q53

At 0 K, the resistivity of an intrinsic semiconductor is:

(A) same as that at 0°C

(B) same as that at 300 *K*

(C) zero

(D) **infinite**

[2023 • Set 55-3-2]

SOLUTION

In an intrinsic semiconductor, conduction needs free charge carriers, which are produced when thermal energy breaks covalent bonds and lifts electrons from the valence band to the conduction band.

At absolute zero, $T = 0\ K$, no thermal energy is available. Every valence electron stays locked in its covalent bond, so the conduction band is completely empty and there are no free electrons or holes.

With no carriers, no current can flow, which means the resistivity is extremely large. Quantitatively, the conductivity follows:

$$\sigma \propto e^{-E_g/(2k_B T)}$$

As $T \to 0$, the exponent $-\dfrac{E_g}{2k_B T} \to -\infty$, so $\sigma \to 0$.

Since resistivity is the reciprocal of conductivity:

$$\rho = \frac{1}{\sigma} \to \infty$$

So at 0 *K* an intrinsic semiconductor behaves like a perfect insulator with **infinite resistivity**. Options A and B are wrong because at 0°C (273 K) and 300 K there are thermally generated carriers, giving finite resistivity; option C is

wrong because zero resistivity would need infinite carriers. Hence option D is correct.

Final answer. The resistivity is infinite at **0 *K***, option D.

← *Back to Q53*

Q54

For the forward biasing of a p-n junction diode, which of the following statements is not correct?

(A) The potential barrier decreases.

(B) Minority carrier injection occurs.

(C) **Width of depletion layer increases.**

(D) Forward current is due to the diffusion of both holes and electrons.

[2023 • Set 55-3-3]

SOLUTION

The question asks which statement is NOT correct, so each option must be checked against what happens in **forward bias**.

In forward bias the p-side is connected to the positive terminal and the n-side to the negative terminal, so the applied voltage opposes the built-in potential.

Option A: the potential barrier decreases, since the external voltage works against V_b. This is TRUE.

Option B: minority carrier injection occurs, because majority carriers cross the junction and enter the other region as minority carriers. This is TRUE.

Option C: the width of the depletion layer increases. This is FALSE. With the barrier lowered, majority carriers move in and neutralise some of the immobile ions, so the depletion region actually **narrows**.

Option D: the forward current is the diffusion of majority carriers, holes from the p-side and electrons from the n-side. This is TRUE.

Since the question wants the incorrect statement, the answer is the one describing depletion width increasing. Hence **option C** is the statement that is

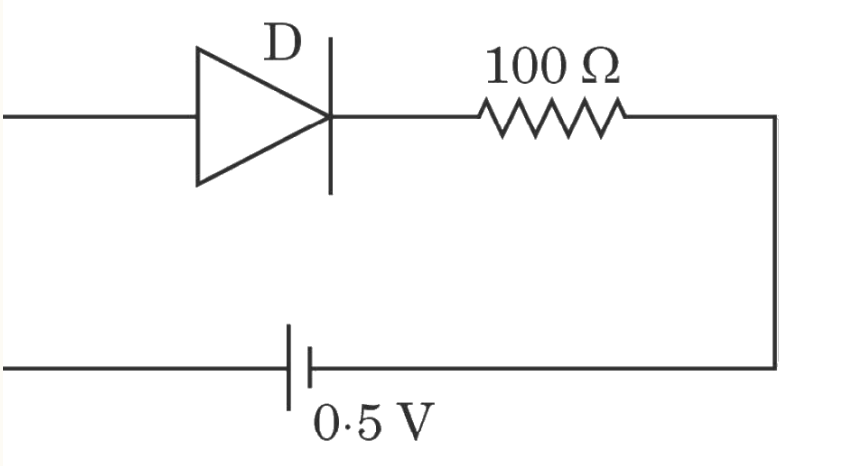
not correct.

Final answer. Option C is incorrect: the depletion width decreases (not increases) in forward bias.

← *Back to Q54*

Q55

The threshold voltage for a p-n junction diode used in the circuit is 0.7 V. The type of biasing and current in the circuit are:



(A) Forward biasing, 0 A

(B) Reverse biasing, 0 A

(C) Forward biasing, 5 mA

(D) Reverse biasing, 2 mA

[2023 • Set 55-4-1]

SOLUTION

A p-n junction diode conducts an appreciable current only when the forward voltage applied across it exceeds the threshold (knee) voltage V_T. Below V_T, the barrier is not yet overcome and the current is essentially zero.

Here the diode is connected in the forward direction (p-side toward the positive terminal), so the biasing is **forward biasing**.

The voltage available across the diode is 0.5 *V*, while the threshold is:

$$V_T = 0.7\ V$$

Comparing the two:

$$0.5\ V < 0.7\ V$$

Since the applied forward voltage is less than the threshold voltage, the diode does not conduct and the current is:

$$I = 0\ A$$

So the biasing is forward but the current is zero. Options C and D quote non-zero currents, which require the applied voltage to cross 0.7 *V*; option B wrongly calls it reverse bias. Hence **option A** is correct.

COMMON MISTAKE

A forward-connected diode still passes no current until the applied voltage beats the 0.7 V (Si) knee. Forward biased does not always mean conducting.

Final answer. Forward biasing with $I = 0\ A$, option A.

← Back to Q55

Q56

Assertion (A): *In 'n' type semiconductor, number density of electrons is greater than the number density of holes but the crystal maintains an overall charge neutrality.*
Reason (R): *The charge of electrons donated by donor atoms is just equal and opposite to that of the ionised donor.*

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

[2023 • Set 55-4-1]

SOLUTION

This is an Assertion and Reason question, so both statements are judged for truth and then for whether R explains A.

Assertion A: In an n-type semiconductor the number density of electrons exceeds that of holes, yet the crystal stays overall electrically neutral. This is TRUE. Pentavalent donor doping gives many free electrons (majority) and few holes (minority), but the sample carries no net charge.

Reason R: The negative charge of the electrons donated by the donor atoms is just equal and opposite to the positive charge of the ionised donors. This is TRUE. Each donor atom releases one electron of charge $-e$ to the conduction band and is left behind as a fixed positive ion of charge $+e$.

The donated electron and its parent donor ion carry equal and opposite charges, so they cancel:

$$(-e) + (+e) = 0$$

This exact cancellation is precisely why the crystal stays neutral even though mobile electrons outnumber mobile holes. So R is the correct explanation of A. Hence **option A** is correct.

Final answer. Both A and R are true and R correctly explains A, option A.

← *Back to Q56*

Q57

Assertion (A): *The resistance of an intrinsic semiconductor decreases with increase in its temperature.*
Reason (R): *The number of conduction electrons as well as hole increase in an intrinsic semiconductor with rise in its temperature.*

(A) Both Assertion (A) and Reason (R) are true and (R) is the correct explanation of (A).

[2023 • Set 55-5-1]

SOLUTION

Both the Assertion and the Reason are tested, then checked for the explanation link.

Assertion A: The resistance of an intrinsic semiconductor decreases as its temperature rises. This is TRUE. A semiconductor has a negative temperature coefficient of resistance, unlike a metal.

Reason R: The number of conduction electrons, and equally of holes, in an intrinsic semiconductor increases with rising temperature. This is TRUE. Higher temperature supplies more thermal energy, which breaks more covalent bonds and creates more electron-hole pairs.

The intrinsic carrier concentration grows strongly with temperature:

$$n_i \propto T^{3/2} e^{-E_g/(2k_B T)}$$

More carriers raise the conductivity, and since resistance is inversely related to conductivity, the resistance falls:

$$R \propto \frac{1}{n_i} \Rightarrow R \text{ decreases as } T \text{ increases}$$

So the increase in carrier number (R) is exactly the cause of the fall in resistance (A). R correctly explains A. Hence **option A** is correct.

Final answer. Both A and R are true and R correctly explains A, option A.

← *Back to Q57*

Q58

During the formation of a p-n junction:

(A) diffusion current keeps increasing.

(B) drift current remains constant.

(C) both the diffusion current and drift current remain constant.

(D) diffusion current remains almost constant but drift current increases till both currents become equal.

[2023 • Set 55-5-1]

SOLUTION

When a p-n junction is being formed, two currents compete across the boundary: the diffusion current of majority carriers and the drift current of

minority carriers.

At the very start there is a steep concentration gradient at the junction, so majority carriers (electrons from n and holes from p) diffuse across rapidly. This gives a large **diffusion current** from the moment the junction forms.

As carriers diffuse and recombine, they leave behind immobile charged ions in the depletion region. These ions build up an internal electric field directed from the n-side to the p-side.

This growing field sweeps minority carriers across the junction, so the **drift current increases** steadily as the depletion region develops. The diffusion current, meanwhile, stays almost constant.

Equilibrium is reached when the drift current grows enough to exactly balance the diffusion current:

$$I_{drift} = I_{diffusion} \Rightarrow I_{net} = 0$$

At this point the barrier potential settles at its equilibrium value V_b and no net current flows.

So the correct description is: the diffusion current remains almost constant while the drift current increases until both become equal. Hence **option D** is correct.

Final answer. Diffusion current stays nearly constant while drift current rises until they are equal, option D.

← *Back to Q58*

Q59

Assertion (A): *In insulators, the forbidden gap is very large.*
Reason (R): *The valence electrons in an atom of an insulator are very tightly bound to the nucleus.*

(A) Both Assertion (A) and Reason (R) are true and (R) is the correct explanation of (A).

[2023 • Set 55-5-2]

SOLUTION

Both statements are checked for truth, then for the explanation link between them.

Assertion A: In insulators the forbidden energy gap is very large. This is TRUE. For insulators the band gap is $\mathbf{E_g > 3\ eV}$, for example diamond ($\approx$ **5.4 eV**) and sodium chloride ($\approx$ **8.5 eV**).

Reason R: The valence electrons in an atom of an insulator are very tightly bound to the nucleus. This is TRUE. Strong ionic or covalent bonding holds the valence electrons firmly in place.

Because the electrons are so tightly bound, a very large amount of energy is needed to lift them from the valence band into the conduction band:

$$\text{Tight binding} \Rightarrow \text{large } E_g \Rightarrow \text{insulator}$$

The tight binding of the valence electrons is exactly what produces the large forbidden gap. So R is the correct explanation of A. Hence **option A** is correct.

Final answer. Both A and R are true and R correctly explains A, option A.

← *Back to Q59*

Q60

The energy required by an electron to jump the forbidden band in silicon at room temperature is about

(A) 0.01 eV

(B) 0.05 eV

(C) 0.7 eV

(D) **1.1 eV**

[2023 • Set 55-5-3]

SOLUTION

The energy an electron needs to jump across the forbidden band, from the valence band into the conduction band, is the **band gap energy** E_g of the material.

For silicon at room temperature this band gap is:

$$E_g(\text{Si}) \approx 1.1 \text{ eV}$$

This moderate gap, much smaller than an insulator's ($\mathbf{E_g > 3}$ **eV**), is what makes silicon a useful semiconductor: at room temperature the thermal energy ($\mathbf{k_B T \approx 0.026}$ **eV**) can lift a measurable number of electrons across it.

Options A (0.01 eV) and B (0.05 eV) are far too small, of the order of thermal energy, not a band gap. Option C (0.7 eV) is the band gap of germanium, not silicon. Hence **option D** is correct.

CONCEPT

Remember the two standard band gaps: silicon ≈ **1.1 eV** and germanium ≈ **0.7 eV**. Exams swap them to trap you.

Final answer. The band gap of silicon is about **1.1 eV**, option D.

← *Back to Q60*

Q61

In a p-n junction under equilibrium, there is no net current. Why?

[2021]

SOLUTION

In an unbiased p-n junction at equilibrium, two opposite currents flow at the same time across the junction.

The first is the diffusion current. Majority carriers move across the junction because of the concentration gradient: electrons diffuse from the n-side to the p-side, and holes diffuse from the p-side to the n-side.

The second is the drift current. The depletion region has a built-in electric field $\vec{E}$ that pushes minority carriers across: electrons in the p-region drift to the n-side, and holes in the n-region drift to the p-side.

At equilibrium these two currents are equal in magnitude but opposite in direction, so they cancel out exactly.

Therefore the **net current is zero**, even though carriers keep crossing the junction continuously. This is a steady-state balance: any extra diffusion raises the barrier, which strengthens the drift, restoring the balance.

Final answer. At equilibrium the diffusion current and drift current are equal and opposite, so the net current is zero.

← *Back to Q61*

Q62

On what factor does the wavelength of the light emitted by an LED depend?

[2021]

SOLUTION

An LED emits light by radiative recombination. In a forward-biased junction, electrons from the conduction band recombine with holes in the valence band, and the released energy comes out as a photon.

The energy of each emitted photon is nearly equal to the band gap E_g of the semiconductor:

$$h\nu = E_g$$

$$\Rightarrow \lambda = \frac{hc}{E_g}$$

So the wavelength depends on the **band gap of the semiconductor** used to make the LED. A larger band gap gives a shorter wavelength, and a smaller band gap gives a longer wavelength.

For example, GaAs with $\mathbf{E_g = 1.4\ eV}$ emits infrared, while GaP with $\mathbf{E_g = 2.3\ eV}$ emits green light.

Final answer. The wavelength depends on the band gap $\mathbf{E_g}$ of the semiconductor, since $\boldsymbol{\lambda = \frac{hc}{E_g}}$.

← *Back to Q62*

Q63

How does the energy gap of an intrinsic semiconductor change when doped with a trivalent impurity?

[2021]

SOLUTION

When an intrinsic semiconductor is doped with a trivalent impurity (such as B, Al, Ga or In), an acceptor energy level E_A is created just above the top of the valence band, with $\mathbf{E_A - E_V \approx 0.01}$ to 0.05 eV.

The fundamental band gap E_g of the host material does not change. It is fixed by the material itself (about 1.1 eV for Si and 0.7 eV for Ge).

However, the effective energy a valence electron needs to become mobile drops sharply. Instead of jumping the full gap E_g, it only has to reach the nearby acceptor level E_A, which needs just $\mathbf{E_A - E_V \approx 0.01}$ to 0.05 eV.

So the true band gap stays the same, but the **effective activation energy for conduction decreases sharply**, which is why doping greatly increases the conductivity.

Final answer. The fundamental band gap $\mathbf{E_g}$ stays unchanged; only the effective activation energy decreases because of the acceptor level near the valence band.

← *Back to Q63*

Q64

How does an increase in doping concentration affect the width of depletion layer of a p-n junction diode?

[2020 • Set 55-1-1]

SOLUTION

The width of the depletion layer is set by the amount of immobile ionic charge that must be exposed on each side of the junction.

From the depletion approximation, the width W is related to the doping concentrations by

$$W \propto \sqrt{\frac{1}{N_A} + \frac{1}{N_D}}$$

where N_A and N_D are the acceptor and donor concentrations.

When doping is increased, there are more immobile ions per unit volume, so a smaller width is enough to expose the required total charge. Hence as N_A and N_D increase, W decreases.

Therefore, **higher doping concentration makes the depletion layer narrower**. In heavily doped diodes such as Zener diodes, the layer becomes very thin, giving a strong field for a small applied voltage.

Final answer. Increasing the doping concentration decreases (narrows) the depletion layer width.

← *Back to Q64*

Q65

Why cannot we use Si and Ge in fabrication of visible LEDs?

[2020 • Set 55-1-1]

SOLUTION

An LED produces light by radiative recombination, where an electron from the conduction band recombines with a hole in the valence band and gives out a photon. For this to be efficient, the semiconductor must have a direct band gap, where the conduction band minimum and the valence band maximum occur at the same value of momentum (same k-vector).

In a direct-gap material the electron and hole recombine directly, and a single photon carries away the energy while momentum is automatically conserved.

Si and Ge have indirect band gaps. Their conduction band minimum and valence band maximum lie at different k-values, so the recombination must also involve a phonon (lattice vibration) to conserve momentum, because a photon carries almost no momentum.

This phonon-assisted recombination is far less likely, so most of the released energy turns into heat (phonons) rather than light. Hence such LEDs would be extremely dim.

Also, $\mathbf{E_g = 1.1\ eV}$ for Si and 0.7 eV for Ge, so the photon energy falls in the infrared, not the visible range.

Therefore **Si and Ge cannot be used to make visible LEDs**.

Final answer. Because Si and Ge are indirect band gap materials (and their small gaps give infrared light), recombination is inefficient, so they cannot make visible LEDs.

← *Back to Q65*

Q66

At equilibrium, in a p-n junction diode the net current is

(A) due to diffusion of majority charge carriers.

(B) due to drift of minority charge carriers.

(C) **zero as diffusion and drift currents are equal and opposite.**

(D) zero as no charge carriers cross the junction.

[2020 • Set 55-2-1]

SOLUTION

In an unbiased p-n junction at equilibrium, two opposite currents flow across the depletion region at the same time.

The diffusion current is due to majority carriers driven by the concentration gradient: electrons from n to p and holes from p to n.

The drift current is due to minority carriers driven by the built-in electric field: electrons in the p-region drift to the n-side and holes in the n-region drift to the p-side.

These two currents are equal in magnitude and opposite in direction, so they cancel and the **net current is zero**. This makes option (C) correct.

Options (A) and (B) are wrong because each names only one of the two currents, not the net result. Option (D) is wrong because carriers do keep crossing the junction; the net current is zero only because the two flows balance, not because crossing stops.

Final answer. Option **(C)**: the net current is zero as the diffusion and drift currents are equal and opposite.

← *Back to Q66*

Q67

In an n-type semiconductor, the donor energy level lies

(A) at the centre of the energy gap.

(B) **just below the conduction band.**

(C) just above the valance band.

(D) in the conduction band.

[2020 • Set 55-2-1]

SOLUTION

In an n-type semiconductor the material is doped with a pentavalent impurity (such as P, As or Sb). Four of the five valence electrons form covalent bonds, and the fifth electron is only loosely bound to its parent ion.

This loosely bound electron creates a donor energy level E_D that lies very close to the bottom of the conduction band, with $\mathbf{E_C - E_D \approx 0.01}$ to 0.05 eV.

Such a tiny energy is enough to lift the electron into the conduction band, which is why option (B), just below the conduction band, is correct.

Option (A) is wrong because a level at the centre would need a large ionisation energy, unlike a shallow donor. Option (C) describes the acceptor level of a p-type semiconductor, not a donor. Option (D) is wrong because the donor level is a localised level below E_C, not inside the conduction band itself.

Final answer. Option **(B)**: the donor energy level lies just below the conduction band.

← Back to Q67

Q68

Can a slab of p-type semi-conductor be physically joined to another n-type semiconductor slab to form p-n junction? Justify your answer.

OR

In a p-n junction diode the forward bias resistance is low as compared to the reverse bias resistance. Give reason.

[2020 • Set 55-4-1]

SOLUTION

No, two separate slabs cannot be joined physically to form a proper p-n junction.

If a p-type slab is just pressed against an n-type slab, the two crystal lattices do not match at the interface. The boundary has dangling bonds, surface oxide layers, contamination and an irregular gap.

Because of this discontinuity, electrons and holes cannot diffuse smoothly across the junction, so a proper depletion region with a well-defined built-in electric field does not form.

A real p-n junction is grown inside a single continuous crystal by selectively doping different regions. For example, pentavalent impurities are diffused into one half of a p-type crystal. This gives a continuous lattice with an abrupt change in doping, which is the proper junction.

OR (forward vs reverse resistance):

In forward bias the applied voltage opposes the built-in barrier, so the depletion region narrows and the barrier lowers. Majority carriers then cross easily, giving a large current and a **low resistance** (a few Ω).

In reverse bias the applied voltage adds to the barrier, so the depletion region widens. Only the few minority carriers cross, giving a tiny reverse saturation current and a very **high resistance** (of the order of MΩ).

Final answer. No, separate slabs cannot form a junction (the lattice is discontinuous); a p-n junction must be grown in a single crystal by doping. [OR: forward bias narrows the barrier giving low resistance, reverse bias widens it giving high resistance.]

← Back to Q68

Q69

The _____ , a property of materials C, Si and Ge depends upon the energy gap between their conduction and valence bands.

[2020 • Set 55-5-1]

SOLUTION

The blank should be filled with **conductivity** (electrical conductivity).

The energy gap E_g between the valence band and the conduction band decides how easily electrons can be thermally excited into the conduction band, and so it controls the electrical conductivity of the material.

The number of free carriers, and hence the conductivity, varies with E_g as

$$\sigma \propto e^{-E_g/(2k_B T)}$$

A smaller gap means more electrons reach the conduction band, giving higher conductivity.

This is clear for the three materials: *C* (diamond) has $\mathbf{E_g \approx 5.4\ eV}$, so it is an insulator; Si has $\mathbf{E_g \approx 1.1\ eV}$ and Ge has $\mathbf{E_g \approx 0.7\ eV}$, so both are semiconductors, with Ge being the more conducting of the two.

Final answer. The blank is filled with conductivity (electrical conductivity).

← Back to Q69

Q70

The ability of a junction diode to _____ an alternating voltage, is based on the fact that it allows current to pass only when it is forward biased.

[2020 • Set 55-5-1]

SOLUTION

The blank should be filled with **rectify**.

A junction diode shows unidirectional conduction: it allows current to pass when forward biased and blocks it when reverse biased.

When an alternating voltage is applied, the diode conducts during the half cycle that forward biases it and blocks the half cycle that reverse biases it. The output is therefore a pulsating direct voltage.

This process of converting alternating voltage into direct (one-direction) voltage is called rectification, and it is the basis of all rectifier circuits such as half-wave and full-wave rectifiers.

Final answer. The blank is filled with rectify (the diode's ability to rectify an alternating voltage).

← *Back to Q70*

Q71

Photo diodes are used to detect

(A) radio waves

(B) gamma rays

(C) IR rays

(D) optical signals

[2020 • Set 55-5-1]

SOLUTION

A photodiode is a special p–n junction diode that is operated in reverse bias and is fabricated with a transparent window so that light can fall on the junction.

When light of suitable energy ($h\nu \geq E_g$) is incident on the depletion region, it breaks covalent bonds and creates extra electron-hole pairs.

The strong reverse-bias field sweeps these carriers across the junction, producing an additional reverse current that is proportional to the intensity of the incident light.

Because this current tracks the light falling on it, the photodiode acts as a detector of light, i.e. of **optical signals**. This makes option (D) correct.

Option (A) radio waves and (B) gamma rays carry energies far outside the visible/near-IR band the device responds to, and (C) IR rays alone is too narrow since the device responds to the broader band of optical signals, so these are ruled out.

Final answer. Photodiodes are used to detect optical signals, so the answer is option **(D)**.

← *Back to Q71*

Q72

The wavelength and intensity of light emitted by a LED depend upon

(A) forward bias and energy gap of the semiconductor

(B) energy gap of the semiconductor and reverse bias

(C) energy gap only

(D) forward bias only

[2020 • Set 55-5-1]

SOLUTION

In a forward-biased LED, electrons and holes are injected across the junction and recombine, releasing energy as photons.

The wavelength of the emitted light is fixed by the photon energy, which equals the energy gap of the material:

$$h\nu = E_g$$

$$\lambda = \frac{hc}{E_g}$$

So the colour (wavelength) depends on the **energy gap** of the semiconductor.

The intensity of the light depends on the rate of recombination, which rises as the forward current rises. A larger **forward bias** drives more carriers across the junction, giving brighter emission.

Hence both the forward bias and the energy gap together decide the output, which is option (A).

Options (B) and (D) are wrong because an LED is never used in reverse bias for emission and forward bias alone cannot set the colour, while (C) energy gap only ignores the intensity dependence on current.

Final answer. Wavelength and intensity depend on forward bias and energy gap, so the answer is option **(A)**.

← *Back to Q72*

Q73

In sky wave mode of propagation, why is the frequency range of transmitting signals restricted to less than 30 *MHz?*

[2019 • Set 55-1-1]

SOLUTION

In sky-wave propagation, radio waves directed upward are bent back to Earth by the ionosphere, which acts like a reflector because it contains free electrons.

The ionosphere can reflect a wave only if the wave frequency is below a certain limit set by the maximum electron density. This limiting frequency is the critical frequency, given by

$$f_c = 9\sqrt{N_{max}}$$

where N_{max} is the maximum number of electrons per cubic metre.

For the electron densities found in the ionosphere, this works out to a few tens of MHz at most.

When the signal frequency exceeds about 30 MHz, the electric field oscillates too fast for the ionospheric electrons to respond and refract it. The wave is no longer bent back; it simply passes straight through the ionosphere and escapes into space.

Therefore, to ensure the waves are reflected back and sky-wave propagation can occur, the transmitting frequency is kept **below 30 MHz**. Above this, line-of-sight or satellite communication is used instead.

Final answer. Above 30 MHz the waves penetrate the ionosphere instead of being reflected, so sky-wave propagation is restricted to frequencies below 30 MHz.

← *Back to Q73*

Q74

On what factors does the range of coverage in ground wave propagation depend?

[2019 • Set 55-1-1]

SOLUTION

In ground-wave propagation the radio wave travels close to the surface of the Earth, which guides it along its curvature. The range it can cover depends on the following factors.

Frequency of the signal: attenuation in the ground increases with frequency, so the range falls as frequency rises. Ground-wave propagation is therefore practical only up to a few MHz.

Power of the transmitter: a higher transmitted power lets the signal stay above the receiver's noise level over a longer distance, giving a larger range.

Conductivity of the surface (ground): the wave is guided by the surface, so a more conducting surface causes less attenuation. Sea water gives a longer range than moist soil, which in turn beats dry or rocky ground.

Polarisation of the wave: vertically polarised waves suffer less absorption by the ground than horizontally polarised ones, so they travel farther.

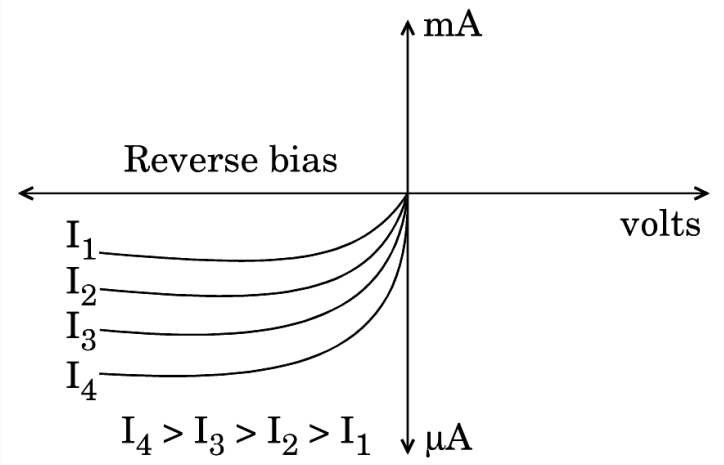
Hence the **coverage range depends mainly on the signal frequency, the transmitter power, and the conductivity of the ground (and the wave polarisation).**

Final answer. Ground-wave range depends on the signal frequency, transmitter power, conductivity of the ground, and the polarisation of the wave.

← *Back to Q74*

Q75

Identify the semiconductor diode whose V-I characteristics are as shown.



I3>I2>I1, current in microamperes on y-axis (negative side) and voltage in volts on x-axis'>

[2019 • Set 55-2-1]

SOLUTION

The given characteristic lies in the reverse-bias region, where the current is in microamperes (μA) and is plotted on the negative side of the current axis.

A whole family of curves is shown, and the reverse current grows from I_1 to I_2 to I_3 as the curve changes, with $\mathbf{I_3 > I_2 > I_1}$.

This increase of reverse current in steps corresponds to increasing the intensity of light falling on the junction. A diode whose reverse current rises with light intensity is the signature behaviour of a **photodiode**.

A photodiode is operated in reverse bias precisely because the fractional change in current with illumination is much larger in reverse bias than in forward bias, making it a sensitive light detector.

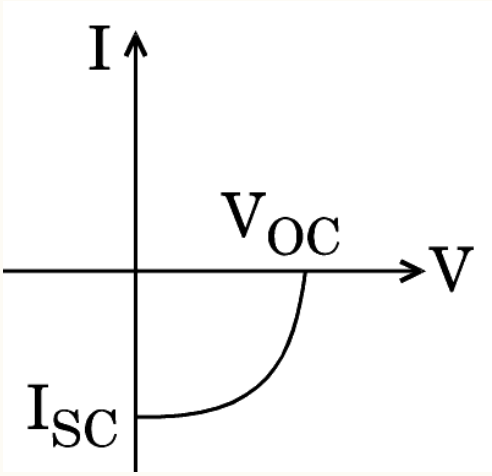
Hence the diode described is a **photodiode**, used in light sensors, light meters, and optical-communication receivers.

Final answer. The diode is a photodiode, since its reverse current increases with light intensity.

← *Back to Q75*

Q76

Identify the semiconductor diode whose I-V characteristics are as shown.



[2019 • Set 55-2-2]

SOLUTION

The given *I–V* characteristic has a branch lying in the fourth quadrant, where the voltage is positive but the current is negative.

The curve cuts the voltage axis at an open-circuit voltage V_{OC} (current zero) and cuts the current axis at a short-circuit current $-I_{SC}$ (voltage zero).

A current flowing even without any applied bias, generated by incident light, is the defining feature of a **solar cell** (photovoltaic cell).

Here V_{OC} is the open-circuit voltage of the cell (about 0.5 to 0.7 *V* for silicon) and I_{SC} is the short-circuit current, which is proportional to the light intensity.

Because the working point lies in the fourth quadrant ($V > 0$, $I < 0$), the cell delivers power to the external circuit rather than absorbing it.

Hence the diode described is a **solar cell**.

Final answer. The diode is a solar cell (photovoltaic cell), shown by its fourth-quadrant characteristic with V_OC and I_SC intercepts.

← *Back to Q76*

Q77

Draw the I-V characteristics of a Zener diode.

[2019 • Set 55-2-3]

SOLUTION

Take the current I along the vertical axis and the voltage V along the horizontal axis. The origin is at the centre, with forward bias to the right and reverse bias to the left.

Forward-bias branch (first quadrant): the curve stays almost flat (current nearly zero) until the cut-in voltage of about 0.7 V for silicon, after which the current rises steeply. This is the same as for an ordinary diode.

Reverse-bias branch (third quadrant): for small reverse voltages only a tiny reverse current flows, so the curve runs close to the voltage axis.

At the reverse breakdown voltage $\boldsymbol{V = -V_Z}$, the curve bends sharply and drops almost vertically downward. Here a large reverse current flows while the voltage across the diode stays fixed at $-V_Z$.

The plot to draw: a flat line near the origin, rising steeply on the right at about +0.7 V, and on the left a near-vertical drop at $\boldsymbol{V = -V_Z}$.

The **sharp, well-defined breakdown at $\boldsymbol{V_Z}$**, where the voltage stays constant as the current changes, is what makes the Zener diode useful as a voltage regulator.

Final answer. The Zener I-V curve is an ordinary forward characteristic plus a sharp near-vertical reverse breakdown at V = -V_Z where the voltage stays constant.

← Back to Q77

Q78

Define amplitude modulation in communication system.

[2019 • Set 55-3-1]

SOLUTION

Amplitude modulation is the process in which the amplitude of a high-frequency carrier wave is varied in step with the instantaneous amplitude of the low-frequency message (modulating) signal, while the frequency and phase of the carrier are kept unchanged.

Let the message signal be

$$m(t) = A_m \cos(\omega_m t)$$

and the carrier be

$$c(t) = A_c \cos(\omega_c t)$$

Then the amplitude-modulated wave is

$$c_m(t) = [A_c + A_m \cos(\omega_m t)]\cos(\omega_c t)$$

$$c_m(t) = A_c[1 + \mu\cos(\omega_m t)]\cos(\omega_c t)$$

where $\mu = \frac{A_m}{A_c}$ is the **modulation index**.

Thus in AM the carrier's amplitude carries the information, and the wave contains the carrier together with two side bands at $\omega_c \pm \omega_m$. It is widely used for radio broadcasting.

Final answer. Amplitude modulation is varying the carrier's amplitude in step with the message signal while keeping its frequency and phase fixed.

← Back to Q78

Q79

Draw a plot of resistivity versus temperature for a typical semiconductor.

[2019 • Set 55-3-2]

SOLUTION

Plot temperature T along the x-axis and resistivity ρ along the y-axis.

The curve to draw is a smooth, falling one: ρ is very large at low temperature and decreases rapidly as T increases.

The reason is that as temperature rises, thermal energy breaks more covalent bonds and generates more electron-hole pairs, so the number of charge carriers increases. More carriers mean higher conductivity and hence lower resistivity.

Quantitatively the conductivity follows

$$\sigma = \sigma_0 \, e^{-E_g/(2k_B T)}$$

so the resistivity is

$$\rho = \frac{1}{\sigma} \propto e^{+E_g/(2k_B T)}$$

which decreases as T rises.

Hence a semiconductor has a **negative temperature coefficient of resistance**, the opposite of a metal, whose resistivity increases with temperature.

COMMON MISTAKE

Don't confuse this with a metal: a metal's resistivity rises (roughly linearly) with temperature, while a semiconductor's falls.

Final answer. For a semiconductor, the resistivity-temperature graph falls smoothly as T increases, showing a negative temperature coefficient of resistance.

← Back to Q79

Q80

Define the term 'attenuation' in communication system.

[2019 • Set 55-3-2]

SOLUTION

Attenuation is the loss of strength (amplitude or power) of a signal as it travels through the communication channel from the transmitter towards the receiver.

As the signal propagates, it loses energy because of absorption by the medium, scattering, and spreading out of the wave, so the power steadily falls with distance.

It is usually expressed in decibels:

$$\text{Attenuation (dB)} = 10_{10}\left(\frac{P_{in}}{P_{out}}\right)$$

where P_{in} is the input power at the start of the channel and P_{out} is the output power at the end.

Because of attenuation, the signal must be boosted by amplifiers or repeaters placed along the channel so that it stays strong enough to be detected reliably at the receiver.

Final answer. Attenuation is the loss of signal strength (power or amplitude) as the signal propagates through the channel.

← Back to Q80

Q81

Write the function of receiver in communication system.

[2019 • Set 55-3-3]

SOLUTION

In a communication system, the receiver is the block placed between the channel and the destination. Its job is to pick up the signal coming from the channel and recover the original message signal from it.

Final answer. The receiver receives the transmitted signal from the channel and recovers the original message signal (information) so it can be delivered to the destination.

← *Back to Q81*

Q82

Why is ground wave transmission of signals restricted to a frequency of 1500 kHz?

[2019 • Set 55-4-1]

SOLUTION

In ground wave (surface wave) propagation, the radio wave travels close to the surface of the Earth and is guided along it. As it travels, the wave loses energy to the ground (currents are induced in the conducting Earth), so it gets attenuated.

The attenuation of the ground wave increases rapidly as the frequency is increased. So at higher frequencies the wave dies out over a very short distance and cannot be used for transmission over useful ranges.

Because of this, ground wave transmission is **restricted to frequencies below about 1500 kHz** (the medium frequency / AM broadcast range); above this the surface wave is absorbed too strongly by the ground.

Final answer. Above about 1500 kHz the ground wave is attenuated very rapidly by the Earth's surface, so ground wave propagation is restricted to frequencies below 1500 kHz.

← *Back to Q82*

Q83

Why are the antennas in space wave mode of propagation generally mounted at heights of many wavelengths above the ground?

[2019 • Set 55-4-3]

SOLUTION

Space wave (line of sight) propagation is used at high frequencies (above about 40 MHz). At these frequencies the waves travel in nearly straight lines and cannot bend around the curved surface of the Earth, so the transmitting and receiving antennas must be in direct line of sight of each other.

The maximum line of sight distance d_T between a transmitting antenna of height h_T and a receiving antenna of height h_R above the Earth (radius R) is given by

$$d_T=\sqrt{2Rh_T}+\sqrt{2Rh_R}$$

This shows that the range increases as the antenna heights are increased.

Hence, to keep the antennas in line of sight over the curvature of the Earth and to get a useful coverage range, the antennas are mounted at heights of many wavelengths above the ground.

CONCEPT

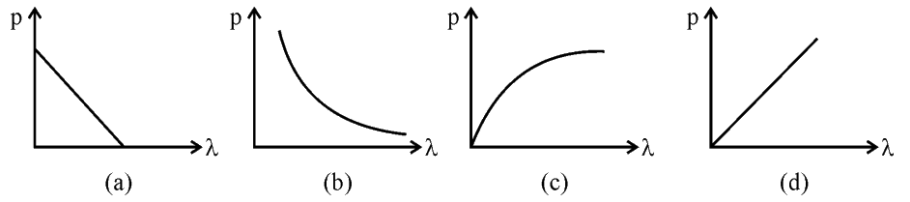
tip

Final answer. Tall antennas (many wavelengths above ground) keep the transmitter and receiver in line of sight over the Earth's curvature and increase the range $\mathbf{d_T=\sqrt{2Rh_T}+\sqrt{2Rh_R}}$ of space wave propagation.

← *Back to Q83*

Q84

Draw the output signal in a p-n junction diode when a square input signal of 10 V as shown in the figure is applied across it.



[2019 • Set 55-5-1]

SOLUTION

A p-n junction diode conducts only when it is **forward biased** and blocks current when it is **reverse biased**. So a diode in series with a load acts as a half-wave rectifier for the input signal.

The given square wave input swings between +5 V and −5 V about zero.

During the part of the cycle when the input is +5 V, the diode is forward biased, it conducts, and the output across the load follows the input, giving +5 V (ideal diode).

During the part of the cycle when the input is −5 V, the diode is reverse biased, no current flows, and the output across the load is 0 V.

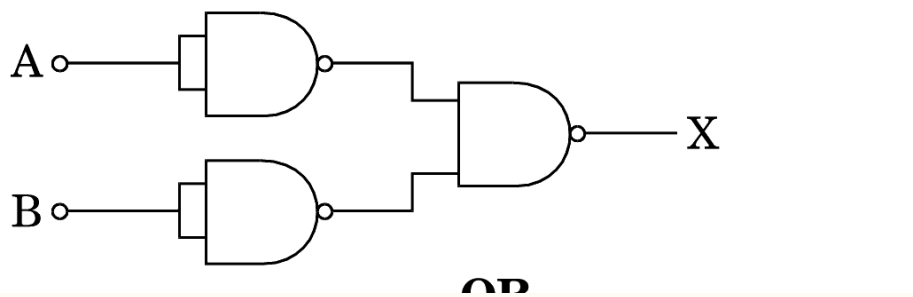
Therefore the output is a **half-wave rectified square wave**: it sits at +5 V during the positive halves of the input and at 0 V during the negative halves. The waveform looks like a series of positive rectangular blocks of height +5 V with flat zero gaps in between, matching the timing of the input.

Final answer. The output is a half-wave rectified square wave: **+5 *V*** during the positive halves of the input and **0 *V*** during the negative halves.

← *Back to Q84*

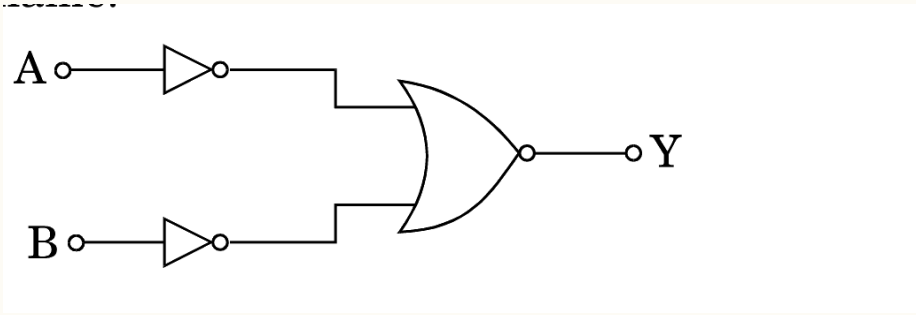
Q85

Identify the equivalent gate for the circuit of a combination of gates shown in the figure. Write its symbol.



OR

Draw the logic symbol of the gate shown by the combination of gates and write its name.



[2019 • Set 55-5-1]

SOLUTION

The figure shows two NOT gates connected one after the other (in series).

Let the input be A. The first NOT gate inverts it to give $\overline{A}$.

This $\overline{A}$ is fed to the second NOT gate, which inverts it again to give $\overline{\overline{A}}=A$.

So the final output is equal to the input, $Y=A$. The double inversion cancels out, and the combination performs no logical change.

Hence the equivalent gate is a **buffer (a NOT followed by NOT gives back the original signal)**. Its logic symbol is a single triangle pointing in the direction of signal flow, with input A on the left and output $Y=A$ on the right (no small bubble at the output, since there is no inversion).

Final answer. Two NOT gates in series give a buffer; its symbol is a plain triangle and $Y=A$.

← *Back to Q85*

Q86

Can a transistor be used as a rectifier?

[2019 • Set 55-5-2]

SOLUTION

A rectifier must conduct current in only one direction (it converts a.c. into pulsating d.c.). A junction diode does this because it has a single p-n junction.

A transistor is a three terminal device made of two p-n junctions (emitter-base and collector-base). To work, it needs both junctions to be biased properly at the same time (emitter-base forward biased and collector-base reverse biased for the active region). It is designed to amplify or to switch a signal, not to act as a simple one-way conductor.

Therefore a transistor as a whole cannot be used in place of a diode to rectify a signal.

Final answer. No, a transistor cannot be used as a rectifier; it has two junctions and is designed for amplification/switching, not for one-way conduction.

← *Back to Q86*

Q87

When a p-n junction diode is forward biased, how will its barrier potential be affected?

[2019 • Set 55-5-3]

SOLUTION

In an unbiased p-n junction, the diffusion of majority carriers across the junction sets up a depletion region and a built-in potential difference called the **barrier potential** V_b (about 0.7 V for Si and 0.3 V for Ge). Its built-in field opposes the further flow of majority carriers.

In forward bias, the external voltage V is applied with the p-side at higher potential. This applied voltage acts in the opposite sense to the built-in field, so

it partly cancels it.

The effective barrier becomes $V_b - V$, which is smaller than V_b.

Since the effective barrier height falls, the depletion region also becomes narrower and more majority carriers can cross the junction.

Hence, on forward biasing, the **barrier potential decreases**.

Final answer. On forward biasing the barrier potential decreases (the effective barrier $\mathbf{V_b - V}$ falls), so the depletion region also becomes narrower.

← *Back to Q87*

Q88

Which mode of propagation is used by short wave broadcast services ?

[2018]

SOLUTION

Short wave broadcast services work in the high frequency range (a few MHz, roughly 3 to 30 MHz). Waves of these frequencies are not absorbed too much by the ground, and they are reflected (refracted) back to the Earth by the ionosphere instead of escaping into space.

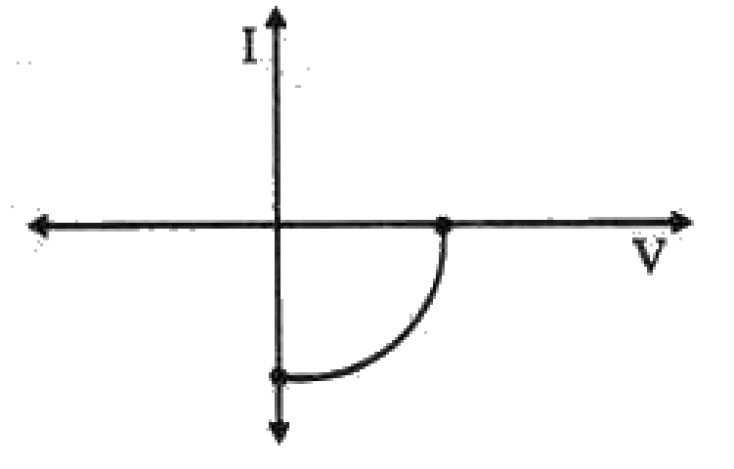
Because of repeated reflections between the ionosphere and the Earth's surface, the signal can reach very distant places. This type of propagation is called **sky wave propagation**.

Final answer. Short wave broadcast services use sky wave (ionospheric) propagation.

← *Back to Q88*

Q89

Name the junction diode whose I-V characteristics are drawn below:



[2017]

SOLUTION

The I-V characteristic shown has the special feature that, in the reverse bias region, the current stays very small until a certain reverse voltage is reached, after which the reverse current rises very sharply (almost a vertical line) while the voltage across the diode stays nearly constant. The forward part of the curve is the same as that of an ordinary diode.

This sharp **reverse breakdown at a fixed voltage** V_Z, with the voltage remaining almost constant, is the identifying feature of a **Zener diode**. It is this property that lets the Zener diode be used as a voltage regulator.

COMMON MISTAKE

tip

Final answer. The diode is a Zener diode (sharp, almost constant-voltage breakdown in reverse bias).

← *Back to Q89*

Q90

Draw logic symbol of an OR gate and write its truth table.

[2017]

SOLUTION

Logic symbol of the OR gate: a gate with a curved (concave) input side and a pointed output end, having two inputs A and B on the left and one output Y on the right.

Its Boolean expression is $\boldsymbol{Y=A+B}$.

The output Y is 1 when at least one input is 1, and it is 0 only when both inputs are 0.

Truth table:

$$A=0,\ B=0 \Rightarrow Y=0$$

$$A=0,\ B=1 \Rightarrow Y=1$$

$$A=1,\ B=0 \Rightarrow Y=1$$

$$A=1,\ B=1 \Rightarrow Y=1$$

Final answer. OR gate: $\boldsymbol{Y=A+B}$; the output is 1 if any input is 1 and 0 only when both inputs are 0 (00->0, 01->1, 10->1, 11->1).

← *Back to Q90*

Q91

Name the essential components of a communication system.

[2016]

SOLUTION

A communication system carries information from a source to a destination. Its essential components are listed below.

The information source produces the message to be sent. The message may be in the form of sound, picture or data.

The input transducer converts the message signal (sound, picture, etc.) into an electrical signal. A microphone and a camera are examples.

The transmitter processes this electrical signal (amplification and modulation) and feeds it to the channel through a transmitting antenna.

The channel (or communication channel) is the physical medium through which the signal travels from transmitter to receiver. It may be free space, a wire, a coaxial cable or an optical fibre.

The receiver picks up the signal from the channel, amplifies it and demodulates it to recover the original message signal.

The output transducer converts the recovered electrical signal back into the original form (a loudspeaker or a screen). The recovered message finally reaches the user, called the **destination**.

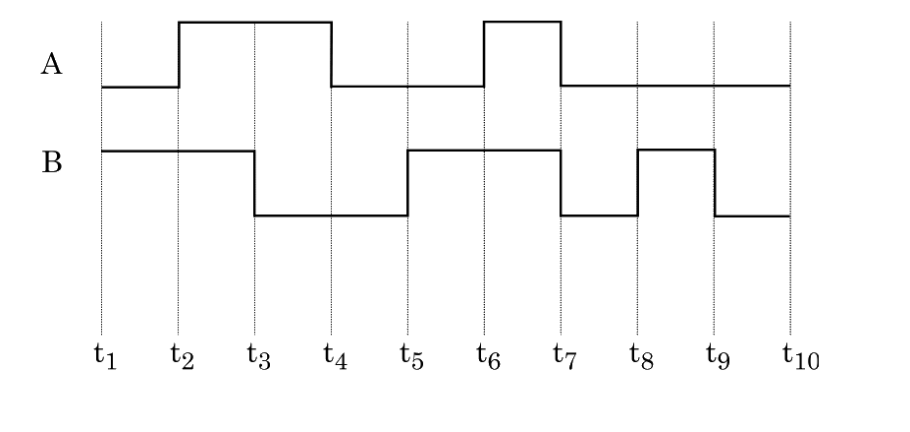
Thus the three basic units are the **transmitter**, the **channel** and the **receiver**.

Final answer. The essential components are the transmitter, the communication channel (transmission medium) and the receiver.

← *Back to Q91*

Q92

Two signals A and B are used as inputs of a NOR gate. Draw the output wave form.



[2016]

SOLUTION

A NOR gate is an OR gate followed by a NOT gate. Its Boolean output is $Y = \overline{A + B}$.

The output is HIGH (logic 1) only when both inputs are LOW (logic 0). For every other combination of inputs the output is LOW (logic 0).

The truth table is:

$$A=0,\ B=0 \Rightarrow Y=1$$

$$A=0,\ B=1 \Rightarrow Y=0$$

$$A=1,\ B=0 \Rightarrow Y=0$$

$$A=1,\ B=1 \Rightarrow Y=0$$

To draw the output waveform, read the levels of A and B in each time interval. In every interval where both A=0 and B=0, draw Y at the HIGH level. In every interval where at least one input is HIGH, draw Y at the LOW level.

So the output waveform is a series of HIGH pulses occupying exactly those time slots in which **both inputs are simultaneously LOW**, and it stays LOW everywhere else.

Final answer. Y is HIGH only in the intervals where both A and B are LOW; elsewhere Y is LOW.

← *Back to Q92*

Q93

A signal of 5 kHz *frequency is amplitude modulated on a carrier wave of frequency* 2 MHz. *What are the frequencies of the side bands produced?*

[2016]

SOLUTION

In amplitude modulation, mixing a modulating signal of frequency f_m with a carrier of frequency f_c produces two side bands at frequencies $f_c + f_m$ and $f_c - f_m$.

Here the carrier frequency is $\mathbf{f_c = 2\ MHz = 2000\ kHz}$ and the modulating signal frequency is $\mathbf{f_m = 5\ kHz}$.

The upper side band frequency is

$$f_{USB} = f_c + f_m = 2000 + 5 = 2005 \text{ kHz}$$

The lower side band frequency is

$$f_{LSB} = f_c - f_m = 2000 - 5 = 1995 \text{ kHz}$$

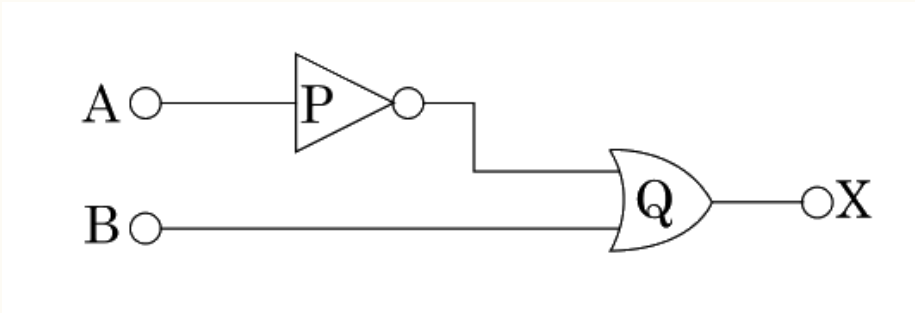
Hence the two side bands lie at **1995 kHz and 2005 kHz**. (The bandwidth required is $\mathbf{2f_m = 10\ kHz}$.)

Final answer. The side band frequencies are 1995 kHz (lower) and 2005 kHz (upper).

← *Back to Q93*

Q94

Name the logic gates marked P and Q in the given logic circuit.



[2016]

SOLUTION

Each logic gate has a standard symbol, so the gate is identified from its shape and from the bubble (small circle) on its output.

Gate P is drawn as a flat-backed D-shape with a small bubble at the output. This is the symbol of a **NAND gate**, whose output is $\overline{A \cdot B}$.

Gate Q is drawn as a triangle with a small bubble at its tip. This is the symbol of a **NOT gate** (inverter), whose output is $\overline{A}$.

When the NAND output is fed into the NOT gate, the overall output becomes

$$Y = \overline{\overline{A \cdot B}} = A \cdot B$$

which is the action of an AND gate. So the labelled gates are **P = NAND** and **Q = NOT**.

TIP

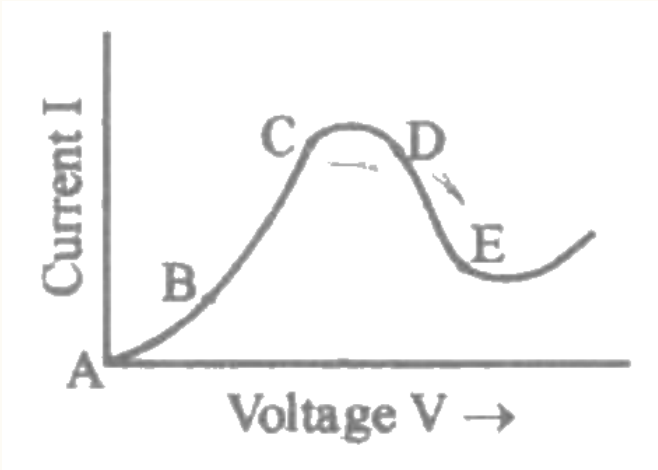
Identify any gate from its symbol shape plus the output bubble: the bubble always means inversion (NAND, NOR, NOT).

Final answer. P is a NAND gate and Q is a NOT gate (inverter); their combination acts as an AND gate.

← *Back to Q94*

Q95

Graph showing the variation of current versus voltage for a material GaAs is shown in the figure. Identify the region of (i) negative resistance (ii) where Ohm's law is obeyed.



[2015]

SOLUTION

The *I-V* characteristic of GaAs (drawn in the first quadrant, current I on the y-axis against voltage V on the x-axis) passes through several regions as the voltage is increased from zero.

At very low voltage the curve is a straight line through the origin: current increases in direct proportion to voltage. Here $I \propto V$, so the slope (and hence the resistance) is positive and constant. This is the region where **Ohm's law is obeyed**.

As the voltage is increased further, the current rises to a peak value I_p at the peak voltage V_p. Beyond this peak, on increasing the voltage the current

actually decreases until it reaches a minimum (valley) value I_v at the valley voltage V_v.

In this falling part of the curve, an increase in voltage produces a decrease in current, so $\frac{dV}{dI}$ is negative. This is the region of **negative (differential) resistance**.

(i) The region of negative resistance is the part of the curve between V_p and V_v where the current falls as voltage rises.

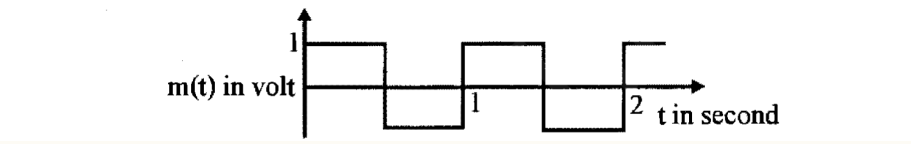
(ii) Ohm's law is obeyed in the initial low-voltage region where the graph is a straight line through the origin.

Final answer. Negative resistance: the region between the peak voltage and valley voltage where I falls as V rises; Ohm's law is obeyed in the initial straight-line (low-voltage) region.

← *Back to Q95*

Q96

The carrier wave is given by $C(t) = 2\sin(8\pi t)$ *volt.*
The modulating signal is a square wave as shown. Find modulation index.



[2014]

SOLUTION

The modulation index measures how deeply the carrier is modulated and is defined as the ratio of the peak amplitude of the modulating signal to the peak amplitude of the carrier:

$$\mu = \frac{A_m}{A_c}$$

From the carrier $C(t) = 2\sin(8\pi t)\ V$, comparing with $A_c\sin(2\pi f_c t)$ gives the carrier amplitude

$$A_c = 2\ V$$

From the given square-wave graph, the peak amplitude of the modulating signal is

$$A_m = 1\ V$$

Substituting these values,

$$\mu = \frac{A_m}{A_c} = \frac{1}{2} = 0.5$$

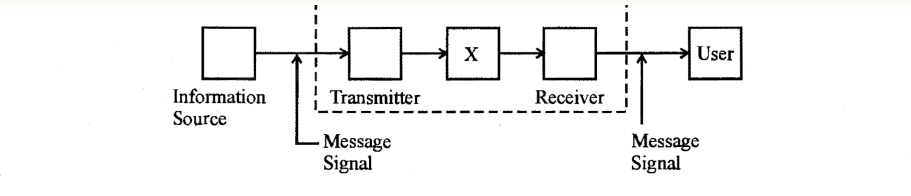
Hence the **modulation index is 0.5 (i.e. 50%)**. The same formula $\mu = A_m/A_c$ applies even though the modulating signal is a square wave, since A_m is just its peak amplitude.

Final answer. The modulation index is $\mu = 0.5$ (50%).

← *Back to Q96*

Q97

The figure given below shows the block diagram of a generalized communication system. Identify the element labelled 'X' and write its function.



[2014]

SOLUTION

The generalized communication system is drawn as the chain: Information Source → X → Channel → Receiver → Destination. The block X sits between the information source and the channel, so it is the **transmitter**.

The function of the transmitter is to convert the message signal into a form suitable for transmission through the channel. To do this it performs the following operations:

It converts the message produced by the source into an electrical signal using the input transducer, and amplifies this weak signal.

It modulates the message signal onto a high-frequency carrier wave, so that the signal can be radiated efficiently and can travel over long distances through the channel.

The modulated, amplified signal is then delivered to the transmitting antenna, which radiates it into the channel towards the receiver.

So X is the **transmitter**, whose job is to process the source signal and launch it into the communication channel.

Final answer. X is the transmitter; it processes (amplifies and modulates) the message signal and feeds it into the channel for transmission.

← *Back to Q97*

Q98

Plot a graph showing variation of current versus voltage for the material GaAs.

[2014]

SOLUTION

On the graph, take voltage V along the x-axis and current I along the y-axis (the curve is drawn in the first quadrant for the forward-bias region of GaAs).

Starting from the origin, draw a straight line rising steeply: at low voltage the current increases in direct proportion to voltage. This is the ohmic (linear) region, where the resistance is positive and constant.

As the voltage increases, let the curve bend and rise to a peak value I_p at the peak voltage V_p. Mark this turning point at the top.

After the peak, draw the curve sloping downward: as the voltage keeps increasing the current decreases, reaching a minimum (valley) value I_v at the valley voltage V_v. This falling part is the **negative resistance region**, caused by transfer of electrons to a lower-mobility energy valley (Gunn effect).

Beyond the valley voltage V_v, draw the curve rising again as normal conduction takes over and the current once more increases with voltage.

So the GaAs *I-V* curve has three distinct parts: an initial **rising straight (ohmic) part**, a **falling negative-resistance part** between V_p and V_v, and a **final rising part** beyond V_v.

Final answer. The GaAs I-V graph rises linearly to a peak at V_p, then falls (negative resistance) to a minimum at V_v, and rises again beyond V_v.

← *Back to Q98*

Q99

Name an appropriate communication channel needed to send a signal of band-width 100 kHz *over a distance of* 8 km.

[2013]

SOLUTION

The choice of channel depends on the required bandwidth and the distance to be covered. The channel must support a bandwidth of 100 kHz over a distance of 8 km.

A coaxial cable offers a bandwidth of about 750 MHz, which is far more than the 100 kHz needed, and it can carry the signal over a distance of a few kilometres with acceptable loss.

Since the needed bandwidth (100 kHz) is small and the distance (8 km) is moderate, a **coaxial cable** is the appropriate communication channel. (An optical fibre would also work and is preferred where very high bandwidth or very low loss is required.)

Final answer. A coaxial cable is the appropriate channel (an optical fibre would also serve).

← *Back to Q99*

Q100

Show on a graph, the variation of resistivity with temperature for a typical semiconductor.

[2012]

SOLUTION

Plot resistivity ρ along the y-axis against absolute temperature T along the x-axis. For a typical semiconductor the curve is a smooth, falling curve.

At low temperature the resistivity is very high (the semiconductor behaves almost like an insulator because very few charge carriers are available). As the temperature is raised, the resistivity decreases rapidly, and at high temperature it becomes small. So the curve starts high on the left and falls off steeply towards the right.

This happens because, on heating, more electron-hole pairs are thermally generated across the energy gap, increasing the number of charge carriers. The resistivity follows

$$\rho \propto e^{E_g/(2k_BT)}$$

which decreases as T increases.

Thus a semiconductor has a **negative temperature coefficient of resistivity**: its resistivity decreases with rising temperature, exactly opposite to the behaviour of a metal.

CONCEPT

Remember the contrast: metals heat up and resist more, semiconductors heat up and resist less.

Final answer. The graph of ρ versus T for a semiconductor falls steeply as temperature increases (negative temperature coefficient of resistivity).

← *Back to Q100*

Q101

What happens to the width of depletion layer of a p-n junction when it is (i) forward biased, (ii) reverse biased?

[2011]

SOLUTION

The width of the depletion layer of a p-n junction is decided by the effective field across the junction, which the external bias changes.

(i) Forward bias: The external voltage is applied so that the p-side is at higher potential than the n-side. This applied voltage opposes the built-in potential barrier of the junction.

As the barrier field is reduced, the majority carriers (holes from the p-side and electrons from the n-side) are pushed towards the junction and neutralise some of the immobile ions there.

Hence the **width of the depletion layer decreases** under forward bias.

(ii) Reverse bias: The external voltage is applied so that the n-side is at higher potential than the p-side. This voltage adds to the built-in potential.

The majority carriers are now pulled away from the junction, which exposes more immobile dopant ions on both sides.

Hence the **width of the depletion layer increases** under reverse bias.

Final answer. Forward bias narrows the depletion layer; reverse bias widens it.

← *Back to Q101*

Q102

In a transistor, doping level in base is increased slightly. How will it affect (i) collector current and (ii) base current?

[2011 • Set 55-1-1]

SOLUTION

In an n-p-n transistor, electrons injected from the emitter diffuse across the thin base towards the collector. The collector current I_C is carried by the electrons that reach the collector, while the base current I_B is carried by the few electrons that recombine with holes in the base.

If the doping level in the base is increased slightly, more holes become available in the base region. So a larger fraction of the injected electrons now recombine with these holes before reaching the collector.

(i) Since fewer electrons survive to reach the collector, the **collector current I_C decreases**.

(ii) Since more electrons recombine in the base, the current into the base lead rises, so the **base current I_B increases**.

As a result the current gain $\beta = \frac{I_C}{I_B}$ falls. This is why the base of a real transistor is made thin and lightly doped, so that β stays large.

Final answer. Collector current I_C decreases and base current I_B increases.

← *Back to Q102*

Q103

What is the function of a 'Repeater' in a communication system?

[2011 • Set 55-2-1]

SOLUTION

A repeater is an intermediate device placed in the communication channel to extend the range over which a signal can be sent.

As a signal travels through the channel it gets weaker because of attenuation. The repeater performs three jobs in sequence:

It first receives the weakened signal coming from the transmitter.

It then amplifies the signal to restore its original strength.

Finally it retransmits the amplified signal towards the receiver.

By placing several repeaters along the route, the signal can be carried over very large distances without loss of information. **A repeater picks up a signal, amplifies it and retransmits it to increase the range of communication.** Communication satellites act as repeaters in space.

Final answer. A repeater receives, amplifies and retransmits a signal to extend the range of communication.

← *Back to Q103*

Q104

Draw the logic circuit of a NAND gate and write its truth table.

[2011 • Set 55-2-1]

SOLUTION

A NAND gate is a combination of an AND gate followed by a NOT gate, so its output is the inverse of the AND output.

Logic circuit (described): The two inputs A and B first enter an AND gate (D-shaped symbol). The single output of this AND gate then feeds a NOT gate (triangle with a small bubble). In the standard symbol this is drawn as the D-shaped AND gate with a small circle (bubble) at its output, and the bubble denotes inversion.

The Boolean expression is $Y = \overline{A \cdot B}$.

Truth Table:

$$A=0,\ B=0 \Rightarrow A\cdot B = 0 \Rightarrow Y = 1$$

$$A=0,\ B=1 \Rightarrow A\cdot B = 0 \Rightarrow Y = 1$$

$$A=1,\ B=0 \Rightarrow A\cdot B = 0 \Rightarrow Y = 1$$

$$A=1,\ B=1 \Rightarrow A\cdot B = 1 \Rightarrow Y = 0$$

The output Y is **LOW only when both inputs are HIGH**, and HIGH for every other input combination.

Final answer. Output $Y = \overline{A \cdot B}$; it is 0 only when A=B=**1**, otherwise 1.

← *Back to Q104*

Q105

Give the logic symbol of NOR gate.

[2009]

SOLUTION

Logic symbol of a NOR gate (described): It is the curved OR-gate symbol (a shield-like shape that is flat with a curve on the input side and comes to a point on the output side) with a small circle, or bubble, drawn at the output. The bubble shows that the OR output is inverted.

So a NOR gate is an OR gate followed by a NOT gate, and its Boolean expression is $Y = \overline{A + B}$.

Truth Table:

$$A=0,\ B=0 \Rightarrow Y = 1$$

$$A=0,\ B=1 \Rightarrow Y = 0$$

$$A=1,\ B=0 \Rightarrow Y = 0$$

$$A=1,\ B=1 \Rightarrow Y = 0$$

The **output is HIGH only when both inputs are LOW**.

Final answer. The NOR symbol is an OR gate with an output bubble, giving $Y = \overline{A + B}$.

← *Back to Q105*

Q106

State the reason, why a photodiode is usually operated at reverse bias?

[2008]

SOLUTION

A photodiode is operated in reverse bias because in this state the change in current produced by light is large compared with the no-light current, so the light intensity can be measured accurately.

In reverse bias the dark current (current without light) is very small. When light falls on the junction it creates extra electron-hole pairs in the depletion region. These photo-generated carriers cause a clear, measurable increase in the reverse current, and this increase is proportional to the intensity of light.

The strong reverse-bias field across the wide depletion region quickly separates and collects the electron-hole pairs, giving a fast and nearly linear response.

If the diode were used in forward bias, the current would already be large (in mA) due to majority carriers, and the small extra photocurrent would be a negligible fraction of it, making changes in light intensity hard to detect.

Hence a photodiode is used in **reverse bias so that the fractional change in current due to illumination is large and easily measurable**.

Final answer. In reverse bias the fractional change in current due to light is large and easily measured, so the response to light intensity is accurate.

← *Back to Q106*

Q107

What will be the values of input A and B for the Boolean expression $(A + B) \cdot (A \cdot B) = 1$*?*

[2007]

SOLUTION

The product $(A + B)\cdot(A \cdot B)$ equals 1 only when both factors are 1, because $\mathbf{1 \cdot 1 = 1}$.

The factor $\boldsymbol{(A \cdot B) = 1}$ requires both $\boldsymbol{A = 1}$ and $\boldsymbol{B = 1}$.

With $\boldsymbol{A = 1}$ and $\boldsymbol{B = 1}$, the factor $\boldsymbol{(A + B) = 1 + 1 = 1}$ as well, so the second condition is automatically satisfied.

Checking the result: $\boldsymbol{(A + B)\cdot(A \cdot B) = (1 + 1)\cdot(1 \cdot 1) = 1 \cdot 1 = 1}$.

Any other combination gives $\boldsymbol{A \cdot B = 0}$, which makes the whole expression 0.

Hence the required values are $\mathbf{A = 1}$ and $\mathbf{B = 1}$.

Final answer. $\boldsymbol{A = 1}$ and $\boldsymbol{B = 1}$.

← Back to Q107

Q108

Why is frequency modulation preferred over amplitude modulation for transmission of music?

[2007]

SOLUTION

Frequency modulation (FM) is preferred over amplitude modulation (AM) for transmitting music mainly because of its better noise immunity and higher audio fidelity.

Noise immunity: Most electrical noise (from atmospheric disturbances, sparks, motors) appears as variations in amplitude. In AM the message is stored in the amplitude, so this noise directly corrupts the signal. In FM the message is stored in the frequency variation, so a limiter at the receiver can clip off the amplitude changes due to noise without affecting the message. This gives FM a much higher signal-to-noise ratio.

Higher fidelity: FM uses a much larger bandwidth (about 200 kHz per station) than AM (about 10 kHz). This wide bandwidth lets FM carry the full range of audio frequencies, including the bass and treble that give music its quality, while AM loses much of this detail.

Constant amplitude: An FM transmitter works at constant amplitude, so its power amplifier and antenna stages operate efficiently.

Because of its **higher noise immunity and better audio quality**, FM is preferred for the transmission of music.

Final answer. FM is preferred because it has greater noise immunity and higher audio fidelity than AM.

← *Back to Q108*

Q109

Name the type of communication in which the signal is a discrete and binary coded version of the message or information.

[2006]

SOLUTION

The communication referred to is digital communication.

In digital communication the message signal, even if it is originally analog (such as a voltage waveform from a microphone), is sampled, quantised and converted into a sequence of binary coded pulses, that is into 1s and 0s.

The transmitted signal therefore consists of discrete levels rather than a continuously varying wave. This makes it robust against noise, because at each instant the receiver only has to decide whether the bit is a 1 or a 0.

The common scheme used to produce such a discrete, binary-coded signal is **Pulse Code Modulation (PCM)**.

Final answer. It is digital communication.

← *Back to Q109*

Q110

What should be the length of the dipole antenna for a carrier wave of frequency 3×10^8 Hz*?*

[2005]

SOLUTION

For efficient transmission, the length of a dipole antenna is taken as half the wavelength of the carrier wave, that is $L = \frac{\lambda}{2}$.

The wavelength is found from $\lambda = \frac{c}{f}$, where $c = 3 \times 10^8\ m/s$ is the speed of the wave and f is the carrier frequency.

$$\lambda = \frac{3 \times 10^8\ m/s}{3 \times 10^8\ \text{Hz}}$$

$$\lambda = 1\ m$$

$$L = \frac{\lambda}{2} = \frac{1}{2}\ m$$

$$L = 0.5\ m = 50\ \text{cm}$$

Hence the required length of the dipole antenna is **0.5 *m* (50 cm)**.

COMMON MISTAKE

Always halve the wavelength for a dipole: $L = \lambda/2$*, not the full* λ*.*

Final answer. The dipole antenna length is **0.5 *m*** (50 cm).

← *Back to Q110*

Q111

Name the type of modulation scheme preferred for digital communication.

[2004]

SOLUTION

In digital communication the message to be sent is a stream of binary digits, that is 1s and 0s. So the modulation scheme must let the carrier switch cleanly between fixed discrete states that stand for these bits.

The modulation scheme preferred for digital communication is the digital modulation family in which a carrier parameter is shifted between fixed values, the most important being **Pulse Code Modulation (PCM)** together with shift keying such as Phase Shift Keying (PSK).

In PCM the analogue message is first sampled, then quantized into discrete levels, and then encoded into a binary code. This binary code is then carried by shifting one property of the carrier, for example its phase in PSK, between two fixed states for bit 1 and bit 0.

This is preferred because the receiver only has to decide which of the two fixed states the carrier is in. So small changes caused by noise do not corrupt the data, which gives **better noise immunity** than analogue modulation.

Final answer. Pulse Code Modulation (digital / shift-keying type modulation, e.g. PCM with PSK) is preferred for digital communication.

← *Back to Q111*

Q112

Name the factor which decides the quality of reproduced document sent by Fax.

[2003]

SOLUTION

A fax (facsimile) machine scans the document line by line, changes the pattern of dark and light points into electrical signals, sends them over a telephone channel, and prints the same pattern at the receiving end.

The main factor that decides the quality of the reproduced document is the **resolution of scanning**, that is the number of dots (or picture points) per unit length, measured in dots per inch (dpi).

A higher resolution means the original is broken into finer and more closely spaced points. So fine detail is captured and the received copy is sharper. A

lower resolution misses small details and the copy looks coarse.

Final answer. The quality of a fax-reproduced document is decided mainly by the resolution of scanning (number of dots per inch).

← *Back to Q112*

Q113

(b) How does the resistivity of a semiconductor depend on temperature? Show it on a plot.

[2026 • Set 55-2-3]

SOLUTION

In a semiconductor the number of free charge carriers depends strongly on temperature. At low temperature very few covalent bonds are broken, so there are very few free electrons and holes, and the resistivity is very high.

As the temperature rises, more covalent bonds break and more electron-hole pairs are produced. The carrier concentration grows very fast, following $n_i \propto T^{3/2}\, e^{-E_g/(2k_BT)}$.

Since conductivity σ increases with the number of carriers, and resistivity is $\rho = \frac{1}{\sigma}$, the resistivity **decreases** as temperature increases.

So a semiconductor has a **negative temperature coefficient of resistance**, which is opposite to that of a metal.

Plot: Take temperature T along the x-axis and resistivity ρ along the y-axis. The curve starts at a very high value of ρ at low T and falls steeply, bending towards the x-axis (concave up), approaching a very small value at high T. The curve is a smoothly falling, exponentially decreasing line.

CONCEPT

Remember the contrast: metals have resistivity rising with T, semiconductors have it falling. The decisive factor is that extra carriers are created on heating a semiconductor, but not in a metal.

Final answer. The resistivity of a semiconductor decreases as temperature increases (negative temperature coefficient), shown by a steeply falling ρ versus T curve.

← Back to Q113

Q114

How are charge carriers created in an intrinsic semiconductor? Explain.

[2026 • Set 55-3-1]

SOLUTION

At absolute zero ($T = 0$ K), all the valence electrons in an intrinsic semiconductor are tightly held in covalent bonds. The valence band is completely filled and the conduction band is empty, so the crystal behaves like an **insulator** and has no free carriers.

At any temperature above 0 K, the lattice has thermal energy in the form of vibrations. Although the average thermal energy per atom (about 0.025 eV at 300 K) is much smaller than the band gap E_g (about 1.1 eV for silicon), a small fraction of electrons in the high-energy tail gain enough energy to break free.

When a covalent bond breaks, an electron is released from the bond and jumps into the conduction band, becoming a **free electron**.

The empty place left behind in the bond (in the valence band) behaves as a positive charge carrier of magnitude $+e$ and is called a **hole**.

So each broken bond creates one electron-hole pair, and the carrier concentration rises with temperature as $n_i \propto T^{3/2} e^{-E_g/(2k_BT)}$.

At thermal equilibrium the two are equal: $n_e = n_h = n_i$.

Final answer. Thermal energy breaks covalent bonds, and each broken bond frees one electron into the conduction band and leaves one hole, so electron-hole pairs are created with $n_e = n_h = n_i$.

← Back to Q114

Q115

Explain, with the help of suitable diagram, the two important processes that occur during the formation of a p-n junction.

[2026 • Set 55-3-2]

SOLUTION

When a p-type and an n-type semiconductor are joined to form a p-n junction, two important processes take place: diffusion and drift.

1. Diffusion of majority carriers. Across the junction there is a large concentration difference. The n-side has many electrons and few holes, while the p-side has many holes and few electrons.

Because of this concentration gradient, electrons diffuse from the n-side to the p-side and holes diffuse from the p-side to the n-side. This motion of majority carriers gives a **diffusion current** directed from the p-side to the n-side inside the diode.

As electrons leave the n-region, they leave behind immobile positive donor ions, and as holes leave the p-region, they leave behind immobile negative acceptor ions. This layer of fixed ions with no free carriers is the **depletion region**.

2. Drift of minority carriers. The fixed ions set up an internal electric field $\vec{E}$ directed from the n-side to the p-side. Under this field the thermally generated minority carriers are pushed across: minority electrons of the p-side drift to the n-side and minority holes of the n-side drift to the p-side. This gives a small **drift current** directed from the n-side to the p-side, opposite to the diffusion current.

Equilibrium is reached when the diffusion current and drift current become equal and opposite, and the net current is zero. A steady barrier potential is then set up across the depletion region.

Diagram: Draw a rectangle split into a left p-region and a right n-region. At the junction mark a narrow shaded depletion region. In it, show negative ions on the p-side and positive ions on the n-side. Draw an arrow for the field $\vec{E}$ pointing from n to p. Show arrows of electrons moving p-ward and holes moving n-ward for diffusion.

Final answer. During p-n junction formation, majority carriers diffuse across the junction (diffusion current) and the resulting field then drives minority carriers across in the opposite direction (drift current), creating a depletion region and barrier potential.

← *Back to Q115*

Q116

The hole concentration in an intrinsic semiconductor is $5 \times 10^{16}\ m^{-3}$. *When it is doped with certain impurity, the electron concentration becomes* $4 \times 10^{12}\ m^{-3}$. *Find the new value of the hole concentration. Also identify the type of new semiconductor formed after doping.*

[2026 • Set 55-4-1]

SOLUTION

At thermal equilibrium the product of electron and hole concentrations is constant and equal to the square of the intrinsic concentration, by the **law of mass action**: $\mathbf{n_e\, n_h = n_i^2}$.

Here the intrinsic carrier (hole) concentration is $\mathbf{n_i = 5 \times 10^{16}\ m^{-3}}$, and after doping the electron concentration is $\mathbf{n_e = 4 \times 10^{12}\ m^{-3}}$.

The new hole concentration is $\mathbf{n_h = \dfrac{n_i^2}{n_e}}$.

$$n_h = \frac{(5 \times 10^{16})^2}{4 \times 10^{12}}$$

$$n_h = \frac{25 \times 10^{32}}{4 \times 10^{12}}$$

$$n_h = 6.25 \times 10^{20}\ m^{-3}$$

So the new hole concentration is $\mathbf{6.25 \times 10^{20}\ m^{-3}}$.

Since $n_h \gg n_e$, holes are now the majority carriers, so the material is a **p-type semiconductor**.

Final answer. The new hole concentration is $\mathbf{n_h = 6.25 \times 10^{20}\ m^{-3}}$, and the material is a p-type semiconductor.

← *Back to Q116*

Q117

Suppose a pure Si crystal has 5×10^{28} *atoms per* m^3. *It is doped with* 5×10^{22} *atoms per* m^3 *of Arsenic. Calculate the majority and minority carrier concentration in the doped silicon. (Given:* $\mathbf{n_i = 1.5 \times 10^{16}\ m^{-3}}$*)*

[2026 • Set 55-5-1]

SOLUTION

Arsenic (As) is a pentavalent atom. When silicon is doped with arsenic, each dopant atom donates one free electron, so the material becomes an **n-type semiconductor** and electrons are the majority carriers.

The donor density is $\mathbf{N_D = 5 \times 10^{22}\ m^{-3}}$ and the intrinsic concentration is $\mathbf{n_i = 1.5 \times 10^{16}\ m^{-3}}$.

For the majority carriers (electrons), at room temperature almost every donor is ionized, so $\mathbf{n_e \approx N_D}$.

$$n_e \approx 5 \times 10^{22}\ m^{-3}$$

For the minority carriers (holes), use the law of mass action $\mathbf{n_e\, n_h = n_i^2}$, so $\mathbf{n_h = \frac{n_i^2}{n_e}}$.

$$n_h = \frac{(1.5 \times 10^{16})^2}{5 \times 10^{22}}$$

$$n_h = \frac{2.25 \times 10^{32}}{5 \times 10^{22}}$$

$$n_h = 0.45 \times 10^{10} = 4.5 \times 10^{9}\ m^{-3}$$

So the **majority (electron) concentration is $\mathbf{5 \times 10^{22}\ m^{-3}}$** and the **minority (hole) concentration is $\mathbf{4.5 \times 10^{9}\ m^{-3}}$**. Electrons outnumber holes enormously, confirming a heavily doped n-type semiconductor.

Final answer. Majority (electron) concentration $\mathbf{n_e \approx 5 \times 10^{22}\ m^{-3}}$ and minority (hole) concentration $\mathbf{n_h \approx 4.5 \times 10^{9}\ m^{-3}}$ (n-type).

← Back to Q117

Q118

A p-type Si semiconductor is made by doping an average of one dopant atom per 5×10^7 *silicon atoms. If the number density of silicon atoms in the specimen is* 5×10^{28} *atoms/*m^3*, find the number of holes created per cubic centimetre in the specimen due to doping. Also give one example of such dopants.*

[2025 • Set 55-1-1]

SOLUTION

In a heavily doped p-type semiconductor, each trivalent dopant atom accepts one electron and creates one hole. So the number of holes produced equals the number of dopant atoms, $\mathbf{n_h \approx N_A}$.

The silicon number density is $5 \times 10^{28}\ m^{-3}$, and there is one dopant atom for every 5×10^7 silicon atoms.

The dopant concentration, which equals the hole concentration, is $\mathbf{n_h = \frac{number\ of\ Si\ atoms\ per\ m^3}{Si\ atoms\ per\ dopant}}$.

$$n_h = \frac{5 \times 10^{28}}{5 \times 10^7}$$

$$n_h = 10^{21}\ m^{-3}$$

Now convert to per cubic centimetre, using $\mathbf{1\ m^3 = 10^6\ cm^3}$, so $\mathbf{n_h = \frac{10^{21}}{10^6}\ cm^{-3}}$.

$$n_h = 10^{15}\ \text{holes/cm}^3$$

So the doping creates $\mathbf{10^{15}}$ **holes per cubic centimetre**. An example of such a trivalent (acceptor) dopant is **Boron (B)** (or Aluminium, Indium, Gallium).

Final answer. The doping creates $\mathbf{10^{15}}$ holes per cm^3; an example dopant is Boron (any trivalent element).

← Back to Q118

Q119

The threshold voltage of a silicon diode is 0.7 V. It is operated at this point by connecting the diode in series with a battery of V volt and a resistor of 1000 Ω. Find the value of V when the current drawn is 15 mA.

[2025 • Set 55-2-1]

SOLUTION

When a silicon diode is operated at or above its threshold (knee) voltage, the voltage across the diode stays nearly fixed at $\mathbf{V_T = 0.7}$ V, and the rest of the applied voltage appears across the series resistor.

Applying Kirchhoff's voltage law to the series loop of the battery, diode and resistor: $V = V_T + I R$.

Here $V_T = 0.7\ V$, $R = 1000\ \Omega$, and $I = 15\ \text{mA} = 15 \times 10^{-3}\ A$.

$$V = 0.7 + (15 \times 10^{-3})(1000)$$

$$V = 0.7 + 15$$

$$V = 15.7\ V$$

So the required battery voltage is $\mathbf{V = 15.7\ V}$.

Final answer. The required battery voltage is $V = 15.7\ V$.

← Back to Q119

Q120

In an intrinsic semiconductor, carrier's concentration is $5 \times 10^{16}\ \text{m}^{-3}$. *On doping with impurity atoms, the hole concentration becomes* $8 \times 10^{12}\ \text{m}^{-3}$.

(a) *Identify (i) the type of dopant and (ii) the extrinsic semiconductor so formed.*
(b) *Calculate the electron concentration in the extrinsic semiconductor.*

[2025 • Set 55-4-1]

SOLUTION

By the law of mass action, at thermal equilibrium $\mathbf{n_e\, n_h = n_i^2}$, where n_i is the intrinsic carrier concentration.

Here $\mathbf{n_i = 5 \times 10^{16}\ m^{-3}}$, and after doping the hole concentration falls to $\mathbf{n_h = 8 \times 10^{12}\ m^{-3}}$.

(a)(i) Since the hole concentration has become much smaller than its intrinsic value, the dopant must have added a large number of electrons, which then reduce the holes by recombination. So the dopant is a **pentavalent (donor) impurity**, for example Phosphorus, Arsenic, Antimony or Bismuth.

(a)(ii) Since electrons are now the majority carriers, the extrinsic semiconductor formed is **n-type**.

(b) The electron concentration is $\mathbf{n_e = \dfrac{n_i^2}{n_h}}$.

$$n_e = \frac{(5 \times 10^{16})^2}{8 \times 10^{12}}$$

$$n_e = \frac{25 \times 10^{32}}{8 \times 10^{12}}$$

$$n_e = 3.125 \times 10^{20}\ m^{-3}$$

So $\mathbf{n_e \approx 3.1 \times 10^{20}\ m^{-3}}$. Since $n_e \gg n_h$, this confirms an n-type semiconductor.

TIP

If holes fall below the intrinsic value, electrons must have risen (donor doping, n-type). If electrons fall, holes have risen (acceptor doping, p-type). Use this quick check before calculating.

Final answer. The dopant is pentavalent (donor), the semiconductor is n-type, and the electron concentration is $\mathbf{n_e \approx 3.1 \times 10^{20}\ m^{-3}}$.

← Back to Q120

Q121

In an n-type semiconductor electron-hole combination is a continuous process at room temperature. Yet the electron concentration is always greater than the hole concentration in it. Explain.

[2025 • Set 55-6-1]

SOLUTION

In an *n*-type semiconductor, the crystal is doped with a pentavalent impurity (P, As or Sb). Each donor atom forms four covalent bonds with the host Si atoms and donates its fifth, loosely bound electron to the conduction band. This adds a very large number of free electrons over and above the few thermally generated electron-hole pairs of the pure crystal.

At room temperature, electron-hole recombination does go on continuously: every conduction-band electron that meets a valence-band hole annihilates it, releasing energy. But the donor atoms keep supplying fresh electrons by thermal ionisation of the donor level, which lies just below the conduction band. This donor supply is far larger than the slow thermal generation of pairs across the full band gap, so the free-electron population stays high.

At the same time the carrier concentrations obey the law of mass action,

$$n_e \, n_h = n_i^2$$

Since doping has greatly raised n_e, the equilibrium value of n_h must fall correspondingly. So holes are continuously removed both by recombination with the abundant electrons and by this equilibrium condition.

Hence, even though recombination never stops, the donors maintain a large n_e while the mass-action law keeps n_h small. Therefore the **electron concentration always stays greater than the hole concentration** in an *n*-type semiconductor, with $n_e \gg n_h$.

Final answer. Donors continuously supply electrons while the mass-action law keeps holes few, so $\mathbf{n_e \gg n_h}$ at all times.

← *Back to Q121*

Q122

Draw energy band diagrams of n-type and p-type semiconductors at temperature $T > 0$ *K. Show the donor/acceptor energy levels with the order of difference of their energies from the bands.*

[2025 • Set 55-7-1]

SOLUTION

For an *n*-type semiconductor at $T > 0$ K, draw a horizontal filled valence band at the bottom and the conduction band above it, separated by the energy gap E_g. Inside the gap, draw the donor level E_D as a short dashed line lying just below the bottom of the conduction band E_C. Label the small spacing between them.

$$E_C - E_D \approx 0.01 \text{ to } 0.05 \text{ eV}$$

Because this gap is so small, at room temperature almost every donor electron at E_D is thermally lifted into the conduction band (shown by upward arrows from E_D to E_C). Electrons are therefore the majority carriers.

For a *p*-type semiconductor at $T > 0$ K, again draw the filled valence band, the empty conduction band and the gap E_g. Now draw the acceptor level E_A as a short dashed line lying just above the top of the valence band E_V. Label the small spacing.

$$E_A - E_V \approx 0.01 \text{ to } 0.05 \text{ eV}$$

Since this gap is small, electrons from the valence band are easily lifted to E_A (upward arrows from E_V to E_A), leaving **holes in the valence band**. Holes are therefore the majority carriers.

In both diagrams the dopant introduces an energy level very close to the relevant band edge, which is why even light doping changes the carrier density so dramatically.

Final answer. Donor level E_D sits about **0.01** to **0.05** eV below E_C (*n*-type); acceptor level E_A sits about **0.01** to **0.05** eV above E_V (*p*-type).

← *Back to Q122*

Q123

Plot a graph showing the variation of current with voltage for the material GaAs. On the graph, mark the region where:
(a) resistance is negative, and
(b) Ohm's law is obeyed.

[2024 • Set 55-1-1]

SOLUTION

Take voltage V on the x-axis and current I on the y-axis. For GaAs the curve shows three distinct parts. Starting from the origin, the current first rises in a straight line with voltage up to a peak current at the threshold voltage V_P (about 3 V). After this peak the current actually decreases as the voltage is increased further, falling to a minimum at the valley voltage V_V (about 4 to 5 V). Beyond V_V the current rises again with voltage.

(a) Region of negative resistance: this is the falling part of the curve, lying between V_P and V_V. Here an increase in voltage gives a decrease in current, so the slope is negative.

$$\frac{dV}{dI} < 0 \Rightarrow R = \frac{dV}{dI} < 0$$

Mark this descending portion (between V_P and V_V) as the negative-resistance region.

(b) Region where Ohm's law is obeyed: this is the initial straight-line portion from the origin up to roughly V_P. Here the current is directly proportional to the voltage with a constant positive slope.

$$I \propto V \Rightarrow R = \frac{V}{I} = \text{constant}$$

Mark this rising straight portion near the origin as the **ohmic region**.

Final answer. Negative resistance is the falling part between $\mathbf{V_P}$ and $\mathbf{V_V}$; Ohm's law holds in the initial straight portion from the origin to about $\mathbf{V_P}$.

← *Back to Q123*

Q124

How does the conductivity of an intrinsic semiconductor vary with temperature? Explain. Show the variation in a plot.

[2024 • Set 55-1-2]

SOLUTION

In an intrinsic semiconductor the number density of charge carriers depends strongly on temperature through

$$n_i \propto T^{3/2} \exp\left(-\frac{E_g}{2kT}\right)$$

where E_g is the band gap and k is Boltzmann's constant. As the temperature rises, more electrons gain enough thermal energy to jump from the valence band to the conduction band, leaving an equal number of holes behind. So both n_e and n_h increase rapidly with T.

The conductivity is given by

$$\sigma = e\,(n_e\,\mu_e + n_h\,\mu_h)$$

where μ_e and μ_h are the mobilities of electrons and holes. With rising temperature the mobilities decrease slightly, but the exponential rise in the carrier number n_i completely dominates this small fall.

Hence the conductivity of an intrinsic semiconductor increases with temperature, and equivalently its resistivity falls. A semiconductor therefore has a **negative temperature coefficient of resistance**.

For the plot, take temperature T on the x-axis and conductivity σ (or $1/\rho$) on the y-axis. The curve starts low at small T, rising gently at first and then climbing very steeply at higher temperatures, following the exponential law.

CONCEPT

concept

Final answer. Conductivity of an intrinsic semiconductor increases (almost exponentially) with temperature, since the carrier number rises far faster than

the mobility falls.

← Back to Q124

Q125

Differentiate between 'diffusion current' and 'drift current'. Explain their role in the formation of p-n junction.

[2024 • Set 55-1-3]

SOLUTION

Both currents flow in a semiconductor but arise from completely different causes.

Diffusion current is caused by the concentration gradient of charge carriers, that is, carriers moving from a region of higher concentration to a region of lower concentration. In a p-n junction it is due to the majority carriers (electrons from the *n*-side and holes from the *p*-side) crossing the junction, and its conventional direction is from the *p*-side to the *n*-side.

Drift current is caused by the electric field set up across the depletion region, which makes carriers drift under that field. In a p-n junction it is due to the minority carriers (electrons from the *p*-side and holes from the *n*-side) being swept across, and its conventional direction is from the *n*-side to the *p*-side.

Role in junction formation: as soon as the junction is made, majority carriers diffuse across it and recombine, leaving behind immobile charged ions. These uncovered ions form the depletion region and set up a built-in electric field directed from the *n*-side to the *p*-side.

This field then drives a drift current of minority carriers in the opposite sense to the diffusion current. The depletion region widens until the two currents become equal and opposite.

$$I_{diffusion} = I_{drift}$$

At this point equilibrium is reached, no net current flows, and the **depletion width and the built-in potential become fixed**.

Final answer. Diffusion current is due to a concentration gradient (majority carriers, $p \to n$); drift current is due to the junction field (minority carriers, $n \to p$); equilibrium fixes the depletion width when the two are equal.

← Back to Q125

Q126

How does the energy gap of an intrinsic semiconductor effectively change when doped with a (a) trivalent impurity, and (b) pentavalent impurity? Justify your answer in each case.

[2024 • Set 55-2-1]

SOLUTION

(a) Doping with a trivalent impurity. A trivalent atom such as boron, aluminium or gallium has only 3 valence electrons. When it replaces a silicon atom (which has 4 valence electrons) in the lattice, one of the four covalent bonds is left incomplete. This vacancy can readily accept an electron, so the impurity is called an acceptor and creates a hole.

The acceptor introduces an energy level E_A just above the top of the valence band, with

$$E_A - E_V \approx 0.045 \text{ eV}$$

Now an electron from the valence band needs to climb only this tiny step (a few meV) to reach E_A, instead of the full gap E_g. So the effective energy gap to be overcome is greatly reduced, and **holes are produced easily in the valence band**.

(b) Doping with a pentavalent impurity. A pentavalent atom such as phosphorus, arsenic or antimony has 5 valence electrons. After forming four covalent bonds with neighbouring Si atoms, its fifth electron is very loosely bound and is easily released. Such an impurity is called a donor.

The donor introduces an energy level E_D just below the bottom of the conduction band, with

$$E_C - E_D \approx 0.045 \text{ eV}$$

The donor electron then needs only this tiny energy (a few meV) to enter the conduction band, again far less than the full gap E_g. So the effective gap is again greatly reduced and many free electrons are produced.

In both cases doping inserts an intermediate level inside the forbidden gap, drastically lowering the effective energy gap that must be crossed to create free carriers.

Final answer. Both trivalent and pentavalent doping insert a level inside the gap (about **0.045** eV from a band edge), so the effective energy gap is drastically reduced in each case.

← *Back to Q126*

Q127

Suppose a pure Si crystal has 5×10^{28} *atoms* m^{-3}. *It is doped by* 1 *ppm concentration of boron. Calculate the concentration of holes and electrons, given that* $\mathbf{n_i = 1.5 \times 10^{16}}$ m^{-3}. *Is the doped crystal n-type or p-type?*

[2024 • Set 55-3-1]

SOLUTION

Boron is a trivalent impurity, so it acts as an acceptor and the doped crystal will have an excess of holes. First find the acceptor (dopant) concentration. A doping of 1 ppm means 1 boron atom per 10^6 silicon atoms.

$$N_A = 5 \times 10^{28} \times 10^{-6}$$

$$N_A = 5 \times 10^{22}\ m^{-3}$$

Each acceptor contributes one hole, so the hole concentration (the majority carriers) is essentially equal to the acceptor concentration.

$$n_h \approx N_A = 5 \times 10^{22}\ m^{-3}$$

The electron concentration follows from the law of mass action.

$$n_e\, n_h = n_i^2$$

$$n_e = \frac{n_i^2}{n_h} = \frac{(1.5 \times 10^{16})^2}{5 \times 10^{22}}$$

$$n_e = \frac{2.25 \times 10^{32}}{5 \times 10^{22}}$$

$$n_e = 4.5 \times 10^{9}\ \text{m}^{-3}$$

Since $n_h \gg n_e$, holes are the majority carriers, so the **doped crystal is *p*-type**.

TIP

tip

Final answer. $n_h = 5 \times 10^{22}\ \text{m}^{-3}$, $n_e = 4.5 \times 10^{9}\ \text{m}^{-3}$; the crystal is *p*-type.

← *Back to Q127*

Q128

Draw the circuit diagram of a p-n junction diode in (i) forward biasing and (ii) reverse biasing. Also draw its I-V characteristics in the two cases.

[2024 • Set 55-4-1]

SOLUTION

(i) Forward-biasing circuit. Connect the *p*-side (anode) of the diode to the positive terminal of a variable DC supply through a current-limiting series resistor, and the *n*-side (cathode) to the negative terminal. Put a milliammeter in series to read the current I and a voltmeter across the diode to read the voltage V, since in forward bias the current is large.

(ii) Reverse-biasing circuit. Use the same arrangement but reverse the battery polarity: the *p*-side is now connected to the negative terminal and the *n*-side to the positive terminal. Here a microammeter is used in series, because the reverse current is very small.

For the *I-V* characteristics, draw the axes with voltage on the x-axis and current on the y-axis.

Forward bias (first quadrant): the current stays almost zero until the applied voltage reaches the cut-in (knee) voltage.

$$V_T \approx 0.7\ V\ (\text{Si}),\ V_T \approx 0.3\ V\ (\text{Ge})$$

Beyond V_T the current rises sharply, almost exponentially, with voltage.

Reverse bias (third quadrant): only a very small, almost constant reverse saturation current (of the order of microamperes, due to thermally generated minority carriers) flows, nearly independent of voltage, until the breakdown voltage V_{Br} is reached, where the current suddenly increases very steeply.

This **asymmetric characteristic** (large forward current, negligible reverse current) is what lets the diode act as a rectifier.

Final answer. Forward bias: ***p*** to + terminal, current rises sharply beyond $\mathbf{V_T \approx 0.7}$ V; reverse bias: polarity reversed, only a tiny saturation current flows until breakdown.

← Back to Q128

Q129

Differentiate between intrinsic and extrinsic semiconductors.

[2023 • Set 55-1-1]

SOLUTION

Intrinsic semiconductor: it is a pure semiconductor with no deliberately added impurity, for example pure silicon or pure germanium. Here the number of free electrons exactly equals the number of holes, both produced by thermal breaking of covalent bonds.

$$n_e = n_h = n_i$$

Its conductivity is small at room temperature and rises strongly as the temperature increases. An intrinsic semiconductor is mainly used as the starting material for doping.

Extrinsic semiconductor: it is a doped semiconductor, made by adding a small, controlled amount (about 1 ppm) of a trivalent impurity (B, Al) or a pentavalent impurity (P, As). The carrier concentrations are now unequal.

$$n_e \neq n_h$$

Depending on the dopant, either electrons dominate (*n*-type, pentavalent donor) or holes dominate (*p*-type, trivalent acceptor). For each type the mass-action law $\mathbf{n_e\, n_h = n_i^2}$ still holds.

Its conductivity is much higher than that of the intrinsic material and is largely controlled by the doping level rather than just by temperature. **Extrinsic semiconductors are used in all practical devices** such as diodes, transistors and integrated circuits.

Final answer. Intrinsic is pure with $\mathbf{n_e = n_h = n_i}$ and low conductivity; extrinsic is doped with $\mathbf{n_e \neq n_h}$, much higher doping-controlled conductivity, and is used in real devices.

← *Back to Q129*

Q130

Draw the circuit arrangement for studying the V-I characteristics of a p-n junction diode in forward bias and reverse bias. Show the plot of V-I characteristic of a silicon diode.

[2023 • Set 55-1-1]

SOLUTION

Circuit for forward bias. Connect the *p*-side of the diode to the positive terminal of a variable DC voltage source through a current-limiting resistor and a milliammeter (in series), and the *n*-side to the negative terminal. Connect a voltmeter directly across the diode to read *V*, while the milliammeter reads the forward current *I*.

Circuit for reverse bias. Keep the same arrangement but reverse the battery polarity, so the *p*-side now goes to the negative terminal and the *n*-side to the

positive terminal. Replace the milliammeter with a microammeter, since the reverse current is very small.

In each case, vary the supply voltage in steps and note the corresponding values of V and I; plotting these gives the full characteristic.

For the silicon-diode I-V plot, take voltage on the x-axis and current on the y-axis.

Forward bias (first quadrant): the current is almost zero until the voltage reaches the cut-in (knee) voltage.

$$V_T \approx 0.7\ V \text{ (for Si)}$$

Beyond this knee, the current rises sharply and nearly exponentially with voltage.

Reverse bias (third quadrant): only a small, almost constant reverse saturation current (of the order of microamperes) flows, nearly independent of the voltage, until the breakdown voltage V_{Br} is reached, where the current shoots up almost vertically.

This **asymmetric shape** is what allows the silicon diode to be used as a rectifier.

Final answer. Forward bias: ***p*** to + terminal, current rises sharply past $\mathbf{V_T \approx 0.7}$ V; reverse bias: polarity reversed, only microampere-level saturation current flows until breakdown.

← Back to Q130

Q131

Briefly explain how the diffusion and drift currents contribute to the formation of potential barrier in a p-n junction diode.

[2023 • Set 55-1-1]

SOLUTION

When a p-type and an n-type semiconductor are joined, two opposite currents set in across the junction and together they build the potential barrier.

Because the n-side is rich in electrons and the p-side is rich in holes, a steep concentration gradient exists at the junction. Driven by this gradient, the majority carriers diffuse across: electrons move from the n-side to the p-side and holes move from the p-side to the n-side. This motion of majority carriers is the **diffusion current** (I_{diff}), whose conventional direction is from the p-side to the n-side.

As the majority carriers cross over and recombine, they leave behind immobile charged ions. Positive donor ions are exposed on the n-side and negative acceptor ions on the p-side. This thin region, free of mobile charges, is the **depletion region**.

These immobile ions set up an electric field $\vec{E}$ directed from the n-side to the p-side, which produces a potential difference across the junction called the **barrier potential** V_b ($\approx 0.7\ V$ for Si).

This built-in field sweeps the minority carriers across the junction: electrons in the p-region drift to the n-side and holes in the n-region drift to the p-side. This minority-carrier flow is the **drift current** (I_{drift}), directed from the n-side to the p-side, that is, opposite to the diffusion current.

The barrier grows until the field is strong enough that the drift current exactly balances the diffusion current.

At equilibrium $|\mathbf{I_{diff}}| = |\mathbf{I_{drift}}|$, so the net current is zero and a stable potential barrier V_b is established across the depletion layer.

Final answer. Diffusion of majority carriers exposes immobile ions whose field drives an opposing drift current; at equilibrium the two balance, leaving a steady barrier potential $\mathbf{V_b \approx 0.7\ V}$.

← *Back to Q131*

Q132

Draw energy band diagram for an n-type and p-type semiconductor at $\boldsymbol{T > 0\ K}$.

[2023 • Set 55-2-1]

SOLUTION

Both diagrams share a filled **valence band (VB)** at lower energy and an empty **conduction band (CB)** at higher energy, separated by the band gap $\mathbf{E_g \approx 1\ eV}$. Draw E_C (bottom edge of CB) and E_V (top edge of VB) as solid horizontal lines.

For the **n-type semiconductor**: mark a donor level E_D as a short dashed line just below the conduction band, with $\mathbf{E_C - E_D \approx 0.01}$ to 0.05 eV. Since this gap is tiny compared with kT at room temperature, at $\boldsymbol{T > 0\ K}$ almost all donor atoms are ionised, sending electrons up into the CB. Show several dots (electrons) in the CB as the **majority carriers** and a few open circles (holes) in the VB as the minority carriers.

For the **p-type semiconductor**: mark an acceptor level E_A as a short dashed line just above the valence band, with $\mathbf{E_A - E_V \approx 0.01}$ to 0.05 eV. At $\boldsymbol{T > 0\ K}$ electrons from the VB jump up to fill E_A, leaving holes behind in the VB. Show several open circles (holes) in the VB as the **majority carriers** and a few dots (electrons) in the CB as the minority carriers.

Label both diagrams clearly with E_C, E_V, E_g, and the dopant level (E_D for n-type, E_A for p-type), keeping the dopant line very close to its respective band edge.

Final answer. n-type: $\mathbf{E_D}$ just below $\mathbf{E_C}$ with electrons in the CB; p-type: $\mathbf{E_A}$ just above $\mathbf{E_V}$ with holes in the VB, both at $\boldsymbol{T > 0\ K}$.

← *Back to Q132*

Q133

Answer the following giving reasons:
(i) A p-n junction diode is damaged by a strong current.
(ii) Impurities are added in intrinsic semiconductors.

[2023 • Set 55-2-1]

SOLUTION

(i) A p-n junction diode is damaged by a strong current.

When a large forward current I flows, the power dissipated as heat is $\boldsymbol{P = V_F I}$, which rises steeply with current.

This heat raises the junction temperature, and a higher temperature thermally generates still more charge carriers, which increases the current further. This self-reinforcing loop is called **thermal runaway**.

The rising temperature can melt the semiconductor crystal and damage the metal contacts, permanently destroying the junction. This is why a current-limiting resistor is always placed in series with a diode.

(ii) Impurities are added in intrinsic semiconductors.

A pure (intrinsic) semiconductor has very few free carriers at room temperature, so its conductivity is too low and too temperature-dependent for practical devices.

Adding a controlled, tiny amount of a suitable impurity, called **doping**, increases the carrier concentration enormously (by a factor of about 10^5 to 10^9).

A pentavalent dopant gives an **n-type** semiconductor (electrons as majority carriers) and a trivalent dopant gives a **p-type** semiconductor (holes as majority carriers). Doping therefore raises conductivity to a useful, controllable level needed to build diodes and transistors.

Final answer. (i) Strong current causes thermal runaway that melts the junction; (ii) doping raises carrier concentration enormously, giving the controlled conductivity needed for devices.

← *Back to Q133*

Q134

With the help of a circuit diagram, explain how a full wave rectifier gives output rectified voltage corresponding to both halves of the input ac voltage.

[2023 • Set 55-2-3]

SOLUTION

Circuit: A centre-tapped step-down transformer has its secondary connected to two diodes D_1 and D_2. The two ends of the secondary connect to the anodes of D_1 and D_2, and their cathodes are joined together. The load resistor R_L is

connected between this common cathode point and the centre tap of the secondary. The output voltage V_o is taken across R_L.

Principle: A diode conducts only when it is forward biased, so each diode conducts on alternate half-cycles, sending current through R_L in the same direction each time.

During the positive half-cycle: the top end of the secondary is positive and the bottom end is negative. So D_1 is forward biased and conducts, while D_2 is reverse biased and stays off. Current flows from the top end through D_1, then down through R_L from top to bottom, and back to the centre tap.

During the negative half-cycle: the polarity of the secondary reverses, making the bottom end positive. Now D_2 is forward biased and conducts, while D_1 is off. Current flows from the bottom end through D_2, then again down through R_L from top to bottom, and back to the centre tap.

In both half-cycles the current through R_L flows in the **same direction**, so the output is a unidirectional (rectified) voltage corresponding to both halves of the input. The output therefore has **twice the frequency** of the input ac.

Final answer. Diodes $\mathbf{D_1}$ and $\mathbf{D_2}$ conduct on alternate half-cycles but pass current through $\mathbf{R_L}$ in the same direction, giving a rectified output of twice the input frequency.

← Back to Q134

Q135

Explain the roles of diffusion current and drift current in the formation of the depletion layer in a p-n junction diode.

[2023 • Set 55-3-1]

SOLUTION

Role of diffusion current (it creates the depletion layer):

At the p-n boundary there is a steep concentration gradient. Driven by it, the majority carriers diffuse across the junction: electrons move from the n-side to the p-side and holes move from the p-side to the n-side.

As these majority carriers leave, they expose immobile dopant ions, positive donor ions on the n-side and negative acceptor ions on the p-side. This thin region, now free of mobile charges, is the **depletion layer**. Thus diffusion is what creates the depletion layer.

Role of drift current (it limits the growth of the depletion layer):

The exposed immobile ions set up an electric field directed from the n-side to the p-side. This field pushes the minority carriers across the junction: electrons in the p-region drift to the n-side and holes in the n-region drift to the p-side. This minority-carrier flow is the **drift current**, opposite in direction to the diffusion current.

As the depletion layer widens, the built-in field strengthens and so does the drift current. The layer stops growing when the drift current exactly balances the diffusion current, that is when $|\mathbf{I_{drift}}| = |\mathbf{I_{diff}}|$.

At this equilibrium the net current is zero and the depletion layer reaches a fixed width.

Final answer. Diffusion of majority carriers builds the depletion layer, while the opposing drift current limits its growth; equilibrium ($|\mathbf{I_{drift}}| = |\mathbf{I_{diff}}|$) fixes its width.

← *Back to Q135*

Q136

Explain the property of a p-n junction which makes it suitable for rectifying alternating voltages. Differentiate between a half-wave and a full-wave rectifier.

[2023 • Set 55-3-1]

SOLUTION

Property used for rectification:

A p-n junction diode offers very **low resistance when forward biased** (it conducts easily) and very **high resistance when reverse biased** (it blocks current). Because of this **unidirectional conduction**, when an alternating voltage is applied, the diode conducts only during the half-cycle for which it is

forward biased and blocks the other half. This converts an alternating voltage into a unidirectional (rectified) voltage.

Difference between half-wave and full-wave rectifier:

A half-wave rectifier uses **one diode**, whereas a full-wave rectifier uses **two diodes** (centre-tap type) or four diodes (bridge type).

In a half-wave rectifier, output is obtained only during the **positive half-cycle** of the input; in a full-wave rectifier, output is obtained during **both half-cycles**.

The output frequency of a half-wave rectifier equals the **input frequency**, while that of a full-wave rectifier is **twice the input frequency**.

A half-wave rectifier has lower efficiency (≈ **40.6%**) with a high ripple factor (≈ **1.21**), whereas a full-wave rectifier has higher efficiency (≈ **81.2%**) with a lower ripple factor (≈ **0.482**), giving a smoother output that is easier to filter.

Final answer. Unidirectional conduction (low forward, high reverse resistance) enables rectification; full-wave (2 diodes, both half-cycles, 2f, higher efficiency) outperforms half-wave (1 diode, one half-cycle, f).

← *Back to Q136*

Q137

How is the width of depletion layer of a p-n junction diode affected when it is (i) forward biased, and (ii) reverse biased? Justify your answers.

[2023 • Set 55-3-2]

SOLUTION

The depletion layer exists because of the immobile ionised dopant charges near the junction, and its width is controlled by the net electric field across the junction, which the external bias can either oppose or assist.

(i) Forward bias: The external voltage is applied with the **p-side positive** and the n-side negative. This applied field opposes the built-in field, so the net barrier is reduced.

With a lower barrier, majority carriers from both sides flood toward the junction and recombine with the exposed immobile ions, neutralising them.

Hence in forward bias the **width of the depletion layer decreases**.

(ii) Reverse bias: The external voltage is applied with the **n-side positive** and the p-side negative. This applied field is in the same direction as the built-in field, so the net barrier increases.

Majority carriers are now pulled away from the junction, exposing still more immobile dopant ions on either side.

Hence in reverse bias the **width of the depletion layer increases**.

Final answer. Forward bias narrows the depletion layer (barrier reduced, ions neutralised); reverse bias widens it (barrier raised, more ions exposed).

← Back to Q137

Q138

Draw a circuit diagram of a p-n junction diode in (a) forward biasing, and (b) reverse biasing. Draw the V-I characteristics for each case.

[2023 • Set 55-3-3]

SOLUTION

(a) Forward-bias circuit: Connect the **p-side of the diode to the positive terminal** and the n-side to the negative terminal of a variable dc supply, with a current-limiting resistor in series. An ammeter (in series) reads the current and a voltmeter (across the diode) reads the voltage.

V-I characteristic in forward bias: The curve lies in the first quadrant. The current stays almost zero until the voltage reaches the **knee (cut-in) voltage** (≈ **0.7** V for Si, ≈ **0.3** V for Ge), after which it rises sharply and almost exponentially with voltage.

(b) Reverse-bias circuit: Use the same arrangement but reverse the supply, so the **p-side goes to the negative terminal** and the n-side to the positive terminal. A microammeter is used because the current is very small.

V-I characteristic in reverse bias: The curve lies in the third quadrant. Only a very small, almost constant **reverse saturation current** (of the order of μA, due to thermally generated minority carriers) flows, nearly independent of voltage. At the **breakdown voltage** V_{Br} the reverse current rises abruptly.

Plotted together on one set of axes, the forward and reverse curves form the characteristic asymmetric V-I graph: a large forward current and a negligible reverse current, which is what makes the diode act as a rectifier.

> **TIP**
>
> Label the knee voltage on the forward curve and V_{Br} on the reverse curve; the markers carry marks even if the shape is rough.

Final answer. Forward bias (p to +) gives a sharply rising curve above the knee voltage; reverse bias (p to -) gives a tiny constant saturation current until breakdown $\mathbf{V_{Br}}$.

← Back to Q138

Q139

Draw energy band diagrams of n-type and p-type semiconductors at temperature $T > 0$ *K, depicting the donor and acceptor energy levels. Mention the significance of these levels.*

[2022 • Set 55-1-1]

SOLUTION

n-type at $T > 0$ K: Draw the valence band (E_V) and conduction band (E_C) separated by $E_g \approx 1.1$ eV for Si. Mark a donor level E_D as a short dashed line just below the conduction band, with $E_C - E_D \approx 0.01$ to 0.05 eV. Since $kT \approx 0.025$ eV at 300 K, the donor electrons are easily promoted into the conduction band, where they act as **majority carriers**.

p-type at $T > 0$ K: Draw the same bands. Mark an acceptor level E_A as a short dashed line just above the valence band, with $E_A - E_V \approx 0.01$ to 0.05 eV. Valence-band electrons easily jump up to E_A, leaving behind **holes** in the valence band that act as the majority carriers.

Significance of these levels: The donor and acceptor levels lie only a few meV from the nearest band edge, so the energy needed to free a carrier is reduced from the full band gap E_g (≈ **1.1 eV**) to just a few meV.

As a result, the dopant atoms are almost fully ionised at room temperature and supply a large carrier density of about 10^{16} to 10^{19} cm^{-3}, which is many orders of magnitude higher than the intrinsic value (≈ $\mathbf{10^{10}\ cm^{-3}}$ for Si at 300 *K*). This is what makes the doped semiconductor highly and controllably conducting.

Final answer. $\mathbf{E_D}$ sits just below $\mathbf{E_C}$ and $\mathbf{E_A}$ just above $\mathbf{E_V}$; their nearness to the band edge lets dopants ionise at room temperature and supply a large carrier density.

← *Back to Q139*

Q140

Write the characteristics of a p-n junction which make it suitable for rectification.

[2022 • Set 55-1-1]

SOLUTION

A p-n junction is suitable for rectification because its conduction depends strongly on the direction of the applied voltage.

Low resistance in forward bias: When the diode is forward biased (p-side positive), the depletion region narrows and the barrier is lowered, so the diode conducts easily. The forward current rises sharply (almost exponentially) once the voltage exceeds the knee voltage (≈ **0.7 *V*** for Si).

Very high resistance in reverse bias: When the diode is reverse biased, the depletion region widens and the barrier rises, so only a tiny reverse saturation current (of the order of μA, due to minority carriers) flows. This is negligible compared with the forward current.

Highly asymmetric (unidirectional) V-I characteristic: Because the diode passes current readily in one direction and almost blocks it in the other, it conducts during the half-cycle for which it is forward biased and blocks during the half-cycle for which it is reverse biased.

Hence, when an alternating voltage is applied, the diode converts it into a **pulsating (unidirectional) dc**, which is the basis of rectification.

Final answer. Low forward resistance, very high reverse resistance, and the resulting asymmetric unidirectional V-I characteristic let a p-n junction convert ac into pulsating dc.

← *Back to Q140*

Q141

Explain the formation of depletion layer and barrier potential in a p-n junction diode.

[2022 • Set 55-1-2]

SOLUTION

When a *p*-type and an *n*-type semiconductor are joined to form a junction inside a single crystal, two processes set up the depletion layer and the barrier potential.

Just after the junction is formed, the *n*-side has a high concentration of free electrons and the *p*-side has a high concentration of holes. Because of this large concentration gradient, electrons diffuse from the *n*-side to the *p*-side and holes diffuse from the *p*-side to the *n*-side.

As an electron leaves the *n*-side, it leaves behind an immobile positively charged donor ion. As a hole leaves the *p*-side, it leaves behind an immobile negatively charged acceptor ion.

Thus a layer of immobile positive ions builds up on the *n*-side of the junction and a layer of immobile negative ions builds up on the *p*-side. This narrow region near the junction, which has no free mobile charge carriers, is called the **depletion layer**. Its width is typically about 0.5 μm.

The immobile charged ions set up an internal electric field $\vec{E}$ directed from the *n*-side towards the *p*-side. This field gives rise to a potential difference across the junction called the **barrier potential** V_0, with the *n*-side at a higher potential than the *p*-side.

The value of the barrier potential is about 0.7 V for silicon and about 0.3 V for germanium.

This barrier potential opposes the further diffusion of majority carriers across the junction. At equilibrium, the diffusion current of majority carriers is exactly balanced by the drift current of minority carriers, so the net current through the junction is zero.

CONCEPT

The depletion layer has no mobile charges, so it acts almost like an insulator; the barrier potential is built-in and cannot be measured with a voltmeter across the diode.

Final answer. Diffusion of majority carriers leaves immobile ions forming a depletion layer about **0.5 µm** wide, whose field gives a barrier potential of about **0.7 *V*** (Si) opposing further diffusion.

← *Back to Q141*

Q142

Distinguish between intrinsic and extrinsic semiconductors. Although in an extrinsic semiconductor $n_e \neq n_h$*, yet it is electrically neutral. Why?*

[2022 • Set 55-1-3]

SOLUTION

An intrinsic semiconductor is a pure semiconductor such as silicon or germanium with no impurity added. In it the number of free electrons equals the number of holes, $\mathbf{n_e = n_h = n_i}$, where n_i is the intrinsic carrier concentration. Its conductivity is low and increases with temperature.

An extrinsic semiconductor is a semiconductor in which a controlled amount of suitable impurity (dopant) has been added by doping. In it the electron and hole concentrations are unequal, $n_e \neq n_h$. It is of two kinds: *n*-type, in which $n_e \gg n_h$ (doped with pentavalent atoms), and *p*-type, in which $n_h \gg n_e$ (doped with trivalent atoms). Its conductivity is much higher and is controlled by the doping level.

Why an extrinsic semiconductor is electrically neutral even though $n_e \neq n_h$:

Doping does not add any net charge to the crystal; the dopant atoms themselves are electrically neutral when added. Doping only changes how the mobile carriers are distributed.

Consider an *n*-type semiconductor. Each pentavalent donor atom gives one free electron of charge $-e$ to the conduction band, and in doing so the donor atom becomes a fixed positive ion of charge $+e$ at its lattice site.

So for every extra mobile electron there is one fixed positive ion, and the charges $-e$ and $+e$ exactly cancel. The total charge of the crystal stays zero.

Similarly, in a *p*-type semiconductor each trivalent acceptor atom creates one mobile hole of charge $+e$ and itself becomes a fixed negative ion of charge $-e$, again cancelling out.

Hence, although $n_e \neq n_h$, the immobile ionised dopant charges balance the excess mobile carriers, so the extrinsic semiconductor remains **electrically neutral**.

Final answer. Intrinsic semiconductors are pure with $\mathbf{n_e = n_h = n_i}$, while extrinsic ones are doped with $\mathbf{n_e \neq n_h}$; the crystal stays neutral because each extra mobile carrier is balanced by an equal and opposite fixed dopant ion.

← *Back to Q142*

Q143

Explain the formation of depletion region in a p-n junction.

[2022 • Set 55-2-1]

SOLUTION

The depletion region is formed by the diffusion of majority charge carriers across the *p-n* junction.

When the *p*-type and *n*-type regions are formed in contact within a single crystal, there is a steep concentration gradient at the boundary: electrons are abundant on the *n*-side but scarce on the *p*-side, while holes are abundant on the *p*-side but scarce on the *n*-side.

Because of this gradient, electrons diffuse from the n-side to the p-side and holes diffuse from the p-side to the n-side.

As the majority carriers leave their regions, they expose immobile charged ions near the junction: positive donor ions (such as P^+, As^+) on the n-side, and negative acceptor ions (such as B^-, Al^-) on the p-side.

This thin region around the junction, which contains only immobile ions and almost no free mobile carriers, is called the **depletion region**. Its width is typically about 1 μm in a moderately doped diode.

The exposed immobile ions set up an electric field $\vec{E}$ directed from the n-side to the p-side, which corresponds to a barrier potential of about 0.7 V in silicon.

This field opposes further diffusion of majority carriers. At equilibrium the diffusion current and the drift current become equal and opposite, so the depletion region attains a steady width.

Final answer. Diffusion of majority carriers across the junction exposes immobile donor and acceptor ions, creating a carrier-free depletion region about **1 μm** wide whose internal field halts further diffusion.

← *Back to Q143*

Q144

How is an emf generated by a solar cell due to the three basic processes involved? Explain.

[2022 • Set 55-2-1]

SOLUTION

A solar cell is a p-n junction that generates an emf when light falls on it. The emf is produced through three basic processes.

First is generation. Photons of incident light with energy greater than the band gap, $\mathbf{h\nu > E_g}$, are absorbed in or very near the depletion region and create electron-hole pairs.

Second is separation. The strong built-in electric field of the depletion region separates the photo-generated pairs before they can recombine: it drives the

electrons towards the *n*-side and the holes towards the *p*-side.

Third is collection. The electrons reach the front contact (*n*-side) and the holes reach the back contact (*p*-side). This accumulation of opposite charges at the two terminals produces a potential difference, which is the photo-emf of the solar cell.

When an external load is connected across the terminals, this emf drives a current through it, so light energy is converted into electrical energy.

For a typical silicon solar cell the open-circuit voltage is about $\mathbf{V_{OC} \approx 0.6\ V}$ and the short-circuit current density is about 30 mA/cm^2.

COMMON MISTAKE

The pairs must be created within or close to the depletion region; pairs formed deep inside the bulk recombine before the junction field can separate them and contribute no emf.

Final answer. Light creates electron-hole pairs (generation), the junction field drives electrons to the ***n***-side and holes to the ***p***-side (separation), and these collect at the terminals (collection), setting up the photo-emf.

← *Back to Q144*

Q145

Briefly explain the working of a light emitting diode. Mention its two uses.

[2022 • Set 55-2-2]

SOLUTION

A light emitting diode (LED) is a heavily doped *p-n* junction diode that emits light when operated under forward bias.

When the junction is forward biased, electrons from the *n*-side and holes from the *p*-side are injected across the junction into the depletion region.

In the junction region, the conduction-band electrons recombine with the valence-band holes. In each such recombination the energy equal to the band gap E_g is released in the form of a photon (light).

The energy and wavelength of the emitted photon are given by:

$$E_g = h\nu = \frac{hc}{\lambda}$$

$$\Rightarrow \lambda = \frac{hc}{E_g}$$

To obtain visible light the band gap must lie roughly in the range $\mathbf{E_g \approx 1.8\ eV}$ to 3.1 eV. Materials such as GaAs, GaAsP and GaN are used, and the colour of the emitted light is decided by the value of E_g.

Two uses of an LED:

(i) As **indicator lamps** in electronic appliances and instrument panels, and in seven-segment numerical displays.

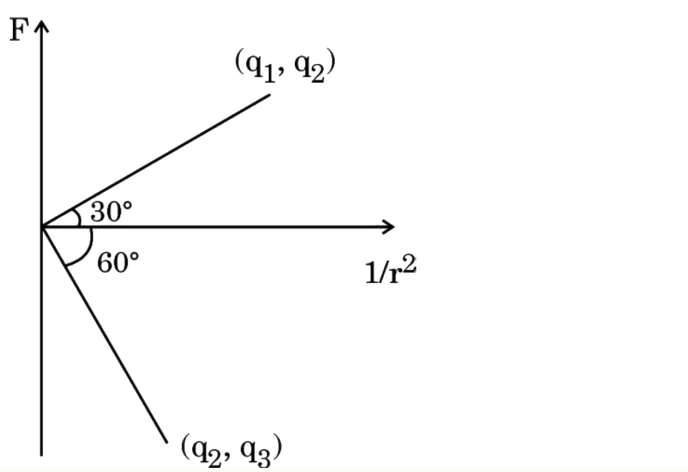
(ii) In **energy-efficient lighting**, such as LED bulbs, traffic signals and automobile lamps.

Final answer. An LED is a forward-biased heavily doped ***p*-*n*** junction in which electron-hole recombination releases band-gap energy as light of wavelength $\boldsymbol{\lambda = \frac{hc}{E_g}}$; used as indicator lamps and in energy-efficient lighting.

← *Back to Q145*

Q146

The V - I characteristics of a solar cell is shown in the figure.



(a) What do the points P and Q represent?
(b) Give the circuit diagram for a solar cell.

[2022 • Set 55-2-3]

SOLUTION

(a) The *V-I* characteristic of an illuminated solar cell lies in the fourth quadrant, because the cell supplies current to the external circuit (the voltage is positive while the current is negative).

Point *P* lies on the voltage axis, where the current is zero. It represents the **open-circuit voltage** V_{OC} of the solar cell, that is, the maximum voltage available when no current is drawn. For a silicon cell $\mathbf{V_{OC} \approx 0.5}$ **to 0.7** ***V***.

Point *Q* lies on the current axis, where the voltage is zero. It represents the **short-circuit current** I_{SC}, that is, the maximum current delivered when the terminals are short-circuited. I_{SC} is proportional to the intensity of the incident light.

(The point of maximum power output lies on the curve somewhere between *P* and *Q*.)

(b) Circuit diagram of a solar cell:

The solar cell is shown by its symbol (a cell/battery symbol with two inward arrows indicating incident light). Its two terminals are connected through connecting wires to an external load resistor R_L, forming a closed loop.

When light falls on the cell, it acts as a source of emf and drives a current through R_L. An ammeter may be connected in series and a voltmeter across the load to read the operating point on the *V-I* curve.

Final answer. ***P*** is the open-circuit voltage $\mathbf{V_{OC}}$ (current zero) and ***Q*** is the short-circuit current $\mathbf{I_{SC}}$ (voltage zero); the cell is drawn connected across a load resistor $\mathbf{R_L}$.

← *Back to Q146*

Q147

What is meant by doping of an intrinsic semiconductor? Name the two types of atoms used for doping of Ge/Si.

[2022 • Set 55-3-1]

SOLUTION

Doping of an intrinsic semiconductor means the deliberate addition of a small, controlled amount of a suitable impurity to a pure (intrinsic) semiconductor crystal in order to greatly increase its electrical conductivity.

The amount of dopant is very small, of the order of one impurity atom per 10^6 to 10^8 host atoms. Even this trace amount raises the carrier concentration, and hence the conductivity, by many orders of magnitude.

The two types of impurity atoms used for doping Ge or Si are:

(i) **Pentavalent (donor) impurities**, which have five valence electrons, such as phosphorus (P), arsenic (As) or antimony (Sb). Each such atom forms four covalent bonds and donates its fifth electron to the conduction band, giving an *n*-type semiconductor in which electrons are the majority carriers.

(ii) **Trivalent (acceptor) impurities**, which have three valence electrons, such as boron (B), aluminium (Al) or indium (In). Each such atom forms only three covalent bonds, leaving one bond incomplete, i.e. a hole. This gives a *p*-type semiconductor in which holes are the majority carriers.

Final answer. Doping is the controlled addition of a tiny amount of impurity to a pure semiconductor to raise its conductivity; the two dopant types for Ge/Si are pentavalent (donor) atoms like P, As and trivalent (acceptor) atoms like B, Al.

← *Back to Q147*

Q148

(a) Draw the circuit diagram of an illuminated photodiode and its I-V characteristics.

(b) How can a photodiode be used to measure the light intensity?

[2022 • Set 55-3-2]

SOLUTION

(a) Circuit diagram and *I-V* characteristics of an illuminated photodiode:

A photodiode is a *p-n* junction diode with a transparent window, operated under reverse bias. In the circuit the *p*-side is connected to the negative terminal and the *n*-side to the positive terminal of the battery, so the diode is reverse biased. Light is allowed to fall on the junction through the window, and an ammeter (microammeter) is connected in series to measure the reverse photocurrent.

The *I-V* characteristics are drawn in the third quadrant (reverse voltage, reverse current). For a fixed reverse voltage the reverse current is almost constant (saturated). A separate, nearly horizontal curve is obtained for each light intensity, and the magnitude of the reverse current increases as the light intensity increases, with $\mathbf{I_3 > I_2 > I_1}$ for intensities $\mathbf{I_3 > I_2 > I_1}$.

(b) Measurement of light intensity using a photodiode:

The reverse saturation photocurrent of a photodiode is directly proportional to the intensity of the incident light:

$$I_{ph} \propto P_{light}$$

So, keeping a fixed reverse bias (in the saturation region), the reverse current is measured with the ammeter. Since this current is proportional to the incident intensity, the reading directly gives a measure of the light intensity.

After calibrating the photodiode with a source of known intensity, the measured reverse current can be converted into the actual incident light intensity.

CONCEPT

A photodiode is operated in reverse bias (not forward) because the fractional change in reverse current with light intensity is much larger and easier to measure accurately.

Final answer. An illuminated photodiode is a reverse-biased junction whose reverse saturation current rises with light intensity ($\mathbf{I_{ph} \propto P_{light}}$); measuring this current at fixed reverse bias gives the light intensity.

← *Back to Q148*

Q149

Name the extrinsic semiconductors formed when a pure germanium is doped with (i) a trivalent and (ii) pentavalent impurity. Draw the energy band diagrams of extrinsic semiconductors so formed.

[2022 • Set 55-3-3]

SOLUTION

(i) When pure germanium is doped with a trivalent impurity (such as B, Al, In or Ga), a ***p*-type semiconductor** is formed.

Each trivalent atom has three valence electrons. It forms three covalent bonds with neighbouring Ge atoms, and the fourth bond remains incomplete, creating a hole. Such an impurity is called an acceptor, and holes are the majority carriers.

Energy band diagram of the *p*-type semiconductor:

Draw the filled valence band (VB) at the bottom and the empty conduction band (CB) at the top, separated by the energy gap $\mathbf{E_g \approx 0.7\ eV}$ for Ge. Just above the top of the valence band, mark an empty **acceptor level** E_A, lying very close to the VB with $\mathbf{E_A - E_V \approx 0.01\ eV}$. Electrons from the valence band are easily excited into this acceptor level, leaving holes in the valence band as the majority carriers.

(ii) When pure germanium is doped with a pentavalent impurity (such as P, As, Sb or Bi), an ***n*-type semiconductor** is formed.

Each pentavalent atom has five valence electrons. After forming four covalent bonds, the fifth electron is loosely bound and is easily set free. Such an impurity is called a donor, and electrons are the majority carriers.

Energy band diagram of the *n*-type semiconductor:

Draw the filled VB at the bottom and the CB at the top, separated by $\mathbf{E_g \approx 0.7\ eV}$. Just below the bottom of the conduction band, mark a filled **donor level** E_D, lying very close to the CB with $\mathbf{E_C - E_D \approx 0.01\ eV}$. At room temperature electrons from E_D are thermally lifted into the conduction band and become the majority carriers.

Final answer. Trivalent doping gives a ***p***-type semiconductor with an acceptor level $\mathbf{E_A}$ just above the valence band; pentavalent doping gives an ***n***-type semiconductor with a donor level $\mathbf{E_D}$ just below the conduction band.

← Back to Q149

Q150

What is meant by energy band gap in a solid? Draw the energy band diagrams for a conductor, an insulator and a semiconductor.

[2022 • Set 55-4-1]

SOLUTION

In a solid, the closely spaced energy levels of the atoms merge into energy bands. The **energy band gap** (E_g) is the forbidden range of energies lying between the top of the highest filled band, called the valence band (VB), and the bottom of the next higher band, called the conduction band (CB).

No electron of the perfect crystal can have an energy lying within this gap. Its width is given by:

$$E_g = E_C - E_V$$

where E_C is the energy of the bottom of the conduction band and E_V is the energy of the top of the valence band. The size of E_g decides whether a solid behaves as a conductor, a semiconductor or an insulator.

Energy band diagrams:

Conductor (metal): Draw the valence band and conduction band overlapping each other (or the conduction band partly filled), so that $\mathbf{E_g = 0}$. Free electrons are available in large numbers, giving high conductivity.

Insulator: Draw a completely filled valence band and an empty conduction band separated by a large gap, $\mathbf{E_g > 3\ eV}$ (for diamond about 5.4 eV). Thermal energy at room temperature is far too small to lift electrons across this large gap, so conductivity is almost zero.

Semiconductor: Draw a filled valence band and an empty conduction band separated by a small gap, $\mathbf{E_g \approx 1\ eV}$ (Si $\approx$ **1.1 eV**, Ge $\approx$ **0.7 eV**). At room temperature some electrons are thermally excited across this small gap into the conduction band, leaving holes in the valence band, so the material has moderate conductivity that rises with temperature.

Final answer. The energy band gap $\mathbf{E_g = E_C - E_V}$ is the forbidden energy range between the valence and conduction bands; it is zero (overlap) for a conductor, large (**>3 eV**) for an insulator and small ($\approx$ **1 eV**) for a semiconductor.

← Back to Q150

Q151

Name the device which converts electrical energy into light energy. Write three advantages of the device.

[2022 • Set 55-4-1]

SOLUTION

The device that converts electrical energy into light energy is a **Light Emitting Diode (LED)**.

It is a heavily doped p-n junction operated under **forward bias**. When forward biased, electrons from the n-side and holes from the p-side are pushed towards the junction, where they recombine. On recombination, the energy released appears as a photon of light.

$$h\nu \approx E_g$$

Here E_g is the band gap of the semiconductor, so the colour of the emitted light depends on the material used (for example GaAs, GaP, GaInN).

Three advantages of an LED are:

1. It works on **low voltage and consumes very little power**, so it is highly efficient (most of the input energy becomes light, not heat).

2. It has a **very long operating life** (of the order of 10^5 hours) and is rugged with no fragile filament.

3. It has **fast switching** (turns on and off in nanoseconds) and is small in size, so it is ideal for indicators and optical communication.

Final answer. The device is a Light Emitting Diode (LED), which is highly efficient, has a long life, and switches very fast.

← *Back to Q151*

Q152

Name the device which converts an ac input signal into a dc output signal. Write the principle of working of the device.

[2022 • Set 55-4-2]

SOLUTION

The device that converts an AC input signal into a DC output signal is called a **rectifier**.

Its working is based on the property that a **p-n junction diode conducts only when it is forward biased** and blocks current when reverse biased.

When AC is applied across the diode in series with a load, the polarity of the input keeps reversing. During the half cycle in which the anode (p-side) is at a higher potential, the diode is forward biased, offers low resistance and conducts, so current flows through the load.

During the opposite half cycle the diode is reverse biased, offers very high resistance and practically stops the current, so no output is obtained for that half.

Because current flows through the load in **only one direction**, the output voltage is unidirectional, that is, pulsating DC. This unidirectional conduction of the p-n junction diode is the **principle of rectification**.

Final answer. The device is a rectifier; it works on the unidirectional conduction property of a p-n junction diode.

← *Back to Q152*

Q153

Define barrier potential. Why does the thickness of the depletion layer in a p-n junction diode vary with increase in reverse bias?

[2022 • Set 55-4-3]

SOLUTION

Barrier potential: When a p-n junction is formed, majority carriers diffuse across the junction and leave behind immobile charged ions, creating a depletion region. The potential difference that develops across this depletion region of an **unbiased** p-n junction is called the **barrier potential** V_b.

$$V_b \approx 0.7\ V\ (\text{Si}),\ V_b \approx 0.3\ V\ (\text{Ge})$$

It opposes the further diffusion of majority carriers across the junction.

Variation of depletion layer thickness with reverse bias: In reverse bias, the external voltage is applied with the n-side at a higher potential than the p-side. This external field is in the **same direction** as the built-in field, so it adds to the barrier potential.

As a result, the majority carriers (holes on the p-side and electrons on the n-side) are pulled away from the junction towards the terminals. This exposes **more immobile ions** on both sides of the junction.

$$W \propto \sqrt{V_b + V_{\text{applied}}}$$

Therefore, as the reverse bias is increased, more ions are uncovered and the **depletion layer becomes thicker (wider)**.

Final answer. Barrier potential is the p.d. across the depletion region of an unbiased junction; reverse bias adds to it, uncovering more ions, so the depletion layer widens.

← Back to Q153

Q154

With the help of a circuit diagram, explain briefly how a p-n junction diode works as a half-wave rectifier.

[2022 • Set 55-5-1]

SOLUTION

A half-wave rectifier uses the property that a p-n junction diode conducts only in forward bias.

Circuit diagram (described): The AC mains is fed to the primary of a step-down transformer. One end of the secondary coil is connected to the **anode (p-side)** of the diode. The **cathode (n-side)** of the diode is connected to one end of the load resistance R_L. The other end of R_L is joined to the other end of the secondary coil. The output DC voltage is taken across R_L.

Working: During the **positive half cycle** of the input, the anode of the diode is at a higher potential, so the diode is **forward biased**. It conducts, and current flows through R_L, giving an output voltage that follows the input.

During the **negative half cycle**, the anode is at a lower potential, so the diode is **reverse biased**. It does not conduct, and the output across R_L is zero.

Hence current flows through the load only during one half of each cycle, so the **output is a pulsating DC**, consisting of positive half-sine humps separated by flat zero gaps. The output has the same frequency as the input.

TIP

Output is taken across the load R_L, not across the diode. Marking the diode polarity and R_L correctly in the circuit carries marks.

Final answer. The diode conducts only in the positive half cycle, giving a unidirectional (pulsating DC) output across the load.

← Back to Q154

Q155

Why a photo-diode is operated in reverse bias whereas current in the forward bias is much larger than that in the reverse bias? Explain. Mention its two uses.

[2022 • Set 55-5-1]

SOLUTION

A photodiode is a light detector. Its purpose is to detect a change in light intensity through the **change in its current**, so what matters is the **fractional change** in current produced by the light.

In reverse bias: The dark current (reverse saturation current) is very small, of the order of μA. When light falls on the junction, the photo-generated carriers add a current that is a **large fraction** of this small dark current. So even a small change in light intensity produces an easily measurable change in current.

In forward bias: The current is already large, of the order of mA, due to the injection of majority carriers. The extra photo-generated current (still only ~ μA) is a **negligible fraction** of this large forward current and gets masked.

Therefore, although the forward current is larger in magnitude, the **fractional change of current with light intensity is much greater in reverse bias**, which makes the device a far more sensitive light detector. Hence a photodiode is operated in reverse bias.

Two uses: (1) As a light intensity sensor in cameras and light meters. (2) As a detector in optical communication and optical fibre systems.

Final answer. Reverse bias is used because the fractional change in current with light is far larger than in forward bias; uses include light sensors and optical communication detectors.

← *Back to Q155*

Q156

Two crystals C_1 and C_2, made of pure silicon, are doped with arsenic and aluminium respectively.

(i) Identify the extrinsic semiconductors so formed.
(ii) Why is doping of intrinsic semiconductors necessary?

[2022 • Set 55-5-2]

SOLUTION

(i) Identifying the extrinsic semiconductors:

Arsenic (As) is a **pentavalent** (Group V) element with 5 valence electrons. When silicon is doped with arsenic, four electrons form covalent bonds and the fifth is loosely bound and becomes free, providing extra electrons. So crystal C_1 (Si + As) is an **n-type semiconductor**.

Aluminium (Al) is a **trivalent** (Group III) element with 3 valence electrons. When silicon is doped with aluminium, one covalent bond remains incomplete, creating a hole. So crystal C_2 (Si + Al) is a **p-type semiconductor**.

(ii) Why doping is necessary:

A pure (intrinsic) semiconductor has very few charge carriers at room temperature ($n_i \sim 10^{16}$ m^{-3} for Si), so its conductivity is very low and strongly temperature dependent. This makes it unsuitable for practical devices.

Doping with a small amount of suitable impurity **greatly increases the number of charge carriers** and lets us control the type (n or p) and the amount of conductivity. This controlled, high conductivity is what makes the material useful for making diodes, transistors and other devices.

Final answer. C_1 (Si + As) is n-type and C_2 (Si + Al) is p-type; doping is done to raise and control the conductivity of the intrinsic semiconductor.

← *Back to Q156*

Q157

Give two differences between a half wave rectifier and a full wave rectifier.

[2022 • Set 55-5-3]

SOLUTION

Two differences between a half-wave rectifier and a full-wave rectifier are:

1. A **half-wave rectifier uses a single diode** and rectifies only **one half** of each AC input cycle, so the other half is wasted. A **full-wave rectifier uses two diodes** (with a centre-tapped transformer) or four diodes (bridge) and rectifies **both halves** of each input cycle.

2. In a half-wave rectifier the **output frequency equals the input frequency** and its efficiency is low (**≈ 40.6%**). In a full-wave rectifier the **output frequency is twice the input frequency**, the output is smoother, and its efficiency is higher (**≈ 81.2%**).

Because of higher efficiency and smoother output, the full-wave rectifier is preferred in practice.

Final answer. A half-wave rectifier uses one diode and rectifies one half cycle (low efficiency); a full-wave rectifier uses two/four diodes and rectifies both half cycles (higher efficiency, double frequency output).

← Back to Q157

Q158

With the help of the circuit diagram, explain the working of a diode as a half-wave rectifier.

[2021]

SOLUTION

A half-wave rectifier converts AC into pulsating DC using a single diode, which conducts only when forward biased.

Circuit diagram (described): The AC supply is connected to the primary of a step-down transformer. One end of the secondary coil goes to the **p-side (anode)** of the diode; the diode's **n-side (cathode)** connects to one end of the load resistance R_L; the other end of R_L returns to the second terminal of the secondary. The output is taken across R_L.

Working during the positive half cycle: The p-side of the diode is at a higher potential, so the diode is **forward biased** and conducts. Current passes through R_L and the output voltage follows the input.

Working during the negative half cycle: The p-side is now at a lower potential, so the diode is **reverse biased** and does not conduct. The output across R_L is zero.

Thus current flows through the load only during alternate half cycles. The output is a **pulsating DC**, a train of positive half-sine humps at the same frequency as the input. Its efficiency is about 40.6%.

Final answer. The diode conducts only in the forward-biased half cycle, so the load receives a unidirectional pulsating DC output.

← Back to Q158

Q159

With the help of the circuit diagram, explain briefly the working of a photodiode.

[2021]

SOLUTION

A photodiode is a special **p-n junction diode operated in reverse bias** that is used to detect light.

Circuit diagram (described): The photodiode is connected in **reverse bias** to a battery through a series load resistor, with a transparent window allowing light to fall on the junction. A microammeter (or a voltmeter across the resistor) is included to read the reverse current.

Working: When light of suitable frequency falls on the junction, photons with energy greater than the band gap are absorbed and create **electron-hole pairs** near the depletion region.

$$h\nu > E_g$$

The strong electric field of the reverse-biased junction sweeps these carriers apart: electrons drift to the n-side and holes to the p-side. This adds to the reverse current.

As the **intensity of incident light increases, the number of electron-hole pairs increases**, so the reverse current increases in proportion to the light intensity.

By measuring this reverse current, the light intensity is determined.

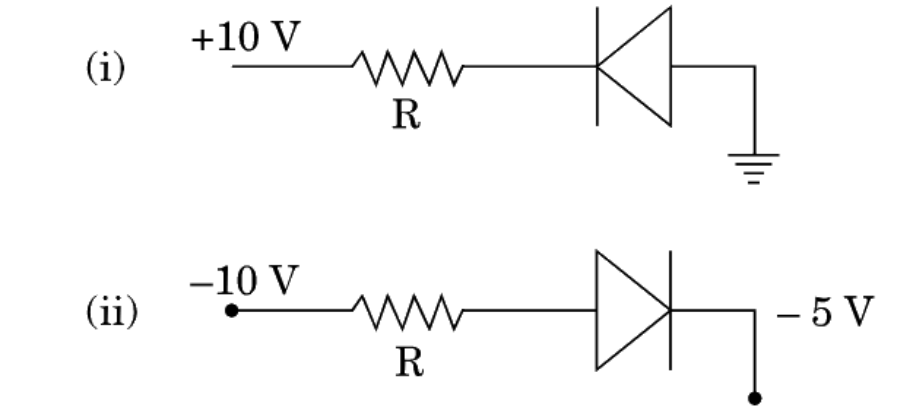
Reverse bias is preferred because the **fractional change in reverse current with light is much larger** than the corresponding change in the (already large) forward current.

Final answer. Incident light generates electron-hole pairs in the reverse-biased junction, increasing the reverse current in proportion to the light intensity.

← *Back to Q159*

Q160

(a) State whether the given ideal diodes are forward or reverse biased:



(b) The current in the forward bias is known to be more () than the current in the reverse bias (μA). What is the reason then to operate the photodiodes in reverse bias?

[2021]

SOLUTION

(a) Bias of each diode: A diode is forward biased when its **p-side is at a higher potential than its n-side**, and reverse biased otherwise.

(i) For the diode whose p-side is connected (through the resistance) towards the +10 *V* supply and whose n-side is towards the lower potential, the p-side is at the higher potential, so it is **forward biased** and conducts.

(ii) For the diode connected between −10 *V* (n-side) and −5 *V* (p-side), the p-side at −5 *V* is at a higher potential than the n-side at −10 *V* (difference +5 *V*), so it is

also **forward biased** and conducts. (If a diode were oriented so that its p-side is at the lower potential, it would be reverse biased.)

(b) Why photodiodes are operated in reverse bias: For a photodiode the useful quantity is the **change in current with light intensity**.

The reverse (dark) current is small (~ μA) but is very sensitive to the carriers generated by incident photons. So the photo-generated current is a **large fraction** of this small reverse current, giving a clearly measurable signal.

In forward bias the current is large (~ mA) and the small photo-current would be **swamped** by it, making the device a poor detector. Hence a photodiode is operated in reverse bias.

CONCEPT

The bias test is always 'is the p-side at a higher potential than the n-side?', not which terminal is positive in isolation.

Final answer. Both diodes are forward biased (p-side at higher potential); photodiodes use reverse bias because the fractional change of the small reverse current with light is much larger than in forward bias.

← *Back to Q160*

Q161

Explain the terms 'depletion layer' and 'potential barrier' in a p-n junction diode. How are the (a) width of depletion layer, and (b) value of potential barrier affected when the p-n junction is forward biased?

[2020 • Set 55-1-1]

SOLUTION

Depletion layer: When a p-type and an n-type semiconductor are joined, the majority electrons of the n-side diffuse into the p-side and the majority holes of the p-side diffuse into the n-side, where they recombine. This leaves behind a narrow region near the junction that contains only immobile ionised dopant atoms (positive donor ions on the n-side and negative acceptor ions on the p-

side) and is empty of mobile charge carriers. This carrier-free region is called the **depletion layer**.

Potential barrier: The immobile ions in the depletion layer set up an internal electric field $\vec{E}$ directed from the n-side towards the p-side. The potential difference that develops across the depletion layer due to this field is called the **potential barrier** V_0. Its value is about 0.7 *V* for silicon and about 0.3 *V* for germanium. It opposes the further diffusion of majority carriers across the junction.

(a) Effect of forward bias on width of depletion layer: In forward bias, the external voltage is applied opposite to the built-in field. It pushes the majority carriers towards the junction, which neutralise some of the uncovered ions. Hence the **width of the depletion layer decreases**.

(b) Effect of forward bias on potential barrier: The applied voltage *V* opposes the built-in potential, so the effective barrier becomes $V_0 - V$. Hence the **potential barrier decreases**, allowing a large forward current to flow.

Final answer. On forward biasing, both the width of the depletion layer and the potential barrier decrease.

← *Back to Q161*

Q162

Draw V-I characteristics of a p-n junction diode. Explain, why the current under reverse bias is almost independent of the applied voltage up to the critical voltage.

[2020 • Set 55-2-1]

SOLUTION

V-I characteristic of a p-n junction diode: Draw two perpendicular axes. The horizontal axis is the applied voltage *V* (positive to the right for forward bias, negative to the left for reverse bias) and the vertical axis is the current *I* (forward current upward in mA, reverse current downward in μA).

In the forward-bias region (first quadrant): the current is almost zero until the applied voltage reaches the knee or cut-in voltage ($\mathbf{V_T \approx 0.7\ V}$ for Si and 0.3 *V* for

Ge). Beyond V_T the curve rises steeply, showing that the forward current increases sharply with voltage.

In the reverse-bias region (third quadrant): a very small, nearly constant reverse saturation current I_S (of the order of μA) flows, staying almost flat until the critical breakdown voltage V_{Br} is reached, after which the reverse current rises very steeply.

Why the reverse current is almost independent of the applied voltage: The reverse current is the **minority-carrier saturation current**. It is produced by minority carriers (electrons in the p-region and holes in the n-region) that are generated thermally in the bulk and then diffuse into the depletion region, where the strong junction field immediately sweeps them across.

The number of such minority carriers depends only on the temperature, not on the applied reverse voltage. So even a small reverse voltage is enough to collect all the available minority carriers. Increasing the voltage cannot collect more carriers than are generated, so the current stays **almost constant (saturated)** until breakdown occurs.

Final answer. The reverse current is the temperature-controlled minority-carrier saturation current, so it stays nearly constant with voltage until breakdown.

← *Back to Q162*

Q163

Briefly explain how a potential barrier is set up across a p-n junction as a result of diffusion and drift of the charge carriers.

[2020 • Set 55-4-1]

SOLUTION

When p-type and n-type semiconductors are joined to form a junction, two opposing processes, diffusion and drift, take place and together set up the potential barrier.

Diffusion of majority carriers: Near the junction there is a steep concentration gradient. Electrons are abundant on the n-side and holes are abundant on the p-side, so the majority carriers diffuse across the junction. Electrons diffuse from the n-region into the p-region, and holes diffuse from the p-region into the n-region, where they recombine. This gives a diffusion current directed from the p-side to the n-side.

As the carriers leave, they uncover immobile ionised dopant atoms: positive donor ions are left on the n-side and negative acceptor ions are left on the p-side. The narrow carrier-free region containing only these ions is the **depletion region**.

These ions set up an internal electric field $\vec{E}$ directed from the n-side to the p-side. The potential difference associated with this field is the **barrier potential** V_b (about 0.7 V for Si).

Drift of minority carriers: The built-in field $\vec{E}$ drives the minority carriers across the junction. Electrons in the p-region drift to the n-side and holes in the n-region drift to the p-side, giving a drift current directed from the n-side to the p-side, opposite to the diffusion current.

Equilibrium: The depletion region keeps growing until the field becomes strong enough that the drift current exactly balances the diffusion current. At this stage the net current is zero and a steady **potential barrier** is established across the junction.

Final answer. Diffusion of majority carriers uncovers immobile ions whose field drives an opposing drift current; at balance a steady potential barrier is set up across the junction.

← *Back to Q163*

Q164

(a) Why is a photo diode operated under reverse bias condition?
(b) Draw V-I characteristic curves of photo diode for incident light of intensities I_1 *and* I_2 **(I_2>I_1)**.
OR

(a) State the level of doping and biasing condition used in light emitting diode (LED).
(b) Write any two advantages of LED over the conventional low power lamps.

[2020 • Set 55-4-1]

SOLUTION

(a) Why a photodiode is operated under reverse bias: A photodiode works by detecting the change in its reverse-saturation current caused by photo-generated minority carriers. When light of energy $\mathbf{h\nu > E_g}$ falls on the junction, extra electron-hole pairs are created and the reverse current increases.

In reverse bias the dark current is very small (of the order of μA), so even a small photo-induced increase is a large fractional change, giving **high sensitivity**. If the diode were forward biased, the large majority-carrier current would mask the small photocurrent and the response to light would be lost. Hence a photodiode is operated in reverse bias.

(b) V-I characteristics for intensities I_1 and I_2 ($\mathbf{I_2 > I_1}$): Draw the curves in the third quadrant (reverse-bias region). Both curves are nearly horizontal, saturating quickly with reverse voltage. The curve for the higher intensity I_2 lies at a larger magnitude of reverse current than the curve for I_1, since brighter light produces more electron-hole pairs and hence a larger photocurrent.

OR

(a) Doping and biasing of an LED: An LED is **heavily doped** on both the p-side and the n-side, and it is operated in **forward bias**. Heavy doping gives a high recombination rate, and forward bias drives the carriers across the junction so that recombination releases photons.

(b) Two advantages of an LED over a conventional low-power lamp: (i) An LED has a **high luminous efficiency**, giving more light output (lumens) per watt of power consumed. (ii) An LED has a **long operating life** and fast on-off switching (no fragile filament to burn out).

Final answer. Reverse bias keeps the dark current small so the photocurrent change is easily detected; the higher-intensity curve carries the larger reverse current.

← *Back to Q164*

Q165

(a) Explain the formation of energy bands in crystalline solids.
(b) Draw the energy band diagrams of (i) a metal and (ii) a semiconductor.

[2020 • Set 55-4-1]

SOLUTION

(a) Formation of energy bands in crystalline solids: In an isolated atom the electrons occupy discrete, sharply defined energy levels. When a large number N of identical atoms are brought together to form a crystal, the atoms are very close to one another.

Because the electron wavefunctions of neighbouring atoms overlap, and because the Pauli exclusion principle forbids any two electrons from occupying the same quantum state, each single atomic energy level splits into N very closely spaced sub-levels.

Since N is enormous in a real crystal (of the order of 10^{23} per cm^3), these sub-levels lie so close together that they merge into a continuous range of allowed energies called an **energy band**.

The highest band filled with electrons at 0 K is the **valence band**, and the next higher allowed band that is empty at 0 K is the **conduction band**. The gap of forbidden energies between them is the energy gap E_g, where no electron states exist.

(b) Energy band diagrams:

(i) Metal: Draw the valence band and the conduction band overlapping each other (or the conduction band shown partly filled), so that $\mathbf{E_g = 0}$. Plenty of empty states lie just above filled states, so electrons conduct freely.

(ii) Semiconductor: Draw a filled valence band and an empty conduction band separated by a small forbidden gap, $\mathbf{E_g \approx 1\ eV}$. At room temperature thermal energy lifts a few electrons across this small gap into the conduction band, giving limited conductivity.

TIP

A clean band diagram needs labels: valence band, conduction band, and the gap E_g marked between them, with $\mathbf{E_g = 0}$ for the metal and $\mathbf{E_g \approx 1\ eV}$ for the semiconductor.

Final answer. Overlap of atomic levels in a crystal forms continuous energy bands; a metal has overlapping (zero-gap) bands while a semiconductor has a small forbidden gap of about **1 eV**.

← *Back to Q165*

Q166

Why a signal transmitted from a TV tower cannot be received beyond a certain distance? Write the expression for the optimum separation between the receiving and the transmitting antenna.

[2019 • Set 55-1-1]

SOLUTION

Why TV signals cannot be received beyond a certain distance: TV broadcasting uses high frequencies in the VHF and UHF range (roughly 50 MHz to 800 MHz). At these frequencies the sky wave fails, because the ionosphere cannot reflect waves above about 30 MHz, and the ground wave is attenuated very rapidly.

Therefore TV signals travel as **space waves**, that is, in a straight line of sight directly from the transmitting antenna to the receiving antenna. Because the Earth is curved, this straight-line path is blocked by the bulge of the Earth beyond the geometric horizon, so the signal cannot be received beyond a certain distance.

For a transmitting antenna of height h on the Earth of radius R, the maximum line-of-sight distance to a receiver at ground level is

$$d_M = \sqrt{2Rh}$$

Optimum separation between transmitting and receiving antennas: If the transmitting antenna has height h_T and the receiving antenna has height h_R,

the maximum distance over which the signal can be received is

$$d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

This is the required **optimum separation**, and it exceeds the single-antenna range because each antenna sees up to its own horizon and the two horizons meet midway.

Final answer. TV uses line-of-sight space waves limited by the Earth's curvature; the optimum antenna separation is $\mathbf{d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}}$.

← *Back to Q166*

Q167

Write the relation between the height of a TV antenna and the maximum range up to which signals transmitted by the antenna can be received. How is this expression modified in the case of line of sight communication by space waves? In which range of frequencies, is this mode of communication used?

[2019 • Set 55-2-1]

SOLUTION

Relation between antenna height and maximum range: TV signals travel as space waves in a straight line of sight. For a transmitting antenna of height h on the Earth of radius R, with the receiver at ground level, the maximum range up to which the signal can be received is

$$d_M = \sqrt{2Rh}$$

This shows that the range increases with the square root of the antenna height.

Modified expression for line-of-sight communication by space waves: In practice both the transmitting antenna (height h_T) and the receiving antenna (height h_R) are raised above the ground. Each antenna contributes its own horizon distance, so the total range becomes

$$d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

This range is larger than that of a single tower, because each antenna sees beyond its own horizon and the two line-of-sight paths meet midway.

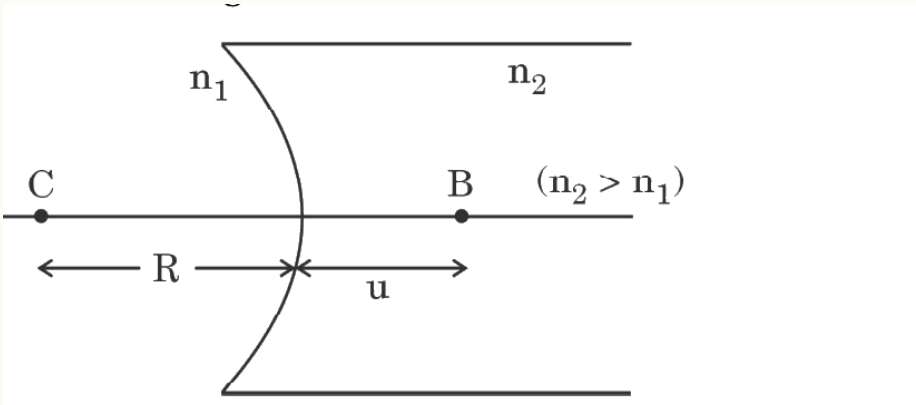
Frequency range of this mode: Space-wave (line-of-sight) propagation is used for frequencies **above about 40 MHz** (the VHF, UHF and microwave bands). At these frequencies the sky wave penetrates the ionosphere and the ground wave dies out quickly, so line-of-sight propagation is used for TV, FM radio, mobile and satellite communication.

Final answer. Single tower: $\mathbf{d_M = \sqrt{2Rh}}$; with both antennas: $\mathbf{d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}}$, used for frequencies above about **40 MHz**.

← *Back to Q167*

Q168

A certain n-p-n transistor has the common emitter output characteristics as shown in the figure.
(a) Find the emitter current at $\mathbf{V_{CE} = 12.5\ V}$ *and* $\mathbf{I_B = 60\ \mu A}$*, and*
(b) Current gain 'β' at this point.



[2019 • Set 55-4-1]

SOLUTION

From the common-emitter output characteristics, locate the operating point at $\mathbf{V_{CE} = 12.5\ V}$ on the curve drawn for $\mathbf{I_B = 60\ \mu A}$, and read off the collector current. From the graph this gives

$$I_C \approx 5.5\ \text{mA}$$

(a) Emitter current: By Kirchhoff's current law applied to the transistor, the emitter current is the sum of the base and collector currents.

$$I_E = I_B + I_C$$

$$I_E = 60\ \mu A + 5.5\ mA$$

$$I_E = 0.06\ mA + 5.5\ mA$$

$$I_E \approx 5.56\ mA$$

(b) Current gain β: The common-emitter current gain is the ratio of the collector current to the base current.

$$\beta = \frac{I_C}{I_B}$$

$$\beta = \frac{5.5 \times 10^{-3}}{60 \times 10^{-6}}$$

$$\beta = \frac{5500}{60}$$

$$\beta \approx 92$$

TIP

The exact numbers depend on the printed graph; the method is fixed: read I_C at the stated V_{CE} and I_B, then use $\mathbf{I_E = I_B + I_C}$ and $\boldsymbol{\beta = I_C/I_B}$.

Final answer. $\mathbf{I_E \approx 5.56\ mA}$ and $\boldsymbol{\beta \approx 92}$ (reading $\mathbf{I_C \approx 5.5\ mA}$ from the graph).

← *Back to Q168*

Q169

The frequencies of two side bands in an amplitude modulated wave are 640 kHz and 660 kHz respectively. Find the frequencies of the carrier and the modulating signals. Also obtain the value of the bandwidth required in amplitude modulation.

OR

A sinusoidal carrier voltage is amplitude modulated by a sinusoidal voltage of 10 kHz

with modulation index 0.3. If the carrier frequency is 10 MHz and its amplitude is 40 V, calculate the frequency and amplitude of the two sidebands.

[2019 • Set 55-5-1]

SOLUTION

In an amplitude-modulated wave the two side bands are the lower side band $\mathbf{f_{LSB} = f_c - f_m}$ and the upper side band $\mathbf{f_{USB} = f_c + f_m}$, where f_c is the carrier frequency and f_m is the modulating (message) frequency. Here $\mathbf{f_{LSB} = 640\ kHz}$ and $\mathbf{f_{USB} = 660\ kHz}$.

Adding the two relations gives the carrier frequency.

$$f_c = \frac{f_{USB} + f_{LSB}}{2} = \frac{660 + 640}{2}$$

$$f_c = 650\ \text{kHz}$$

Subtracting the two relations gives the modulating frequency.

$$f_m = \frac{f_{USB} - f_{LSB}}{2} = \frac{660 - 640}{2}$$

$$f_m = 10\ \text{kHz}$$

The bandwidth required in amplitude modulation is twice the modulating frequency.

$$BW = 2\, f_m = 2 \times 10\ \text{kHz}$$

$$BW = 20\ \text{kHz}$$

OR

Given $\mathbf{f_c = 10\ MHz}$, $\mathbf{f_m = 10\ kHz}$, carrier amplitude $\mathbf{A_c = 40\ V}$ and modulation index $\boldsymbol{\mu = 0.3}$.

The side-band frequencies are $f_c \pm f_m$.

$$f_{USB} = 10\ \text{MHz} + 10\ \text{kHz} = 10.01\ \text{MHz}$$

$$f_{LSB} = 10\ \text{MHz} - 10\ \text{kHz} = 9.99\ \text{MHz}$$

The amplitude of each side band is $\frac{\mu A_c}{2}$.

$$A_{SB} = \frac{\mu A_c}{2} = \frac{0.3 \times 40}{2}$$

$A_{SB} = 6\ V$ for each side band.

Final answer. $f_c = 650$ kHz, $f_m = 10$ kHz, BW = 20 kHz; (OR) $f_{LSB} = 9.99$ MHz, $f_{USB} = 10.01$ MHz, each of amplitude $6\ V$.

← *Back to Q169*

Q170

Draw a block diagram of a generalized communication system and write the functions of (i) a transmitter, and (ii) a receiver.

[2019 • Set 55-5-1]

SOLUTION

Block diagram of a generalised communication system: Draw a horizontal chain of labelled boxes connected by arrows in this order.

Information Source → Input Transducer → Transmitter → Channel → Receiver → Output Transducer → Destination

A small upward arrow labelled Noise is shown entering the Channel, since noise adds to the signal during transmission.

(i) Functions of a transmitter: A transmitter processes the message signal so that it can travel efficiently over the channel. It modulates the low-frequency message signal onto a high-frequency carrier wave, amplifies the modulated signal to a suitable power level, and radiates it into the channel through the transmitting antenna. Its main parts are the modulator, the carrier oscillator and the power amplifier with antenna.

(ii) Functions of a receiver: A receiver recovers the original message from the signal coming out of the channel. It picks up the weak, noisy modulated signal through the receiving antenna, amplifies and tunes to the desired carrier, demodulates it to extract the original message signal, and finally amplifies the recovered message to drive the output transducer. Its main parts are the

receiving antenna with tuner-amplifier, the detector or demodulator and the audio amplifier.

Final answer. The transmitter modulates, amplifies and radiates the signal; the receiver picks up, amplifies, demodulates and delivers the recovered message to the output transducer.

← *Back to Q170*

Q171

A carrier wave of peak voltage 15 *V is used to transmit a message signal. Find the peak voltage of the modulating signal in order to have a modulation index of* 60%.

[2018]

SOLUTION

The modulation index of an amplitude-modulated wave is defined as the ratio of the peak amplitude of the modulating (message) signal to the peak amplitude of the carrier signal.

$$\mu = \frac{V_m}{V_c}$$

where V_m is the peak voltage of the modulating signal and V_c is the peak voltage of the carrier wave.

Here the carrier peak voltage is $\mathbf{V_c = 15\ V}$ and the required modulation index is $\mathbf{\mu = 60\% = 0.60}$.

Rearranging the formula to find the modulating-signal voltage,

$$V_m = \mu \times V_c$$

$$V_m = 0.60 \times 15\ V$$

$$V_m = 9\ V$$

Hence, to obtain a modulation index of 60% with a carrier of peak voltage 15 *V*, the peak voltage of the modulating signal must be **9 V**.

TIP

Keep $\mu \leq 1$. If $V_m > V_c$ the modulation index exceeds 1, causing over-modulation and signal distortion at the receiver.

Final answer. The peak voltage of the modulating signal is **9 *V***.

← *Back to Q171*

Q172

Which basic mode of communication is used in satellite communication? What type of wave propagation is used in this mode? Write, giving reason, the frequency range used in this mode of propagation.

[2017]

SOLUTION

The basic mode of communication used in satellite communication is the **point-to-point (and broadcast) mode through a repeater in space**. A ground station beams the signal up to the satellite (the uplink); the satellite acts as a repeater, amplifying the weak received signal and retransmitting it back to one or many ground stations (the downlink).

The type of wave propagation used is **space-wave (line-of-sight) propagation**. The signal travels in a nearly straight line directly between the ground station antenna and the satellite, since there is a clear, obstacle-free path from the Earth's surface up to the satellite.

The frequency range used lies in the microwave (GHz) band, typically above about 30 MHz, and in practice in the range of a few GHz (for example, uplink near 6 GHz and downlink near 4 GHz).

The reason for using this high frequency range is that frequencies above the critical frequency (about 30 MHz) are not reflected back by the ionosphere; instead they penetrate it and travel straight to the satellite. Such high frequencies also allow the use of small, highly directional dish antennas and provide a wide bandwidth for carrying large amounts of information.

Hence satellite communication uses **space-wave propagation in the GHz range** because only these frequencies can pass through the ionosphere to reach the satellite.

Final answer. Point-to-point mode using space-wave (line-of-sight) propagation, in the GHz frequency range (above **30 MHz**), because such waves penetrate the ionosphere and reach the satellite.

← *Back to Q172*

Q173

Distinguish between a transducer and a repeater.

[2017]

SOLUTION

A transducer and a repeater perform very different jobs in a communication system, as set out below.

A transducer is a device that converts one form of energy or signal into another, usually converting a non-electrical signal into an electrical signal or vice versa. It is placed at the source or the destination of the communication system, that is, at the very beginning (where the message is generated) or at the very end (where it is finally received). It does not amplify the signal; it only changes its form. For example, a microphone converts sound into an electrical signal, and a loudspeaker converts an electrical signal back into sound.

A repeater is a combination of a receiver and a transmitter. It picks up the weak signal arriving along the channel, amplifies it, and then retransmits it onward. It is located in between the transmitter and the receiver, along the communication channel, to extend the range of communication. For example, a communication satellite acts as a repeater, receiving a weak uplink signal and retransmitting an amplified downlink signal.

Thus a transducer **converts the form of a signal at the ends** of the system, while a repeater **boosts and relays the signal along the channel**.

Final answer. A transducer converts one form of energy into another at the source/destination, whereas a repeater amplifies and retransmits a weakened signal along the channel to extend range.

← *Back to Q173*

Q174

(i) What is the line of sight communication?
(ii) Why is it not possible to use sky waves for transmission of TV signals? Upto what distance can a signal be transmitted using an antenna of height 'h'?

[2017]

SOLUTION

(i) Line-of-sight (LOS) communication is the mode of communication in which the electromagnetic signal travels in a nearly straight line directly from the transmitting antenna to the receiving antenna, like a ray of light. It needs the two antennas to be in direct view of each other, with no obstruction blocking the path. It is used at frequencies above about 40 MHz, for example in TV broadcasting, microwave links and mobile communication.

(ii) TV signals lie in the VHF/UHF band, that is, at frequencies well above 30 MHz. Sky-wave propagation works by reflection of the wave from the ionosphere, but the ionosphere can reflect waves only below about 30 MHz. Frequencies above this critical value are not reflected; they simply penetrate the ionosphere and escape into space. Since TV frequencies exceed this limit, the waves cannot be returned to Earth by the ionosphere, and therefore sky waves cannot be used for transmitting TV signals.

For an antenna of height h, the maximum distance up to which the signal can be transmitted to a receiver at ground level is given by

$$d = \sqrt{2\,R\,h}$$

where R is the radius of the Earth. This is the line-of-sight range over the curved surface of the Earth.

Hence TV signals must be sent by **line-of-sight (space-wave) propagation**, and a transmitting antenna of height h can cover a ground distance of $d = \sqrt{2Rh}$.

Final answer. LOS is straight-line antenna-to-antenna transmission; sky waves fail for TV because the high TV frequencies are not reflected by the ionosphere; the range is $d = \sqrt{2Rh}$.

← *Back to Q174*

Q175

Write two points of difference between intrinsic and extrinsic semiconductors.

[2017]

SOLUTION

The two main points of difference between intrinsic and extrinsic semiconductors are given below.

Purity and composition: An intrinsic semiconductor is a pure semiconductor, such as pure silicon or germanium, with no impurity added. An extrinsic semiconductor is a doped semiconductor, in which a small amount of a trivalent or pentavalent impurity is added to the pure crystal to increase its conductivity.

Carrier concentration and conductivity: In an intrinsic semiconductor the number of free electrons equals the number of holes, that is

$$n_e = n_h = n_i$$

and the conductivity is low and depends only on temperature. In an extrinsic semiconductor the two carrier concentrations are unequal: in an n-type $n_e \gg n_h$, and in a p-type $n_h \gg n_e$. The conductivity is therefore much higher and is controlled mainly by the amount of doping.

Hence an intrinsic semiconductor is **pure with equal electrons and holes**, while an extrinsic semiconductor is **doped with unequal carriers and far higher conductivity**.

Final answer. Intrinsic semiconductors are pure with $n_e = n_h = n_i$ and low conductivity; extrinsic semiconductors are doped with unequal carrier concentrations and much higher conductivity.

← *Back to Q175*

Q176

Distinguish between broadcast mode and point-to-point mode of communication and give one example for each.

[2017]

SOLUTION

The difference between the broadcast mode and the point-to-point mode of communication is as follows.

In the broadcast mode, a single transmitter sends information out to a large number of receivers spread over a wide area. The communication is one-way: the many receivers simply pick up the signal, and there is no individual addressing of each receiver. This mode is used for mass dissemination of information. An example of broadcast mode is **radio or television broadcasting**, where one transmitting tower serves a huge number of listeners or viewers at the same time.

In the point-to-point mode, communication takes place between a single transmitter and a single receiver, that is, between two specific points only. The link is private and is usually two-way. An example of point-to-point mode is a **telephone call**, where only the two parties on the line are connected to each other.

Thus broadcast mode is **one transmitter to many receivers**, whereas point-to-point mode is **one transmitter to one specific receiver**.

Final answer. Broadcast mode: one transmitter to many receivers (example, TV/radio broadcasting); point-to-point mode: one transmitter to one specific receiver (example, a telephone call).

← Back to Q176

Q177

Describe, with the help of a circuit diagram, the working of a photodiode.

[2017]

SOLUTION

A photodiode is a special junction diode that detects light. It is a p-n junction made with a transparent window so that light can fall on the junction, and it is always operated under reverse bias.

For the circuit diagram, draw the p-n photodiode connected so that its p-side goes to the negative terminal and its n-side to the positive terminal of a battery, that is, in reverse bias. A milliammeter is connected in series in the circuit to read the reverse current, and light (shown by arrows) is allowed to fall on the junction through the transparent window. Label the diode as photodiode, the cell as the biasing battery, and the meter as the current-measuring ammeter.

Working: When light of frequency ν such that the photon energy exceeds the band gap, that is

$$h\nu > E_g$$

falls on the junction, it is absorbed near the depletion region and creates electron-hole pairs. The strong electric field already present in the depletion region (because of the reverse bias) immediately sweeps these charges apart: electrons are pushed to the n-side and holes to the p-side. This flow of photo-generated carriers adds to the reverse current of the diode.

The greater the intensity of the incident light, the more electron-hole pairs are produced, and the larger the reverse current. Thus the reverse current is a measure of the light intensity, which is why the photodiode is used as a light detector.

The diode is kept in reverse bias because the fractional change in the reverse current with a given change in light intensity is much larger than the

corresponding change would be in forward bias, making the photodiode a more sensitive and reliable light sensor in reverse bias.

Final answer. A reverse-biased p-n photodiode: incident light ($\mathbf{h\nu > E_g}$) generates electron-hole pairs in the depletion region, which the junction field sweeps apart, increasing the reverse current in proportion to the light intensity.

← *Back to Q177*

Q178

Define modulation index. Why is it kept low? What is the role of a bandpass filter?

[2016]

SOLUTION

The modulation index of an amplitude-modulated wave is defined as the ratio of the peak amplitude of the modulating signal to the peak amplitude of the carrier signal.

$$\mu = \frac{A_m}{A_c}$$

where A_m is the amplitude of the modulating signal and A_c is the amplitude of the carrier. It tells us the depth of modulation, that is, how strongly the carrier is modulated by the message.

The modulation index is kept low, that is $\mu \leq 1$, to avoid distortion of the signal. If $\mathbf{\mu > 1}$ (over-modulation), the shape of the wave's envelope no longer matches the original message and the envelope gets distorted. The receiver then cannot recover the message faithfully, leading to loss of information. So μ is kept at or below 1 for distortion-free transmission and faithful recovery of the message.

The role of a band-pass filter is to allow only a chosen band of frequencies to pass through while blocking all frequencies outside that band. In a communication system it is used to select the desired range of frequencies (the wanted signal or sidebands) and to reject unwanted frequencies and noise, so that only the useful part of the signal is passed on for further processing.

Hence modulation index is $\mu = A_m/A_c$, it is kept ≤ 1 to **avoid distortion**, and a band-pass filter **passes the wanted frequency band and rejects the rest**.

Final answer. Modulation index $\mu = A_m/A_c$ is the depth of modulation; it is kept ≤ 1 to avoid distortion (over-modulation); a band-pass filter passes only the desired frequency band and blocks the rest.

← *Back to Q178*

Q179

State the concept of mobile telephony and explain its working.

[2016]

SOLUTION

Mobile telephony is the concept of providing voice and data communication to people who are on the move, allowing them to make and receive calls from anywhere within the network's coverage area using lightweight handheld phones that work over wireless (radio) links.

The working is based on a cellular system. The whole coverage area is divided into a number of small regions called cells. Each cell has a base station with antennas and radio equipment, and all the base stations are connected to a central office.

When a user makes a call, the mobile phone sends a radio signal to the base station of the cell it is currently in. That base station passes the call to the central switching office, which routes it onward, either to another base station (for a call to another mobile) or to the ordinary telephone network (for a call to a landline).

As the user moves from one cell to another, the call is automatically handed over from the base station of the old cell to that of the new cell, without breaking the conversation. This automatic transfer is called handoff.

A key feature is frequency reuse: cells that are far apart can use the same set of frequencies without interfering with one another, so that a large number of users can be served at the same time with a limited band of frequencies.

Hence mobile telephony lets users **communicate on the move through a network of cells, base stations and a central switching office, using handoff and frequency reuse**.

Final answer. Mobile telephony gives wireless voice/data service to users on the move, using a network of small cells, each with a base station linked to a central switching office, with automatic handoff between cells and frequency reuse to serve many users.

← Back to Q179

Q180

Why is base band signal not transmitted directly? Give any two reasons.

[2016]

SOLUTION

A base-band (low-frequency message) signal is not transmitted directly for the following two reasons.

Impractically large antenna size: For efficient radiation, the size of the transmitting antenna must be of the order of the wavelength of the signal, at least about λ/4. An audio (base-band) signal has a very low frequency and hence a very large wavelength. For example, for a 5 kHz audio signal,

$$\lambda = \frac{c}{f} = \frac{3 \times 10^8}{5 \times 10^3} = 6 \times 10^4\ m = 60\ \text{km}$$

so the antenna would need to be about 15 km long, which is impossible to build. By modulating the message onto a high-frequency carrier, the wavelength becomes small and a practical, short antenna can radiate it efficiently.

Mixing up of signals (no channel separation): If all transmitters sent their base-band signals directly, every audio signal would occupy the same low-frequency band, so the signals from different stations would overlap and mix together, and the receiver could not separate one from another. By shifting each message

to a different carrier frequency through modulation, many signals can be sent over the same medium at the same time without interfering.

Hence the base-band signal is not transmitted directly because it would need an **impractically large antenna** and because **different signals would overlap and mix**.

Final answer. Because (1) the low-frequency base-band signal needs an impractically large antenna (**~ 15 km** for **5 kHz**), and (2) all signals would occupy the same band and mix; modulation onto high-frequency carriers solves both.

← *Back to Q180*

Q181

What is ground wave communication? Explain why this mode cannot be used for long distance communication using high frequencies.

[2015]

SOLUTION

In ground wave (also called surface wave) communication, the radio waves travel directly along the surface of the Earth from the transmitting antenna to the receiving antenna. The wave glides over the ground and the ground acts as a guide for it.

As the wave moves along the surface, it induces currents in the conducting Earth. These induced currents take energy from the wave, so the wave is progressively absorbed and attenuated as it travels. Ground waves bend slightly with the curvature of the Earth and can reach receivers that are just below the horizon. This mode is used for amplitude-modulated broadcast in the medium frequency band.

The reason this mode cannot be used for long-distance communication at high frequencies is that the absorption of energy by the Earth increases as the frequency of the wave increases. The attenuation of a ground wave rises rapidly with frequency.

At high frequencies, the wave loses its energy very quickly to the ground, so the signal dies out after travelling only a short distance. Because of this heavy energy loss at high frequencies, ground wave propagation is **limited to low frequencies and short ranges**, and cannot carry high-frequency signals over long distances.

Final answer. Ground wave is surface propagation along the Earth; at high frequency the ground absorbs the energy too rapidly, so the signal attenuates over a short range and cannot reach far.

← *Back to Q181*

Q182

Distinguish between 'intrinsic' and 'extrinsic' semiconductors.

[2015]

SOLUTION

An intrinsic semiconductor is a pure semiconductor with no impurity added to it, for example pure silicon (Si) or pure germanium (Ge). An extrinsic semiconductor is one in which a small, controlled amount of suitable impurity (a dopant) has been deliberately added to a pure semiconductor, for example Si doped with phosphorus or with boron.

In an intrinsic semiconductor the charge carriers are produced only by thermal breaking of covalent bonds, which creates electron-hole pairs. Therefore the number of free electrons exactly equals the number of holes:

$$n_e = n_h = n_i$$

In an extrinsic semiconductor the doping adds extra carriers, so the electron and hole concentrations are no longer equal:

$$n_e \neq n_h$$

In an n-type extrinsic semiconductor $n_e \gg n_h$ (electrons are majority carriers), while in a p-type extrinsic semiconductor $n_h \gg n_e$ (holes are majority carriers).

The conductivity of an intrinsic semiconductor is **very small** at room temperature and increases only with rise in temperature. The conductivity of an extrinsic semiconductor is **much larger** and is controlled mainly by the amount of dopant added.

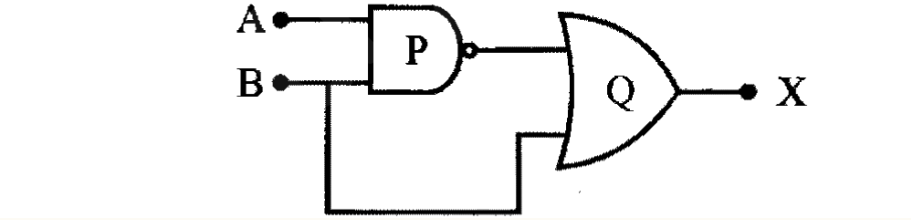
Thus the key distinction is purity and carrier balance: intrinsic is pure with equal electrons and holes, while extrinsic is doped with unequal carrier concentrations and higher conductivity.

Final answer. Intrinsic is a pure semiconductor with equal electrons and holes and low conductivity; extrinsic is a doped semiconductor with unequal carriers and much higher controlled conductivity.

← *Back to Q182*

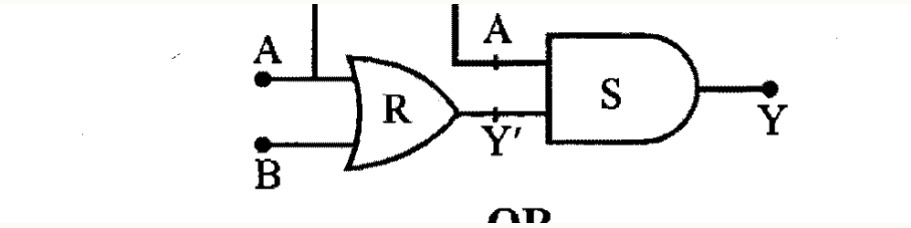
Q183

Write the truth table for the combination of the gates shown. Name the gates used.



OR

Identify the logic gates marked 'P' and 'Q' in the given circuit. Write the truth table for the combination.



[2014]

SOLUTION

The circuit shown is a combination of two NAND gates connected in series. The first NAND gate has the two inputs A and B. Its output is then fed to both input

terminals of the second NAND gate, which therefore acts as a NOT gate (a NAND with both inputs joined together behaves as an inverter).

The output of the first NAND gate is the NAND of A and B:

$$Y_1 = \overline{A \cdot B}$$

The second NAND gate inverts this output because both of its inputs are Y_1:

$$Y = \overline{Y_1 \cdot Y_1} = \overline{Y_1}$$

$$Y = \overline{A \cdot B} = A \cdot B$$

So the overall combination behaves exactly like an **AND gate**. The gates used are NAND and NAND.

The truth table of the combination is:

$$A = 0, B = 0 \Rightarrow Y_1 = 1, Y = 0$$

$$A = 0, B = 1 \Rightarrow Y_1 = 1, Y = 0$$

$$A = 1, B = 0 \Rightarrow Y_1 = 1, Y = 0$$

$$A = 1, B = 1 \Rightarrow Y_1 = 0, Y = 1$$

Hence the output Y is 1 only when both inputs are 1, which is the AND operation.

> **TIP**
>
> A NAND gate with its two inputs tied together always works as a NOT gate, which is the trick to recognise these gate combinations quickly.

Final answer. Both gates are NAND; the series combination gives $\boldsymbol{Y = A \cdot B}$, so it works as an AND gate (output 1 only when both inputs are 1).

← *Back to Q183*

Q184

Explain, with the help of a circuit diagram, the working of a p-n junction diode as a half-wave rectifier.

[2014]

SOLUTION

A half-wave rectifier uses a single p-n junction diode to convert alternating current into pulsating direct current.

Circuit diagram (described): The secondary coil of a step-down transformer supplies the AC input. One end of the secondary is joined to the p-side (anode) of the diode. The n-side (cathode) of the diode is connected to one end of a load resistor R_L. The other end of R_L goes back to the second terminal of the secondary, completing the loop. The output voltage V_{out} is taken across the load resistor R_L.

Working is based on the property that a p-n junction diode conducts only when it is forward biased and offers almost infinite resistance when reverse biased.

During the positive half cycle of the AC input, the p-side of the diode becomes positive, so the diode is forward biased. It conducts and a current flows through R_L, producing an output voltage across R_L that follows the input.

During the negative half cycle of the AC input, the p-side becomes negative, so the diode is reverse biased. It does not conduct, almost no current flows, and the output voltage across R_L is zero.

Therefore the diode allows current to pass only during the positive half cycles. The output across R_L consists of only the positive halves of the input wave, appearing as separate humps with gaps in between during the negative half cycles.

Hence the AC input is converted into a **unidirectional pulsating DC output**. Since only one half of each cycle is used, it is called a half-wave rectifier.

Final answer. A single diode conducts only in the positive half cycle and blocks the negative half, so the output across $\mathbf{R_L}$ is unidirectional pulsating DC: a half-wave rectifier.

← Back to Q184

Q185

Draw a circuit diagram of n-p-n transistor amplifier in CE configuration. Under what condition does the transistor act as an amplifier ?

[2014]

SOLUTION

Circuit diagram (described): In a common-emitter (CE) n-p-n transistor amplifier, the emitter terminal is common to both the input and the output, and is connected to the negative line (earth). The base-emitter input loop is supplied by a battery V_{BB} through a base resistor R_B, which forward biases the emitter-base junction. The collector is connected to the positive terminal of a supply battery V_{CC} through a load resistor R_L (or R_C), which reverse biases the collector-base junction. The AC input signal v_i is applied between base and emitter through an input coupling capacitor C_1, and the amplified AC output v_o is taken across the collector and emitter through an output coupling capacitor C_2.

Condition for amplification: A transistor acts as an amplifier only when it is working in its **active region**. For this, the two junctions must be biased as follows.

The emitter-base junction must be forward biased.

The collector-base junction must be reverse biased.

When biased this way, a small change in the input voltage V_{BE} causes a small change in the base current, which in turn produces a much larger change in the collector current (related by the current gain). This large collector current passing through R_L produces a large output voltage, so the weak input signal is amplified. (The output is also 180° out of phase with the input in CE configuration.)

Final answer. The transistor amplifies only in the active region, that is when the emitter-base junction is forward biased and the collector-base junction is

reverse biased.

← *Back to Q185*

Q186

Write the functions of the following in communication systems :
(i) Receiver
(ii) Demodulator

[2014]

SOLUTION

(i) Receiver: The receiver is the unit placed at the destination end of a communication system. Its function is to receive the modulated signal that has travelled through the channel and to process and recover the original message signal from it.

The received signal is usually very weak and mixed with noise, so the receiver first picks it up through the receiving antenna, then amplifies the weak signal, selects the desired carrier frequency, and finally recovers the message signal so that it can be delivered to the output device. In short, the receiver reconstructs a usable form of the original information signal from the received corrupted signal.

(ii) Demodulator: A demodulator is the part of the receiver that performs the reverse of modulation. Its function is to extract (separate out) the original message signal from the received modulated carrier wave.

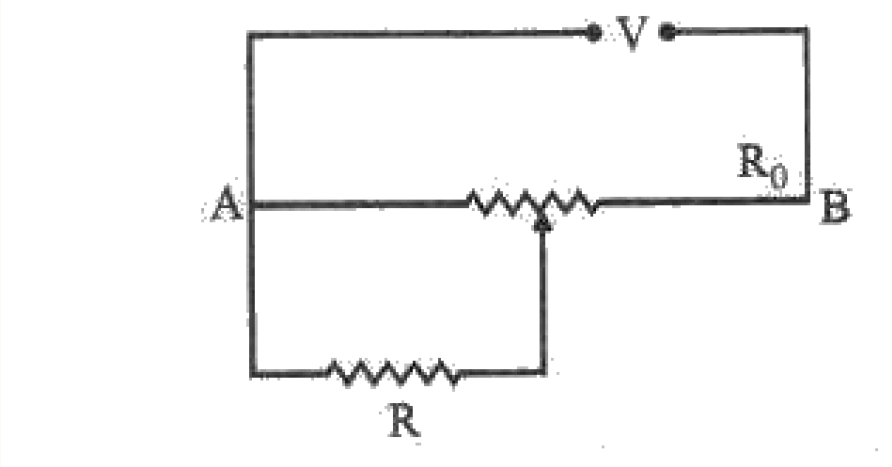
During transmission the message signal had been superimposed on a high-frequency carrier; the demodulator removes the carrier and returns the low-frequency message signal in its original form. Thus the **demodulator recovers the information signal** from the modulated wave, completing the inverse of the modulation done at the transmitter.

Final answer. The receiver picks up, amplifies and processes the received modulated signal to recover the message; the demodulator extracts the original message signal from the modulated carrier (reverse of modulation).

← Back to Q186

Q187

Block diagram of a receiver is shown in the figure:



(a) Identify 'X' and 'Y'.
(b) Write their functions.

[2013]

SOLUTION

The standard CBSE receiver block diagram is: Receiving Antenna goes into the Amplifier (X), then into the Detector (demodulator), then into the Amplifier (Y), and finally to the Output. Here X is the receiver-side amplifier that handles the incoming modulated signal, and Y is the amplifier that handles the recovered message signal.

(a) Identification:

X = Amplifier (the receiving amplifier that amplifies the weak modulated signal picked up by the antenna).

Y = Amplifier (the amplifier that strengthens the demodulated message signal before it reaches the output).

(b) Functions:

Function of X: The signal picked up by the receiving antenna is very weak. X amplifies this weak received (modulated) signal and raises it to a level suitable for the detector to work on.

Function of Y: After the detector recovers the original message signal, that signal is still weak. Y amplifies this demodulated message signal so that it is strong enough to drive the output device, such as a loudspeaker. Thus **X amplifies the received modulated signal and Y amplifies the demodulated message signal** before it is delivered to the output.

Final answer. X is the receiving amplifier that boosts the weak modulated signal from the antenna; Y is the amplifier that strengthens the demodulated message signal before the output.

← *Back to Q187*

Q188

Explain, with the help of a circuit diagram, the working of a photo-diode. Write briefly how it is used to detect the optical signals.
OR
Mention the important considerations required while fabricating a p-n junction diode to be used as a Light Emitting Diode (LED). What should be the order of band gap of an LED if it is required to emit light in the visible range?

[2013]

SOLUTION

Photodiode: A photodiode is a special p-n junction diode that is operated in reverse bias and is fabricated with a transparent window so that light can fall on the junction.

Circuit diagram (described): The photodiode is connected in reverse bias in series with a battery and a load resistor R. The p-side is joined to the negative terminal and the n-side to the positive terminal of the battery. Light is allowed to fall on the junction through the transparent window, and the output is taken across the resistor R (which measures the reverse current).

Working: When light photons of energy greater than or equal to the band gap fall on the junction, they are absorbed and create extra electron-hole pairs near the depletion region. This condition is:

$$h\nu \geq E_g$$

Because the diode is reverse biased, the strong electric field across the depletion region sweeps these newly generated electrons to the n-side and holes to the p-side. This adds to the reverse current of the diode.

The magnitude of this photo-generated reverse current is directly proportional to the intensity of the incident light. So a brighter light gives a larger reverse current.

Detection of optical signals: When an optical signal whose intensity varies with the information falls on the photodiode, the reverse current of the diode varies in step with the intensity of the light. These current variations, measured as the changing voltage across R, reproduce the original signal. Hence the photodiode **converts the optical signal into an equivalent electrical signal**, which can then be amplified and processed.

Final answer. A reverse-biased photodiode generates electron-hole pairs when light ($\mathbf{h\nu \geq E_g}$) falls on its junction; the reverse current is proportional to light intensity, so intensity variations of an optical signal appear as current variations, detecting the signal.

← Back to Q188

Q189

Write two factors justifying the need of modulating a signal.
A carrier wave of peak voltage 12 *V is used to transmit a message signal. What should be the peak voltage of the modulating signal in order to have a modulation index of* 75%*?*

[2012]

SOLUTION

Two factors that justify the need for modulating a signal:

1. Size of the antenna: For efficient radiation, the length of the transmitting antenna should be of the order of the wavelength of the signal (about $\lambda/4$). A low-frequency message signal has a very large wavelength, so it would need an impractically long antenna. By modulating the message onto a high-frequency carrier, the wavelength becomes small and the antenna size becomes practical.

2. Effective power radiated and avoiding mixing of signals: The power radiated by an antenna is much larger at high frequencies, so modulation improves the strength of the transmitted signal. Also, by giving different carrier frequencies to different transmitters, many message signals can be sent simultaneously through the same medium without mixing with one another.

Numerical: For a modulation index (modulation factor) μ, the relation between the peak voltage of the modulating signal V_m and the peak voltage of the carrier V_c is:

$$\mu = \frac{V_m}{V_c}$$

$$V_m = \mu\, V_c$$

$$V_m = 0.75 \times 12\ V$$

$$V_m = 9\ V$$

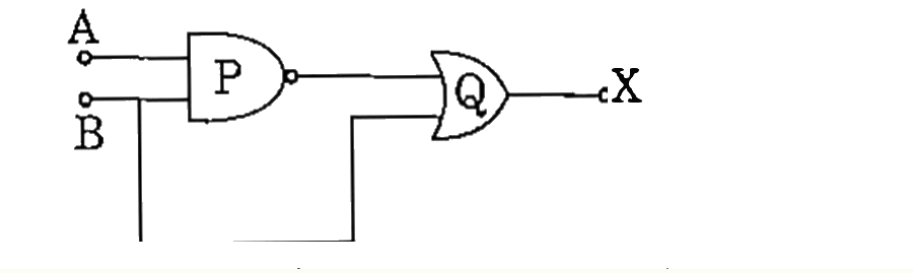
Hence the peak voltage of the modulating signal must be **9 *V***.

Final answer. Modulation is needed to keep the antenna size practical and to avoid mixing of different signals; the required peak modulating voltage is $\mathbf{V_m = \mu\, V_c = 0.75 \times 12 = 9\ V}$.

← *Back to Q189*

Q190

(i) Identify the logic gates marked P and Q in the given logic circuit.
*(ii) Write down the output at X for the inputs **A**=0, **B**=0 and **A**=1, **B**=1.*



[2012]

SOLUTION

(i) Identifying the gates P and Q: In the given circuit, gate P is a NAND gate having the two inputs A and B, and gate Q is a NOT gate that inverts the output of P. So the message passes first through a NAND gate and then through a NOT gate.

The output of NAND gate P is:

$$P = \overline{A \cdot B}$$

The NOT gate Q inverts this:

$$X = \overline{P} = \overline{\overline{A \cdot B}} = A \cdot B$$

So the combination of NAND followed by NOT works as an **AND gate**, and the final output is $\boldsymbol{X = A \cdot B}$.

(ii) Output X for the given inputs:

For $\boldsymbol{A = 0, B = 0}$:

$$X = 0 \cdot 0 = 0$$

For $\boldsymbol{A = 1, B = 1}$:

$$X = 1 \cdot 1 = 1$$

Hence the outputs are $\boldsymbol{X = 0}$ when $\boldsymbol{A = 0, B = 0}$ and $\boldsymbol{X = 1}$ when $\boldsymbol{A = 1, B = 1}$.

Final answer. P is a NAND gate and Q is a NOT gate, so the combination acts as AND ($\boldsymbol{X = A \cdot B}$); thus $\boldsymbol{X = 0}$ for $\boldsymbol{A=0, B=0}$ and $\boldsymbol{X = 1}$ for $\boldsymbol{A=1, B=1}$.

← Back to Q190

Q191

Describe briefly with the help of a circuit diagram, how the flow of current carriers in a p-n-p transistor is regulated with emitter-base junction forward biased and base-collector junction reverse biased.

[2012]

SOLUTION

In a p-n-p transistor, the heavily doped p-region is the emitter (E), the thin lightly doped n-region in the middle is the base (B), and the moderately doped p-region is the collector (C).

Biasing: the emitter-base junction is forward biased by a battery V_{EE}, and the base-collector junction is reverse biased by a battery V_{CC}. In the circuit, the emitter is connected to the positive terminal of V_{EE} and the base to its negative terminal, while the collector is connected to the negative terminal of V_{CC} and the base to its positive terminal. A milliammeter is placed in the collector lead and a microammeter in the base lead.

Because the emitter-base junction is forward biased, the potential barrier there is lowered, so the majority carriers of the emitter, which are holes, are injected (emitted) from the p-type emitter into the n-type base. This injection of holes constitutes the emitter current I_E.

The base is made very thin and lightly doped, so it has very few free electrons. Hence only a small fraction (about 1 to 5 percent) of the injected holes recombine with electrons in the base, giving a small base current I_B.

The remaining holes (about 95 to 99 percent) diffuse across the thin base and reach the reverse-biased base-collector junction. The strong electric field at this reverse-biased junction sweeps these holes into the collector, constituting the collector current I_C.

By Kirchhoff's current law, the emitter current equals the sum of the base and collector currents.

$$I_E = I_B + I_C$$

Since the base is thin and lightly doped, almost all the injected holes reach the collector, so $I_C \gg I_B$. Thus a small change in base current controls a much larger

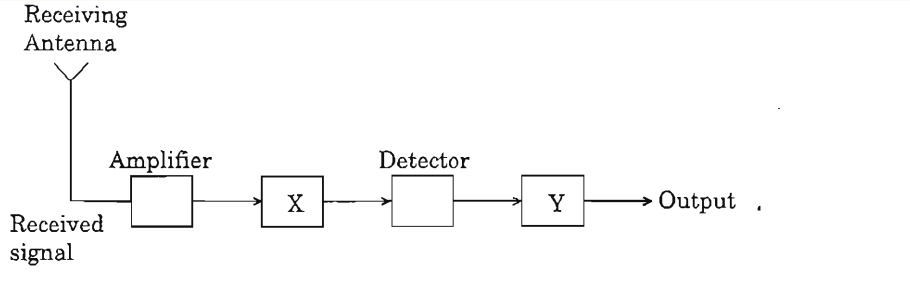
collector current, which is the basis of transistor action.

Final answer. Forward-biased emitter injects holes into the thin base; most cross to the reverse-biased collector giving $\mathbf{I_C}$, a few recombine giving $\mathbf{I_B}$, with $\mathbf{I_E = I_B + I_C}$.

← *Back to Q191*

Q192

In the given block diagram of a receiver, identify the boxes labelled as X and Y and write their functions.



X -> Amplifier -> Detector -> Y -> Output'>

[2012]

SOLUTION

The receiver block diagram is: Receiving Antenna (X) goes to Amplifier, then to Detector, then to Amplifier (Y), then to Output.

Box X is the **Receiving Antenna**. Its function is to intercept the modulated electromagnetic waves travelling through space and convert them into a corresponding feeble modulated electrical signal, which is then passed on to the amplifier stage.

Box Y is the **Amplifier**. It comes after the detector. Its function is to amplify the weak message (audio) signal recovered by the detector so that it has enough power to operate the output transducer such as a loudspeaker.

Summary of functions: X (Receiving Antenna) picks up the transmitted modulated signal from the channel; Y (Amplifier) strengthens the detected message signal to drive the output device.

Final answer. X is the Receiving Antenna that picks up the modulated electromagnetic waves; Y is the Amplifier that strengthens the detected message signal to drive the output device.

← Back to Q192

Q193

Distinguish between 'Analog and Digital signals'.
OR
Mention the function of any two of the following used in communication system:
(i) Transducer
(ii) Repeater
(iii) Transmitter
(iv) Bandpass Filter

[2012]

SOLUTION

Analog and Digital signals are distinguished as follows.

An analog signal is a continuous signal that varies smoothly with time and can take any value within a given range. Examples are the human voice, the output of a microphone, and the temperature read by a sensor.

A digital signal is a discontinuous signal that takes only a few discrete values, usually two levels written as 0 (low) and 1 (high). Examples are the output of a computer and the binary pulses on a data line.

Analog signals get corrupted by noise easily and the noise accumulates along the channel, whereas digital signals are far more immune to noise because each bit can be regenerated cleanly even if it is partly distorted.

OR (functions of any two communication-system components):

(i) Transducer: a device that converts energy from one form into another, for example sound or pressure into a corresponding electrical signal, or an electrical signal back into sound.

(ii) Repeater: a combination of a receiver and a transmitter that picks up a weak signal, amplifies and reshapes (regenerates) it, and retransmits it, so that the signal can travel over long distances.

(iii) Transmitter: it processes the incoming message signal (modulates and amplifies it) so that it is suitable for transmission and radiation through the channel by the antenna.

(iv) Bandpass filter: it allows only a desired band of frequencies to pass and blocks all other frequencies, and is used in tuning and in reducing noise.

Final answer. Analog signals vary continuously and are noise-prone; digital signals take discrete 0/1 values and are noise-immune (or: any two component functions as stated above).

← *Back to Q193*

Q194

The current in the forward bias is known to be more () than the current in the reverse bias (μA). What is the reason, then, to operate the photodiode in reverse bias?

[2012]

SOLUTION

A photodiode is used to detect light by measuring the change in its current caused by incident light, so what matters is not the absolute current but how much the current changes when light falls on it.

Consider a photodiode under illumination. Let the carrier concentrations in reverse bias be n (electrons) and p (holes), and under illumination let them increase to n' and p'.

When light of suitable energy falls on the junction, it generates electron-hole pairs. The number of extra (minority) carriers generated is comparable to the small number of minority carriers already present, so the fractional change in current is large in reverse bias.

In reverse bias the current is carried mainly by minority carriers, whose number is small (current of the order of μA). When light creates additional

electron-hole pairs, the increase in minority carriers is a large fraction of the small dark current, so the change in current is easily measurable.

In forward bias the current is large (of the order of mA) and is dominated by the majority-carrier injection. The small photo-generated change due to light is then a tiny fraction of this large current and is masked by it, so it cannot be measured reliably.

Hence, even though the forward current is larger, the photodiode is operated in reverse bias because there the fractional change in current produced by light is large, making the device a sensitive and accurate detector of light intensity.

TIP

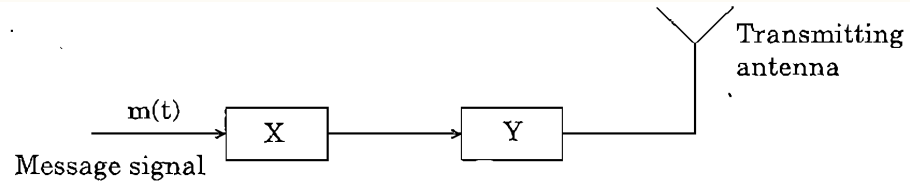
The exam-worthy idea is the fractional change in current, not the absolute current; state it explicitly to earn full marks.

Final answer. Reverse bias is used because the light-induced fractional change in current is large there, giving a sensitive measure of light intensity, whereas in forward bias this change is swamped by the large current.

← *Back to Q194*

Q195

Figure shows a block diagram of a transmitter. Identify the boxes 'X' and 'Y' and write their functions.



X -> Y -> Transmitting antenna'>

[2012]

SOLUTION

The block diagram of the transmitter is: Message signal goes to X, then to Y, then to the Transmitting antenna.

The box X is the Modulator. The audio (message) signal is first amplified and then fed to the modulator along with a high-frequency carrier wave produced by an oscillator. The modulator superimposes the low-frequency message on the high-frequency carrier and produces a modulated wave (such as an amplitude-modulated wave).

Function of X (Modulator): it combines the message signal with the high-frequency carrier so that the message is shifted to the carrier frequency band, which can be radiated efficiently and transmitted over long distances.

The box Y is the Power Amplifier. The modulated wave coming from the modulator is at low power, so it is amplified to a high power level.

Function of Y (Power Amplifier): it raises the power of the modulated signal to a level high enough to drive the transmitting antenna, so that the antenna can radiate the signal effectively into space.

Final answer. X is the Modulator (superimposes the message on the carrier); Y is the Power Amplifier (boosts the modulated wave's power to drive the transmitting antenna).

← *Back to Q195*

Q196

Write two characteristic features to distinguish between n-type and p-type semiconductors.
OR
How does a light emitting diode (LED) work? Give two advantages of LED's over the conventional incandescent lamps.

[2012]

SOLUTION

Two characteristic features that distinguish n-type and p-type semiconductors:

(1) Type of dopant and majority carriers: an n-type semiconductor is made by doping pure silicon or germanium with a pentavalent impurity such as phosphorus, arsenic or antimony; its majority carriers are electrons and

minority carriers are holes. A p-type semiconductor is made by doping with a trivalent impurity such as boron, aluminium, gallium or indium; its majority carriers are holes and minority carriers are electrons.

(2) Position of the impurity energy level: in an n-type semiconductor the donor energy level E_D lies just below the conduction band, so electrons are easily excited into the conduction band. In a p-type semiconductor the acceptor energy level E_A lies just above the valence band, so electrons from the valence band are easily accepted, creating holes.

OR (LED working and advantages):

A light emitting diode (LED) is a heavily doped p-n junction operated under forward bias. When forward biased, electrons from the n-side and holes from the p-side are pushed towards the junction and recombine near it.

During each electron-hole recombination, energy nearly equal to the band gap E_g is released in the form of a photon.

$$h\nu \approx E_g$$

If E_g lies in the range of about 1.8 eV to 3 eV, the emitted photons are in the visible region and the LED glows with visible light whose colour depends on E_g.

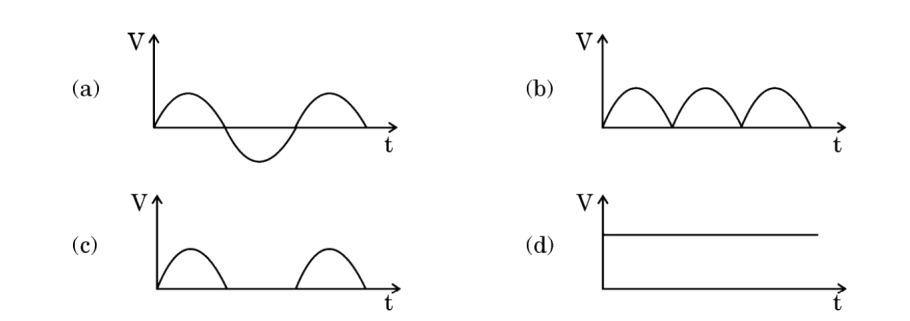
Two advantages of LEDs over incandescent lamps: (1) LEDs have much higher efficiency, giving more light output for the same power consumed; (2) LEDs have a much longer operating life and faster response (switching) time.

Final answer. n-type uses pentavalent dopant with electrons as majority carriers ($\mathbf{E_D}$ near conduction band); p-type uses trivalent dopant with holes as majority carriers ($\mathbf{E_A}$ near valence band). (OR LED: forward-biased junction emitting photons of energy $\approx \mathbf{E_g}$; advantages are higher efficiency and longer life.)

← *Back to Q196*

Q197

Draw the output waveform at X, using the given inputs A and B for the logic circuit shown below. Also, identify the logic operation performed by this circuit.



[2011 • Set 55-1-1]

SOLUTION

The given logic circuit produces a single output X from the two inputs A and B. Working out its Boolean output, the circuit gives:

$$X = A \cdot B$$

So the circuit performs the **AND operation**.

Truth table: when A=0, B=0 then X=0; when A=0, B=1 then X=0; when A=1, B=0 then X=0; when A=1, B=1 then X=1.

Output waveform at X: along the common time axis, X stays LOW (0) at every instant, and goes HIGH (1) only during the time intervals in which inputs A and B are simultaneously HIGH. Thus the X pulse appears only where the high regions of A and B overlap, and is low everywhere else.

Hence the logic operation performed by this circuit is the AND operation.

Final answer. The output X is HIGH only when A and B are both HIGH (LOW otherwise), so the circuit performs the AND operation.

← Back to Q197

Q198

Name the semiconductor device that can be used to regulate an unregulated dc power supply. With the help of I-V characteristics of this device, explain its working principle.

[2011 • Set 55-1-1]

SOLUTION

The semiconductor device used to regulate an unregulated DC power supply is the Zener diode.

It is a specially designed, heavily doped p-n junction diode that is operated in the reverse breakdown region.

I-V characteristics: in forward bias the Zener diode behaves like an ordinary junction diode, with the current remaining small until the knee voltage is reached and then rising rapidly. In reverse bias only a very small reverse saturation current flows as the reverse voltage is increased, until the reverse voltage reaches the Zener (breakdown) voltage V_Z. At V_Z, the reverse current increases very sharply (the curve becomes almost vertical) while the voltage across the diode stays practically constant at V_Z.

Thus on the I-V graph, the reverse-bias portion shows an almost vertical line at $\boldsymbol{V = V_Z}$, meaning a large change in current produces almost no change in voltage.

Working principle of regulation: the unregulated DC input voltage V_{in} is connected to the Zener diode through a series current-limiting resistor R_S, and the load R_L is connected in parallel with the Zener (Zener in reverse bias).

When V_{in} increases, the extra voltage is dropped across R_S and the Zener draws more current, but the voltage across it stays fixed at V_Z. When V_{in} decreases, the Zener draws less current, and again the voltage across it stays at V_Z.

Hence the output voltage across the load remains constant.

$$V_{out} = V_Z$$

In this way the Zener diode keeps the output voltage steady at V_Z even when the input supply voltage or the load current fluctuates, thus acting as a voltage regulator.

Final answer. The Zener diode regulates the supply: operated in reverse breakdown, it holds the voltage constant at $\mathbf{V_Z}$ across the load despite input or load fluctuations, so $\mathbf{V_{out} = V_Z}$.

← Back to Q198

Q199

Draw the transfer characteristic curve of a base biased transistor in CE configuration. Explain clearly how the active region of the V_o versus V_i curve in a transistor is used as an amplifier.

[2011 • Set 55-1-1]

SOLUTION

The transfer characteristic of a base-biased transistor in common-emitter (CE) configuration is a graph of the output voltage V_o (equal to V_{CE}) along the y-axis against the input voltage V_i (applied at the base) along the x-axis.

The curve has three regions. For small V_i the curve is flat and high near $\mathbf{V_o = V_{CC}}$ (cut-off region). As V_i increases past the point where the emitter-base junction is forward biased, the curve falls steeply and almost linearly (active region). For large V_i the curve flattens again near $\mathbf{V_o \approx 0}$ (saturation region).

Cut-off region: V_i is too small to forward bias the emitter-base junction, so the transistor is OFF, $\mathbf{I_C \approx 0}$ and $\mathbf{V_o \approx V_{CC}}$ (transistor acts as an open switch).

Active region: the emitter-base junction is forward biased and the base-collector junction is reverse biased. Here V_o decreases almost linearly as V_i increases, and the slope of this region is steep and negative.

Saturation region: V_i is large, both junctions are forward biased, the transistor is fully ON, and $\mathbf{V_o \approx 0.2\ V}$ (transistor acts as a closed switch).

Use of the active region as an amplifier: in the active region the output voltage V_o changes by a large amount for a small change in input voltage V_i, and the relationship is nearly linear with a large negative slope. The magnitude of this slope is the voltage gain.

$$A_v = \frac{\Delta V_o}{\Delta V_i}$$

The transistor is first biased at a suitable operating point (Q-point) in the middle of the active region. A small AC input signal is then superimposed on this DC bias at the base.

Because the active region is linear with a large negative slope, this small AC input produces a large AC output voltage of the same waveform but opposite

(reversed) phase. This faithful, amplified output is how the active region is used for amplification.

Final answer. The transfer curve shows cut-off, active and saturation regions; in the linear active region a small change in $\mathbf{V_i}$ gives a large, phase-reversed change in $\mathbf{V_o}$, so a transistor biased there amplifies an AC signal.

← *Back to Q199*

Q200

(i) Define modulation index.
(ii) Why is the amplitude of modulating signal kept less than the amplitude of carrier wave?

[2011 • Set 55-1-1]

SOLUTION

(i) Modulation index: the modulation index μ (also called the modulation factor) is defined as the ratio of the amplitude of the modulating (message) signal to the amplitude of the carrier wave.

$$\mu = \frac{A_m}{A_c}$$

Here A_m is the amplitude of the modulating signal and A_c is the amplitude of the carrier wave. It can also be written in terms of the maximum and minimum amplitudes of the modulated wave's envelope.

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

It is a dimensionless quantity, and for distortion-free amplitude modulation it lies between 0 and 1.

(ii) Reason for keeping the modulating signal amplitude less than the carrier amplitude: keeping $\mathbf{A_m < A_c}$ keeps the modulation index $\boldsymbol{\mu < 1}$.

If $\mathbf{A_m > A_c}$, then $\boldsymbol{\mu > 1}$, which is called overmodulation. In overmodulation the envelope of the modulated wave gets distorted and the wave crosses zero, causing the shape of the original message to be lost.

As a result the message signal cannot be recovered faithfully at the receiver during demodulation. To avoid such distortion and to ensure faithful recovery of the message, the amplitude of the modulating signal is kept less than the amplitude of the carrier wave.

Final answer. (i) Modulation index $\mu = A_m/A_c$. (ii) Keeping $A_m < A_c$ ensures $\mu < 1$, avoiding overmodulation and distortion so the message can be faithfully recovered.

← *Back to Q200*

Q201

Plot a graph showing temperature dependence of resistivity for a typical semiconductor. How is this behaviour explained?

[2011 • Set 55-2-1]

SOLUTION

The resistivity of a semiconductor decreases as its temperature increases, which is exactly opposite to the behaviour of a metal.

The graph of resistivity ρ against temperature T is a smoothly falling curve. It starts from a high value of ρ at low temperature, on the left of the graph, and falls steeply (almost exponentially) as T increases towards the right. The slope is negative throughout, so the curve always goes downhill.

The conductivity of a semiconductor is given by

$$\sigma = e(n_e\mu_e + n_h\mu_h)$$

where n_e, n_h are the electron and hole concentrations and μ_e, μ_h are their mobilities, and $\rho = \frac{1}{\sigma}$.

As the temperature rises, many more covalent bonds break, so the number of free charge carriers increases very rapidly. The intrinsic carrier concentration follows

$$n_i \propto T^{3/2}\exp\left(-\frac{E_g}{2k_B T}\right)$$

so n_e and n_h rise almost exponentially with T.

Although the mobilities μ_e, μ_h decrease slightly due to increased lattice vibrations, this fall is very small. The huge exponential rise in carrier concentration completely dominates.

As a result σ increases and ρ falls sharply with rising temperature. A semiconductor therefore has a **negative temperature coefficient of resistivity**.

Final answer. Resistivity falls steeply (exponentially) with rising temperature because the number of charge carriers increases rapidly; semiconductors have a negative temperature coefficient of resistivity.

← *Back to Q201*

Q202

For an amplitude modulated wave, the maximum amplitude is found to be 10 V while the minimum amplitude is 2 V. Calculate the modulation index. Why is modulation index generally kept less than one?

[2011 • Set 55-2-1]

SOLUTION

For a sinusoidal amplitude modulated wave, the maximum and minimum amplitudes are related to the carrier amplitude A_c and the modulating amplitude A_m by $\mathbf{A_{max} = A_c + A_m}$ and $\mathbf{A_{min} = A_c - A_m}$.

The modulation index is

$$\mu = \frac{A_m}{A_c} = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

Here $\mathbf{A_{max} = 10\ V}$ and $\mathbf{A_{min} = 2\ V}$.

Substituting the values,

$$\mu = \frac{10 - 2}{10 + 2} = \frac{8}{12}$$

$$\mu = 0.667$$

So the **modulation index $\mu \approx 0.67$** (about 67%).

The modulation index is generally kept less than one to avoid **overmodulation**. If $\mu > 1$, the modulating amplitude A_m exceeds the carrier amplitude A_c, so the envelope of the AM wave becomes negative during part of each cycle. This distorts the wave and the original message cannot be recovered faithfully at the receiver. Keeping $\mu \le 1$ ensures the envelope stays positive and the signal is reproduced without distortion.

Final answer. Modulation index $\mu = 0.67$; it is kept below one to prevent overmodulation and distortion of the recovered signal.

← Back to Q202

Q203

Draw a block diagram showing the important components in a communication system. What is the function of a transducer?

[2011 • Set 55-2-1]

SOLUTION

A communication system has three basic units, the transmitter, the channel and the receiver, with transducers at the two ends and noise acting on the channel.

The block diagram, drawn left to right as a chain of boxes joined by arrows, is

Information Source → Input Transducer → Transmitter → Communication Channel → Receiver → Output Transducer → User of Information.

A separate block labelled Noise points an arrow onto the Communication Channel, since noise enters the signal while it travels through the channel.

The role of each block is as follows. The transmitter processes and modulates the signal and sends it into the channel. The channel is the physical medium, such as wires, free space or optical fibre, that carries the signal. The receiver picks up the weak incoming signal, amplifies it and demodulates it to recover the message.

A **transducer** is a device that converts one form of energy into another. In communication it usually converts a non-electrical signal such as sound, light, pressure or temperature into a corresponding electrical signal (input transducer), or converts an electrical signal back into the original form (output transducer).

For example, a microphone converts sound into an electrical signal, a loudspeaker converts an electrical signal back into sound, a photodiode converts light into current, and an LED converts current into light.

Final answer. A transducer converts one form of energy (such as sound or light) into a corresponding electrical signal, or vice versa.

← *Back to Q203*

Q204

Draw the circuit diagram of an illuminated photodiode in reverse bias. How is photodiode used to measure light intensity?

[2010]

SOLUTION

A photodiode is a p-n junction diode that is always operated in **reverse bias** and is illuminated through a transparent window over its junction.

In the circuit diagram, the p-side of the diode is connected to the negative terminal and the n-side to the positive terminal of a battery, so the junction is reverse biased. A milliammeter and a load resistor R_L are connected in series in the circuit to read the reverse current. Arrows are shown falling on the junction to indicate incident light of photon energy $\mathbf{h\nu > E_g}$.

When light of photon energy greater than the band gap, $\mathbf{h\nu > E_g}$, falls on or near the depletion region, the photons are absorbed and generate **electron-hole pairs**.

The strong electric field across the reverse-biased depletion region immediately separates these pairs. Electrons drift to the n-side and holes drift

to the p-side, and this adds an extra current, the photocurrent I_{ph}, to the small reverse saturation current.

The photocurrent is directly proportional to the incident light intensity,

$$I_{ph} \propto I$$

So by measuring the reverse current, for example through the voltage drop across the load resistor R_L, the intensity of the incident light can be determined. This makes the photodiode useful in light meters and optical detectors.

Final answer. The reverse photocurrent of an illuminated photodiode is directly proportional to the incident light intensity, so measuring this current gives the light intensity.

← *Back to Q204*

Q205

Explain the function of a repeater in a communication system.

[2010]

SOLUTION

A repeater is an intermediate device placed along a long-distance communication link to restore a signal that has weakened while travelling through the channel.

It performs three operations in sequence. First, it receives the attenuated and possibly noisy signal arriving from the source. Second, it amplifies this signal back to a useful level and, in digital links, reshapes or regenerates the pulses so that accumulated noise is removed. Third, it retransmits the cleaned and amplified signal onward to the next stage of the link.

By placing several repeaters one after another along the channel, the signal strength is repeatedly restored, so the information can be carried over **very large distances** that would otherwise be impossible due to attenuation.

A common example is a **communication satellite** in geostationary orbit, which acts as a repeater in the sky: it receives a signal from the ground, amplifies it and retransmits it back to a distant location on Earth.

Final answer. A repeater receives a weakened signal, amplifies and reshapes it, and retransmits it, allowing communication over very large distances.

← *Back to Q205*

Q206

By what percentage will the transmission range of a TV tower be affected when the height of the tower is increased by 21%*?*

[2009]

SOLUTION

The maximum line-of-sight range of a TV tower of height h is

$$d = \sqrt{2Rh}$$

where R is the radius of the Earth. Since R is constant, the range depends on the height as $d \propto \sqrt{h}$.

The height is increased by 21%, so the new height is

$$h' = h + 0.21h = 1.21\ h$$

The new range is

$$d' = \sqrt{2Rh'} = \sqrt{2R \times 1.21\ h}$$

$$d' = \sqrt{1.21} \times \sqrt{2Rh} = 1.10\ d$$

The percentage increase in range is

$$\frac{d' - d}{d} \times 100 = (1.10 - 1) \times 100$$

$$= 0.10 \times 100 = 10\%$$

So the transmission range **increases by 10%**.

TIP

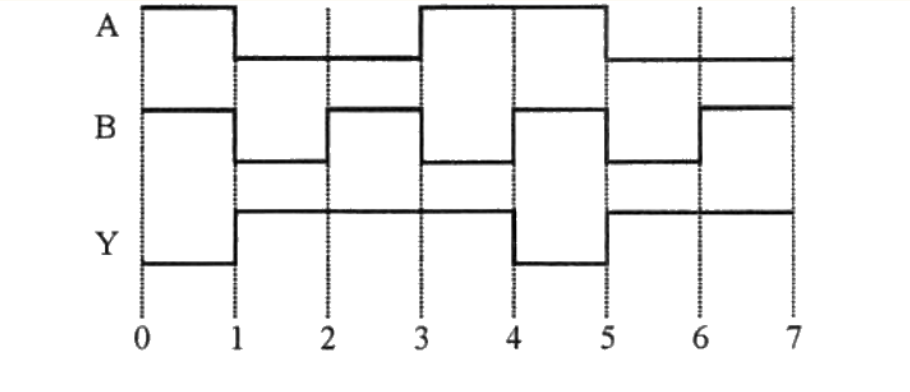
Because $d \propto \sqrt{h}$, a 21% rise in height gives a $\sqrt{1.21} = 1.1$ factor, i.e. only a 10% rise in range, not 21%.

Final answer. The transmission range increases by **10%**.

← *Back to Q206*

Q207

The following figure shows the input waveforms (A, B) and the output waveform (Y) of a gate. Identify the gate, write its truth table and draw its logic symbol.



[2009]

SOLUTION

To identify the gate, read off the input values A and B and the output Y for each time interval of the waveform, and tabulate them.

From the waveform, the output Y is HIGH for every input combination except when both A and B are HIGH, where Y is LOW. This gives the truth table

$$A=0,\ B=0 \Rightarrow Y=1$$

$$A=0,\ B=1 \Rightarrow Y=1$$

$$A=1,\ B=0 \Rightarrow Y=1$$

$$A=1,\ B=1 \Rightarrow Y=0$$

Since the output is LOW only when both inputs are HIGH, this is a **NAND gate**.

Its Boolean expression is

$$Y = \overline{A \cdot B}$$

The logic symbol is the D-shaped AND-gate symbol (flat on the left with the two inputs A and B, curved on the right) with a small circle, the inversion bubble, drawn at its output before the line carrying Y.

Final answer. The gate is a NAND gate, with output $\boldsymbol{Y = \overline{A \cdot B}}$ that is LOW only when both inputs are HIGH.

← *Back to Q207*

Q208

Why are high frequency carrier waves used for transmission?

[2009]

SOLUTION

High-frequency carrier waves are used for transmission for several reasons.

Practical antenna size. For efficient radiation the antenna length must be of the order of λ/4. A high-frequency carrier has a small wavelength λ, so the required antenna is short and easy to build. A low-frequency signal would need an impractically tall antenna.

Effective power radiated. The power radiated by an antenna of length l varies as

$$P \propto (\frac{l}{\lambda})^2$$

so for a given antenna length, a smaller λ (higher frequency) radiates far more power, giving stronger transmission.

Better directivity. High-frequency antennas can be made sharply directional with a reasonable physical size, which focuses the transmitted energy towards the receiver.

Multiplexing of channels. Many different users can be assigned separate high-frequency carriers, so a large number of signals can share the spectrum

without overlapping.

Faithful transmission also requires the carrier frequency to be much greater than the message frequency, $f_c \gg f_m$, which only high carrier frequencies can satisfy.

Final answer. High carrier frequencies are used because they need small, practical antennas, radiate more power, give better directivity, and allow many channels to be multiplexed.

← *Back to Q208*

Q209

What is meant by term 'modulation'? Draw a block diagram of a simple modulator for obtaining an AM signal.

[2009]

SOLUTION

Modulation is the process of varying one of the parameters (amplitude, frequency or phase) of a high-frequency sinusoidal carrier wave in step with the instantaneous amplitude of the low-frequency message signal. This allows the low-frequency information to be carried efficiently over long distances.

When the amplitude of the carrier is varied in accordance with the message, the process is called amplitude modulation (AM).

The block diagram of a simple AM modulator is a chain of boxes joined by arrows. The message signal $m(t)$ enters one input of a Multiplier (or modulator) block, while the carrier $A_c\omega_c t$ from an Oscillator enters the second input of the same multiplier. The output of the multiplier feeds a Band-pass Filter centred on ω_c, whose output is the AM Signal, which then passes through a Power Amplifier and finally to the Antenna.

In the multiplier, a non-linear device mixes the message and the carrier, producing sum and difference frequencies. The band-pass filter selects the band $\omega_c \pm \omega_m$ around the carrier, giving the amplitude modulated wave

$$c_{AM}(t) = (A_c + m(t))\omega_c t$$

Final answer. Modulation is varying a parameter of a high-frequency carrier in accordance with the message signal; the AM modulator mixes message and carrier in a multiplier and selects the band $\omega_c \pm \omega_m$ with a band-pass filter.

← *Back to Q209*

Q210

Write the function of (i) Transducer and (ii) Repeater in the context of communication system.

[2009]

SOLUTION

(i) **Transducer.** A transducer is a device that converts one form of energy into another. In a communication system it usually converts a non-electrical signal such as sound, light, pressure or temperature into a corresponding electrical signal (input transducer), or converts an electrical signal back into the original form (output transducer).

At the source, an input transducer changes the message into an electrical signal suitable for the transmitter, and at the destination an output transducer changes the recovered electrical signal back to the original form for the user. Examples are a microphone (sound to electrical), a loudspeaker (electrical to sound), a photodiode (light to current) and an LED (current to light).

(ii) **Repeater.** A repeater is an intermediate device placed along a long-distance communication link. It receives the weakened (attenuated) signal, amplifies it back to a useful level, and retransmits it onward, at the same or a different carrier frequency.

By placing repeaters at intervals along the channel, the signal strength is restored repeatedly, so communication becomes possible over a much greater range. A communication satellite acts essentially as a repeater placed in space.

Final answer. A transducer converts one form of energy into another (usually a signal to or from electrical form); a repeater receives, amplifies and retransmits a weakened signal to extend the communication range.

← Back to Q210

Q211

Write two factors justifying the need of modulation for transmission of a signal.

[2009]

SOLUTION

Modulation means superposing a low-frequency message signal onto a high-frequency carrier wave before transmission. It is needed for the following two reasons.

Practical size of the transmitting antenna: For an antenna to radiate power efficiently, its length must be at least of the order of $\frac{\lambda}{4}$, where λ is the wavelength of the signal being radiated. An audio (message) signal has a low frequency, around $f \sim 5$ kHz, so its wavelength is $\lambda = \frac{c}{f} = \frac{3\times 10^8}{5\times 10^3} = 6\times 10^4\ m = 60$ **km**.

This would need an antenna about 15 km long, which is impossible to build. By shifting the signal onto a high-frequency carrier (~ MHz), λ falls to a few hundred metres, so the required antenna length reduces to only a few metres, which is practical.

Avoiding overlap of signals (channel sharing): If all stations transmitted their messages directly in the same low audio-frequency band, the signals from different transmitters would overlap and could not be separated at the receiver. Modulation places each station's message on a **different carrier frequency**, so many signals can travel through the same medium at the same time without mutual interference, and the receiver can pick out any one by tuning.

Thus, the need for a small, efficient **antenna** and the need to allow **many signals to share the same channel** both justify modulation.

Final answer. Modulation is needed (i) to keep the antenna size practical and (ii) to let many signals share the channel without overlapping.

← Back to Q211

Q212

Distinguish between an intrinsic semiconductor and P-type semiconductor. Give reason, why a P-type semiconductor crystal is electrically neutral, although $n_h \gg n_e$.

[2008]

SOLUTION

The distinction between an intrinsic semiconductor and a P-type semiconductor is as follows.

An **intrinsic semiconductor** is a pure semiconductor (like pure Si or Ge) with no impurity added. In it, free electrons are produced only by thermal breaking of covalent bonds, so the number of free electrons equals the number of holes: $\mathbf{n_e = n_h = n_i}$. Its conductivity is very small and depends only on temperature.

A **P-type semiconductor** is made by doping a pure semiconductor with a trivalent impurity such as boron (B), aluminium (Al), gallium (Ga) or indium (In). Each trivalent atom creates a hole, so holes are the majority carriers and electrons the minority carriers: $n_h \gg n_e$. Its conductivity is much higher and is controlled mainly by the holes supplied by the dopant.

Reason why a P-type crystal is electrically neutral although $n_h \gg n_e$: When a trivalent impurity atom replaces a tetravalent host atom, it forms only three covalent bonds and is short of one electron. To complete the fourth bond it accepts an electron from a neighbouring bond and becomes a fixed, negatively-charged **acceptor ion**. The electron it took away leaves behind a free hole.

Hence every free (mobile) hole that is created is accompanied by one equal and opposite, immobile negative acceptor ion fixed in the lattice. The positive charge of the mobile holes is exactly balanced by the negative charge of these fixed ions at every point.

Therefore the total charge of the crystal is zero. Even though the mobile-hole density n_h is far greater than the mobile-electron density n_e, the crystal as a whole remains **electrically neutral**.

Final answer. Intrinsic is pure with $\mathbf{n_e = n_h}$; P-type is trivalent-doped with $\mathbf{n_h \gg n_e}$; it stays neutral because each free hole is balanced by a fixed negative acceptor ion.

← Back to Q212

Q213

The given inputs A, B are fed to a 2-input NOR gate. Draw the output wave form of the gate.

[2008]

SOLUTION

A NOR gate is an OR gate followed by a NOT gate. Its Boolean expression is $Y = \overline{A + B}$.

The rule for its output is: the output Y is HIGH (1) only when both inputs A and B are LOW (0). If either input or both inputs are HIGH, the output Y is LOW (0).

The truth table is:

$$A = 0, B = 0 \Rightarrow Y = 1$$

$$A = 0, B = 1 \Rightarrow Y = 0$$

$$A = 1, B = 0 \Rightarrow Y = 0$$

$$A = 1, B = 1 \Rightarrow Y = 0$$

To draw the output waveform, line it up directly below the given input waveforms of A and B on the same time axis. For each time interval, check the levels of A and B in that interval:

In every interval where both A and B are LOW, draw Y at the HIGH level.

In every interval where at least one of A or B is HIGH, draw Y at the LOW level.

The result is a square output waveform that is **HIGH only during the time slots when both inputs are simultaneously LOW**, and LOW everywhere else, with sharp vertical transitions at the instants where the input levels change.

Final answer. The output $Y=\overline{A+B}$ is HIGH only in the intervals where both A and B are LOW, and LOW in every other interval.

← *Back to Q213*

Q214

A transmitting antenna at the top of a tower has a height of 36 *m and the height of the receiving antenna is* 49 *m. What is the maximum distance between them, for satisfactory communication in the LOS mode? (Radius of earth* = 6400 km*)*

[2008]

SOLUTION

In line-of-sight (LOS) propagation, the maximum distance for satisfactory communication between a transmitting antenna of height h_T and a receiving antenna of height h_R is given by

$$d_M = \sqrt{2R\,h_T} + \sqrt{2R\,h_R}$$

where R is the radius of the Earth.

Here $\mathbf{h_T = 36\ m}$, $\mathbf{h_R = 49\ m}$ and $\mathbf{R = 6400\ km = 6.4\times 10^6\ m}$.

First term: $\sqrt{2R\,h_T} = \sqrt{2\times 6.4\times 10^6\times 36}$

$$= \sqrt{4.608\times 10^8} = 2.146\times 10^4\ m \approx 21.46\ \text{km}$$

Second term: $\sqrt{2R\,h_R} = \sqrt{2\times 6.4\times 10^6\times 49}$

$$= \sqrt{6.272\times 10^8} = 2.504\times 10^4\ m \approx 25.04\ \text{km}$$

Adding the two contributions:

$$d_M = 21.46\ \text{km} + 25.04\ \text{km} = 46.5\ \text{km}$$

The maximum distance for satisfactory LOS communication is $\mathbf{d_M \approx 46.5\ km}$.

COMMON MISTAKE

Remember the formula uses $\sqrt{2Rh}$ *for each antenna separately and then adds them; do not use* $\sqrt{2R(h_T+h_R)}$.

Final answer. The maximum LOS distance is $\mathbf{d_M \approx 46.5\ km}$.

← *Back to Q214*

Q215

Why do we need carrier waves of very high frequency in the modulation of signals? A carrier wave of peak voltage 20 *V is used to transmit a message signal. What should be the peak voltage of the modulating signal, in order to have a modulation index of* 80%*?*

[2008]

SOLUTION

Need for a very high-frequency carrier wave in modulation:

Size of antenna: An antenna radiates efficiently only when its length is of the order of $\frac{\lambda}{4}$. A low-frequency audio signal ($f \sim 1$ kHz) has $\boldsymbol{\lambda = \frac{c}{f} = \frac{3\times 10^8}{10^3} = 3\times 10^5\ m = 300\ km}$, needing an antenna about 75 km long, which is impossible. A high-frequency carrier ($f \sim 1$ MHz) has $\boldsymbol{\lambda = 300\ m}$, so a practical antenna of a few tens of metres is enough.

Effective power radiated: The power radiated by an antenna varies as $(\frac{L}{\lambda})^2$, where L is its length. For a fixed antenna length, a smaller λ (higher carrier frequency) gives much greater radiated power, so the signal travels farther. (High carriers also allow many stations to use different frequencies without overlap.)

Numerical: Peak carrier voltage $\mathbf{V_c = 20\ V}$ and required modulation index $\boldsymbol{\mu = 80\% = 0.80}$.

The modulation index is defined as $\boldsymbol{\mu = \frac{V_m}{V_c}}$, where V_m is the peak voltage of the modulating signal.

$$\therefore V_m = \mu\, V_c$$

$$V_m = 0.80 \times 20$$

$$V_m = 16\ V$$

The peak voltage of the modulating signal must be $\mathbf{V_m = 16\ V}$.

Final answer. High-frequency carriers are needed for a practical antenna and for greater radiated power; the peak modulating voltage is $\mathbf{V_m = 16\ V}$.

← *Back to Q215*

Q216

The output of an OR gate is connected to both the inputs of a NAND gate. Draw the logic circuit of this combination of gates and write its truth table.

[2007]

SOLUTION

Logic circuit: First take a two-input OR gate with inputs A and B; let its output be X. Connect this single output X to both input terminals of a two-input NAND gate, and let the NAND gate's output be Y. So the OR gate feeds the NAND gate, whose two inputs are joined together and tied to X.

Working out the logic:

The OR gate gives $\boldsymbol{X = A + B}$.

The NAND gate has both inputs equal to X, so $\boldsymbol{Y = \overline{X \cdot X}}$.

Since $\boldsymbol{X \cdot X = X}$, this becomes $\boldsymbol{Y = \overline{X} = \overline{A + B}}$.

Thus the combination behaves exactly like a single **NOR gate**.

Truth table:

$$A = 0, B = 0 \Rightarrow X = 0, Y = 1$$

$$A = 0, B = 1 \Rightarrow X = 1, Y = 0$$

$$A = 1, B = 0 \Rightarrow X = 1, Y = 0$$

$$A = 1, B = 1 \Rightarrow X = 1, Y = 0$$

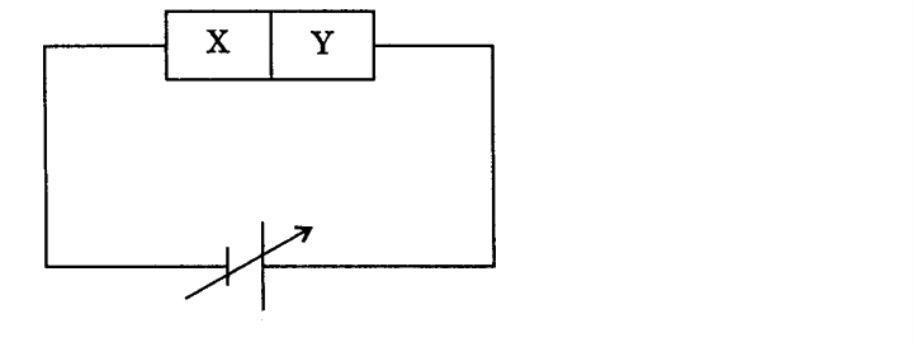
The output Y is HIGH only when both A and B are LOW, confirming that the circuit is equivalent to a NOR gate.

Final answer. The combination gives $Y=\overline{A+B}$, i.e. it works as a NOR gate (output 1 only when both inputs are 0).

← *Back to Q216*

Q217

Two semiconductor materials X and Y shown in the given figure, are made by doping germanium crystal with indium and arsenic respectively. The two are joined end to end and connected to a battery as shown.



(i) Will the junction be forward biased or reverse biased?
(ii) Sketch a V-I graph for this arrangement.

[2007]

SOLUTION

Identifying the two materials:

Indium is trivalent, so germanium doped with indium becomes a **p-type** semiconductor. Hence material *X* is p-type.

Arsenic is pentavalent, so germanium doped with arsenic becomes an **n-type** semiconductor. Hence material *Y* is n-type.

(i) Nature of biasing: In the given arrangement the battery's positive terminal is connected to the p-side (*X*) and its negative terminal to the n-side (*Y*). When the positive terminal is joined to the p-region and the negative terminal to the n-region, the external voltage opposes the junction's built-in potential barrier, the depletion layer narrows, and a large current flows. Therefore the junction is **forward biased**.

(ii) V-I graph: Plot the applied voltage V on the x-axis and the current I on the y-axis.

In the first quadrant (forward bias), the curve stays almost flat with negligible current up to the threshold (knee) voltage, which is about 0.3 V for germanium. Beyond this knee voltage the current rises very steeply, giving a sharply increasing curve.

In the third quadrant (reverse bias, shown only for comparison), there is a very small, almost constant **reverse saturation current** that stays nearly flat until breakdown.

So the characteristic is the standard p-n junction diode curve: little current until the $\approx 0.3\ V$ knee, then a sharp rise, marking the forward-biased Ge diode.

Final answer. ***X*** is p-type and ***Y*** is n-type; the junction is forward biased, and the V-I graph is the usual diode curve with a sharp current rise past the $\approx 0.3\ V$ knee.

← *Back to Q217*

Q218

Draw and explain the output waveform across the load resistor R, if the input waveform is as shown in the given figure.



[2006]

SOLUTION

This is a half-wave rectifier action of a single p-n junction diode. A diode conducts only when its anode (p-side) is at a higher potential than its cathode (n-side), i.e. only when it is forward biased.

During the positive half-cycle of the input v_{in}, the anode is positive with respect to the cathode, so the diode is **forward biased** and conducts. The current flows

through the load R, and the output voltage across R follows the positive part of the input.

During the negative half-cycle of v_{in}, the anode is negative with respect to the cathode, so the diode is **reverse biased** and does not conduct. Practically no current flows, so the voltage across R is zero during this interval.

Output waveform across R: It is drawn directly below the input on the same time axis. It reproduces only the **positive halves** of the input, while every negative half is cut off and the output stays flat at 0 V in those intervals.

If the input is a sinusoid, the output is a train of positive sine humps separated by flat zero gaps. If the input is a square wave swinging between $+V$ and $-V$, the output is a unipolar square wave that switches between $+V$ and 0. This is the characteristic **half-wave rectified output**.

Final answer. The output across R contains only the positive halves of the input (half-wave rectified); the negative halves are blocked and read **0 V**.

← *Back to Q218*

Q219

Explain how the width of depletion layer in a p-n junction diode changes when the junction is (i) forward biased (ii) reverse biased.

[2006]

SOLUTION

The depletion layer is the region near the p-n junction that is free of mobile charge carriers and contains only fixed donor and acceptor ions; it sets up a built-in barrier potential V_b. Its width changes with the applied bias as follows.

(i) Forward bias: The p-side is connected to the positive terminal and the n-side to the negative terminal of the external battery. The applied voltage V opposes the built-in barrier. Majority carriers, holes from the p-side and electrons from the n-side, are pushed towards the junction, where they neutralise some of the fixed ions.

As a result the depletion-layer width **decreases**, and the effective barrier potential falls to $V_b - V$. A large forward current can then flow.

(ii) Reverse bias: The p-side is connected to the negative terminal and the n-side to the positive terminal. Now the applied voltage V aids the built-in field. Majority carriers are pulled away from the junction, uncovering more fixed donor and acceptor ions.

Hence the depletion-layer width **increases**, and the effective barrier potential rises to $V_b + V$. Only a very small minority-carrier current (reverse saturation current) flows.

In short, forward bias narrows the depletion layer and reverse bias widens it.

Final answer. Forward bias decreases the depletion-layer width (barrier $\mathbf{V_b - V}$); reverse bias increases it (barrier $\mathbf{V_b + V}$).

← *Back to Q219*

Q220

With the help of a diagram, show the biasing of a light emitting diode (LED). Give its two advantages over conventional incandescent lamps.

[2004]

SOLUTION

Biasing of an LED (diagram in words): A light-emitting diode is a heavily-doped p-n junction that is always operated in **forward bias**. In the circuit, connect the **p-side of the LED to the positive terminal** of the battery and the **n-side to the negative terminal**, with a current-limiting resistor R in series to keep the current safe. The battery voltage is kept above the LED's threshold (about 1.8 to 3.0 V, depending on the colour). The diagram shows: battery → resistor R → p-side of LED → junction → n-side → back to battery, with the LED symbol (a diode with two small arrows pointing outward to indicate emitted light).

Working: In forward bias the depletion region is narrow. Electrons from the n-side and holes from the p-side are pushed across the junction and recombine in the junction region. Each recombination releases energy nearly equal to the

band gap E_g, which is emitted as a photon of frequency $\nu = \frac{E_g}{h}$, so the LED glows.

Two advantages over a conventional incandescent lamp:

1. It is far more **efficient and consumes much less power**, because it converts electrical energy almost directly into light with no filament-heating losses.

2. It has a **very long operating life** (of the order of 10^5 hours) and switches on and off instantly with no warm-up.

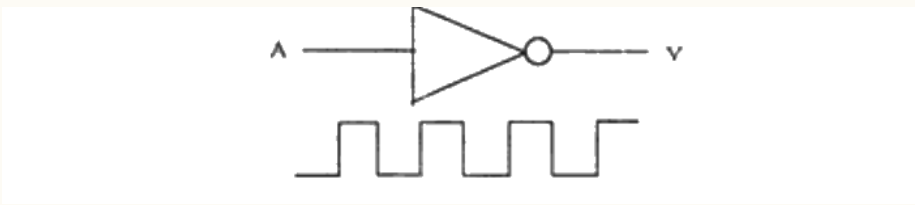
(Other merits: small size, ruggedness, and availability in many colours.)

Final answer. An LED is forward biased (p to +, n to -) with a series resistor; advantages over an incandescent lamp are higher efficiency/low power and a very long life with instant switching.

← *Back to Q220*

Q221

In the figure below, circuit symbol of a logic gate and input wave form is shown.
(i) Name the logic gate,
(ii) write its truth table and
(iii) give the output wave form.



[2003]

SOLUTION

The symbol shown is a single curved-back gate body with a small circle (bubble) at its output, and the input waveform applied alternates between the LOW (0) and HIGH (1) levels over successive time intervals. A bubble on the output of a gate means inversion, so this gate inverts whatever it receives.

(i) The logic gate is a NOT gate (an inverter). It has a single input A and a single output Y, and its action is $\boldsymbol{Y = \overline{A}}$, that is, the output is the logical complement of the input.

(ii) Truth table of the NOT gate:

$$\text{array}|c|c|\ A\ \&\ Y{=}\overline{A}\ 0\ \&\ 1\ 1\ \&\ 0\ \text{array}$$

When the input is LOW (0) the output is HIGH (1), and when the input is HIGH (1) the output is LOW (0).

(iii) Output waveform: draw the output on the same time axis as the input, directly below it. Wherever the input A is HIGH (1), the output Y is drawn LOW (0); wherever the input A is LOW (0), the output Y is drawn HIGH (1). So the output waveform is the exact mirror image (inversion) of the input waveform: every HIGH band of the input becomes a LOW band of the output and every LOW band becomes a HIGH band, with the transitions occurring at the same instants as in the input.

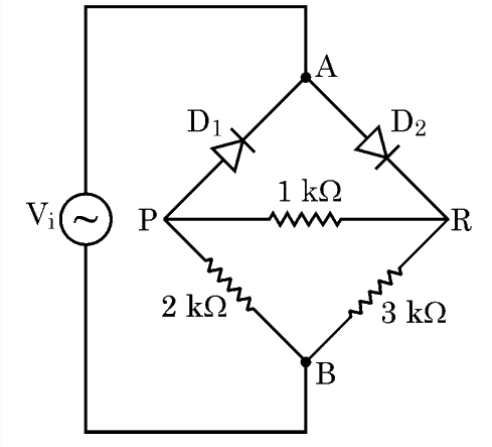
> **TIP**
>
> A bubble at the gate output always means inversion: a triangle with a bubble is NOT, a D-shape with a bubble is NAND, and a curved-back OR with a bubble is NOR.

Final answer. The gate is a NOT gate (inverter); its output waveform is the inverted (mirror) image of the input waveform.

← *Back to Q221*

Q222

An ac voltage $\mathbf{V_i = 12 \sin (100\pi\, t)}$ *V is applied between points A and B in a network of two ideal diodes and three resistors as shown in figure.*



During the positive half-cycle of the input voltage V_i *supplied to the network:*
(a) Identify which of the two diodes will conduct and why?
(b) Redraw an equivalent circuit diagram to show the flow of current.
(c) Calculate the output voltage drops V_o *across the three resistors when the input voltage attains its peak value.*

[2026 • Set 55-1-1]

SOLUTION

An ideal diode behaves as a closed switch (zero resistance) when forward biased and as an open switch (infinite resistance) when reverse biased. The peak value of the applied voltage is $\mathbf{V_p = 12\ V}$, since $\mathbf{V_i = 12\sin(100\pi\ t)\ V}$.

(a) During the positive half cycle, point *A* is at a higher potential than point *B*. The diode whose anode (p-side) faces the high-potential point *A* gets forward biased and conducts, while the diode whose anode faces the low-potential side is reverse biased and does not conduct. In the given network diode D_1 conducts (it is forward biased because current can flow through it from A towards B), and diode D_2 is reverse biased and stays OFF.

(b) Equivalent circuit: redraw the network replacing the conducting diode D_1 by a plain wire (short circuit) and removing the non-conducting diode D_2 (leave that branch open). The current then flows from *A*, through the D_1 branch, through the series combination of the resistors, and back to *B*. With D_2 open, its resistor carries no current.

(c) Taking the standard printed values $\mathbf{R_1 = R_2 = R_3 = 4\ \Omega}$, with D_1 ON and D_2 OFF the conducting path places two resistors in series across the peak source

voltage while the third (in the cut-off D_2 branch) carries no current. The current in the conducting loop at peak is

$$I = \frac{V_p}{R_1 + R_2} = \frac{12}{4 + 4} = 1.5\ A$$

The voltage drop across each conducting resistor is

$$V_{R_1} = I\,R_1 = 1.5 \times 4 = 6\ V$$

$$V_{R_2} = I\,R_2 = 1.5 \times 4 = 6\ V$$

The resistor in the open (D_2 OFF) branch carries no current, so

$$V_{R_3} = I'\,R_3 = 0 \times 4 = 0\ V$$

Hence at the peak of the positive half cycle the drops are 6 *V*, 6 *V* and 0 *V* across the two conducting resistors and the resistor in the cut-off branch respectively.

Final answer. D1 conducts (D2 off); at peak the drops are 6 V and 6 V across the two conducting resistors and 0 V across the resistor in the cut-off branch.

← *Back to Q222*

Q223

Briefly explain the two important processes that occur during the formation of a p-n junction.

[2026 • Set 55-1-1]

SOLUTION

When a p-n junction is formed, two opposing processes set up the depletion region and the barrier potential: diffusion and drift.

1. Diffusion: Across the junction there is a large concentration gradient. The n-side has a high concentration of free electrons while the p-side has a high concentration of holes. Because of this gradient, the majority carriers diffuse across the junction: electrons diffuse from the n-side to the p-side, and holes diffuse from the p-side to the n-side. This motion of majority carriers gives rise to a diffusion current directed from the p-side to the n-side.

As electrons leave the n-side, they expose immobile positively charged donor ions there; as holes leave the p-side, they expose immobile negatively charged acceptor ions. A narrow region near the junction, containing only these immobile charged ions and no free carriers, is thus created. This is the depletion region.

2. Drift: The immobile ions of the depletion region set up a built-in electric field $\vec{E}$ directed from the n-side (positive ions) to the p-side (negative ions). This field exerts a force on the minority carriers near the junction: it sweeps electrons (minority in the p-region) towards the n-side and holes (minority in the n-region) towards the p-side. This motion of minority carriers under the built-in field constitutes the drift current, which is directed from the n-side to the p-side, that is, opposite to the diffusion current.

In equilibrium (no external bias) the diffusion current and the drift current become equal in magnitude and opposite in direction, so the net current across the junction is zero.

Final answer. The two processes are diffusion of majority carriers (giving a diffusion current p to n) and drift of minority carriers under the built-in field (giving a drift current n to p), which balance at equilibrium.

← *Back to Q223*

Q224

A semiconductor has equal electron and hole concentration of $3 \times 10^6\ m^{-3}$. *On doping with a certain impurity, the hole concentration increases to* $6 \times 10^{10}\ m^{-3}$.
(a) What type of semiconductor is obtained on doping?
(b) Calculate the new electron concentration of the semiconductor.
(c) How does the energy gap of semiconductor change with doping? Draw the energy band diagram for it.

[2026 • Set 55-1-2]

SOLUTION

The product of electron and hole concentrations in a semiconductor at a given temperature is constant and equal to the square of the intrinsic concentration

(law of mass action):

$$n_e\, n_h = n_i^2$$

Here the intrinsic concentration is $\mathbf{n_i = 3 \times 10^6\ m^{-3}}$ and, after doping, the hole concentration is $\mathbf{n_h = 6 \times 10^{10}\ m^{-3}}$.

(a) Since doping has greatly increased the hole concentration above its intrinsic value, the impurity has supplied extra holes. This happens when a trivalent (acceptor) impurity is added, so a *p*–type semiconductor is obtained, with holes as the majority carriers.

(b) Using the law of mass action, the new electron concentration is

$$n_e = \frac{n_i^2}{n_h} = \frac{(3 \times 10^6)^2}{6 \times 10^{10}}$$

$$n_e = \frac{9 \times 10^{12}}{6 \times 10^{10}}$$

$$n_e = 1.5 \times 10^2\ m^{-3} = 150\ m^{-3}$$

This very small value confirms that in the p-type sample electrons are the minority carriers and holes the majority carriers.

(c) Doping does not change the fundamental energy gap E_g of the host semiconductor. Instead, a trivalent dopant introduces an extra discrete acceptor energy level E_A just above the top of the valence band (within about 0.01 to 0.05 eV of E_V).

Energy band diagram: draw the valence band (VB) at the bottom (shaded, filled) and the conduction band (CB) at the top (empty), separated by the gap $\mathbf{E_g \approx 1.1}$ **eV** (for Si). Just above the top edge E_V of the valence band, draw a short dashed acceptor level E_A. Electrons from the valence band are easily excited to this nearby acceptor level, leaving behind holes in the valence band; these holes are the majority carriers of the p-type material. The new electron concentration is $\mathbf{150\ m^{-3}}$.

COMMON MISTAKE

The energy gap E_g is a property of the host crystal and is not altered by doping; doping only adds donor or acceptor levels inside the gap close to a band edge.

Final answer. (a) p-type semiconductor; (b) new electron concentration = 150 m^-3; (c) the band gap is unchanged but a trivalent dopant adds an acceptor level E_A just above the valence band.

← Back to Q224

Q225

(a) Draw the V-I characteristics of silicon diode.
(b) Explain the following terms:
(i) minority carrier injection in forward bias
(ii) breakdown voltage in reverse bias

[2026 • Set 55-1-3]

SOLUTION

(a) V-I characteristics of a silicon diode: plot the diode current I on the y-axis against the applied voltage V on the x-axis.

In the forward-bias region (first quadrant, positive V), the current is almost zero until the applied voltage reaches the threshold (knee or cut-in) voltage $\mathbf{V_T \approx 0.7}$ $\boldsymbol{V}$ for silicon; beyond V_T the current rises steeply and almost exponentially with voltage.

In the reverse-bias region (third quadrant, negative V), only a very small, nearly constant reverse saturation current (of the order of microamperes) flows over a wide range of reverse voltages. When the reverse voltage reaches the breakdown voltage V_{Br}, the reverse current increases very sharply (the curve drops almost vertically).

(b)(i) Minority carrier injection in forward bias: When the diode is forward biased, the barrier potential and the width of the depletion region are reduced. The majority carriers then cross the junction and enter the opposite region, where they become minority carriers: electrons crossing from the n-side into the p-side become excess minority electrons in the p-region, and holes crossing from the p-side into the n-side become excess minority holes in the n-region. This injection of carriers as minority carriers across the junction is called

minority carrier injection, and it is responsible for the large forward current of the diode.

(b)(ii) Breakdown voltage in reverse bias: The breakdown voltage is the minimum reverse-bias voltage at which the reverse current through the diode suddenly increases to a very large value. Below this voltage the reverse current is the small, almost constant saturation current; at the breakdown voltage V_{Br}, the strong reverse field releases or accelerates carriers (by avalanche or Zener mechanisms), so the current rises very sharply even though the voltage hardly changes.

Final answer. The Si diode shows a 0.7 V forward knee and a small constant reverse current until breakdown; minority carrier injection is the crossing of majority carriers to become minority carriers in forward bias, and breakdown voltage is the reverse voltage at which the reverse current rises sharply.

← *Back to Q225*

Q226

With the help of circuit diagrams, briefly explain the forward biasing and the reverse biasing of a p-n junction diode.

[2026 • Set 55-3-1]

SOLUTION

Forward biasing: Connect the external battery so that its positive terminal goes to the p-side and its negative terminal to the n-side of the diode, in series with a current-limiting resistor and an ammeter. Circuit diagram: battery (+) to p-side, battery (−) to n-side, with the resistor R and ammeter in the loop and a voltmeter across the diode.

The applied voltage V opposes the built-in barrier potential. Its effects are:

The effective barrier height decreases from V_b to $(V_b - V)$.

The width of the depletion region decreases, because the supplied majority carriers neutralise some of the immobile ions.

The majority carriers (holes from the p-side and electrons from the n-side) now cross the junction easily, giving a large forward current that increases almost exponentially with V. The conventional current flows from the p-side (anode) to the n-side (cathode) through the diode.

Reverse biasing: Reverse the battery: connect its positive terminal to the n-side and its negative terminal to the p-side, in series with a microammeter (the current is very small). Circuit diagram: battery (+) to n-side, battery (−) to p-side.

The applied voltage V now adds to the built-in barrier potential. Its effects are:

The effective barrier height increases to $(V_b + V)$.

The width of the depletion region increases, since more immobile ions are exposed.

The majority carriers are pulled away from the junction, so they cannot cross it. Only a tiny reverse saturation current (of the order of microamperes), due to minority carriers, flows; it remains almost constant with voltage until breakdown is reached at V_{Br}.

Thus the diode conducts strongly in forward bias and almost not at all in reverse bias, which is the property that makes it useful as a rectifier.

Final answer. In forward bias (+ to p, - to n) the barrier and depletion width fall and a large forward current flows; in reverse bias (+ to n, - to p) the barrier and depletion width grow and only a tiny reverse saturation current flows.

← *Back to Q226*

Q227

With the help of a circuit diagram, explain the working of a full wave rectifier. Depict the input and output waveforms.

[2026 • Set 55-4-1]

SOLUTION

Circuit: A full-wave rectifier uses a step-down transformer with a centre–tapped secondary and two p-n junction diodes D_1 and D_2. The anodes of

D_1 and D_2 are connected to the two outer ends of the secondary winding, and their cathodes are joined together. The load resistor R_L is connected between this common cathode point and the centre-tap of the secondary; the output (rectified) voltage is taken across R_L.

Working: The two halves of the secondary supply equal voltages of opposite phase with respect to the centre-tap, so in each half cycle one diode is forward biased and the other reverse biased.

During the positive half cycle, suppose the top end of the secondary is positive and the bottom end negative. Then D_1 is forward biased and conducts, while D_2 is reverse biased and does not conduct. Current flows from the top end of the secondary, through D_1, then through R_L (from top to bottom), and back to the centre-tap.

During the negative half cycle, the secondary polarity reverses: the bottom end becomes positive and the top end negative. Now D_2 is forward biased and conducts, while D_1 is OFF. Current flows from the bottom end of the secondary, through D_2, then through R_L (again from top to bottom, in the same direction), and back to the centre-tap.

Since the current through R_L is in the same direction during both half cycles, both halves of the input are rectified, and the output is unidirectional.

Waveforms: Draw the input as a full sine wave (alternating positive and negative humps). Draw the output directly below as a series of positive half-sine humps with no gaps between them: every negative hump of the input is flipped up to become a positive hump. The output therefore has a ripple frequency that is twice the input frequency.

Final answer. Using a centre-tapped transformer and two diodes that conduct alternately, both half cycles drive current through R_L in the same direction, giving a unidirectional output whose ripple frequency is twice the input frequency.

← Back to Q227

Q228

(a) *Draw circuit arrangement for studying V-I characteristics of a p-n junction diode.*
(b) *Show the shape of the characteristics of a diode.*
(c) *Mention two information that you can get from these characteristics.*

[2025 • Set 55-1-1]

SOLUTION

(a) Circuit arrangement: To study the V-I characteristics, connect the p-n junction diode in series with a battery, a current-limiting (rheostat or fixed) resistor and an ammeter, with a voltmeter across the diode.

For the forward-bias measurement, connect the p-side of the diode to the positive terminal of the supply and use a milliammeter (since the forward current is relatively large).

For the reverse-bias measurement, reverse the diode (p-side to the negative terminal) and use a microammeter (since the reverse current is very small). The applied voltage is varied with the rheostat and the corresponding current is recorded.

(b) Shape of the characteristics: Plot I (y-axis) against V (x-axis).

In the forward-bias region (first quadrant), the current is nearly zero up to the threshold voltage (about 0.7 V for silicon and about 0.3 V for germanium), after which it rises sharply and almost exponentially with voltage.

In the reverse-bias region (third quadrant), a small and nearly constant reverse saturation current (of the order of microamperes) flows, until the reverse voltage reaches the breakdown voltage, beyond which the reverse current increases very steeply.

(c) Two pieces of information obtained from the characteristics:

The threshold (cut–in) voltage, that is, the forward voltage at which the current begins to rise rapidly; it indicates the semiconductor material (silicon or germanium).

The reverse breakdown voltage, that is, the reverse voltage beyond which the reverse current increases sharply; it is important for the safe voltage rating and

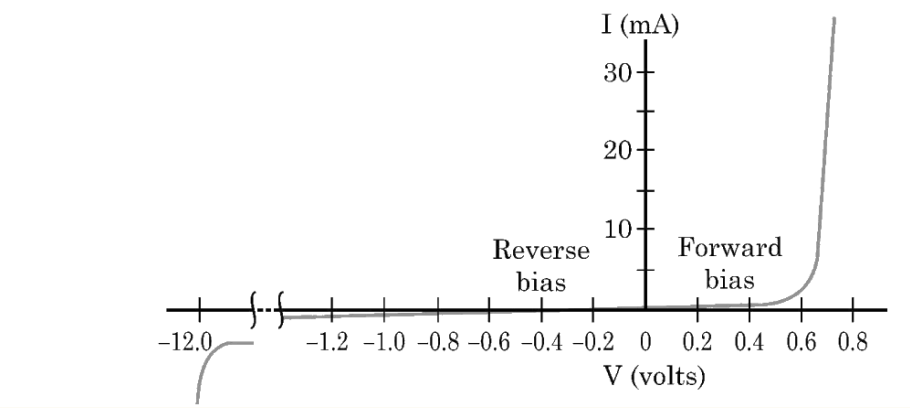
for Zener applications. (The dynamic resistance $\mathbf{r_d = \Delta V / \Delta I}$ can also be obtained from the slope.)

Final answer. Connect the diode in series with a supply, resistor and ammeter (voltmeter across the diode); the curve shows a forward knee near 0.7 V and a small reverse current until breakdown, from which the threshold voltage and the reverse breakdown voltage can be read.

← *Back to Q228*

Q229

(a) What are majority and minority charge carriers in an extrinsic semiconductor?
(b) A p-n junction is forward biased. Describe the movement of the charge carriers which produce current in it.
(c) The graph shows the variation of current with voltage for a p-n junction diode. Estimate the dynamic resistance of diode at $V = -0.6$ volt.



[2025 • Set 55-2-1]

SOLUTION

(a) In an extrinsic semiconductor the carrier present in larger number is the majority carrier and the one present in smaller number is the minority carrier.

In a p-type semiconductor (trivalent dopant), holes are the majority carriers and electrons are the minority carriers ($n_h \gg n_e$).

In an n-type semiconductor (pentavalent dopant), electrons are the majority carriers and holes are the minority carriers ($n_e \gg n_h$). In both cases $\mathbf{n_e\, n_h = n_i^2}$.

(b) Movement of charge carriers in forward bias: The applied voltage opposes the built-in junction potential, so the barrier height and the depletion-region width are reduced. The majority carriers can then cross the junction: holes from the p-side move towards and across the junction into the n-side, and electrons from the n-side move towards and across the junction into the p-side. Inside the p-region the current is mainly a hole current, and inside the n-region it is mainly an electron current; in the external circuit these combine to give a conventional current flowing from the positive terminal of the battery through the diode and back to the battery. The continuous crossing of holes one way and electrons the other constitutes the forward current.

(c) **Dynamic resistance at $V = -0.6\ V$:**

$$r_d = \frac{\Delta V}{\Delta I}$$

At $V = -0.6\ V$ the diode is reverse biased, and in this region the current is the almost constant reverse saturation current, so a change in voltage produces an extremely small change in current. Reading the graph near $V = -0.6\ V$, a voltage change of, say, $\Delta V = 0.1\ V$ produces a current change of only about $\Delta I = 1\ \mu A = 1 \times 10^{-6}\ A$. Hence

$$r_d = \frac{0.1}{1 \times 10^{-6}} = 1 \times 10^5\ \Omega = 10^5\ \Omega$$

So the dynamic resistance in reverse bias is very high (of the order of $10^5\ \Omega$), as expected, because the reverse current barely changes with voltage. (The exact figure must be read off the slope of the given graph at $V = -0.6\ V$.)

CONCEPT

Dynamic resistance is r_d = ΔV/ΔI, the reciprocal slope of the V-I curve at the operating point: it is small in forward bias (steep curve) and very large in reverse bias (almost flat curve).

Final answer. Holes/electrons are majority/minority in p-type and electrons/holes in n-type; in forward bias holes and electrons cross the narrowed junction to give the current, and the dynamic resistance at V = −0.6 V is very high, of the order of 10^5 ohm.

← Back to Q229

Q230

(a) *What are majority and minority charge carriers of p-type and n-type semiconductors ?*
(b) *Explain briefly the formation of diffusion current and drift current in a p-n junction diode.*

[2025 • Set 55-4-1]

SOLUTION

(a) Majority and minority charge carriers:

In a p-type semiconductor (formed by doping with a trivalent impurity such as B, Al, In or Ga), the holes are the majority carriers and the electrons are the minority carriers ($n_h \gg n_e$).

In an n-type semiconductor (formed by doping with a pentavalent impurity such as P, As, Sb or Bi), the electrons are the majority carriers and the holes are the minority carriers ($n_e \gg n_h$).

(b) Diffusion current: At the junction there is a large concentration gradient, since holes are abundant on the p-side and electrons abundant on the n-side. Because of this gradient, holes diffuse from the p-side to the n-side and electrons diffuse from the n-side to the p-side. This diffusion of the majority carriers across the junction constitutes the diffusion current, directed from the p-side to the n-side.

Drift current: As the diffusing carriers leave, they expose immobile positive donor ions on the n-side and immobile negative acceptor ions on the p-side. These ions set up a built-in electric field directed from the n-side to the p-side. This field sweeps the minority carriers across the junction, electrons (minority in the p-region) towards the n-side and holes (minority in the n-region) towards the p-side. This motion of minority carriers constitutes the drift current, directed from the n-side to the p-side, that is, opposite to the diffusion current.

At equilibrium, with no external bias, the diffusion current and the drift current are equal in magnitude and opposite in direction, so the net current

across the junction is zero.

Final answer. Holes/electrons are majority/minority in p-type and electrons/holes in n-type; diffusion current (p to n) is due to majority carriers crossing the gradient, drift current (n to p) is due to minority carriers swept by the built-in field, and the two balance at equilibrium.

← *Back to Q230*

Q231

(a) *Draw the energy-band diagrams for conductors, semiconductors and insulators at* $T = 0\ K$. *How is an electron-hole pair formed in a semiconductor at room temperature?*
(b) *Carbon and silicon both, are members of IV group of periodic table and have the same lattice structure. Carbon is an insulator whereas silicon is a semiconductor. Explain.*

[2025 • Set 55-5-1]

SOLUTION

(a) Energy-band diagrams at $T = 0\ K$:

For a conductor, the valence band (VB) and conduction band (CB) either overlap, or the conduction band is partially filled. The energy gap is $E_g = 0$. Because empty allowed states lie just above the filled states, even a tiny applied field sets the electrons moving, so conduction occurs at all temperatures.

For a semiconductor, the valence band is completely filled and the conduction band is completely empty at 0 K, separated by a small forbidden energy gap $E_g \approx 1\ eV$. At $T=0\ K$ no electron has enough energy to cross this gap, so it behaves like an insulator.

For an insulator, the valence band is full and the conduction band is empty, but they are separated by a large gap $E_g > 3\ eV$. This gap is too large to be bridged thermally, so conduction is practically impossible.

Formation of an electron-hole pair at room temperature:

At room temperature ($T \approx 300\ K$) the available thermal energy is $k_B T \approx 0.025\ eV$. Although this average value is smaller than E_g, a fraction of electrons gain

enough energy (aided by lattice vibrations / phonons) to jump from the valence band across the gap into the conduction band.

The electron that reaches the conduction band becomes a **free electron** and acts as a negative charge carrier. The vacancy left behind in the valence band is called a **hole**, which behaves as a positive charge carrier. Thus each thermal excitation creates one **electron-hole pair**.

(b) Why carbon is an insulator and silicon is a semiconductor:

Both carbon (diamond) and silicon belong to group IV and form the same tetrahedral covalent lattice. However, the carbon atom is much smaller than the silicon atom, so the inter-atomic spacing in diamond is much smaller and the *C–C* bonds are shorter and far stronger than the Si–Si bonds.

Because the bonding is stronger, the forbidden energy gap of diamond is large ($\mathbf{E_g \approx 5.4\ eV}$), so electrons cannot be excited across it at room temperature and carbon behaves as an **insulator**. In silicon the bonds are weaker, giving a small gap ($\mathbf{E_g \approx 1.1\ eV}$), so electrons are easily excited and silicon behaves as a **semiconductor**.

CONCEPT

At $T=0\ K$ a semiconductor and an insulator look the same (full VB, empty CB); only the size of E_g distinguishes them. Always state the gap value to earn the mark.

Final answer. Conduction band gap decides the class: $\mathbf{E_g = 0}$ gives a conductor, $\mathbf{E_g \approx 1\ eV}$ a semiconductor, $\mathbf{E_g > 3\ eV}$ an insulator; diamond's large gap ($\mathbf{\approx 5.4\ eV}$) makes carbon an insulator while silicon's small gap ($\mathbf{\approx 1.1\ eV}$) makes it a semiconductor.

← *Back to Q231*

Q232

Differentiate between half-wave and full-wave rectification. With the help of a circuit diagram, explain the working of a full-wave rectifier.

[2025 • Set 55-5-1]

SOLUTION

Difference between half-wave and full-wave rectification:

In half-wave rectification only one diode is used and it conducts during only one half cycle of the input AC, so the other half cycle is lost. The output frequency equals the input frequency, the rectification efficiency is low (≈ **40.6%**) and the ripple factor is high (≈ **1.21**).

In full-wave rectification two diodes (with a centre-tapped transformer) or four diodes (bridge) are used and the circuit conducts during both half cycles of the input AC. The output frequency is twice the input frequency, the efficiency is higher (≈ **81.2%**) and the ripple factor is lower (≈ **0.48**), giving a smoother output.

Working of a full-wave rectifier (circuit description):

The circuit uses a centre-tapped step-down transformer. The anodes of the two diodes D_1 and D_2 are connected to the two ends of the secondary winding, and their cathodes are joined together. The load resistance R_L is connected between this common cathode point and the centre tap of the secondary. The output is taken across R_L.

During the positive half cycle, the upper end of the secondary is positive and the lower end negative. Hence D_1 is forward biased and conducts, while D_2 is reverse biased and does not conduct. Current flows from the top of the secondary through D_1, then through R_L from top to bottom, and back to the centre tap.

During the negative half cycle, the polarity reverses; now the lower end is positive. D_2 becomes forward biased and conducts while D_1 is reverse biased and is OFF. Current flows from the bottom of the secondary through D_2, then through R_L again from top to bottom, and back to the centre tap.

In both half cycles the current through R_L flows in the **same direction**, so the output across R_L is a **unidirectional pulsating DC**, with the output frequency equal to twice the input frequency.

Final answer. Half-wave uses one diode and rectifies one half cycle; full-wave uses two diodes that conduct in alternate half cycles so current through $\mathbf{R_L}$ stays unidirectional, giving smoother pulsating DC at twice the input frequency.

← Back to Q232

Q233

Explain the process of formation of 'depletion layer' and 'potential barrier' in a p-n junction region of a diode, with the help of a suitable diagram. Which feature of junction diode makes it suitable for its use as a rectifier?

[2025 • Set 55-7-2]

SOLUTION

Formation of the depletion layer:

When a p-type and an n-type semiconductor are formed in a single crystal, a sharp concentration gradient of carriers exists at the junction. The n-side has a high concentration of electrons and the p-side a high concentration of holes.

Because of this gradient, electrons diffuse from the n-side to the p-side and holes diffuse from the p-side to the n-side. As an electron leaves the n-side it exposes a fixed, **positively-charged immobile donor ion**, and as a hole leaves the p-side it exposes a fixed, **negatively-charged immobile acceptor ion**.

The diffusing electrons combine with holes near the junction, so a thin region around the junction is left with only immobile ions and no free charge carriers. This charge-free region is called the **depletion layer**, of typical width about 10^{-6} m.

Formation of the potential barrier:

The layer of negative acceptor ions on the p-side and positive donor ions on the n-side sets up an electric field directed from the n-side to the p-side across the junction. This field opposes further diffusion of majority carriers.

The potential difference developed across the depletion region due to this field is called the **potential barrier** V_b (about 0.7 V for silicon and 0.3 V for germanium). At equilibrium, this barrier field drives minority carriers in the direction opposite to diffusion, and the net current across the junction becomes zero.

Diagram (described): Draw a rectangle split into a left p-region and a right n-region. At the centre mark a narrow band labelled 'depletion layer'. Inside it, place a column of '–' (negative acceptor) ions on the p-side and a column of '+' (positive donor) ions on the n-side. Draw an arrow from the n-side towards the p-side inside this band labelled $\vec{E}$ (internal field), with the barrier potential V_b marked across the layer.

Feature making it suitable as a rectifier:

A junction diode offers **very low resistance in forward bias** (the barrier is lowered and a large current flows) and **very high resistance in reverse bias** (the barrier is raised and only a negligible current flows). This one-way or unidirectional conduction lets it pass current in only one direction, which is exactly what is required to convert AC into DC.

Final answer. Diffusion of majority carriers leaves immobile ions that form the carrier-free depletion layer and a potential barrier $\mathbf{V_b}$ (**≈ 0.7 *V*** for Si); the diode's unidirectional conduction (low resistance forward, high reverse) makes it suitable as a rectifier.

← *Back to Q233*

Q234

Draw the circuit diagrams for obtaining the V – I characteristics of a p-n junction diode. Explain briefly the salient features of the V – I characteristics in (i) forward biasing, and (ii) reverse biasing.

[2024 • Set 55-3-1]

SOLUTION

Circuit diagrams for the *V-I* characteristics:

Forward-bias circuit (described): A variable DC source is connected so that its positive terminal goes through a series ammeter (mA) to the p-side of the diode, and the n-side returns to the negative terminal; a voltmeter is connected across the diode to read the forward voltage. The p-side is at the higher potential.

Reverse-bias circuit (described): The same arrangement is used but the battery polarity is reversed, so the p-side is connected to the negative terminal and the n-side to the positive terminal; here a microammeter (μA) reads the small reverse current and the voltmeter reads the reverse voltage.

(i) Salient features in forward bias:

Up to a certain threshold (knee or cut-in) voltage V_T, the forward current is negligibly small because the applied voltage is opposing the barrier potential. For silicon $\mathbf{V_T \approx 0.7\ V}$ and for germanium $\mathbf{V_T \approx 0.3\ V}$.

Beyond V_T the barrier is largely overcome, and the forward current increases sharply and almost exponentially with a small increase in voltage. The **dynamic (a.c.) resistance** $r = \frac{\Delta V}{\Delta I}$ is very small in this region (a few ohms).

(ii) Salient features in reverse bias:

The current is a very small, almost constant **reverse saturation current** (of the order of a few μA), because it is carried only by thermally generated minority carriers, whose number is independent of the applied voltage.

This small current stays nearly constant until the **breakdown voltage** is reached, beyond which the reverse current suddenly increases sharply. In normal operation the diode is essentially non-conducting in reverse bias, and the dynamic resistance here is very large (of the order of MΩ).

This **asymmetric** behaviour, conducting strongly in forward bias and almost blocking in reverse bias, is what gives the diode its rectifying property.

TIP

Dynamic resistance is $r = \frac{\Delta V}{\Delta I}$ from the slope of the curve, not V/I. It is small (forward) and huge (reverse) and is a favourite follow-up mark.

Final answer. Forward bias gives negligible current below $\mathbf{V_T}$ ($\mathbf{\approx 0.7\ V}$ for Si) then a steep rise with small dynamic resistance; reverse bias gives a tiny constant saturation current (μA) until breakdown, the asymmetry making the diode a rectifier.

← *Back to Q234*

Q235

On the basis of energy band diagrams, distinguish between (i) an insulator, (ii) a semiconductor, and (iii) a conductor.

[2024 • Set 55-3-1]

SOLUTION

The electrical behaviour of a solid is decided by the width of the forbidden energy gap E_g between its valence band (VB) and conduction band (CB).

(i) Insulator:

The valence band is completely filled and the conduction band is completely empty, and they are separated by a **large** energy gap, $\mathbf{E_g > 3\ eV}$ (for example diamond, $\mathbf{E_g \approx 6\ eV}$). At ordinary temperatures electrons cannot be excited across this large gap, so the conduction band stays empty and the conductivity is extremely small (of the order of $10^{-12}\ S\ m^{-1}$). Essentially no conduction takes place.

(ii) Semiconductor:

At $\boldsymbol{T = 0\ K}$ the valence band is full and the conduction band is empty, but the gap between them is **small**, $\mathbf{E_g \approx 0.7}$ to 1.5 eV (Si $\approx$ **1.1 eV**, Ge $\approx$ **0.7 eV**). At room temperature some electrons gain enough thermal energy to cross into the conduction band, leaving holes behind. So it has a small but appreciable conductivity that **increases rapidly** with rise of temperature or with doping.

(iii) Conductor:

The valence band and conduction band **overlap**, or the conduction band is partially filled, so effectively $\mathbf{E_g = 0}$. A very large number of free electrons are always available in the conduction band, so even a small applied field produces a large current. The conductivity is very high (of the order of $10^7\ S\ m^{-1}$).

Thus conductivity is governed by E_g: zero gap gives a conductor, a small gap gives a semiconductor, and a large gap gives an insulator.

Final answer. Insulator: full VB, empty CB, large gap $\mathbf{E_g > 3\ eV}$; semiconductor: small gap $\mathbf{E_g \approx 1\ eV}$; conductor: overlapping bands, $\mathbf{E_g = 0}$; the gap size decides

the conductivity.

← Back to Q235

Q236

(a) Explain the characteristics of a p-n junction diode that makes it suitable for its use as a rectifier.
(b) With the help of a circuit diagram, explain the working of a full wave rectifier.

[2024 • Set 55-5-1]

SOLUTION

(a) Characteristics that make a p-n junction diode suitable as a rectifier:

Unidirectional conduction: A junction diode offers **very low resistance** (of the order of a few ohms) when it is forward biased and **very high resistance** (of the order of MΩ) when it is reverse biased. Therefore an appreciable current flows only during the forward-biased half cycle, while the reverse-biased half cycle is almost completely blocked.

Sharp cut-in (threshold) voltage: Below the cut-in voltage (≈ **0.7 *V*** for silicon) the forward current is negligible, so only voltages above this threshold produce a significant output, giving clean rectification.

Fast switching: The diode can switch between its conducting and non-conducting states very rapidly, so it works effectively even at the usual AC supply frequencies. Together these properties let the diode pass current in only one direction, converting AC into pulsating DC.

(b) Working of a full-wave rectifier:

Circuit (described): A centre-tapped step-down transformer feeds two diodes D_1 and D_2. The anodes of D_1 and D_2 are connected to the two ends of the secondary winding and their cathodes are joined together. The load resistance R_L is connected between this common cathode point and the centre tap, and the output V_o is taken across R_L.

During the positive half cycle, the top end of the secondary is positive and the bottom negative. Hence D_1 is forward biased and conducts while D_2 is reverse

biased and is OFF. Current flows from the top through D_1 and then through R_L from top to bottom.

During the negative half cycle, the polarity reverses; the bottom end becomes positive. Now D_2 is forward biased and conducts while D_1 is OFF. Current flows from the bottom through D_2 and then through R_L again from top to bottom.

Since the current through R_L flows in the **same direction in both half cycles**, the output is a **unidirectional pulsating DC** whose frequency is twice the input frequency.

Final answer. Low forward / high reverse resistance plus a sharp cut-in voltage gives the diode its rectifying property; in a centre-tap full-wave rectifier $\mathbf{D_1}$ and $\mathbf{D_2}$ conduct in alternate half cycles so $\mathbf{R_L}$ carries unidirectional pulsating DC at twice the input frequency.

← Back to Q236

Q237

Explain the following, giving reasons:
(a) A doped semiconductor is electrically neutral.
(b) In a p-n junction under equilibrium, there is no net current.
(c) In a diode, the reverse current is practically not dependent on the applied voltage.

[2024 • Set 55-5-1]

SOLUTION

(a) A doped semiconductor is electrically neutral:

Doping does not add any net charge to the crystal. A neutral host atom (such as Si) is replaced by an equally neutral dopant atom (such as pentavalent P or trivalent B). Each dopant atom supplies one extra charge carrier (a free electron in n-type, or a hole in p-type) but at the same time it becomes a **fixed ion of equal and opposite charge** locked in the lattice.

For example, a phosphorus atom donates one free electron and itself becomes a fixed positive ion of $+e$. The free electron's negative charge is exactly balanced

by this positive donor ion. Hence the total charge of the crystal remains zero, and the doped semiconductor is electrically neutral.

(b) No net current in a p-n junction at equilibrium:

At an unbiased junction two equal and opposite currents flow. A **diffusion current** arises because majority carriers diffuse across the junction down their concentration gradients (electrons n to p, holes p to n). A **drift current** arises because the built-in field of the depletion region sweeps the thermally generated minority carriers in the opposite direction.

In equilibrium the barrier potential adjusts itself so that the diffusion current and the drift current become exactly equal in magnitude but opposite in direction. They cancel out, so the **net current across the junction is zero**.

(c) Reverse current is nearly independent of the applied voltage:

In reverse bias the current is carried only by **minority carriers** (electrons in the p-region and holes in the n-region) that are generated thermally and are swept across by the junction field. The rate at which these minority carriers are produced depends only on the temperature, not on the applied reverse voltage.

Increasing the reverse voltage only widens the depletion region; it does not increase the limited supply of minority carriers. Hence almost all the available minority carriers are already being collected, and the reverse (saturation) current stays practically constant, independent of the applied voltage, until breakdown.

Final answer. (a) Each neutral dopant adds a carrier and an equal fixed ion, so net charge stays zero; (b) at equilibrium the diffusion and drift currents cancel; (c) the reverse current is set by the temperature-limited supply of minority carriers, so it is nearly independent of voltage.

← *Back to Q237*

Q238

State the working principle of an LED. Write any two important advantages and two

disadvantages of LED.

[2022 • Set 55-1-1]

SOLUTION

Working principle of an LED (Light Emitting Diode):

An LED is a **heavily doped p-n junction diode operated under forward bias**. When it is forward biased, the barrier potential is lowered and majority carriers are injected across the junction, so that there is a large concentration of electrons in the conduction band and holes in the valence band near the junction.

Close to the junction these electrons and holes **recombine**. In each recombination an electron falls from the conduction band into a hole in the valence band, releasing energy roughly equal to the band gap E_g in the form of a **photon**.

The energy and hence the frequency of the emitted photon is given by

$$h\nu = E_g$$

$$\Rightarrow \nu = \frac{E_g}{h}$$

The semiconductor material is chosen so that E_g corresponds to a photon in the **visible** (or infrared) region, giving the desired colour of light.

Two advantages of an LED: (i) It has a **low operating voltage** ($\approx 2\ V$) and a high luminous efficiency, so it consumes very little power. (ii) It has a **long working life** (**> 50000** hours), turns on almost instantly, and is rugged and small in size with fast on/off switching.

Two disadvantages of an LED: (i) Each LED gives nearly **monochromatic (single-colour) light** over a narrow band, so producing pure white light needs a phosphor coating or a combination of several LEDs. (ii) Its performance degrades at high temperatures, and it is **sensitive to over-voltage and reverse bias**, so it needs a current-limiting circuit for safe operation.

Final answer. An LED is a heavily doped forward-biased p-n junction in which electron-hole recombination releases photons of frequency $\boldsymbol{\nu = E_g/h}$;

advantages: low voltage and long life, disadvantages: monochromatic output and sensitivity to over-voltage/temperature.

← *Back to Q238*

Q239

With the help of a circuit diagram, explain the working of a p-n junction diode as a full-wave rectifier. Also draw its input and output waveforms.

[2022 • Set 55-2-1]

SOLUTION

Working of a p-n junction diode as a full-wave rectifier:

Circuit (described): A centre-tapped step-down transformer feeds two p-n junction diodes D_1 and D_2. The anodes of the diodes are joined to the two ends of the secondary winding, while their cathodes are connected together. The load resistance R_L is connected between this common cathode point and the centre tap of the secondary, and the output V_{out} is taken across R_L.

During the positive half cycle, the top end of the secondary is positive and the bottom end negative. Hence the anode of D_1 is positive with respect to the centre tap, so D_1 is **forward biased and conducts**, while D_2 is reverse biased and is OFF. The current flows from the top end through D_1, then through R_L from top to bottom, and back to the centre tap.

During the negative half cycle, the polarity of the secondary reverses, so the bottom end is now positive. Therefore D_2 is **forward biased and conducts**, while D_1 is OFF. The current now flows from the bottom end through D_2, then through R_L again from top to bottom, and back to the centre tap.

In both half cycles the current through R_L flows in the **same direction**, so the output is a **unidirectional pulsating DC**, and its frequency is twice the frequency of the input AC.

Input and output waveforms (described): The input waveform is a smooth sinusoidal AC, alternating above and below the time axis. The output waveform is a series of positive half-sine humps that follow one after another with no gaps between them (one hump for every half cycle of the input), all lying above the

time axis. This is smoother than the half-wave output, where alternate humps are missing. With a capacitor filter the output becomes almost steady DC.

Final answer. In a centre-tap full-wave rectifier $\mathbf{D_1}$ and $\mathbf{D_2}$ conduct in alternate half cycles, keeping the current through $\mathbf{R_L}$ unidirectional; the input is a sine wave and the output is a continuous train of positive half-sine humps at twice the input frequency.

← *Back to Q239*

Q240

Briefly explain how emf is generated in a solar cell. Draw its I-V characteristics.

[2022 • Set 55-3-1]

SOLUTION

Generation of emf in a solar cell:

A solar cell is a p-n junction that generates an emf when light falls on it, through three basic processes.

Generation: When photons of energy $\mathbf{h\nu > E_g}$ are incident on the junction, they are absorbed and create **electron-hole pairs** in and near the depletion region.

$$h\nu > E_g \Rightarrow \text{electron–hole pair created}$$

Separation: The built-in electric field of the depletion region separates these carriers. It sweeps the **electrons towards the n-side** and the **holes towards the p-side**.

Collection: The electrons collected on the n-side and the holes collected on the p-side accumulate at the contacts. This builds up a potential difference, the **photo-emf**, across the junction. When an external load is connected, this emf drives a current through it, so the cell delivers electrical energy.

I-V characteristic (described): The current-voltage curve of an illuminated solar cell lies in the **fourth quadrant** (positive voltage, negative current), because the cell delivers power rather than absorbing it. The curve cuts the voltage axis at the **open-circuit voltage** V_{OC} (the voltage when $I = 0$) and cuts the current axis

at the **short-circuit current** I_{SC} (the current when $V = 0$). The curve runs nearly horizontally at I_{SC} for low voltages and then bends down sharply near V_{OC}; the operating point is chosen near this knee so that the output power $P = VI$ is maximum.

CONCEPT

Remember the order Generation -> Separation -> Collection, and that the solar cell I-V curve sits in the fourth quadrant because the cell supplies power instead of consuming it.

Final answer. Light generates electron-hole pairs that the junction field separates (electrons to n-side, holes to p-side), building a photo-emf; the I-V curve lies in the fourth quadrant with intercepts $\mathbf{V_{OC}}$ (at $I=0$) and $\mathbf{I_{SC}}$ (at $V=0$).

← *Back to Q240*

Q241

Draw the energy band diagrams for conductors, semiconductors and insulators. Which band determines the electrical conductivity of a solid? How is the electrical conductivity of a semiconductor affected with rise in its temperature? Explain.

[2022 • Set 55-3-2]

SOLUTION

On the basis of the energy gap between the valence band and the conduction band, solids are classified into conductors, semiconductors and insulators.

In a conductor (metal), the valence band and the conduction band overlap, or the conduction band is partially filled even at $T = 0\ K$. There is no forbidden energy gap, so a large number of free electrons are always available. In the band diagram, the shaded valence band touches or overlaps the conduction band ($\mathbf{E_g = 0}$).

In an insulator, a wide forbidden energy gap separates the two bands, with $\mathbf{E_g > 3\ eV}$. The valence band is completely full and the conduction band is completely empty at room temperature. In the band diagram, a large blank gap is shown between the full valence band and the empty conduction band.

In a semiconductor, the forbidden energy gap is small, $E_g \sim 1$ eV (about 1.1 eV for Si, 0.7 eV for Ge). At $T = 0\ K$ the conduction band is empty, but at room temperature a few electrons are thermally excited across this narrow gap. In the band diagram, a small gap is shown between the valence band and the conduction band.

The conduction band determines the electrical conductivity of a solid, because only the electrons that reach the conduction band (and the holes they leave behind in the valence band) are free to move and carry current under an applied field.

As the temperature of a semiconductor rises, more electron-hole pairs are generated thermally, so the carrier concentration increases. The intrinsic carrier density follows

$$n_i \propto T^{3/2}\, e^{-E_g/2kT}$$

so n_i increases rapidly (almost exponentially) with temperature. Hence the conductivity of a semiconductor increases and its resistance decreases as temperature rises. A semiconductor therefore has a **negative temperature coefficient of resistance**, which is opposite to the behaviour of a metal.

Final answer. The conduction band determines conductivity; a semiconductor's conductivity increases with temperature because the carrier density rises as $n_i \propto T^{3/2}\, e^{-E_g/2kT}$.

← *Back to Q241*

Q242

Answer the following, giving reason:
(a) The resistance of a p-n junction is low when it is forward biased and is high when it is reverse biased.
(b) Doping of intrinsic semiconductors is a necessity for making electronic devices.
(c) Photodiodes are operated in reverse bias.

[2022 • Set 55-4-1]

SOLUTION

(a) In forward bias, the external voltage opposes the built-in potential barrier, so the barrier height and the width of the depletion region both decrease. The large population of majority carriers can then easily cross the junction, giving a large forward current. A large current for a small applied voltage means the junction offers a **low resistance** (a few ohms).

In reverse bias, the external voltage adds to the built-in barrier, so the barrier height and depletion-region width both increase. Majority carriers cannot cross; only the very few minority carriers contribute, giving a tiny reverse saturation current (~ μA). A very small current for a large applied voltage means a **high resistance** (of the order of MΩ).

(b) An intrinsic (pure) semiconductor has a very small carrier concentration at room temperature ($n_i \sim 10^{16}$ m^{-3} for Si), so its conductivity is too low to be useful in devices. Adding a small amount of dopant (even a few parts per million) raises the carrier concentration by several orders of magnitude and lets us decide whether electrons (n-type) or holes (p-type) are the majority carriers. This **control of carrier type and number** is the basis of every junction device, so doping is a necessity.

(c) In a photodiode the change in current with illumination is detected. Under reverse bias the dark current (current without light) is extremely small, so the additional current produced when light falls on the junction is a large fractional change and is easily measured. Hence the photodiode is more sensitive to changes in light intensity in reverse bias. In forward bias the large majority-carrier current would swamp this small photo-current, making the change hard to detect. Therefore **photodiodes are operated in reverse bias**.

Final answer. Forward bias narrows the barrier (low R), reverse bias widens it (high R); doping controls carrier type and number; reverse-biased photodiodes give a detectable fractional current change.

← *Back to Q242*

Q243

(i) Draw V-I characteristics of a p-n junction diode.
(ii) Differentiate between the threshold voltage and the breakdown voltage for a diode.

(iii) Write the property of a junction diode which makes it suitable for rectification of ac voltages.

[2022 • Set 55-5-1]

SOLUTION

(i) The V-I characteristic of a p-n junction diode is plotted with the applied voltage on the x-axis and the current on the y-axis. In the forward-bias region (first quadrant), the current is almost zero until the voltage reaches the knee (threshold) voltage V_k (~ 0.7 *V* for Si and ~ 0.3 *V* for Ge), after which the current rises steeply. In the reverse-bias region (third quadrant), only a very small reverse saturation current (~ μA) flows, shown as a nearly flat line close to the axis, until the breakdown voltage is reached, where the reverse current suddenly increases sharply.

(ii) Threshold (knee) voltage and breakdown voltage differ as follows.

Threshold voltage is a forward-bias quantity. It is the minimum forward voltage above which the forward current starts to rise sharply (~ 0.3 *V* for Ge, ~ 0.7 *V* for Si).

Breakdown voltage is a reverse-bias quantity. It is the reverse voltage at which a large reverse current suddenly flows (by the Zener or avalanche mechanism). Its value ranges from a few volts to hundreds of volts depending on doping.

(iii) The property used for rectification is the **unidirectional (one-way) conduction** of the diode: it conducts easily in forward bias but blocks current in reverse bias. This allows it to convert alternating current into pulsating direct current.

Final answer. Threshold voltage (**~ 0.7 *V*** Si) is a forward-bias turn-on value; breakdown voltage is the reverse-bias value of sudden current rise; the diode's unidirectional conduction enables rectification.

← *Back to Q243*

Q244

Draw circuit diagram and explain the working of a zener diode as a dc voltage regulator with the help of its I-V characteristic.

[2020 • Set 55-1-1]

SOLUTION

A Zener diode is a heavily doped diode operated in its reverse breakdown region, where the voltage across it stays nearly constant.

Circuit diagram: The unregulated DC input V_{in} is connected through a series current-limiting resistor R_S. The Zener diode is connected after R_S in **reverse bias**, in parallel with the load R_L. The regulated output V_{out} is taken across the Zener diode (and hence across the load). The cathode of the Zener goes to the positive line and the anode to ground.

I-V characteristic: In forward bias the Zener behaves like an ordinary diode. In reverse bias a small saturation current flows until the voltage reaches V_Z; at $\boldsymbol{V = -V_Z}$ the curve becomes almost vertical, meaning a large change in current produces almost no change in voltage. This near-vertical breakdown region is where the diode is operated, and here the voltage stays clamped at V_Z.

Working: The Zener works in the breakdown region where the voltage across it is constant at V_Z regardless of the current through it. Therefore

$$V_{out} = V_Z = \text{constant}$$

If V_{in} increases, the extra voltage appears across R_S (since $\mathbf{V_{in} = I\,R_S + V_Z}$). The current rises, but the Zener takes the extra current while V_Z stays fixed, so V_{out} is unchanged.

If the load current I_L changes, the Zener current I_Z adjusts so that the total current $\boldsymbol{I = I_Z + I_L}$ through R_S stays nearly constant, again keeping $\mathbf{V_{out} = V_Z}$.

Thus the Zener diode supplies a **constant regulated output voltage** V_Z even when the input voltage or the load current varies.

Final answer. A reverse-biased Zener in parallel with the load (through series $\mathbf{R_S}$) clamps the output at $\mathbf{V_{out} = V_Z}$, absorbing input and load variations to give a regulated DC voltage.

← Back to Q244

Q245

What is the purpose of heavy doping of p- and n-sides of a zener diode?

[2020 • Set 55-1-1]

SOLUTION

In a Zener diode both the p-side and the n-side are doped very heavily, much more than in an ordinary p-n junction diode. The purpose of this heavy doping is as follows.

Heavy doping puts a large number of immobile dopant ions per unit volume on each side of the junction. To create the required space-charge, only a very thin region is needed, so the **depletion region becomes extremely narrow** (of the order of 10^{-6} cm, about 10 nm).

Because the depletion layer is so thin, even a small reverse voltage produces a very strong electric field across it:

$$E = \frac{V}{d} \sim 10^8\ V/m$$

This intense field is strong enough to pull valence electrons directly from the p-side into the conduction band of the n-side by quantum-mechanical tunnelling. This is the **Zener effect**, which causes breakdown at a low, sharply defined reverse voltage V_Z.

By choosing the doping concentration during manufacture, the breakdown voltage V_Z can be fixed precisely (from a few volts to about 100 V), and it remains nearly constant over a wide range of currents. This makes the heavily doped Zener diode suitable as a stable voltage regulator and voltage reference.

Final answer. Heavy doping makes the depletion region very thin, so even a small reverse voltage gives a huge field (**~ 10^8 V/m**) that produces a sharp, well-defined Zener breakdown at **V_Z**.

← Back to Q245

Q246

With the help of a simple diagram, explain the working of a silicon solar cell, giving all three basic processes involved. Draw its I-V characteristic.

[2020 • Set 55-1-2]

SOLUTION

A silicon solar cell is a large-area p-n junction that converts sunlight directly into electrical energy.

Diagram: A thin n-type layer is formed on top of a thicker p-type silicon base, so the depletion region lies just below the illuminated surface. The top surface carries an anti-reflection coating and a metal grid (front contact) that lets light pass through, while the bottom has a full metal contact (back contact). The two contacts are joined through an external load.

The working involves three basic processes.

Generation: Photons of sunlight with energy $\mathbf{h\nu > E_g}$ are absorbed near the junction and create electron-hole pairs.

Separation: The built-in electric field of the depletion region separates these pairs. The electrons are swept towards the n-side and the holes towards the p-side. This separation of charge sets up a photo-voltage (photo-EMF) across the junction.

Collection: The front grid contact (n-side) and the back contact (p-side) collect the separated charges. When an external load is connected, electrons flow through it from the n-side to the p-side, delivering electrical power.

I-V characteristic: The characteristic of an illuminated solar cell lies in the fourth quadrant, because the cell drives current through the external circuit (positive voltage, negative current). The curve meets the current axis at the short-circuit current I_{SC} (at $\boldsymbol{V = 0}$, proportional to light intensity) and meets the voltage axis at the open-circuit voltage V_{OC} (at $\boldsymbol{I = 0}$, about 0.5 to 0.6 V for Si). The knee of the curve marks the **maximum power point**, where the product $|V \times I|$ is greatest.

Final answer. A solar cell works by generation, separation and collection of photo-carriers; its illuminated I-V curve lies in the fourth quadrant, cutting the axes at $\mathbf{I_{SC}}$ and $\mathbf{V_{OC}}$.

← Back to Q246

Q247

Draw the circuit diagram of a full wave rectifier using two p-n junction diodes. Explain its working and show input and output voltage variations.

[2020 • Set 55-1-3]

SOLUTION

A full-wave rectifier converts both halves of an AC cycle into a unidirectional (DC) output using two diodes.

Circuit diagram: A centre-tapped step-down transformer is used. The two ends of the secondary winding are connected to the anodes of the two diodes D_1 and D_2. The cathodes of both diodes are joined together, and the load resistance R_L is connected between this common cathode point and the centre tap of the secondary. The output voltage V_{out} is taken across R_L.

Working during the positive half cycle: The top end of the secondary is positive and the bottom end negative. The anode of D_1 is positive with respect to the centre tap, so D_1 is forward biased and conducts, while D_2 is reverse biased and does not conduct. Current flows from the top end through D_1, then through R_L from top to bottom, and back to the centre tap.

Working during the negative half cycle: The polarity reverses, so the bottom end of the secondary becomes positive. Now D_2 is forward biased and conducts while D_1 is off. Current flows from the bottom end through D_2, then through R_L in the same direction (top to bottom), and back to the centre tap.

Since the current through R_L is in the same direction in both half cycles, the output is a **unidirectional (DC) voltage**. The output frequency is twice the input frequency.

Waveforms: The input is a pure sinusoidal AC wave. The output across R_L is a series of positive half-sine humps with no gaps between them (both half cycles

appear as positive humps), which is much smoother than the half-wave output. A capacitor filter can make it almost steady DC.

Final answer. Two diodes fed by a centre-tapped transformer conduct alternately, so $\mathbf{R_L}$ carries current in the same direction in both half cycles, giving a unidirectional output of double the input frequency.

← *Back to Q247*

Q248

What is the function of a solar cell? Briefly explain its working and draw its I-V characteristic curve.

[2020 • Set 55-2-1]

SOLUTION

Function: A solar cell (photovoltaic cell) is a semiconductor device that converts solar (light) energy directly into electrical energy, producing DC power from sunlight with no moving parts.

Construction: It is a large-area p-n junction with a thin transparent n-type layer formed on a thicker p-type substrate, so the depletion region lies just under the lit surface. The front has a metal grid contact (to admit light) and the back has a full metal contact, both joined through an external load.

Working involves three processes.

Generation: Photons of sunlight with energy $\mathbf{h\nu \geq E_g}$ are absorbed near the junction, creating electron-hole pairs.

Separation: The built-in field of the depletion region drives the electrons to the n-side and the holes to the p-side, setting up a photo-EMF across the junction.

Collection: The front grid and back contact collect the separated charges, so current flows through the external load and the cell delivers electrical power.

I-V characteristic: The characteristic of the illuminated cell lies in the fourth quadrant (positive voltage, negative current), because the cell acts as a source. The curve cuts the current axis at the short-circuit current I_{SC} (at $\boldsymbol{V = 0}$, proportional to light intensity) and cuts the voltage axis at the open-circuit

voltage V_{OC} (at $I = 0$, about 0.5 to 0.7 V for Si). The knee of the curve is the **maximum power point** P_{max}, where $|V \times I|$ is largest.

Final answer. A solar cell converts light energy directly into electrical energy by generation, separation and collection of carriers; its illuminated I-V curve lies in the fourth quadrant with intercepts $\mathbf{I_{SC}}$ and $\mathbf{V_{OC}}$.

← *Back to Q248*

Q249

(a) Plot V-I characteristics for an illuminated photodiode under reverse bias for three different illumination intensities $\mathbf{I_1 > I_2 > I_3}$.
(b) In a zener regulated power supply, a zener diode with $\mathbf{V_Z = 6}$ *V is used for regulation. The load current should be* 4.0 *mA and zener current is five times the load current for unregulated input of* 10 *V. Find the series resistor* R_s *used in the power supply.*

[2020 • Set 55-2-2]

SOLUTION

(a) The reverse-biased characteristic of an illuminated photodiode is plotted with reverse voltage on the negative x-axis and reverse current on the negative y-axis (third quadrant). For each intensity the curve is almost flat (saturated) once a small reverse voltage is applied. Three separate parallel curves are drawn, one for each intensity. Since the reverse photo-current increases with light intensity, the curve for I_1 lies at the largest reverse current, the curve for I_2 in the middle, and the curve for I_3 closest to the voltage axis (smallest reverse current), matching $\mathbf{I_1 > I_2 > I_3}$.

(b) Given data: $\mathbf{V_{in} = 10}$ *V*, $\mathbf{V_Z = 6}$ *V*, load current $\mathbf{I_L = 4.0\ mA}$, and Zener current $\mathbf{I_Z = 5\ I_L}$.

$$I_Z = 5 \times 4.0 = 20 \text{ mA}$$

The total current through the series resistor is the sum of the Zener and load currents.

$$I = I_Z + I_L = 20 + 4.0 = 24 \text{ mA}$$

The voltage dropped across R_S is the difference between the input and the regulated output.

$$V_{R_S} = V_{in} - V_Z = 10 - 6 = 4\ V$$

Applying Ohm's law to the series resistor,

$$R_S = \frac{V_{R_S}}{I}$$

$$R_S = \frac{4}{24 \times 10^{-3}}$$

$$R_S = 166.7\ \Omega \approx 167\ \Omega$$

Hence the required series resistance is $\mathbf{R_S \approx 167\ \Omega}$.

COMMON MISTAKE

Always add the Zener current and the load current to get the current through the series resistor; a common mistake is to use only the load current.

Final answer. $R_S = \frac{V_{in} - V_Z}{I_Z + I_L} = \frac{4}{2410^{-3}} \approx 167\ \Omega$.

← Back to Q249

Q250

Name the diode which can act as a voltage regulator. Explain its working with the help of its labelled circuit diagram. Draw its V-I characteristic.

[2020 • Set 55-3-1]

SOLUTION

The diode that acts as a voltage regulator is the **Zener diode**. It is heavily doped and is operated in its reverse breakdown region, where its voltage stays nearly constant.

Circuit diagram: The unregulated DC input V_{in} is connected through a series current-limiting resistor R_S. The Zener diode is connected in **reverse bias** in parallel with the load R_L, with its cathode towards the positive line and its

anode to ground. The regulated output V_{out} is taken across the Zener diode (and hence across the load).

V-I characteristic: In forward bias the Zener behaves like an ordinary diode. In reverse bias only a small saturation current flows until the voltage reaches V_Z; at $\boldsymbol{V = -V_Z}$ the curve becomes almost vertical (the breakdown region), meaning the voltage stays clamped at V_Z even for a large change in current.

Working: The Zener is operated in the breakdown region, where the voltage across it remains constant at V_Z irrespective of the current through it, so

$$V_{out} = V_Z = \text{constant}$$

If V_{in} increases, the extra voltage drops across R_S (since $\mathbf{V_{in} = I\,R_S + V_Z}$); the Zener simply draws more current while V_Z stays fixed, so V_{out} does not change.

If the load current I_L changes, the Zener current adjusts so that the total current through R_S stays nearly constant, again keeping $\mathbf{V_{out} = V_Z}$.

Thus the Zener diode delivers a **constant regulated output voltage** V_Z even when the input voltage or load current varies.

Final answer. The Zener diode, reverse-biased in its breakdown region in parallel with the load, clamps the output at $\mathbf{V_{out} = V_Z}$ and so acts as a voltage regulator.

← *Back to Q250*

Q251

(a) Why is an intrinsic semiconductor deliberately converted into an extrinsic semiconductor by adding impurity atoms?
(b) Explain briefly the two processes that occur in p-n junction region to create a potential barrier.

[2020 • Set 55-3-1]

SOLUTION

(a) An intrinsic semiconductor at room temperature has very few free charge carriers. For pure silicon the intrinsic carrier concentration is only about $n_i \sim$

10^{16} m^{-3}, which gives an extremely low conductivity that is useless for practical devices.

Also, this carrier concentration depends strongly on temperature, $n_i \propto e^{-E_g/2k_BT}$, so the behaviour of an intrinsic semiconductor changes a lot with temperature and is hard to control.

By adding a tiny, controlled amount of impurity (called **doping**), we add either a pentavalent donor (giving extra free electrons, *n*-type) or a trivalent acceptor (giving extra holes, *p*-type). Even about 1 part per million of impurity raises the carrier concentration by a factor of 10^5 to 10^7.

So doping is done to (i) greatly increase conductivity, (ii) fix the majority carrier type (electrons or holes) as we choose, and (iii) make the conductivity stable and controllable rather than temperature dependent. This makes diodes, transistors and other devices possible.

(b) Two processes create the potential barrier across a *p*-*n* junction:

Diffusion: Because the hole concentration is high on the *p*-side and the electron concentration is high on the *n*-side, holes diffuse from $p \to n$ and electrons diffuse from $n \to p$ due to the concentration gradient. This leaves behind immobile negative acceptor ions on the *p*-side and immobile positive donor ions on the *n*-side, forming a charge-free **depletion region**.

Drift: The uncovered ions set up an electric field $\vec{E}$ directed from the *n*-side to the *p*-side. This field pushes the minority carriers, causing a drift current opposite to the diffusion current. Equilibrium is reached when the drift current exactly balances the diffusion current, and the built-in field corresponds to a **potential barrier** $V_0 \approx 0.7\ V$ for silicon.

Final answer. Doping converts an intrinsic semiconductor to extrinsic to greatly raise and control conductivity; diffusion of majority carriers and the opposing drift due to the field of uncovered ions together set up the potential barrier.

← *Back to Q251*

Q252

With the help of a circuit diagram, explain how two p-n junction diodes along with a centre tapped transformer can be used as a full wave rectifier.

[2020 • Set 55-3-2]

SOLUTION

A full-wave rectifier converts both halves of an AC input into a unidirectional (DC) output using two diodes and a centre-tapped transformer.

Circuit diagram (described): A centre-tapped step-down transformer has its primary connected to the AC mains. The two ends *A* and *B* of the secondary winding are connected to the anodes of two diodes D_1 and D_2 respectively. The cathodes of D_1 and D_2 are joined together. The load resistor R_L is connected between this common cathode point and the centre tap *C* of the secondary. The output voltage V_o is taken across R_L.

Working principle: The two halves of the secondary winding produce voltages that are always in opposite phase with respect to the centre tap, so when end *A* is positive, end *B* is negative, and vice versa. The diodes act as one-way valves and conduct only when forward biased.

During the positive half cycle: end *A* is positive and end *B* is negative with respect to centre tap *C*. So D_1 is forward biased and conducts, while D_2 is reverse biased and stays off. Current flows $\boldsymbol{A \rightarrow D_1}$ → through R_L (from top to bottom) → back to centre tap *C*.

During the negative half cycle: the polarity reverses, end *B* becomes positive and end *A* negative. Now D_2 is forward biased and conducts, while D_1 is off. Current flows $\boldsymbol{B \rightarrow D_2}$ → through R_L (again from top to bottom) → back to centre tap *C*.

In both half cycles the current through R_L flows in the **same direction**, so the output is a **unidirectional (DC) voltage**. Since the load receives a pulse in each half cycle, the output frequency is **twice** the input frequency.

Waveforms: The input is a full sine wave (positive and negative humps). The output across R_L is a series of positive humps with no gaps, one for each half cycle of the input.

TIP

Remember the key advantage: a full-wave rectifier uses both half cycles, so its output is smoother and its ripple frequency is double that of a half-wave rectifier.

Final answer. Two diodes and a centre-tapped transformer conduct on alternate half cycles, driving current through $\mathbf{R_L}$ in the same direction in both halves, giving a unidirectional output of twice the input frequency.

← *Back to Q252*

Q253

(a) How are energy bands formed in a crystalline solid?
(b) Draw the energy band diagrams for p-type and n-type semiconductors. Depict the donor/acceptor energy levels in these diagrams and write their significance.

[2020 • Set 55-3-3]

SOLUTION

(a) Formation of energy bands in a crystalline solid:

In an isolated atom, electrons occupy discrete, sharply defined energy levels (such as 1s, 2s, 2p, 3s). When a large number N of identical atoms are brought close together to form a crystal, the atoms are so near that the outer electron orbitals of neighbouring atoms overlap and interact.

By the **Pauli exclusion principle**, no two electrons in the crystal can occupy exactly the same quantum state. So each single atomic energy level must split into N very closely spaced sub-levels.

Since N is enormous (of the order of 10^{23} atoms), these N sub-levels are spaced so closely (separation ~ 10^{-23} eV) that they merge into a virtually continuous range of allowed energies called an **energy band**. Between two such allowed bands there may be a range of energies with no allowed states, called the **forbidden energy gap** E_g.

The highest band that is completely or nearly filled with electrons is the **valence band**, the next higher (mostly empty) band is the **conduction band**,

and the gap between them is the band gap E_g.

(b) Energy band diagrams (described):

n-type semiconductor: Draw the valence band at the bottom (filled, top edge E_V) and the conduction band at the top (empty, bottom edge E_C), separated by the gap E_g. Just below the conduction band, draw a discrete **donor energy level** E_D, very close to E_C (typically $\mathbf{E_C - E_D \approx 0.01–0.05\ eV}$).

Significance: This donor level lies so close to E_C that, even at room temperature, the small thermal energy easily excites the donor electrons into the conduction band, providing plenty of free electrons (majority carriers) for conduction.

p-type semiconductor: Again draw the valence band and conduction band separated by E_g. Just above the valence band, draw a discrete **acceptor energy level** E_A, very close to E_V (typically $\mathbf{E_A - E_V \approx 0.01–0.05\ eV}$).

Significance: This acceptor level lies so close to E_V that electrons from the valence band are easily thermally excited into it, leaving behind **holes** in the valence band. These holes act as the majority carriers for conduction.

Final answer. Atomic levels split and merge into continuous energy bands when N atoms form a crystal; in ***n***-type a donor level $\mathbf{E_D}$ sits just below $\mathbf{E_C}$ supplying free electrons, while in ***p***-type an acceptor level $\mathbf{E_A}$ sits just above $\mathbf{E_V}$ supplying holes.

← *Back to Q253*

Q254

What is a solar cell ? Draw its V-I characteristics. Explain the three processes involved in its working.

OR

Draw the circuit diagram of a full wave rectifier. Explain its working showing its input and output waveforms.

[2020 • Set 55-5-1]

SOLUTION

A **solar cell** is a specially designed *p*-*n* junction diode that converts solar light energy directly into electrical energy, based on the **photovoltaic effect**.

Construction (described): It has a very thin, lightly doped *p*-type (or *n*-type) layer on top so that light can reach near the junction, over a thicker base layer of the opposite type. A metallic grid (finger electrode) is placed on the top illuminated surface and a full metallic contact on the back, so that current can be drawn from an external load.

V-*I* characteristic (described): The curve lies in the **fourth quadrant** of the *I* versus *V* plane (positive voltage, negative current), because the cell delivers power rather than absorbing it. It cuts the voltage axis at the **open-circuit voltage** V_{OC} (current zero) and cuts the current axis at the **short-circuit current** I_{SC} (voltage zero). The bend or knee between these two points is the maximum power point.

Three processes in the working of a solar cell:

Generation: Photons of incident sunlight with energy $\mathbf{h\nu > E_g}$ are absorbed near the junction and create **electron-hole pairs**.

Separation: The built-in electric field of the depletion region separates these pairs, sweeping electrons to the *n*-side and holes to the *p*-side.

Collection: The separated charges are collected by the front and back electrodes, building up a photo-voltage; when connected to an external load, a current flows and electrical power is delivered.

OR

Full-wave rectifier (described): A centre-tapped transformer has the two ends *A*, *B* of its secondary connected to the anodes of diodes D_1 and D_2. Their cathodes are joined and the load R_L is connected between this common point and the centre tap *C*; output is taken across R_L.

Working: During the positive half cycle, end *A* is positive, so D_1 is forward biased and conducts while D_2 is off; current passes through R_L from top to bottom. During the negative half cycle, end *B* is positive, so D_2 conducts and D_1 is off; current again passes through R_L from top to bottom. Thus current through R_L is unidirectional in both halves.

Waveforms: The input is a full sine wave; the output across R_L is a continuous series of positive humps (no negative part), with a ripple frequency twice that

of the input.

Final answer. A solar cell is a photovoltaic ***p-n*** junction whose ***V-I*** curve lies in the fourth quadrant ($\mathbf{V_{OC}}$, $\mathbf{I_{SC}}$); it works by generation, separation and collection of light-induced electron-hole pairs.

← *Back to Q254*

Q255

Explain the formation of potential barrier and depletion region in a p-n junction diode. What is effect of applying forward bias on the width of depletion region ?
OR
What is photo diode ? Briefly explain its working and draw its V-I characteristics.

[2020 • Set 55-5-3]

SOLUTION

Formation of the depletion region and potential barrier:

When *p*-type and *n*-type semiconductors are joined to form a junction, there is a large concentration difference of carriers across it. So holes diffuse from the *p*-side to the *n*-side and electrons diffuse from the *n*-side to the *p*-side.

As majority carriers cross over and recombine near the junction, they leave behind **immobile ions**: negative acceptor ions on the *p*-side and positive donor ions on the *n*-side. This narrow region near the junction, having only immobile charged ions and no free mobile carriers, is called the **depletion region**.

These uncovered ions set up an electric field $\vec{E}$ directed from the *n*-side to the *p*-side. This field opposes the further diffusion of majority carriers and is equivalent to a potential difference across the junction called the **potential barrier** V_0 (about 0.7 *V* for silicon).

Effect of forward bias on the depletion width:

In forward bias the *p*-side is connected to the positive terminal and the *n*-side to the negative terminal of the battery. The applied voltage opposes the built-in field, so the effective barrier reduces to $V_0 - V$.

As a result the **depletion region becomes narrower** and the internal field weakens, allowing majority carriers to cross the junction easily, producing a large forward current.

OR

A **photodiode** is a special reverse-biased *p-n* junction diode, made with a transparent window, used to detect light.

Working: When light of photon energy $\mathbf{h\nu > E_g}$ falls on or near the depletion region, it is absorbed and creates additional electron-hole pairs. The strong field of the reverse-biased junction quickly separates these carriers, so the reverse (minority-carrier) current increases. As the light intensity increases, more electron-hole pairs are generated and the reverse current increases in proportion, so the photodiode measures light intensity.

V-I characteristic (described): It is drawn entirely in the **third quadrant** (reverse bias: negative voltage on the *x*-axis, negative reverse current on the *y*-axis). For each higher value of illumination intensity the reverse-current curve shifts further down (larger reverse current), and the curves for increasing intensities $\mathbf{I_1 < I_2 < I_3}$ lie one below the other, the spacing showing the linear rise of photocurrent with intensity.

Final answer. Diffusion of majority carriers leaves immobile ions forming the depletion region and a barrier $\mathbf{V_0 \approx 0.7\ V}$; forward bias reduces the barrier and narrows the depletion region. (OR: a photodiode is a reverse-biased junction whose reverse-current V-I characteristic lies in the third quadrant and grows with light intensity.)

← *Back to Q255*

Q256

Three photo diodes D_1, D_2 *and* D_3 *are made of semiconductors having band gaps of* 2.5 eV, 2 eV *and* 3 eV *respectively. Which of them will not be able to detect light of wavelength* 600 nm*?*

[2019 • Set 55-1-1]

SOLUTION

A photodiode can detect light only if the energy of the incident photon is at least equal to the band gap of its semiconductor, that is $E_{photon} \geq E_g$. If the photon energy is less than E_g, no electron-hole pairs are created and the light cannot be detected.

Energy of a photon of wavelength $\lambda = 600$ **nm**:

$$E_{photon} = \frac{hc}{\lambda}$$

Using $hc = 1240$ **eV**:

$$E_{photon} = \frac{1240 \text{ eV}}{600 \text{ nm}}$$

$$E_{photon} = 2.07 \text{ eV}$$

Now compare this photon energy with the band gap of each diode:

For D_1: $E_g = 2.5$ **eV** > 2.07 **eV**, so the photon energy is too small. D_1 **cannot detect**.

For D_2: $E_g = 2.0$ **eV** < 2.07 **eV**, so the photon energy is enough. D_2 **can detect**.

For D_3: $E_g = 3.0$ **eV** > 2.07 **eV**, so the photon energy is too small. D_3 **cannot detect**.

Hence **D_1 and D_3** cannot detect light of wavelength 600 nm, because for both of them the band gap is greater than the photon energy of 2.07 eV.

TIP

Always convert wavelength to photon energy with $E = 1240/\lambda$ (with λ in nm giving E in eV) and compare with E_g; a diode detects only when $E \geq E_g$.

Final answer. D_1 ($E_g = 2.5$ **eV**) and D_3 ($E_g = 3$ **eV**) cannot detect **600 nm** light, since its photon energy (**2.07 eV**) is less than their band gaps.

← *Back to Q256*

Q257

Why photodiodes are required to operate in reverse bias? Explain.

[2019 • Set 55-1-1]

SOLUTION

A photodiode is used to detect light by measuring the small change in current produced when light falls on the junction. The key reason for operating it in reverse bias is the way the **fractional change in current** behaves, which can be understood from the carrier concentrations.

When light of energy $\mathbf{h\nu > E_g}$ falls on the depletion region, it generates extra electron-hole pairs. Let the increase in electron and hole concentrations due to illumination be Δn and Δp. The same number of extra carriers is generated whether the diode is forward or reverse biased.

In reverse bias, the dark current is due only to the few **minority carriers**, so the current is very small. The photo-generated carriers add a comparatively large fraction to this small minority-carrier current. Therefore the **fractional change in current** with light is large and easy to measure accurately.

In forward bias, the current is large because it is dominated by the huge number of **majority carriers**. The same extra Δn and Δp from light then form only a tiny fraction of this large current, so the change is very hard to detect.

In short, although the same amount of extra carriers is generated, the fractional change in current produced by the light is much more significant in reverse bias. So it is **easier to detect the change in light intensity** when the photodiode is operated in reverse bias.

Final answer. Photodiodes are operated in reverse bias because the fractional change in the (small) reverse minority-carrier current produced by light is large and easily measurable, unlike the tiny fractional change in the large forward current.

← *Back to Q257*

Q258

Describe briefly the functions of the three segments of n-p-n transistor.

[2019 • Set 55-1-1]

SOLUTION

An *n-p-n* transistor has three doped regions, each with a specific size, doping level and function.

Emitter: It is the left-hand *n*-region. It is of **moderate size and heavily doped**. Its function is to **emit (inject) the majority carriers (electrons)** into the base. The emitter-base junction is forward biased so that a large number of electrons are supplied into the base, giving the emitter current I_E.

Base: It is the central *p*-region. It is made **very thin and lightly doped**. Its function is to pass on most of the electrons injected by the emitter to the collector. Because it is thin and lightly doped, only a very small fraction of the electrons recombine with holes here, giving a small base current I_B; the rest move on to the collector. This thin, lightly doped base is essential for a high current gain.

Collector: It is the right-hand *n*-region. It is **moderately doped and physically the largest** (so that it can dissipate the heat generated). Its function is to **collect** the electrons that pass through the base. The collector-base junction is reverse biased, and its field sweeps the arriving electrons across, giving the collector current I_C.

The three currents are related by Kirchhoff's law: $\mathbf{I_E = I_B + I_C}$, with $\mathbf{I_C \approx I_E}$ because I_B is very small.

Final answer. Emitter (heavily doped) injects electrons, the thin lightly doped base passes most of them on with only a small $\mathbf{I_B}$, and the larger collector collects them under reverse bias, with $\mathbf{I_E = I_B + I_C}$.

← *Back to Q258*

Q259

Draw the circuit arrangement for studying the output characteristics of n-p-n transistor in CE configuration. Explain how the output characteristics is obtained.

[2019 • Set 55-1-1]

SOLUTION

Circuit arrangement (described): The *n-p-n* transistor is connected in **common-emitter (CE) configuration**, so the emitter is common to both the input and output circuits.

Input side (base-emitter): The base-emitter junction is forward biased by a variable DC supply V_{BB} through a series resistor R_B. A microammeter in the base lead reads the base current I_B.

Output side (collector-emitter): A separate variable DC supply V_{CC} is connected between collector and emitter (collector positive), making the collector-base junction reverse biased. A milliammeter reads the collector current I_C and a voltmeter connected across collector and emitter reads V_{CE}.

How the output characteristics are obtained:

The **output characteristic** is the graph of collector current I_C against collector-emitter voltage V_{CE}, keeping the base current I_B fixed.

First, fix the base current at a chosen value (for example $\mathbf{I_B = 10\ \mu A}$) by adjusting V_{BB} and R_B.

Then increase V_{CC} in small steps; at each step note the corresponding values of V_{CE} and I_C.

Plot I_C versus V_{CE} for this fixed I_B. Repeat the whole process for several other fixed values of I_B (such as 20 μA, 30 μA, ...) to get a **family of output characteristic curves**.

Shape of curves: For each curve, I_C first rises sharply for small V_{CE} (the saturation region), and then becomes almost flat (the active region) where I_C depends mainly on I_B and very little on V_{CE}. The vertical spacing between successive curves at a fixed V_{CE} gives a measure of the current gain $\beta_{ac} = \dfrac{\Delta I_C}{\Delta I_B}$.

Final answer. In CE configuration, fix $\mathbf{I_B}$, vary $\mathbf{V_{CC}}$ and record $\mathbf{I_C}$ versus $\mathbf{V_{CE}}$; repeating for several $\mathbf{I_B}$ values gives the family of output characteristic curves with a saturation region followed by a flat active region.

← *Back to Q259*

Q260

Draw the circuit diagram of a full wave rectifier and explain its working. Also, give the input and output waveforms.

[2019 • Set 55-1-1]

SOLUTION

A full-wave rectifier uses two diodes and a centre-tapped transformer to convert both halves of an AC input into a unidirectional DC output.

Circuit diagram (described): A centre-tapped step-down transformer has the two ends A and B of its secondary connected to the anodes of two diodes D_1 and D_2 respectively. The cathodes of the two diodes are joined together. The load resistor R_L is connected between this common cathode point and the centre tap C. The output voltage V_{out} is measured across R_L.

Working: With respect to the centre tap, the voltages at A and B are always opposite in phase, so when one end is positive the other is negative. A diode conducts only when forward biased.

During the positive half cycle: end A is positive and end B is negative with respect to C. So D_1 is forward biased and conducts, while D_2 is reverse biased and remains off. Current flows $\boldsymbol{A \rightarrow D_1}$ → through R_L (top to bottom) → back to C.

During the negative half cycle: end B becomes positive and end A negative. Now D_2 is forward biased and conducts, while D_1 stays off. Current flows $\boldsymbol{B \rightarrow D_2}$ → through R_L (again top to bottom) → back to C.

Since in both half cycles the current passes through R_L in the **same direction**, the output is unidirectional (DC). The load receives a current pulse in every half cycle, so the output ripple frequency is **twice** the input frequency.

Waveforms (described): The input waveform is a full sine wave with positive and negative humps. The output waveform across R_L is a continuous train of positive humps with no gaps, one for each half cycle of the input.

Final answer. Diodes $\mathbf{D_1}$ and $\mathbf{D_2}$ conduct on alternate half cycles of the centre-tapped supply, sending current through $\mathbf{R_L}$ in the same direction in both halves, giving a unidirectional output at twice the input frequency.

← *Back to Q260*

Q261

If A and B represent the maximum and minimum amplitudes of an amplitude modulated wave, write the expression for the modulation index in terms of A & B.

[2019 • Set 55-1-1]

SOLUTION

In an amplitude modulated (AM) wave, the carrier wave of amplitude A_c is modulated by a message signal of amplitude A_m. The amplitude of the resultant wave varies (swings) about A_c between a maximum and a minimum value.

The maximum amplitude occurs when the message adds fully to the carrier:

$$A_{max} = A_c + A_m$$

The minimum amplitude occurs when the message subtracts fully from the carrier:

$$A_{min} = A_c - A_m$$

Here $\mathbf{A_{max}} = \boldsymbol{A}$ and $\mathbf{A_{min}} = \boldsymbol{B}$. Adding the two equations gives the carrier amplitude:

$$A + B = 2A_c \Rightarrow A_c = \frac{A + B}{2}$$

Subtracting the two equations gives the message amplitude:

$$A - B = 2A_m \Rightarrow A_m = \frac{A - B}{2}$$

The modulation index is defined as the ratio of the message amplitude to the carrier amplitude:

$$\mu = \frac{A_m}{A_c}$$

Substituting the values of A_m and A_c:

$$\mu = \frac{(A - B)/2}{(A + B)/2}$$

$$\mu = \frac{A - B}{A + B}$$

This ratio is a measure of the **depth of modulation**. For full (100%) modulation, $\mu = 1$, which means $B = 0$, i.e. the minimum amplitude of the AM wave falls to zero.

TIP

Always express the modulation index in terms of the carrier amplitude (not A_{max}): $\mu = A_m/A_c$, where A_c is the average of A_{max} and A_{min}.

Final answer. The modulation index is $\mu = \dfrac{A - B}{A + B}$.

← *Back to Q261*

Q262

A message signal of frequency 20 kHz *and peak voltage* 10 *V is used to modulate a carrier of frequency* 2 MHz *and peak voltage of* 15 *V. Calculate the modulation index. Why the modulation index is generally kept less than one?*

[2019 • Set 55-1-1]

SOLUTION

The modulation index of an amplitude modulated wave is the ratio of the peak amplitude (voltage) of the message signal to the peak amplitude (voltage) of the carrier wave:

$$\mu = \frac{V_m}{V_c}$$

Here the peak voltage of the modulating (message) signal is $\mathbf{V_m = 10\ V}$ and the peak voltage of the carrier is $\mathbf{V_c = 15\ V}$. The frequencies are not needed to find the modulation index.

Substituting the values:

$$\mu = \frac{10\ V}{15\ V}$$

$$\mu = \frac{2}{3} \approx 0.67$$

So the **modulation index is 0.67** (about 67%).

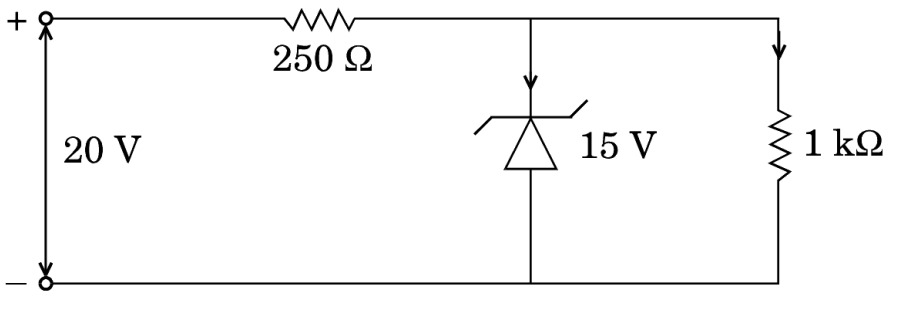
The modulation index is generally kept less than one (i.e. $\mu \leq 1$) to avoid **overmodulation**. If $\mu > 1$, the envelope of the AM wave no longer faithfully follows the shape of the message signal. During the deep troughs the envelope tends to go negative and reverses sign, which distorts the wave. A simple envelope detector at the receiver then cannot recover the original message correctly. Keeping $\mu \leq 1$ ensures the envelope always remains positive and exactly reproduces the modulating signal, so the message is received without distortion.

Final answer. The modulation index is $\mu \approx 0.67$; it is kept below 1 to avoid overmodulation and the resulting signal distortion.

← *Back to Q262*

Q263

Give reason to explain why n and p regions of a Zener diode are heavily doped. Find the current through the Zener diode in the circuit given below: (Zener breakdown voltage is 15 *V)*



[2019 • Set 55-2-1]

SOLUTION

Why the *n* and *p* regions of a Zener diode are heavily doped: Heavy doping makes the depletion region of the junction extremely **thin (narrow)**. When the diode is reverse biased, even a small reverse voltage produces a very strong electric field across this narrow region. This intense field is able to pull electrons out of their covalent bonds (Zener effect), so breakdown occurs sharply at a low, well-defined reverse voltage V_Z. After breakdown the voltage across the diode stays almost constant at V_Z even though the current changes a lot, which is why a Zener diode is used as a **voltage regulator**.

Finding the current through the Zener diode (breakdown voltage $\mathbf{V_Z = 15\ V}$, input 20 *V*, series resistance 250 Ω, load 1 kΩ):

In breakdown the Zener clamps the voltage across the parallel load at $\mathbf{V_Z = 15\ V}$. The current through the load resistor:

$$I_L = \frac{V_Z}{R_L} = \frac{15}{1000}$$

$$I_L = 0.015\ A = 15\ \text{mA}$$

The voltage dropped across the series 250 Ω resistor is the difference between the input voltage and the Zener voltage:

$$V_R = V_{in} - V_Z = 20 - 15 = 5\ V$$

The total current drawn from the supply through the series resistor:

$$I = \frac{V_R}{R} = \frac{5}{250}$$

$$I = 0.020\ A = 20\ \text{mA}$$

By Kirchhoff's current law, this current splits into the Zener current and the load current, so the Zener current is:

$$I_Z = I - I_L = 20 - 15$$

$$I_Z = 5 \text{ mA}$$

Final answer. Heavy doping narrows the depletion layer for sharp low-voltage breakdown; the Zener current is $\mathbf{I_Z = 5\ mA}$.

← *Back to Q263*

Q264

A signal of low frequency f_m is to be transmitted using a carrier wave of frequency f_c. Derive the expression for the amplitude modulated wave and deduce expressions for the lower and upper sidebands produced. Hence, obtain the expression for modulation index.

[2019 • Set 55-2-1]

SOLUTION

Let the carrier wave have amplitude A_c and angular frequency $\omega_c = 2\pi f_c$, and the low-frequency message signal have amplitude A_m and angular frequency $\omega_m = 2\pi f_m$:

$$c(t) = A_c \sin(\omega_c t)$$

$$m(t) = A_m \sin(\omega_m t)$$

In amplitude modulation the amplitude of the carrier is made to vary in step with the message signal, so the carrier amplitude A_c is replaced by $[A_c + A_m \sin(\omega_m t)]$. The modulated wave is:

$$s(t) = [A_c + A_m \sin(\omega_m t)] \sin(\omega_c t)$$

Taking A_c common and writing $\mu = \frac{A_m}{A_c}$:

$$s(t) = A_c [1 + \mu \sin(\omega_m t)] \sin(\omega_c t)$$

Expanding the bracket:

$$s(t) = A_c \sin(\omega_c t) + \mu A_c \sin(\omega_m t)\sin(\omega_c t)$$

Using the product-to-sum identity $\sin X \sin Y = \frac{1}{2}[\cos(X-Y) - \cos(X+Y)]$ on the second term:

$$s(t) = A_c \sin(\omega_c t) + \frac{\mu A_c}{2}\cos((\omega_c - \omega_m)t) - \frac{\mu A_c}{2}\cos((\omega_c + \omega_m)t)$$

This shows the AM wave contains three frequencies. The two cosine terms are the **side bands**:

Lower side band: frequency $f_c - f_m$, amplitude $\frac{\mu A_c}{2}$.

Upper side band: frequency $f_c + f_m$, amplitude $\frac{\mu A_c}{2}$.

The modulation index is the ratio of the message amplitude to the carrier amplitude:

$$\mu = \frac{A_m}{A_c}$$

In terms of the envelope extrema $A_{max} = A_c + A_m$ and $A_{min} = A_c - A_m$:

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

Final answer. $s(t) = A_c \sin(\omega_c t) + \frac{\mu A_c}{2}\cos((\omega_c-\omega_m)t) - \frac{\mu A_c}{2}\cos((\omega_c+\omega_m)t)$, with sidebands at $f_c \mp f_m$ and $\mu = A_m/A_c$.

← *Back to Q264*

Q265

Prove that in a common-emitter amplifier, the output and input differ in phase by 180°.
In a transistor, the change of base current by 30 μA *produces change of* 0.02 *V in the base-emitter voltage and a change of* 4 mA *in the collector current. Calculate the*

current amplification factor and the load resistance used, if the voltage gain of the amplifier is 400.

[2019 • Set 55-2-1]

SOLUTION

Phase difference of 180° in a common-emitter (CE) amplifier: In the CE configuration the input signal is applied across the base-emitter junction and the output is taken across the collector. The collector voltage is given by the output (collector) loop equation:

$$V_o = V_{CC} - I_C R_L$$

When the input voltage increases during the positive half of the signal, the base-emitter voltage V_{BE} increases, so the base current I_B increases. Since $\mathbf{I_C = \beta I_B}$, the collector current I_C also increases.

From $\mathbf{V_o = V_{CC} - I_C R_L}$, an increase in I_C makes the drop $I_C R_L$ larger, so the output voltage V_o **decreases**. Thus when the input rises, the output falls, and vice versa. The output is therefore exactly out of step with the input, i.e. the output and input differ in phase by **180°**.

Numerical part. Given: $\mathbf{\Delta I_B = 30\ \mu A = 30\times 10^{-6}\ A}$, $\mathbf{\Delta V_{BE} = 0.02\ V}$, $\mathbf{\Delta I_C = 4\ mA = 4\times 10^{-3}\ A}$, voltage gain $\mathbf{A_V = 400}$.

The current amplification factor (common-emitter current gain) is:

$$\beta = \frac{\Delta I_C}{\Delta I_B} = \frac{4\times 10^{-3}}{30\times 10^{-6}}$$

$$\beta \approx 133$$

The input resistance is:

$$r_i = \frac{\Delta V_{BE}}{\Delta I_B} = \frac{0.02}{30\times 10^{-6}}$$

$$r_i \approx 667\ \Omega$$

The voltage gain is related to the load resistance by $\mathbf{A_V = \beta \frac{R_L}{r_i}}$. Solving for R_L:

$$R_L = \frac{A_V r_i}{\beta} = \frac{400 \times 667}{133}$$

$$R_L \approx 2006\ \Omega \approx 2\ k\Omega$$

Final answer. The CE output is **180°** out of phase with the input; **$\beta \approx 133$** and **$R_L \approx 2\ k\Omega$.**

← *Back to Q265*

Q266

What is the reason to operate photodiodes in reverse bias?
A p-n photodiode is fabricated from a semiconductor with a band gap of range of 2.5 *to* 2.8 eV. *Calculate the range of wavelengths of the radiation which can be detected by the photodiode.*

[2019 • Set 55-2-2]

SOLUTION

Reason for operating a photodiode in reverse bias: A photodiode is used to measure light intensity by sensing the change in its current when light falls on it. It is operated in reverse bias because the **fractional change in current** caused by illumination is then very large and easy to measure.

In reverse bias, without light, only a small dark current flows (due to thermally generated minority carriers, in the μA range). When light of energy $h\nu \geq E_g$ falls on the depletion region, it creates additional electron-hole pairs, and the strong reverse-bias field quickly sweeps them across, adding a sizeable photocurrent. Since the dark current is small, even a small added photocurrent produces a large relative change that can be measured reliably.

In forward bias, the diode current is already large (in the mA range) due to majority-carrier injection. The tiny photo-generated current (μA) would be a negligible fraction of this large forward current and could not be detected. Hence reverse bias is preferred.

Range of detectable wavelengths. A photon is absorbed (detected) only if its energy is at least equal to the band gap:

$$E_{\text{photon}} = \frac{hc}{\lambda} \geq E_g$$

The maximum detectable wavelength corresponds to the minimum band gap, and is given by:

$$\lambda_{max} = \frac{hc}{E_{g,\,min}}$$

Using $\mathbf{hc = 6.63\times 10^{-34}\times 3\times 10^{8}\ J\,m}$ and $\mathbf{1\ eV = 1.6\times 10^{-19}\ J}$, so $\mathbf{hc = 1240\ eV\ nm}$. For $\mathbf{E_{g,\,min} = 2.5\ eV}$:

$$\lambda_{max} = \frac{1240\ \text{eV nm}}{2.5\ \text{eV}}$$

$$\lambda_{max} = 496\ \text{nm}$$

For $\mathbf{E_{g,\,max} = 2.8\ eV}$:

$$\lambda_{min} = \frac{1240\ \text{eV nm}}{2.8\ \text{eV}}$$

$$\lambda_{min} = 443\ \text{nm}$$

So the photodiode can detect radiation in the wavelength range **443 nm to 496 nm** (i.e. about 4430 Å to 4960 Å).

TIP

Use **hc = 1240 eV nm** to go straight from band gap in eV to wavelength in nm: a smaller E_g gives the longer λ_{max}.

Final answer. Reverse bias gives a large measurable fractional change in current; the detectable wavelength range is about **443 nm** to **496 nm**.

← *Back to Q266*

Q267

Draw the circuit diagram of a full wave rectifier. Explain its working principle. Show the input waveforms given to the diodes D_1 and D_2 and the corresponding output waveforms obtained at the load connected to the circuit.

[2019 • Set 55-2-3]

SOLUTION

Circuit diagram of a full-wave rectifier: Draw an AC source connected to the primary of a **centre-tapped step-down transformer**. The secondary has three terminals: the top end, the bottom end, and the centre tap. Connect the anode of diode D_1 to the top end of the secondary and the anode of diode D_2 to the bottom end. Join the cathodes of D_1 and D_2 together to form one common output point. Connect the load resistor R_L between this common cathode point and the centre tap of the secondary. The output voltage is taken across R_L.

Working principle: A p-n junction diode conducts only when it is forward biased. In this circuit the two diodes conduct in alternate half cycles, so that current always flows through the load in the same direction.

During the positive half cycle: the top end of the secondary is positive and the bottom end is negative. Diode D_1 is forward biased and conducts, while D_2 is reverse biased and stays off. Current flows from the top end, through D_1, down through R_L (from its top to its bottom), and back through the centre tap.

During the negative half cycle: the polarity of the secondary reverses, so the bottom end is positive. Now D_2 is forward biased and conducts, while D_1 is off. Current flows from the bottom end, through D_2, and again down through R_L in the **same direction** (top to bottom), returning through the centre tap.

Since the current through R_L is in the same direction during both half cycles, the output is a **unidirectional pulsating DC**, and its frequency is twice that of the input AC.

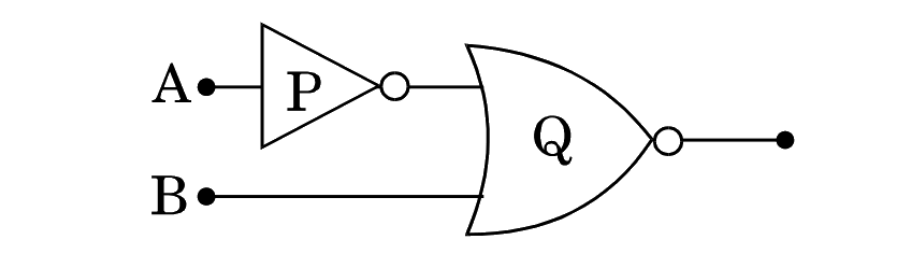
Waveforms (described): The input is a sinusoidal AC wave. The current through D_1 appears as positive half-sine humps only during the positive half cycles (gaps in between). The current through D_2 appears as positive half-sine humps only during the negative half cycles, filling the gaps left by D_1. The output across R_L is the sum of these: a continuous train of positive half-sine humps (one hump per half cycle of input), i.e. pulsating DC at twice the input frequency.

Final answer. A centre-tapped transformer with two diodes conducting in alternate half cycles gives a unidirectional pulsating DC output at twice the input frequency.

← Back to Q267

Q268

(a) (i) Write the truth tables of the logic gates marked P and Q in the given circuit.
(ii) Write the truth table for the circuit.



(b) Why are NOR gates considered as universal gates?

OR

(a) Explain how a potential barrier is developed in a p-n junction diode.
(b) Draw the circuit arrangement for studying the V-I characteristics of a p-n junction diode in reverse bias. Plot the V-I characteristics in this case.

[2019 • Set 55-3-1]

SOLUTION

This question has two unrelated demands that must be handled separately.

(a)(i) Truth tables of gates P and Q. The specific gate types P and Q, and how their outputs feed the final output Y, can only be read off the circuit figure printed in the paper, which is not reproduced here. So the truth table must be written for the exact gates shown in the figure; do not assume them. The general method is: (1) write the truth table of gate P for all input combinations, (2) write the truth table of gate Q for all input combinations, (3) feed P and Q outputs into the final gate to get Y.

A very common version of this CBSE circuit uses two NAND (or two NOR) gates feeding a third gate. For a worked illustration, suppose P is a NAND of A and B and the second stage is an OR with B, giving $Y = \overline{A \cdot B} + B$:

$$A=0,\ B=0:\ P=\overline{00}=1,\ Y=1+0=1$$

$$A=0,\ B=1:\ P=1,\ Y=1+1=1$$

$$A=1,\ B=0:\ P=1,\ Y=1+0=1$$

$$A=1,\ B=1:\ P=\overline{1}=0,\ Y=0+1=1$$

but in the exam the gates and wiring of the printed figure govern the answer, so read them directly.

(b) Why NOR gates are universal: every basic gate, and hence any logic circuit, can be built using only NOR gates.

NOT: $\overline{A}=\mathbf{NOR}(A,A)$

OR: $A+B=\mathbf{NOR}(\mathbf{NOR}(A,B),\ \mathbf{NOR}(A,B))$

AND: $A\cdot B=\mathbf{NOR}(\mathbf{NOR}(A,A),\ \mathbf{NOR}(B,B))$

Since NOT, OR and AND can all be realised from NOR alone, the **NOR gate is a universal gate**.

OR (alternative) (a) Formation of the potential barrier in a p-n junction: on forming the junction, majority carriers diffuse across it (holes from p to n, electrons from n to p) and recombine, leaving immobile negative acceptor ions on the p-side and positive donor ions on the n-side. This carrier-free depletion region sets up an electric field from n-side to p-side, which is the potential barrier (about 0.7 V for Si, 0.3 V for Ge). Equilibrium is reached when this barrier just stops further net diffusion.

(b) Circuit and V-I in reverse bias: connect the diode with the n-side to the positive terminal and the p-side to the negative terminal of a variable DC supply, a voltmeter across the diode and a microammeter in series. The reverse-bias V-I curve (reverse voltage on the negative x-axis, reverse current on the negative y-axis) shows a tiny, almost constant reverse-saturation current of the order of μA until the breakdown voltage, where the reverse current rises sharply.

Final answer. The truth tables of P, Q and Y must be read from the printed circuit figure (not reproduced here); the NOR gate is universal because NOT, OR and AND can all be built from NOR gates alone.

← *Back to Q268*

Q269

(a) What do you mean by bandwidth of a signal? Give its importance.
(b) Differentiate between Analog and Digital communication.
(c) Write the functions of transducer and repeaters.

[2019 • Set 55-3-1]

SOLUTION

(a) Bandwidth of a signal: The bandwidth of a signal is the **range of frequencies** present in it, i.e. the difference between its highest and lowest frequency components:

$$\text{Bandwidth} = f_{max} - f_{min}$$

Importance: The bandwidth decides the minimum capacity that the communication channel must provide. A signal can be transmitted faithfully only if the channel bandwidth is at least equal to the signal bandwidth. A larger bandwidth allows a higher rate of information transfer and better signal fidelity. Typical values are: speech **≈ 3 kHz**, music **≈ 20 kHz**, and a TV video signal **≈ 6 MHz**.

(b) Difference between analog and digital communication:

Analog communication uses signals that are continuous in both time and amplitude (the signal can take any value). Digital communication uses signals that have only discrete values, usually the two binary levels 0 and 1.

Analog signals are more easily affected by noise and the quality falls off with distance. Digital signals are far more immune to noise, and the original bits can be regenerated exactly, so quality is preserved over long distances.

Analog is used in older systems such as AM/FM radio and analog telephony, while digital is used in modern systems such as the internet, mobile phones and all data networks.

(c) Functions of transducer and repeater:

A **transducer** converts one form of energy or signal into another form. For example, a microphone converts sound energy into an electrical signal, and an

antenna converts an electrical signal into electromagnetic waves (and vice versa).

A **repeater** receives a weak signal, amplifies it and re-transmits it. It is used to extend the range of communication over long distances (for example in long-distance links and cellular/satellite networks).

Final answer. Bandwidth is $\mathbf{f_{max}-f_{min}}$ and fixes the channel capacity; analog uses continuous signals while digital uses discrete (binary) signals; a transducer converts one signal form to another and a repeater amplifies and retransmits a weak signal.

← *Back to Q269*

Q270

Draw the energy band diagram of (i) n-type, and (ii) p-type semiconductors at temperature $\boldsymbol{T > 0\ K}$.
In the case of n-type Si-semiconductor, the donor energy level is slightly below the bottom of conduction band whereas in p-type semiconductor, the acceptor energy level is slightly above the top of valence band. Explain, giving examples, what role do these energy levels play in conduction and valence bands.

[2019 • Set 55-4-1]

SOLUTION

(i) Energy band diagram of an n-type semiconductor at $\boldsymbol{T > 0\ K}$: Draw the **valence band** at the bottom and the **conduction band** at the top, separated by the forbidden energy gap E_g (**≈ 1.1 eV** for silicon). Just below the bottom edge of the conduction band, mark a discrete **donor energy level** E_D (the gap $E_C - E_D$ is very small, only about 0.01 to 0.05 eV). At $\boldsymbol{T > 0\ K}$, thermal energy easily lifts the electrons from E_D into the conduction band, shown by arrows from E_D pointing up into the conduction band. Donor elements are pentavalent, e.g. phosphorus, arsenic or antimony added to silicon.

(ii) Energy band diagram of a p-type semiconductor at $\boldsymbol{T > 0\ K}$: Again draw the valence and conduction bands separated by E_g. Just above the top edge of the valence band, mark a discrete **acceptor energy level** E_A (the gap $E_A - E_V$ is very

small, about 0.01 to 0.05 eV). At $T > 0\ K$, electrons from the valence band easily jump up to E_A, shown by arrows from the valence band up to E_A, leaving behind **holes** in the valence band. Acceptor elements are trivalent, e.g. boron, aluminium or gallium added to silicon.

Role of these energy levels in conduction and valence bands: The donor and acceptor levels lie very close to the band edges, so they greatly reduce the energy needed to create free charge carriers. Instead of needing the full band-gap energy of about 1.1 eV to be excited:

In the n-type, an electron at the donor level E_D needs only a few meV to be promoted into the **conduction band**, where it becomes a free electron (the majority carrier). This greatly increases the number of conduction electrons.

In the p-type, an electron in the **valence band** needs only a few meV to jump up to the acceptor level E_A. This leaves a hole behind in the valence band, and these holes are the majority carriers.

Because only a tiny energy is required, the dopant atoms are almost fully ionised at room temperature, so the carrier concentration (and hence conductivity) is much larger than in an intrinsic semiconductor.

Final answer. The donor level sits just below the conduction band and supplies free electrons in n-type; the acceptor level sits just above the valence band and creates holes in p-type, both needing only a few meV so conductivity rises sharply at room temperature.

← *Back to Q270*

Q271

With what considerations in view is a photodiode fabricated? Explain its working with the help of a suitable diagram. With the help of V-I characteristics, state how photodiode is used to detect optical signals.

[2019 • Set 55-4-1]

SOLUTION

A photodiode is a special *p-n* junction diode fabricated to detect light, and it is always operated under reverse bias.

Considerations kept in view during fabrication: (i) The semiconductor material is chosen so that its band gap E_g matches the photons to be detected. For a photon to create an electron-hole pair, the incident photon energy must satisfy $h\nu \geq E_g$. For visible light detection this needs a material with E_g of roughly 1.8 to 3.0 eV.

(ii) The junction is fabricated with a transparent window or a thin transparent layer on the illuminated side so that light can reach the depletion region instead of being absorbed in the bulk material.

(iii) Doping is arranged so that the depletion region is reasonably wide, allowing more photons to be absorbed and more electron-hole pairs to be generated near the junction.

Working: A photodiode is illuminated through its transparent window while kept in reverse bias. When light of energy $h\nu \geq E_g$ falls on the depletion region, it generates extra electron-hole pairs.

The strong reverse-bias field present in the depletion region sweeps these carriers apart: electrons move towards the *n*-side and holes towards the *p*-side.

This adds to the reverse (minority-carrier) current. The magnitude of this photocurrent depends on the intensity of the incident light.

Diagram description: Draw a *p-n* junction with the *p*-region on top and *n*-region below, a battery connected for reverse bias (negative terminal to *p*, positive terminal to *n*), light arrows incident on the junction through a transparent window, and a microammeter in series to read the reverse current.

V-I characteristics and detection of optical signals: Draw the reverse-bias part of the characteristic in the third quadrant. For different light intensities $\mathbf{I_1 < I_2 < I_3 < I_4}$, the reverse current increases. Higher illumination gives a larger reverse current, and these curves run nearly parallel, lying further from the voltage axis as intensity rises.

Since the reverse current depends almost linearly on light intensity, by measuring the reverse photocurrent we can directly measure the intensity of the **optical signal**. This is how a photodiode is used as a light detector.

Reverse bias is preferred because the change in current produced by the same change in light intensity is much more easily measured in reverse bias than in forward bias.

TIP

Remember: in the V-I graph the higher-intensity curve always sits further below the voltage axis (larger reverse current).

Final answer. A photodiode is a reverse-biased junction whose reverse current rises with light intensity, so the photocurrent measures the optical signal.

← *Back to Q271*

Q272

(a) What is amplitude modulation? Draw a diagram showing an amplitude modulated wave obtained by modulation of a carrier sinusoidal wave on a modulating signal.
(b) Define the terms (i) modulation index, and (ii) side bands. Mention the significance of side bands.

[2019 • Set 55-4-1]

SOLUTION

(a) Amplitude modulation (AM) is the process in which the amplitude of a high-frequency carrier wave is varied in accordance with the instantaneous value (amplitude) of the message or modulating signal, while the frequency and phase of the carrier remain unchanged.

Let the message signal be $m(t) = A_m \cos(\omega_m t)$ and the carrier be $c(t) = A_c \cos(\omega_c t)$.

The amplitude-modulated wave is given by $c_m(t) = [A_c + A_m \cos(\omega_m t)]\cos(\omega_c t)$.

Diagram description: Draw three stacked graphs against a common time axis. The modulating signal is a slow low-frequency sine wave. The carrier is a fast constant-amplitude sine wave. The AM wave is a fast carrier oscillation whose

envelope (the line joining the peaks) traces out the shape of the modulating signal, so the peaks swell and shrink in step with the message.

(b)(i) Modulation index (μ): It is the ratio of the amplitude of the modulating signal to the amplitude of the carrier wave.

$$\mu = \frac{A_m}{A_c}.$$

In terms of the maximum and minimum amplitudes of the AM wave, $\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$. To avoid distortion, μ is kept ≤ 1.

(b)(ii) Side bands: When the AM expression $c_m(t)=[A_c+A_m\cos(\omega_m t)]\cos(\omega_c t)$ is expanded, it gives three frequency components.

$$c_m(t) = A_c\cos(\omega_c t) + \frac{A_m}{2}\cos(\omega_c + \omega_m)t + \frac{A_m}{2}\cos(\omega_c - \omega_m)t.$$

Besides the carrier frequency ω_c, two new frequencies appear: the upper side band (USB) at $(\omega_c + \omega_m)$ and the lower side band (LSB) at $(\omega_c - \omega_m)$. These two are called the **side bands**.

Significance: The side bands are the components that actually carry the information of the message signal; the carrier by itself contains no information. Hence the side bands are essential for transmitting the message.

Final answer. AM varies carrier amplitude with the message; $\mu = A_m/A_c$, and the information-carrying side bands lie at $(\omega_c \pm \omega_m)$.

← *Back to Q272*

Q273

How is a light emitting diode fabricated? Briefly explain the basic processes involved in the emission of spontaneous radiation from it. Write two advantages of LED lamps over the incandescent low power lamps.

[2019 • Set 55-4-2]

SOLUTION

A light emitting diode (LED) is a heavily doped *p-n* junction diode that emits spontaneous radiation when forward biased.

Fabrication: An LED is made from a direct band-gap semiconductor (such as GaAs, GaP or GaAsP) whose band gap E_g lies in or near the visible range. Both sides of the junction are heavily doped. Layers of the doped semiconductor are grown on a substrate, metal contacts are provided on both faces, and the junction is encased in a transparent cover so that the light can escape. The colour of the emitted light depends on the band gap, and hence on the choice of material.

Basic process of spontaneous emission: The LED is connected in forward bias. Under forward bias, electrons are injected from the *n*-side and holes from the *p*-side into the junction region. The diode is heavily doped, so the concentration of injected minority carriers is large.

Near the junction these injected electrons and holes recombine. When an electron in the conduction band recombines with a hole in the valence band, energy of about E_g is released.

This energy is given out in the form of a photon of frequency $\nu = \dfrac{E_g}{h}$ and wavelength $\lambda = \dfrac{hc}{E_g}$.

Because the recombination occurs on its own, without any external trigger, the radiation is called **spontaneous emission**. The forward voltage of the LED is therefore close to E_g/e.

Two advantages of LED lamps over incandescent low-power lamps: (i) LEDs have low operating voltage and consume much less power for the same brightness, so they are far more energy efficient. (ii) LEDs have a much longer working life, fast switching (no warm-up time) and emit nearly monochromatic light.

Final answer. An LED is a heavily doped forward-biased junction that emits a photon of frequency $\nu = E_g/h$ on electron-hole recombination; it is more efficient and longer lasting than incandescent lamps.

← *Back to Q273*

Q274

How is Zener diode fabricated? Draw its I-V characteristic and use this to explain the working of a Zener diode as a voltage regulator.

[2019 • Set 55-4-3]

SOLUTION

Fabrication of a Zener diode: A Zener diode is a specially designed *p-n* junction diode in which both the *p*-side and the *n*-side are heavily doped. This heavy doping makes the depletion region very narrow (of the order of 10^{-6} m or less).

Because the depletion region is so thin, even a small reverse voltage sets up a very strong electric field across the junction. At a sharply defined reverse voltage V_Z (the Zener / breakdown voltage), this field causes a large reverse current, but the voltage across the diode stays almost constant. The value of V_Z is fixed by the doping level.

I-V characteristic (diagram description): In the first quadrant (forward bias) the curve is like an ordinary diode, rising after the knee voltage. In the third quadrant (reverse bias) the current stays very small (saturation current) until the reverse voltage reaches V_Z, where the curve drops almost vertically, showing a sharp increase in reverse current while the voltage remains nearly fixed at V_Z.

Working as a voltage regulator: Connect the unregulated input voltage V_{in} through a series resistor R_S to the Zener diode, which is in reverse bias and placed in parallel with the load R_L. The output is taken across the Zener (and load).

As long as V_{in} exceeds V_Z, the diode operates in the breakdown region, so the voltage across it (and hence across the load) stays constant at V_Z.

If V_{in} increases, the extra voltage is dropped across R_S and the Zener current increases, but the diode voltage remains V_Z. If the load current changes, the Zener current adjusts in the opposite sense so that the total current and the output voltage stay steady.

Thus the output remains constant at $\mathbf{V_Z}$ in spite of changes in the input voltage or the load. This is the working of a Zener diode as a **voltage regulator**.

COMMON MISTAKE

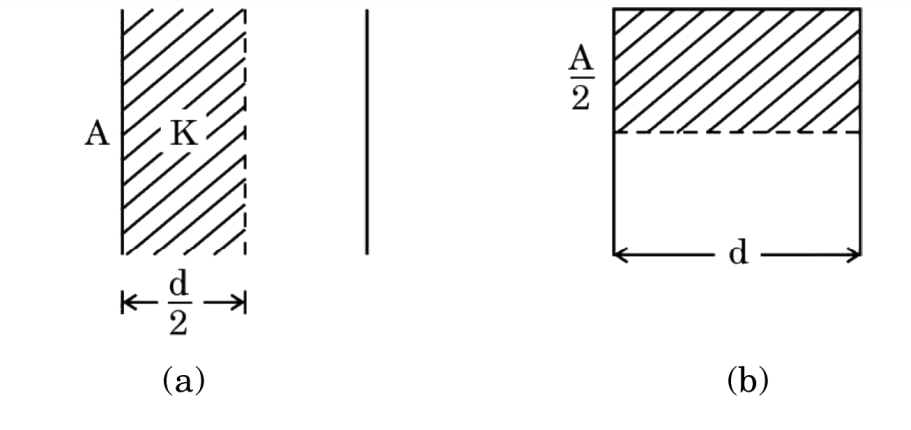
The series resistor R_S is essential: it absorbs the extra voltage and limits the Zener current, preventing the diode from burning out.

Final answer. A heavily doped Zener diode breaks down at a fixed reverse voltage $\mathbf{V_Z}$ and, in reverse bias across the load, holds the output constant at $\mathbf{V_Z}$.

← *Back to Q274*

Q275

The figure shows the V-I characteristic of a semiconductor diode designed to operate under reverse bias.



(a) Identify the semiconductor diode used.
(b) Draw the circuit diagram to obtain the given characteristics of this device.
(c) Briefly explain one use of this device.

[2019 • Set 55-5-1]

SOLUTION

(a) Identification of the device: The given characteristic shows an ordinary forward-bias curve in the first quadrant, but in the reverse-bias direction the curve falls almost vertically at a particular reverse voltage V_Z, after which the current increases sharply while the voltage stays nearly constant. This sharp breakdown at a fixed reverse voltage is the signature of a **Zener diode**, which is designed to work in the reverse breakdown region.

(b) Circuit diagram to obtain the characteristic (description): Connect the Zener diode in reverse bias to a variable DC supply through a current-limiting (series) resistor R. Place a milliammeter in series with the diode to read the reverse current I, and a voltmeter across the diode to read the reverse voltage V. Increase the supply voltage in small steps, note V and I at each step, and plot I against V to obtain the reverse characteristic.

(c) One use of the device, as a voltage regulator: When operated in its reverse breakdown region, the Zener diode keeps a constant voltage V_Z across itself even though the current through it changes considerably.

By connecting it in reverse bias in parallel with the load and in series with a current-limiting resistor across the unregulated DC supply, the voltage across the load is held steady at V_Z. Hence it provides a constant (regulated) output voltage despite fluctuations in the input voltage or in the load current.

Final answer. The device is a Zener diode; obtained from a reverse-biased variable supply through a series resistor, it is used as a voltage regulator giving a constant output $\mathbf{V_Z}$.

← *Back to Q275*

Q276

(a) Differentiate between three segments of an n-p-n transistor on the basis of their size and level of doping.
(b) Draw a plot of transfer characteristic and show which portion of the characteristic is used in amplification and why.

[2019 • Set 55-5-1]

SOLUTION

(a) The three segments of an *n-p-n* transistor differ in size and in level of doping as follows.

Emitter (n-type): It is of moderate size and is heavily doped. Being heavily doped, it supplies a large number of majority carriers (electrons) for the current flow.

Base (p-type): It is very thin and lightly (weakly) doped. Because it is thin and lightly doped, most of the electrons injected by the emitter pass through it to the collector, with very little recombination in the base.

Collector (n-type): It is the largest in size and is moderately doped (doping more than the base but less than the emitter). Its larger size lets it collect most of the carriers and dissipate the heat generated.

(b) Transfer characteristic: This is a plot of the output voltage V_o (i.e. V_{CE}) against the input voltage V_i (i.e. V_{BE}), for an *n-p-n* transistor in CE configuration.

Diagram description: Starting from low V_i, the curve is almost flat and high (output nearly V_{CC}): this is the cut-off region. As V_i increases past the knee, the curve falls steeply with a nearly straight slope: this is the active region. At high V_i the curve flattens again at a low value of V_o: this is the saturation region.

Portion used for amplification: The middle, steeply sloping **active region** is used for amplification.

Reason: In this region the output voltage varies almost linearly with the input voltage, so a small change in the input produces a proportional (faithful) change in the output. This gives distortion-free amplification. In the cut-off and saturation regions the curve is flat, so the output does not respond to input changes and no amplification is possible there.

Final answer. Emitter (moderate, heavily doped), base (thin, lightly doped), collector (largest, moderately doped); amplification uses the linear active region of the transfer characteristic.

← *Back to Q276*

Q277

(a) A student wants to use two p-n junction diodes to convert alternating current into direct current. Draw the labelled circuit diagram she would use and explain how it works.
(b) Give the truth table and circuit symbol for NAND gate.

[2018]

SOLUTION

(a) Two-diode full-wave rectifier: The student should use a full-wave rectifier built with a centre-tapped transformer and two *p-n* junction diodes.

Circuit description: A centre-tapped step-down transformer has its two secondary ends connected to the anodes of diodes D_1 and D_2. The cathodes of D_1 and D_2 are joined together. The load resistor R_L is connected between this common cathode point and the centre tap of the transformer, and the DC output is taken across R_L.

Working: During the positive half cycle the top end of the secondary is positive, so D_1 is forward biased and conducts while D_2 is reverse biased and does not conduct. Current passes through D_1 and R_L back to the centre tap.

During the negative half cycle the polarity reverses, so D_2 is forward biased and conducts while D_1 is off. Current now passes through D_2 and R_L.

In both half cycles the current through R_L flows in the same direction, so the output is a unidirectional (DC) voltage. Hence both halves of the AC input are rectified, and the output frequency is twice the input frequency.

(b) NAND gate: Circuit symbol description: It is the AND-gate symbol (a flat back with a curved front, D-shape) with a small circle (bubble) at the output, the bubble indicating inversion. Its Boolean expression is $\boldsymbol{Y = \overline{A \cdot B}}$.

Truth table of the NAND gate:

$$A=0,\ B=0 \Rightarrow Y=1$$

$$A=0,\ B=1 \Rightarrow Y=1$$

$$A=1,\ B=0 \Rightarrow Y=1$$

$$A=1,\ B=1 \Rightarrow Y=0$$

The output is LOW (0) only when both inputs are HIGH; in all other cases it is HIGH (1).

Final answer. A centre-tapped two-diode full-wave rectifier gives unidirectional output in both half cycles; the NAND gate gives $\boldsymbol{Y=\overline{A\cdot B}}$, which is 0 only when both inputs are 1.

← Back to Q277

Q278

Draw the typical input and output characteristics of an n-p-n transistor in CE configuration. Show how these characteristics can be used to determine (a) the input resistance (r_i)*, and (b) current amplification factor* (β).

[2018]

SOLUTION

For an *n-p-n* transistor in the common-emitter (CE) configuration, two sets of characteristics are drawn.

Input characteristic: This is a plot of the base current I_B against the base-emitter voltage V_{BE}, keeping the collector-emitter voltage V_{CE} constant. The curve resembles that of a forward-biased diode: I_B is almost zero until V_{BE} reaches about 0.7 V (for Si), after which I_B rises sharply with V_{BE}.

Output characteristic: This is a family of curves of collector current I_C against V_{CE}, each curve drawn for a fixed value of I_B. For each curve, I_C rises steeply at small V_{CE} (saturation region) and then becomes almost flat (active region) at larger V_{CE}. Higher I_B curves lie above lower I_B curves.

(a) Input resistance r_i: It is found from the input characteristic at a chosen operating point as the ratio of a small change in V_{BE} to the resulting change in I_B, keeping V_{CE} constant.

$$r_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}\ =\ \text{constant}}.$$

So r_i is the reciprocal of the slope of the input characteristic at the operating point. Its typical value is of the order of a few kΩ.

(b) Current amplification factor β: It is found from the output characteristics. At a fixed value of V_{CE} in the active region, take two curves of different I_B, read the corresponding I_C values, and compute the ratio of the change in I_C to the change in I_B.

$$\beta = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}\ =\ \text{constant}}.$$

So β is the ratio of the change in collector current to the change in base current at constant V_{CE}; its typical value lies between about 50 and 500.

COMMON MISTAKE

Keep the held-constant quantity straight: V_{CE} is fixed for both r_i and β, since both are read at one operating point.

Final answer. Input resistance $r_i=(\Delta V_{BE}/\Delta I_B)_{V_{CE}}$ from the input curve; current amplification factor $\beta=(\Delta I_C/\Delta I_B)_{V_{CE}}$ from the output curves.

← *Back to Q278*

Q279

(a) Give three reasons why modulation of a message signal is necessary for long distance transmission.
(b) Show graphically an audio signal, a carrier wave and an amplitude modulated wave.

[2018]

SOLUTION

(a) Three reasons why modulation of a message signal is necessary for long-distance transmission:

(i) Size of the antenna: For efficient transmission and reception, the height of the antenna should be comparable to the wavelength, at least about λ/4. An audio signal of frequency around 5 kHz has wavelength $\lambda = c/\nu = (310^8)/(510^3) = 610^4$ m, needing an antenna kilometres long, which is impractical. Using a high-frequency carrier reduces λ to a few metres, so the antenna becomes a manageable size.

(ii) Effective power radiated by the antenna: The power radiated by an antenna of length l varies as $(l/\lambda)^2$. A low-frequency (large λ) audio signal radiates very little power, whereas a high-frequency carrier (small λ) radiates much more power, giving stronger transmission.

(iii) Mixing up (overlap) of signals: If all sources transmitted the original low audio band directly, the signals from different transmitters would overlap and could not be separated. By modulating each message on a different carrier frequency, many signals can be sent over the same channel without interference, and the receiver can pick out the wanted station.

(b) Graphical illustration (description): Draw three graphs one below the other against a common time axis.

The audio (message) signal: a slow, low-frequency sinusoid of varying amplitude.

The carrier wave $c(t) = A_c\cos(\omega_c t)$: a fast, high-frequency sinusoid of constant amplitude.

The amplitude-modulated wave: a fast carrier oscillation whose envelope (the curve joining its peaks) follows the shape of the audio signal, so the amplitude swells and shrinks in step with the message.

Final answer. Modulation is needed for practical antenna size, larger radiated power, and to prevent signal overlap; the AM wave is the fast carrier with an envelope shaped like the audio signal.

← *Back to Q279*

Q280

Write the two processes that take place in the formation of a p-n junction. Explain with the help of a diagram, the formation of depletion region and barrier potential in a p-n junction.

[2017]

SOLUTION

Two processes that take place during the formation of a *p-n* junction:

(i) Diffusion: Because of the difference in carrier concentration across the junction, the majority carriers diffuse: holes move from the *p*-side to the *n*-side and electrons move from the *n*-side to the *p*-side. This sets up a diffusion current.

(ii) Drift: The motion of the carriers due to the built-in electric field set up at the junction is called drift. Under this field minority carriers are swept across the junction (electrons from p to n and holes from n to p), giving a drift current opposite in direction to the diffusion current.

Formation of the depletion region and barrier potential (with diagram description): Draw a bar with the p-region on the left and the n-region on the right. Near the junction, mark a row of negative (acceptor) ions on the p-side and a row of positive (donor) ions on the n-side, with an arrow for the internal field E pointing from the n-side to the p-side.

When electrons diffuse from n to p and holes from p to n, they recombine near the junction and leave behind immobile charged ions. The n-side near the junction is left with positive donor ions and the p-side with negative acceptor ions.

This thin region near the junction, which has only immobile ions and no free charge carriers, is called the **depletion region** (typically about 1 μm wide).

These immobile ions set up an electric field directed from the n-side to the p-side. The potential difference developed across the depletion region due to this field is called the **barrier potential** V_b (about 0.7 V for Si, 0.3 V for Ge).

This barrier opposes the further diffusion of majority carriers. Equilibrium is reached when the diffusion current is exactly balanced by the drift current, and no further net flow of charge takes place across the junction.

Final answer. The two processes are diffusion and drift; diffusion leaves immobile ions forming the depletion region, whose field gives the barrier potential $\mathbf{V_b}$ (≈ **0.7** V for Si).

← *Back to Q280*

Q281

Define the term 'amplitude modulation'. Explain any two factors which justify the need for modulating a low frequency base-band signal.

[2017]

SOLUTION

Amplitude Modulation (AM) is the process in which the amplitude of a high-frequency carrier wave is varied in accordance with the instantaneous value of the low-frequency message (modulating) signal, while the frequency and phase of the carrier are kept constant.

Let the carrier wave be $c(t)=A_c\omega_c t$ and the message signal be $m(t)=A_m\omega_m t$, where $\omega_c \gg \omega_m$.

The amplitude-modulated wave is then

$$c_m(t)=(A_c+A_m\omega_m t)\omega_c t$$

The amplitude of the carrier now follows the message, so the envelope of the AM wave traces out the message signal and can be recovered at the receiver.

Two factors that justify the need for modulating a low-frequency base-band signal:

1. Size of the antenna: For efficient radiation, the antenna length should be of the order of $\frac{\lambda}{4}$. An audio signal of frequency $f_m \approx 5\ \text{kHz}$ has wavelength $\lambda=\frac{c}{f_m}=\frac{3\times 10^8}{5\times 10^3}=6\times 10^4\ m=60\ \text{km}$, needing an antenna about 15 km long, which is impossible. When this signal is modulated onto a carrier of a few MHz, λ falls to a few hundred metres and the antenna becomes only a few metres long, which is practical.

2. Effective power radiated and multiplexing: The power radiated by an antenna of length l varies as $(\frac{l}{\lambda})^2$, so radiation at low frequencies (large λ) is extremely weak. High-frequency carriers radiate efficiently. Also, by assigning different carrier frequencies to different stations, many signals can share the same medium without overlapping, which would be impossible if all transmitted in the same audio band.

Final answer. AM varies carrier amplitude with the message; modulation is needed for a practical antenna size and for efficient radiation/multiplexing.

← *Back to Q281*

Q282

For a CE-transistor amplifier, the audio signal voltage across the collector resistance of 2 kΩ *is* 2 *V. Given the current amplification factor of the transistor is 100, find the input signal voltage and base current, if the base resistance is* 1 kΩ.

[2017]

SOLUTION

Given: output voltage across collector resistance $\mathbf{V_o=2\ V}$, collector resistance $\mathbf{R_C=2\ k\Omega}$, current amplification factor $\mathbf{\beta=100}$, base resistance $\mathbf{R_B=1\ k\Omega}$.

The voltage gain of a CE amplifier is

$$A_V=\beta\ \frac{R_C}{R_B}$$

$$A_V=100\times\frac{2\ k\Omega}{1\ k\Omega}$$

$$A_V=200$$

The input signal voltage is obtained from $\mathbf{A_V=\frac{V_o}{V_i}}$, so

$$V_i=\frac{V_o}{A_V}$$

$$V_i=\frac{2}{200}$$

$$V_i=0.01\ V=\mathbf{10\ mV}$$

The base current is found using Ohm's law across the base resistance,

$$I_B=\frac{V_i}{R_B}$$

$$I_B=\frac{0.01}{1000}$$

$$I_B=1\times 10^{-5}\ A=\mathbf{10\ \mu A}$$

Hence the input signal voltage is 10 mV and the base current is 10 μA.

Final answer. Input signal voltage $\mathbf{V_i = 10\ mV}$ and base current $\mathbf{I_B = 10\ \mu A}$.

← *Back to Q282*

Q283

A zener diode is fabricated by heavily doping both p- and n- sides of the junction. Explain, why? Briefly explain the use of zener diode as a dc voltage regulator with the help of a circuit diagram.

[2017]

SOLUTION

Why a Zener diode is heavily doped on both sides: Heavy doping of the p-side and n-side makes the depletion region extremely thin (of the order of $10^{-6}\ m$ or less), much narrower than in an ordinary diode.

Because the electric field across the junction is $E=\frac{V}{d}$, a very small width d produces a very strong field even at a low reverse voltage.

This strong field gives a sharp, well-defined breakdown (Zener breakdown by tunnelling) at a low and precisely controlled reverse voltage V_Z, which is exactly what a reference/regulator element needs.

Use of a Zener diode as a DC voltage regulator:

Circuit diagram (described): The unregulated DC input voltage V_i is connected through a series resistance R_S. The Zener diode is connected in reverse bias across the output, that is, its cathode (n-side) to the + line and anode (p-side) to the ground line. The load resistance R_L is connected in parallel with the Zener diode, and the regulated output V_o is taken across this parallel combination.

Working: The Zener is operated in its breakdown region, where the voltage across it stays constant at V_Z even when the current through it changes. Therefore

$$V_o=V_Z=\text{constant}$$

If the input voltage V_i increases, the current through R_S and the Zener increases, so the extra voltage is dropped across R_S and V_o stays at V_Z.

If V_i decreases, the Zener current decreases, the drop across R_S decreases, and again V_o remains V_Z.

Thus the output voltage is regulated and kept steady at V_Z in spite of changes in input voltage (or load current).

Final answer. Heavy doping gives a thin junction and a sharp low Zener breakdown; in the breakdown region the Zener holds $\mathbf{V_o = V_Z}$ constant, acting as a voltage regulator.

← *Back to Q283*

Q284

Explain briefly with the help of necessary diagrams, the forward and the reverse biasing of a p-n junction diode. Also draw their characteristic curves in the two cases.

[2017]

SOLUTION

Forward biasing of a p-n junction:

Diagram (described): The positive terminal of the external battery is connected to the p-side and the negative terminal to the n-side of the diode (through a resistor/key).

The applied voltage V opposes the built-in potential barrier V_0, so the effective barrier becomes $(V_0 - V)$ and the width of the depletion region decreases.

When V exceeds the cut-in (knee) voltage (about 0.7 V for Si and 0.3 V for Ge), the majority charge carriers cross the junction easily and a large forward current flows, increasing rapidly (almost exponentially) with voltage.

Reverse biasing of a p-n junction:

Diagram (described): The positive terminal of the battery is connected to the n-side and the negative terminal to the p-side.

Now the applied voltage adds to the barrier, making it $(V_0 + V)$, so the depletion region becomes wider and the junction resistance becomes very high.

Majority carriers cannot cross; only a very small reverse saturation current (of the order of μA) flows due to the minority carriers, and it is almost independent of the applied voltage until breakdown.

Characteristic curves (described): The V-I graph is drawn with voltage on the x-axis and current on the y-axis. In the first quadrant (forward bias) the current stays nearly zero up to the knee voltage and then rises steeply. In the third quadrant (reverse bias) there is only a tiny, almost constant reverse current until the breakdown voltage V_{Br}, where the current suddenly increases sharply. This asymmetry between the two halves is what makes the diode act as a rectifier.

Final answer. In forward bias the barrier lowers and a large current flows above the knee voltage; in reverse bias the barrier widens and only a small saturation current flows till breakdown, as shown by the V-I curves.

← *Back to Q284*

Q285

(a) Define the term 'modulation index,' used in communication system. Why is its value kept less than or equal to one?
(b) A message signal of frequency 10 kHz *and peak voltage of* 10 *V is used to modulate a carrier frequency* 1 MHz *and peak voltage* 10 *V. Determine the (i) modulation index, and (ii) side bands produced.*

[2017]

SOLUTION

(a) Modulation index μ is defined as the ratio of the peak amplitude (or change in amplitude) of the message signal to the peak amplitude of the carrier wave:

$$\mu = \frac{A_m}{A_c}$$

Its value is kept less than or equal to one because if **μ>1** (over-modulation), the envelope of the AM wave no longer follows the message faithfully; the envelope gets distorted and clipped where it would cross zero. This causes distortion of

the recovered signal at the receiver. For $\mu \leq 1$, the envelope is an undistorted, faithful replica of the message signal.

(b) Given: message peak voltage $\mathbf{A_m = 10\ V}$, carrier peak voltage $\mathbf{A_c = 10\ V}$, message frequency $\mathbf{f_m = 10\ kHz}$, carrier frequency $\mathbf{f_c = 1\ MHz = 1000\ kHz}$.

(i) Modulation index:

$$\mu = \frac{A_m}{A_c} = \frac{10}{10}$$

$\mu = 1$ (i.e. 100% modulation).

(ii) The sidebands are produced at $f_c \pm f_m$.

Upper side band: $\mathbf{f_{USB} = f_c + f_m = 1000 + 10 = 1010\ kHz}$

Lower side band: $\mathbf{f_{LSB} = f_c - f_m = 1000 - 10 = 990\ kHz}$

Hence the **modulation index is 1** and the **side bands are at 1010 kHz and 990 kHz**.

Final answer. $\mu = 1$; side bands are produced at **1010 kHz** (USB) and **990 kHz** (LSB).

← *Back to Q285*

Q286

What is sky wave propagation? Which frequency range is suitable for sky wave propagation and why? Over which range of frequencies can communication through free space using radio waves take place?

[2017]

SOLUTION

Sky-wave propagation: In sky-wave propagation, the radio waves sent obliquely (at an angle) from the transmitting antenna travel upward, get refracted back (effectively reflected) by the ionosphere, and return to a distant point on the Earth's surface. The ionosphere is the layer of ionised gases lying between heights of about 65 km and 400 km. By repeated reflections between the Earth

and the ionosphere, these waves can reach far-off places, which is used for long-distance and intercontinental short-wave broadcasting.

Suitable frequency range: Sky-wave propagation is suitable for frequencies in the range of about 3 MHz to 30 MHz (the high-frequency band).

Why this range: The ionosphere can refract (bend back) waves only up to a certain maximum frequency. Waves in the 3 to 30 MHz range are bent back effectively by the ionised layers and return to Earth. Waves of frequency below this range are mostly absorbed/handled by the ground wave, while waves above about 30 MHz have too high a frequency to be refracted; they penetrate the ionosphere and escape into outer space, so they cannot be used for sky-wave propagation.

Free-space communication using radio waves: Communication through free space (space-wave and satellite communication) using radio waves takes place over a wide range of frequencies, roughly from about 40 MHz up to about 300 GHz.

Final answer. Sky-wave is ionospheric reflection, suitable for **3** to **30 MHz** (higher frequencies escape the ionosphere); free-space radio communication ranges up to about **300 GHz**.

← *Back to Q286*

Q287

What is space wave propagation? Which systems of communication use space waves? What is 'radio horizon' of a transmitting antenna of height h? Why is space wave propagation suitable for frequencies above 40 MHz*?*

[2017]

SOLUTION

Space-wave propagation: In space-wave propagation the electromagnetic wave travels in a straight line (line of sight) from the transmitting antenna directly to the receiving antenna through the lower atmosphere (troposphere), much like a ray of light.

Systems of communication that use space waves: Television (TV) broadcasting, FM radio broadcasting, microwave (point-to-point) links, mobile/cellular telephony, and satellite communication all use space-wave (line-of-sight) propagation.

Radio horizon of a transmitting antenna of height h: It is the largest distance up to which the space wave can reach the receiving antenna at ground level, limited by the curvature of the Earth. For a transmitting antenna of height h,

$$d_T = \sqrt{2Rh}$$

where R is the radius of the Earth **($\approx 6.4\times 10^6\ m$)**. If the receiving antenna also has a height h_R, the maximum line-of-sight distance becomes

$$d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Why space-wave propagation is suitable for frequencies above 40 MHz: For frequencies above about 40 MHz, the waves are no longer reflected back by the ionosphere (sky-wave fails as they penetrate it), and the ground wave is attenuated very quickly. Therefore these high-frequency waves can be sent only along the line of sight as space waves. Moreover, at such high frequencies the wavelength is small, so compact and highly directional antennas of size about $\frac{\lambda}{4}$ can be built for efficient transmission.

Final answer. Space-wave is line-of-sight propagation used by TV/FM/microwave/satellite links; radio horizon $\mathbf{d_T = \sqrt{2Rh}}$, and it suits frequencies above **40 MHz** because such waves penetrate the ionosphere.

← *Back to Q287*

Q288

(a) How is amplitude modulation achieved?
(b) The frequencies of two side bands in an AM wave are 640 kHz *and* 660 kHz *respectively. Find the frequencies of carrier and modulating signal. What is the bandwidth required for amplitude modulation?*

[2017]

SOLUTION

(a) How amplitude modulation is achieved: The message (modulating) signal $m(t)=A_m\omega_m t$ is added to the high-frequency carrier $c(t)=A_c\omega_c t$ and the sum is passed through a non-linear (square-law) device, such as a transistor biased in its non-linear region.

The output of the square-law device contains a term proportional to the square of the input, which generates new frequencies, namely the sum and difference frequencies (f_c+f_m) and (f_c-f_m) along with f_c, f_m and $2f_m$.

A band-pass filter centred at f_c is then used to pass only the carrier and the two side bands, giving the amplitude-modulated wave

$$c_m(t)=(A_c+A_m\omega_m t)\omega_c t$$

(b) Given: the two side-band frequencies are $\mathbf{f_{LSB}=640\ kHz}$ and $\mathbf{f_{USB}=660\ kHz}$.

Since the side bands are at $f_c \pm f_m$, the carrier frequency is their average:

$$f_c=\frac{f_{USB}+f_{LSB}}{2}=\frac{660+640}{2}$$

$$f_c=650\ \text{kHz}$$

The modulating-signal frequency is half their difference:

$$f_m=\frac{f_{USB}-f_{LSB}}{2}=\frac{660-640}{2}$$

$$f_m=10\ \text{kHz}$$

The bandwidth required for amplitude modulation is the difference between the two side bands:

$$BW=f_{USB}-f_{LSB}=2f_m=660-640$$

$$BW=20\ \text{kHz}$$

Hence the **carrier frequency is 650 kHz, the modulating frequency is 10 kHz, and the required bandwidth is 20 kHz**.

TIP

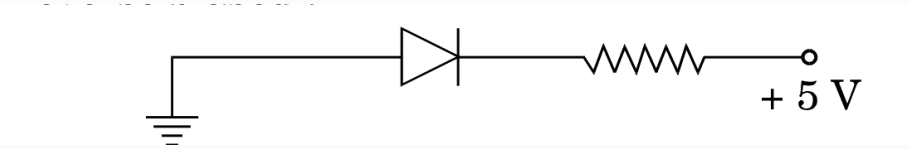
Bandwidth of an AM wave is always twice the highest modulating frequency, **BW = 2f_m**.

Final answer. Carrier **f_c=650 kHz**, modulating signal **f_m=10 kHz**, and bandwidth **BW=20 kHz.**

← *Back to Q288*

Q289

(a) In the following diagram, is the junction diode forward biased or reverse biased?



(b) Draw the circuit diagram of a full wave rectifier and state how it works.

[2017]

SOLUTION

(a) Identifying the bias: In the given diagram, the positive terminal (+5 *V*) of the battery is connected to the p-side (anode) of the diode and the negative terminal to the n-side (cathode). Since the anode is at higher potential than the cathode, the applied voltage opposes the barrier potential, so the junction diode is **forward biased** and it conducts with low resistance.

(b) Full-wave rectifier:

Circuit diagram (described): A centre-tapped step-down transformer is used. The two ends of the secondary winding are connected to the anodes (p-sides) of two diodes D_1 and D_2. The cathodes (n-sides) of both diodes are joined together. The load resistance R_L is connected between this common cathode point and the centre tap of the secondary winding. The output is taken across R_L.

Working: During the positive half-cycle of the AC input, the upper end of the secondary is positive. Diode D_1 becomes forward biased and conducts, while D_2

is reverse biased and does not conduct. Current flows through D_1, then through R_L (in a fixed direction), and back to the centre tap.

During the negative half-cycle, the lower end of the secondary becomes positive. Now diode D_2 becomes forward biased and conducts, while D_1 is reverse biased. Current flows through D_2 and again through R_L in the same direction as before.

Thus in both half-cycles the current through the load R_L flows in the same direction, so the output is a unidirectional (DC) voltage. The output consists of a continuous series of positive humps and has a ripple frequency twice that of the input.

Final answer. (a) The diode is forward biased; (b) in a centre-tapped full-wave rectifier $\mathbf{D_1}$ and $\mathbf{D_2}$ conduct in alternate half-cycles, giving unidirectional current through $\mathbf{R_L}$.

← *Back to Q289*

Q290

Draw a block diagram of a generalized communication system. Write the functions of each of the following:
(a) Transmitter
(b) Channel
(c) Receiver

[2017]

SOLUTION

Block diagram of a generalized communication system (described): The blocks are connected in a straight line as:

Information source → Transmitter → Channel → Receiver → User of information.

An additional input of Noise enters the system at the Channel block, where it corrupts the signal during transmission.

(a) Transmitter: It processes the message signal coming from the information source and converts it into a form suitable for transmission through the

channel. This usually includes modulating the message onto a high-frequency carrier and amplifying it, and then feeding it to the transmitting antenna so it can be sent over the channel.

(b) Channel: It is the physical medium that carries the signal from the transmitter to the receiver, for example free space, the atmosphere, optical fibre, or a transmission cable. While passing through the channel the signal gets attenuated (weakened) and noise/distortion is added to it.

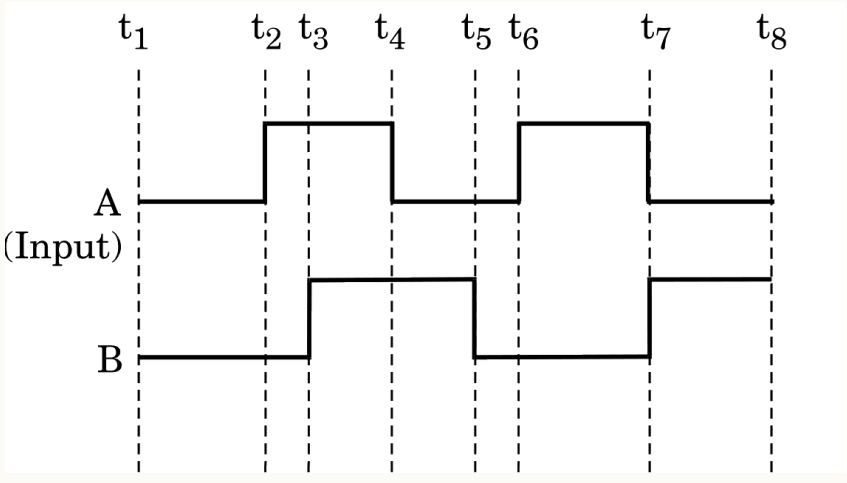
(c) Receiver: It receives the signal coming through the channel and recovers the original message signal from it. The receiver performs amplification (to make up for channel attenuation), filtering (to remove unwanted noise), and demodulation (to separate the message signal from the carrier), and then delivers it to the user of information.

Final answer. The chain is source → transmitter → channel → receiver → user; the transmitter prepares/modulates the signal, the channel carries it (adding noise), and the receiver demodulates and recovers the message.

← Back to Q290

Q291

(a) Write the functions of the three segments of a transistor.
(b) The figure shows the input waveforms A and B for 'AND' gate. Draw the output waveform and write the truth table for this logic gate.



[2017]

SOLUTION

(a) A transistor has three doped regions, each with a definite job.

The emitter is heavily doped and of moderate size. Its function is to supply the majority charge carriers (electrons in an n-p-n transistor, holes in a p-n-p transistor) in large numbers to the base.

The base is the central region. It is very thin and lightly doped. Because it is thin and lightly doped, very few carriers recombine here, so almost all the carriers injected by the emitter pass on to the collector. The base controls the flow of carriers from emitter to collector and so controls I_C through the small current I_B.

The collector is moderately doped and is the largest in size. Its function is to collect the majority carriers that come across from the emitter through the base. Its large size helps it to dissipate the heat produced.

(b) For an AND gate the output is $\boldsymbol{Y = A \cdot B}$.

The output Y is HIGH (1) only in those time intervals where BOTH inputs A and B are HIGH at the same time. In every other interval (where either input or both are LOW) the output is LOW (0).

So on the output waveform, draw Y at the HIGH level only during the intervals where the A waveform and the B waveform are simultaneously HIGH, and keep Y at the LOW level everywhere else.

The **truth table for the AND gate** is:

$$A = 0, B = 0 \Rightarrow Y = 0$$

$$A = 0, B = 1 \Rightarrow Y = 0$$

$$A = 1, B = 0 \Rightarrow Y = 0$$

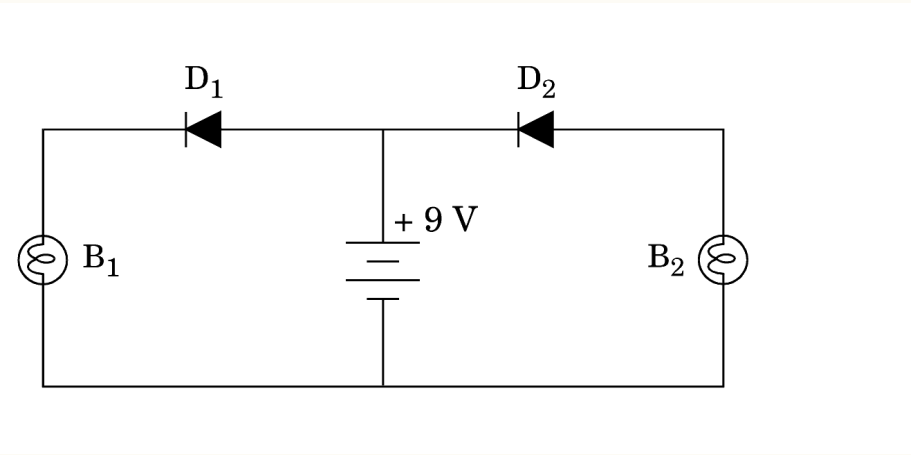
$$A = 1, B = 1 \Rightarrow Y = 1$$

Final answer. Emitter supplies carriers, thin base controls and passes them, collector collects them; AND output is HIGH only when both A and B are HIGH.

← Back to Q291

Q292

(a) In the following diagram, which bulb out of B_1 *and* B_2 *will glow and why?*



(b) Draw a diagram of an illuminated p-n junction solar cell.
(c) Explain briefly the three processes due to which generation of emf takes place in a solar cell.

[2017]

SOLUTION

(a) A diode conducts only when it is forward biased, that is when its p-side is at a higher potential than its n-side. In the given circuit, the diode whose p-side is connected to the higher (positive) potential of the supply gets forward biased and conducts, so the bulb in series with it glows. The other diode is reverse biased and blocks the current, so the bulb in its branch stays off.

Checking the polarity of the cell against each diode shows that the diode satisfying the forward-bias condition (p at higher potential) carries the current, and only the bulb in series with that forward-biased diode glows. The bulb in series with the reverse-biased diode does not glow because that branch carries practically no current.

(b) **Diagram of an illuminated p-n junction solar cell:** Draw a p-n junction with a very thin, heavily doped top p-layer and a thicker n-layer below it, with the depletion region at the junction. On the top surface draw a metallic finger (grid) electrode that leaves most of the surface open so that sunlight can enter, and draw a continuous metal contact at the bottom (back contact). Show arrows

of incident light (photons) falling on the top surface. The top and bottom contacts are the two terminals across which the photo-emf appears.

(c) The generation of emf in a solar cell takes place because of three processes.

Generation: Photons of incident light with energy $\mathbf{h\nu > E_g}$ are absorbed near the junction and create electron-hole pairs.

Separation: The built-in electric field of the depletion region separates these pairs. It drives the electrons towards the n-side and the holes towards the p-side.

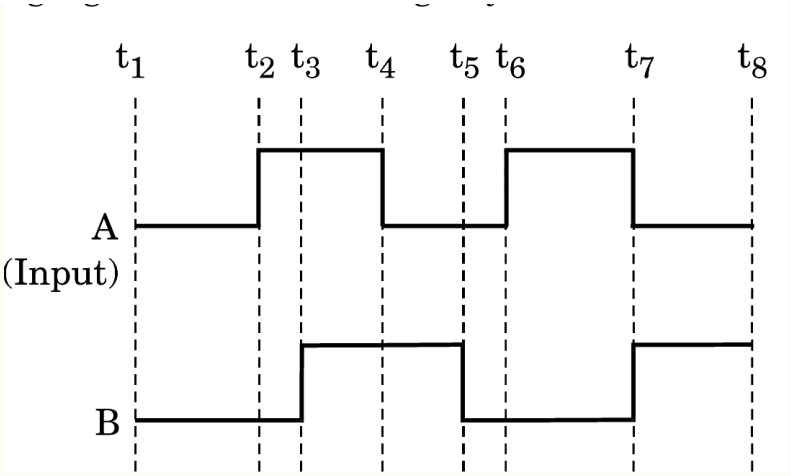
Collection: The electrons collected at the n-side and the holes collected at the p-side reach the front and back contacts. This separation of charge produces a potential difference (the photo-emf) across the terminals, which drives a current through an external load.

Final answer. The bulb in series with the forward-biased diode glows; the solar cell emf arises from generation, separation, and collection of photo-generated carriers.

← *Back to Q292*

Q293

(a) Draw the circuit diagram for studying the characteristics of a transistor in common emitter configuration. Explain briefly and show how input and output characteristics are drawn.
(b) The figure shows input waveforms A and B to a logic gate. Draw the output waveform for an OR gate. Write the truth table for this logic gate and draw its logic symbol.



[2017]

SOLUTION

(a) **Circuit for CE characteristics:** Draw an n-p-n transistor with the emitter at the bottom connected to the common (ground) line. The base is connected through a resistor R_B to a variable supply V_{BB}, which forward biases the emitter-base junction. The collector is connected through a load resistor R_L to a variable supply V_{CC}, which reverse biases the collector-base junction. A milliammeter in the base lead reads I_B and a voltmeter across base-emitter reads V_{BE}; a milliammeter in the collector lead reads I_C and a voltmeter across collector-emitter reads V_{CE}.

To draw the **input characteristic**, keep V_{CE} fixed at a chosen value. Vary V_{BE} in small steps using V_{BB} and note the corresponding I_B. The plot of I_B against V_{BE} rises slowly at first and then sharply after about 0.7 *V*, like a forward-biased diode curve.

To draw the **output characteristic**, keep I_B fixed at a chosen value. Vary V_{CE} in small steps using V_{CC} and note the corresponding I_C. The plot of I_C against V_{CE} rises quickly and then becomes almost flat (saturation). Repeating for several values of I_B gives a family of curves.

(b) For an OR gate the output is $Y = A + B$.

The output *Y* is HIGH (1) in every interval where at least one of the inputs is HIGH, and is LOW (0) only when both inputs are LOW at the same time. So on the output waveform, draw *Y* at the HIGH level during every interval where *A* or *B* (or both) is HIGH, and keep *Y* LOW only where both *A* and *B* are LOW.

The **truth table for the OR gate** is:

$$A = 0, B = 0 \Rightarrow Y = 0$$

$$A = 0, B = 1 \Rightarrow Y = 1$$

$$A = 1, B = 0 \Rightarrow Y = 1$$

$$A = 1, B = 1 \Rightarrow Y = 1$$

Logic symbol of OR gate: a shield-shaped body with a curved (concave) input side and a pointed output tip, with two input lines A and B entering the curved side and one output line Y leaving the pointed side.

Final answer. CE input curve is diode-like ($\mathbf{I_B}$ vs $\mathbf{V_{BE}}$), output curves saturate ($\mathbf{I_C}$ vs $\mathbf{V_{CE}}$); OR output is HIGH whenever any input is HIGH.

← *Back to Q293*

Q294

(a) Draw the circuit diagram of an n-p-n transistor amplifier in common emitter configuration.
(b) Derive an expression for voltage gain of the amplifier and hence show that the output voltage is in opposite phase with the input voltage.

[2017]

SOLUTION

(a) **Circuit of n-p-n CE amplifier:** Draw an n-p-n transistor with its emitter connected to the common (earth) line. The base is biased into the active region through a base resistor R_B from the supply V_{BB}, so the emitter-base junction is forward biased. The collector is connected through a load resistor R_L to the supply V_{CC}, so the collector-base junction is reverse biased. The ac input signal v_i is fed between base and emitter (through a coupling capacitor), and the amplified ac output v_o is taken between collector and emitter.

(b) Let the input resistance of the transistor be r_i. A small change in the input signal voltage v_i produces a change in base current:

$$\Delta I_B = \frac{v_i}{r_i}$$

This change in base current produces a much larger change in collector current, related by the ac current gain β_{ac}:

$$\Delta I_C = \beta_{ac}\,\Delta I_B = \frac{\beta_{ac}\, v_i}{r_i}$$

The output voltage is the change in voltage across the collector. Applying Kirchhoff's law to the output loop, $V_{CE} = V_{CC} - I_C R_L$, so a change gives:

$$v_o = \Delta V_{CE} = -\Delta I_C \cdot R_L$$

$$v_o = -\frac{\beta_{ac} R_L}{r_i} v_i$$

The voltage gain is the ratio of output to input voltage:

$$A_V = \frac{v_o}{v_i} = -\frac{\beta_{ac} R_L}{r_i}$$

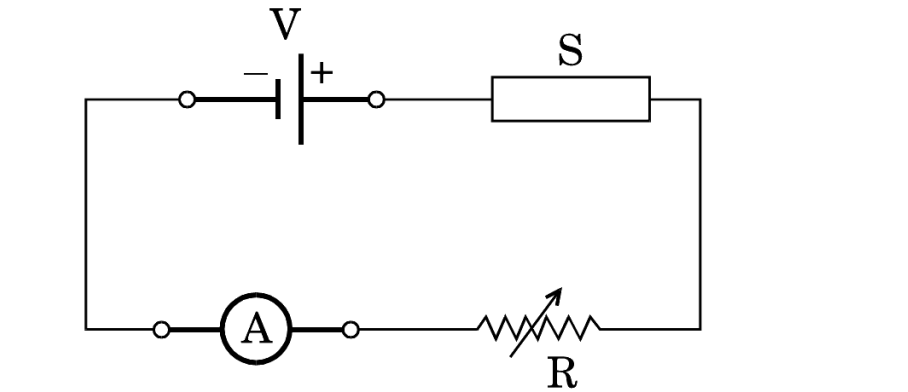
The **negative sign** shows that when the input voltage increases, the collector current increases, so the voltage across R_L increases and V_{CE} decreases. Thus the output voltage falls when the input rises. Hence the output voltage is in **opposite phase**, that is, there is a phase difference of 180° between the input and output of a CE amplifier.

Final answer. Voltage gain $A_V = -\frac{\beta_{ac} R_L}{r_i}$; the negative sign means the output is **180°** out of phase with the input.

← Back to Q294

Q295

(a) In the following diagram 'S' is a semiconductor. Would you increase or decrease the value of R to keep the reading of the ammeter A constant when S is heated? Give reason for your answer.



(b) Draw the circuit diagram of a photodiode and explain its working. Draw its I – V characteristics.

[2017]

SOLUTION

(a) A semiconductor has a negative temperature coefficient of resistance. When the semiconductor S is heated, more electron-hole pairs are generated thermally, so its conductivity increases and its resistance R_S decreases.

The semiconductor and the resistance R are in series across the source, so the ammeter reading is:

$$I = \frac{V}{R_S + R}$$

For the current I to stay constant when V is fixed, the total resistance $(R_S + R)$ must stay the same. Since heating makes R_S decrease, the value of R must be **increased** by the same amount to keep the sum, and hence the ammeter reading, constant.

(b) **Circuit diagram of a photodiode:** Draw a p-n junction diode connected in reverse bias (p-side to the negative terminal of the battery, n-side to the positive terminal) in series with a load and a microammeter. The junction has a transparent window so that light can fall on the depletion region.

Working: When light of energy $h\nu \geq E_g$ falls on the depletion region, it generates electron-hole pairs there. The built-in field of the reverse-biased junction quickly separates these carriers and sweeps them across, adding to the reverse current. As the intensity of light increases, more electron-hole pairs are produced, so the reverse current increases. Thus the reverse current is a

measure of the light intensity, and the photodiode is operated in reverse bias because the fractional change in reverse current with intensity is easily measurable.

I-V characteristics: The curves lie in the third quadrant (reverse-bias region). For each higher light intensity the reverse (saturation) current is larger, giving a set of nearly parallel curves with $\mathbf{I_4 > I_3 > I_2 > I_1}$ for increasing illumination intensities.

> **COMMON MISTAKE**
>
> *A common slip is thinking a semiconductor behaves like a metal. Remember: metals heat up to MORE resistance, semiconductors heat up to LESS resistance.*

Final answer. Increase R, because heating lowers the semiconductor's resistance and R must rise to keep $\mathbf{R_S + R}$ (and hence I) constant.

← *Back to Q295*

Q296

(i) Explain with the help of a diagram the formation of depletion region and barrier potential in a pn *junction.*
(ii) Draw the circuit diagram of a half wave rectifier and explain its working.

[2016]

SOLUTION

(i) When a p-type and an n-type semiconductor are joined to form a p-n junction, there is a large concentration difference of carriers across the junction. Because of this, electrons diffuse from the n-region to the p-region and holes diffuse from the p-region to the n-region.

As the electrons leave the n-side and the holes leave the p-side, they leave behind immobile charged ions near the junction: positive donor ions on the n-side and negative acceptor ions on the p-side. The thin region near the junction that now has only these immobile ions and no free charge carriers is called the **depletion region** (width of the order of 1 μm).

These immobile ions set up an electric field directed from the n-side to the p-side. The potential difference associated with this field is the **barrier potential** V_b. It is about 0.7 V for silicon and about 0.3 V for germanium. This field opposes the further diffusion of majority carriers. At equilibrium the diffusion current is just balanced by the opposite drift current of minority carriers, so the net current through the junction is zero.

Diagram: Draw the p-region on the left and n-region on the right; mark a row of − (negative acceptor ions) just left of the junction and a row of + (positive donor ions) just right of it; mark this central strip as the depletion region; draw an arrow for the internal field E pointing from n-side to p-side.

(ii) **Half-wave rectifier circuit:** The secondary of a step-down transformer feeds the ac input. One end of the secondary is connected through a diode to the load resistance R_L; the other end returns to the transformer. The output is taken across R_L.

Working: During the positive half cycle of the ac input, the diode is forward biased, so it conducts and current flows through R_L, producing an output voltage across it.

During the negative half cycle of the ac input, the diode is reverse biased, so it does not conduct and there is almost no current through R_L, giving no output during that half.

Thus current flows through R_L only during the positive half cycles, so the output is a unidirectional (dc) but pulsating voltage present only for one half of each cycle. This is why it is called a **half-wave rectifier**.

Final answer. Diffusion of carriers leaves immobile ions forming the depletion region and barrier potential; the half-wave rectifier conducts only on positive half cycles to give pulsating dc.

← *Back to Q296*

Q297

(i) Which mode of propagation is used by shortwave broadcast services having frequency range from a few MHz upto 30 MHz*? Explain diagrammatically how long*

distance communication can be achieved by this mode.
(ii) Why is there an upper limit to frequency of waves used in this mode?

[2016]

SOLUTION

(i) Short-wave broadcast services in the range of a few MHz up to about 30 MHz use **sky-wave propagation**.

In this mode, the transmitting antenna radiates the signal obliquely upward towards the sky. The wave travels up to the ionosphere, a layer of the atmosphere (about 60 to 400 km high) that contains a large number of free electrons and ions. There the wave is gradually bent (refracted) more and more until it is effectively reflected back to the Earth's surface at a far-off place.

By repeated reflections between the ionosphere and the Earth's surface, the signal can hop large distances and reach receivers thousands of kilometres away, allowing long-distance and intercontinental communication.

Diagram: Draw the curved Earth surface with a transmitting antenna; draw a ray going up at an angle from it to the ionosphere layer drawn above; show the ray bending and coming back down to a distant point on Earth; you may show a second hop, that is, the ray going up again and returning to Earth still farther away.

(ii) There is an upper limit to the frequency used in this mode because the ionosphere can reflect a wave only up to a certain maximum (critical) frequency. This critical frequency depends on the maximum electron density of the ionosphere through:

$$f_c = 9\sqrt{N_{max}}$$

where N_{max} is the maximum number density of electrons. For the actual electron densities of the ionosphere, this limit works out to be only a few tens of MHz (around 30 MHz).

If the frequency of the wave is greater than this critical frequency, the free electrons cannot respond fast enough to refract the wave back. The wave then penetrates straight through the ionosphere and escapes into outer space instead of returning to Earth. **Hence sky-wave propagation is limited to frequencies below about 30 MHz.**

Final answer. Sky-wave (ionospheric) propagation is used; above about 30 MHz the wave penetrates the ionosphere instead of being reflected, setting the upper frequency limit.

← *Back to Q297*

Q298

For a CE-transistor amplifier, the audio signal voltage across the collector resistance of 2 kΩ *is* 2 *V. Suppose the current amplification factor of the transistor is* 100*, find the input signal voltage and base current, if the base resistance is* 1 kΩ*.*

[2016]

SOLUTION

Here the output (collector) signal voltage is $\mathbf{V_o = 2\ V}$ across $\mathbf{R_C = 2\ k\Omega}$, the current amplification factor is $\boldsymbol{\beta = 100}$, and the base resistance is $\mathbf{R_B = 1\ k\Omega}$.

The voltage gain of a CE amplifier is:

$$A_V = \beta \times \frac{R_C}{R_B}$$

$$A_V = 100 \times \frac{2\ \text{k}\Omega}{1\ \text{k}\Omega}$$

$$A_V = 100 \times 2 = 200$$

The voltage gain is also the ratio of output to input voltage, $\mathbf{A_V = \frac{V_o}{V_i}}$. So the input signal voltage is:

$$V_i = \frac{V_o}{A_V}$$

$$V_i = \frac{2}{200}$$

$$V_i = 0.01\ V = 10\ \text{mV}$$

The base current is found from the input voltage across the base resistance:

$$I_B = \frac{V_i}{R_B}$$

$$I_B = \frac{0.01\ V}{1 \times 10^3\ \Omega}$$

$$I_B = 1 \times 10^{-5}\ A = 10\ \mu A$$

So the **input signal voltage is 10 mV** and the **base current is 10 µA**.

Final answer. $V_i = 10$ **mV** and $I_B = 10$ **µA**.

← *Back to Q298*

Q299

What is Global Positioning System? Explain its working principle in brief.

[2016]

SOLUTION

The Global Positioning System (GPS) is a satellite-based navigation system that lets a receiver find its exact three-dimensional position (latitude, longitude and altitude) and the accurate time, at any place on or near the Earth, at any time and in any weather.

It uses a constellation of about 24 satellites orbiting the Earth at a height of roughly 20,000 km in several orbital planes. They are arranged so that at least four satellites are visible from any point on the Earth at all times. Each satellite carries a very precise atomic clock.

Working principle: Each satellite continuously broadcasts a radio signal that carries its own precise position and the exact time at which the signal was sent.

A GPS receiver on the ground picks up the signals from at least four satellites at the same time. From the difference between the time the signal was sent and the time it was received, it finds the time delay Δt, and so the distance to each satellite:

$$d = c \times \Delta t$$

where c is the speed of the radio (electromagnetic) wave.

Knowing its distance from at least three satellites of known positions, the receiver locates itself by **trilateration**, that is, it is at the common intersection of spheres of those radii centred on the satellites. A fourth satellite is used to correct the small error in the receiver's own clock. By solving these range equations the receiver finds its exact 3D position and time.

Final answer. GPS is a satellite-based navigation system that fixes a receiver's 3D position by measuring distances ($d = c\,\Delta t$) to at least four satellites and using trilateration.

← *Back to Q299*

Q300

Explain briefly, with the help of a circuit diagram, the transistor action of npn transistor in CE configuration. Draw the typical shapes of input and output characteristics.

[2016]

SOLUTION

An n-p-n transistor has three regions: a heavily doped n-type emitter, a thin and lightly doped p-type base, and a moderately doped n-type collector.

Circuit (CE configuration): Draw the n-p-n transistor with the emitter connected to the common (earth) line. The base is connected through a resistor to a supply V_{BB} so that the emitter-base junction is forward biased. The collector is connected through a load to a supply V_{CC} so that the collector-base junction is reverse biased. A milliammeter reads I_B in the base lead and I_C in the collector lead.

Transistor action: Because the emitter-base junction is forward biased, the emitter injects a large number of electrons (its majority carriers) from the n-type emitter into the p-type base. Since the base is very thin and lightly doped, only a very small fraction (about 1 to 5%) of these electrons recombine with holes in the base; this small recombination current is the base current I_B.

The remaining electrons (about 95 to 99%) reach the collector-base junction. The strong electric field of the reverse-biased collector-base junction sweeps these electrons across into the collector, making up the collector current I_C.

By the conservation of charge, the currents are related as:

$$I_E = I_B + I_C$$

with $I_C \gg I_B$. The current amplification factor is $\beta = \frac{I_C}{I_B}$, which is large (about 50 to 500). This is the **transistor action** that makes amplification possible.

Input characteristic: A graph of I_B against V_{BE} at constant V_{CE}. It is like a forward-biased diode curve, rising slowly at first and then steeply after the threshold (about 0.7 *V*).

Output characteristic: A graph of I_C against V_{CE} for different fixed values of I_B. Each curve rises sharply at low V_{CE} and then becomes almost flat (saturation), giving a family of nearly horizontal parallel curves, one for each value of I_B.

Final answer. The forward-biased emitter injects electrons into the thin base; most reach the reverse-biased collector to give $\mathbf{I_C \gg I_B}$ with $\mathbf{I_E = I_B + I_C}$, the input curve being diode-like and the output curves saturating.

← *Back to Q300*

Q301

(i) Describe the working of photodiode by drawing the circuit diagram.
(ii) Draw the characteristics of a photodiode for different illumination intensities.
(iii) Why is photodiode operated in reverse bias even though the reverse bias current is much weaker than the current in forward bias?

[2016]

SOLUTION

(i) Working of a photodiode: A photodiode is a special *p-n* junction diode fabricated with a transparent window that allows light to fall on the junction. It is always connected in reverse bias through a series resistor *R* and a battery, as shown in the circuit diagram.

Circuit diagram description: A p-n junction diode is drawn with its p-side connected to the negative terminal and n-side to the positive terminal of a battery (reverse bias). A series resistor R is connected in the loop, and a milliammeter/microammeter reads the reverse current. Light (photons of energy hν) is shown by arrows falling on the junction through the transparent window.

When light of energy $\mathbf{h\nu > E_g}$ falls on the junction, it generates electron-hole pairs near or in the depletion region. The strong reverse-bias electric field sweeps the electrons towards the n-side and holes towards the p-side, adding to the reverse current. The magnitude of this photocurrent increases with the intensity of incident light.

(ii) Characteristics for different illumination intensities: Draw the reverse-bias I-V characteristic in the third quadrant (V negative on the x-axis, reverse current I negative on the y-axis). For zero illumination, a small dark current flows. For increasing intensities $\mathbf{I_1 < I_2 < I_3 < I_4}$, a family of parallel curves is obtained, each lying further below the previous one, showing that the reverse photocurrent increases with intensity at a given reverse voltage.

(iii) Why reverse bias is preferred: The purpose of a photodiode is to detect changes in current caused by light. In reverse bias, the dark current (due to minority carriers) is very small, so the photo-generated current produces a large fractional change in the total reverse current, giving high sensitivity. On illumination, the change in reverse current with change in light intensity is much more measurable in reverse bias than the corresponding fractional change in the large forward current. Hence, even though the reverse current is smaller in magnitude, the relative change is far easier to detect, so the **photodiode is operated in reverse bias**.

Final answer. A reverse-biased photodiode detects light because its photocurrent gives a large fractional change in the small reverse (dark) current, which is far easier to measure than in forward bias.

← *Back to Q301*

Q302

(i) Name two important processes that occur during the formation of a pn junction.
(ii) Draw the circuit diagram of a full wave rectifier along with the input and output waveforms. Briefly explain how the output voltage/current is unidirectional.

[2016]

SOLUTION

(i) Two important processes during *p*-*n* junction formation are **diffusion** and **drift**.

Diffusion: Due to the large concentration gradient across the junction, majority carriers move from the region of higher to lower concentration. Holes diffuse from the *p*-side to the *n*-side and electrons diffuse from the *n*-side to the *p*-side, setting up a diffusion current.

Drift: The diffusion leaves behind immobile ionised donor and acceptor atoms, creating a depletion region and a built-in electric field directed from the *n*-side to the *p*-side. This field drives the minority carriers (electrons in *p*, holes in *n*) across the junction, constituting a drift current opposite to the diffusion current. At equilibrium, the drift current exactly balances the diffusion current, giving zero net current.

(ii) Full-wave rectifier circuit description: A centre-tapped secondary transformer is used. The two ends of the secondary are connected to the *p*-sides of two diodes D_1 and D_2. The *n*-sides of both diodes are joined together and connected to one end of the load resistance R_L. The other end of R_L is connected to the centre tap of the transformer. The output is taken across R_L.

Input and output waveforms: The input is a full sinusoidal AC voltage (both positive and negative half cycles). The output across R_L is a series of positive half-cycle humps for every half cycle of input (a unidirectional pulsating DC, with no gaps between humps).

Explanation of unidirectional output: During the positive half cycle, the top end of the secondary is positive, so D_1 is forward biased and conducts while D_2 is reverse biased. During the negative half cycle, the bottom end is positive, so D_2 conducts while D_1 is reverse biased. In both half cycles, the current through the

load R_L flows in the same direction (from the joined n-sides through R_L to the centre tap).

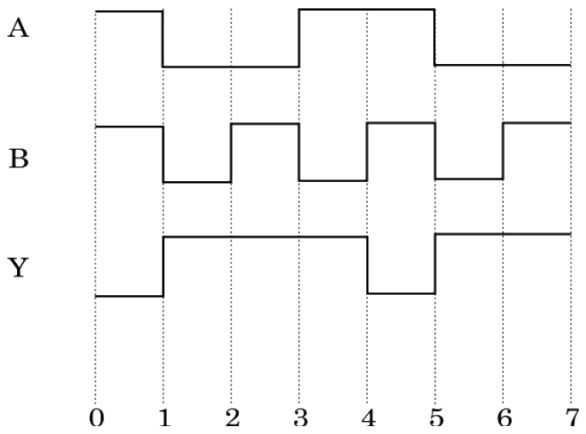
Hence the **output voltage/current across the load is unidirectional**, since whichever diode conducts always sends current through R_L in the same direction.

Final answer. Diffusion and drift form the junction; in a full-wave rectifier each diode conducts on alternate half cycles but drives current through the load in the same direction, giving a unidirectional output.

← *Back to Q302*

Q303

(i) Distinguish between a conductor and a semi conductor on the basis of energy band diagram.
(ii) The following figure shows the input waveforms (A, B) and the output waveform (Y) of a gate. Identify the gate, write its truth table and draw its logic symbol.



[2016]

SOLUTION

(i) Distinction between a conductor and a semiconductor (energy band diagram):

Conductor: The valence band and conduction band overlap, or the conduction band is partially filled, so the energy gap $\mathbf{E_g = 0}$. A large number of free electrons is available even at room temperature, giving very high conductivity.

In the band diagram, the shaded valence band and conduction band touch or overlap with no forbidden gap between them.

Semiconductor: The valence band and conduction band are separated by a small forbidden energy gap, $\mathbf{E_g \approx 1\ eV}$ (Si: 1.1 eV, Ge: 0.7 eV). At 0 K it behaves like an insulator (CB empty), but at room temperature thermal energy excites a few electrons across the gap, giving moderate conductivity. In the band diagram, a filled valence band lies below an empty conduction band, separated by a narrow gap E_g.

(ii) Identification of the gate: From the given waveforms, the output Y is high (1) only when both inputs A and B are simultaneously high (1); in every other interval Y is low (0). This is the behaviour of an **AND gate**, $Y = A \cdot B$.

Truth table of the AND gate:

$$A = 0, B = 0 \Rightarrow Y = 0$$

$$A = 0, B = 1 \Rightarrow Y = 0$$

$$A = 1, B = 0 \Rightarrow Y = 0$$

$$A = 1, B = 1 \Rightarrow Y = 1$$

Logic symbol description: A D-shaped symbol (flat vertical left edge, rounded right edge) with two input lines A and B entering the flat side and a single output line Y leaving the rounded side. There is no bubble at the output (a bubble would make it a NAND gate).

TIP

Output high only when all inputs are high is the signature of AND; output high when any input is high is OR. Adding an output bubble inverts each to NAND and NOR respectively.

Final answer. Conductors have overlapping bands ($\mathbf{E_g = 0}$) while semiconductors have a small gap ($\mathbf{E_g \approx 1\ eV}$); the given waveforms correspond to an AND gate, $\boldsymbol{Y = A \cdot B}$.

← *Back to Q303*

Q304

What is space wave propagation? State the factors which limit its range of propagation. Derive an expression for the maximum line of sight distance between two antennas for space wave propagation.

[2016]

SOLUTION

Space wave propagation: For frequencies above about 40 MHz (VHF, UHF and microwave), the radio wave travels in a nearly straight line directly from the transmitting antenna to the receiving antenna through the space near the Earth's surface. This is called **space wave (line-of-sight) propagation**. Such waves are not reflected by the ionosphere and do not follow the curvature of the Earth.

Factors limiting the range of space wave propagation:

(a) Curvature of the Earth, since the straight-line wave cannot bend along the Earth's surface.

(b) Height of the transmitting and receiving antennas, as greater heights increase the line-of-sight range.

(c) Obstacles such as buildings, hills and trees lying in the path that block or attenuate the signal.

Derivation of maximum line-of-sight distance: Consider a transmitting antenna of height h_T standing on a spherical Earth of radius R. The geometry of the right triangle formed by the Earth's centre, the antenna top, and the tangent point on the surface gives:

$$(R + h_T)^2 = R^2 + d_T^2$$

$$R^2 + 2Rh_T + h_T^2 = R^2 + d_T^2$$

$$d_T^2 = 2Rh_T + h_T^2$$

Since $h_T \ll R$, the term h_T^2 is negligible, so

$$d_T = \sqrt{2Rh_T}$$

This is the distance from the transmitting antenna to the horizon. Similarly, for a receiving antenna of height h_R, the horizon distance is

$$d_R = \sqrt{2Rh_R}$$

The maximum line-of-sight distance between the two antennas is the sum of these two horizon distances:

$$d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

Final answer. Space wave propagation is straight-line travel above ~ **40 MHz**, limited by Earth's curvature, antenna heights and obstacles, with maximum range $\mathbf{d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}}$.

← *Back to Q304*

Q305

(i) Explain briefly the process of emission of light by a Light Emitting Diode (LED).
(ii) Which semiconductors are preferred to make LEDs and why?
(iii) Give two advantages of using LEDs over conventional incandescent lamps.

[2016]

SOLUTION

(i) Emission of light by an LED: A Light Emitting Diode is a heavily doped *p-n* junction diode operated in forward bias. Under forward bias, electrons from the *n*-side and holes from the *p*-side are injected across the junction. Near the junction, the minority carriers increase, and these injected electrons recombine with holes. Each recombination of an electron in the conduction band with a hole in the valence band releases energy nearly equal to the band gap, which is emitted in the form of a photon:

$$h\nu \approx E_g$$

so the wavelength of the emitted light is

$$\lambda = \frac{hc}{E_g}$$

For visible emission ($\lambda \approx 400$ to 700 nm), the band gap must be about $E_g \approx 1.8$ to 3.0 eV.

(ii) Preferred semiconductors: LEDs are made from **direct band-gap compound semiconductors** such as gallium arsenide phosphide ($GaAs_{1-x}P_x$), gallium arsenide (GaAs) and gallium phosphide (GaP). These are preferred because in a direct band-gap material the electron-hole recombination is radiative, releasing the energy directly as a photon. In indirect band-gap materials like Si and Ge, the recombination energy is mostly given to the lattice as heat (phonons) rather than light, so they are unsuitable for LEDs. The band gap is also chosen so that the emitted photon lies in the visible (or required) region.

(iii) Two advantages of LEDs over incandescent lamps:

(a) LEDs have much higher efficiency and consume far less power (no energy wasted as filament heat), so they are far more energy efficient.

(b) LEDs have a long operating life, fast response time, and ruggedness, working at low operating voltage with negligible warm-up time. (Either two of these are acceptable.)

Final answer. An LED emits light when forward-bias recombination releases photons of energy $h\nu \approx E_g$; direct band-gap compounds like GaAsP are used because their recombination is radiative, and LEDs are more efficient and longer-lasting than incandescent lamps.

← *Back to Q305*

Q306

Draw a circuit diagram of a common emitter amplifier using n-p-n transistor. Derive an expression for the current gain β_{ac}.

[2016]

SOLUTION

Circuit diagram of a common-emitter (CE) amplifier using an *n-p-n* transistor: The emitter is common to both input and output and is grounded. The base-emitter junction is forward biased by a supply V_{BB} through a base resistor R_B.

The collector-base junction is reverse biased by a supply V_{CC} through a load (collector) resistance R_L (or R_C). The AC input signal v_i is applied between base and emitter through a coupling capacitor, and the amplified AC output v_o is taken between collector and emitter.

Description: *n-p-n* transistor with emitter at the bottom connected to the common (ground) line; V_{BB} and R_B in the base loop with input v_i in series; V_{CC} and R_L in the collector loop with output v_o taken across collector-emitter.

Derivation of the AC current gain β_{ac}: The current amplification factor in the common-emitter configuration is defined as the ratio of the small change in collector current to the small change in base current, keeping the collector-emitter voltage constant:

$$\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}\ =\ \text{constant}}$$

This is obtained from the output characteristics (a family of I_C versus V_{CE} curves for different base currents I_B). At a fixed value of V_{CE}, a change in base current ΔI_B from one curve to the next produces a corresponding change in collector current ΔI_C, measured as the vertical spacing between the curves:

$$\beta_{ac} = \frac{\Delta I_C}{\Delta I_B}\Big|_{V_{CE}}$$

Since the emitter current $\mathbf{I_E = I_B + I_C}$ and $I_C \gg I_B$, the value of β_{ac} is large, **typically between 20 and a few hundred** for a silicon transistor.

The corresponding voltage gain of the amplifier is $\mathbf{A_V = -\frac{\beta_{ac} R_L}{r_i}}$, where r_i is the input resistance, and the negative sign indicates a phase difference of 180° between input and output.

Final answer. In a CE ***n-p-n*** amplifier the AC current gain is $\mathbf{\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}}}$, read from the output characteristics, and is typically 20 to a few hundred.

← *Back to Q306*

Q307

Mention any three applications of the internet. Explain one of these in detail.

[2016]

SOLUTION

Three applications of the internet:

(a) Electronic mail (Email) and instant messaging: exchanging text and multimedia messages between users anywhere in the world.

(b) E-commerce and online banking: online shopping, payments, booking of tickets and money transfer.

(c) Video conferencing and media streaming: real-time audio-video communication and on-demand streaming of audio and video content.

(Other valid applications: cloud storage, social networking, distance/online education, and online news/information retrieval.)

Detailed explanation of Email (electronic mail): Email is a method of exchanging digital messages between users over the internet. The sender composes a message using an email client and addresses it to the recipient's unique email address of the form user@example.com. On clicking send, the message is carried by the Simple Mail Transfer Protocol (SMTP) from the sender's outgoing mail server, possibly through intermediate relay servers, to the recipient's incoming mail server, where it is stored in the recipient's mailbox.

The recipient later retrieves the message from the mail server using a retrieval protocol such as POP3 or IMAP. Email is **asynchronous**, meaning the recipient need not be online at the moment the message is sent; it waits in the mailbox until read. It can carry text along with attachments such as images, audio, video and documents, making it a fast, low-cost and globally accessible mode of communication.

Final answer. Three internet applications are email, e-commerce/online banking and video conferencing; email exchanges digital messages over SMTP to the recipient's server, retrieved later via POP3/IMAP, and is asynchronous.

← Back to Q307

Q308

Write two important considerations used while fabricating a Zener diode. Explain, with the help of a circuit diagram, the principle and working of a Zener diode as voltage regulator.

[2015]

SOLUTION

Two important fabrication considerations for a Zener diode:

(a) Both the *p*-side and the *n*-side are **heavily doped**, much more than an ordinary diode. This makes the depletion region very thin ($\sim 10^{-8}\ m$), so that even a small reverse voltage produces a very high electric field at the junction.

(b) The diode is designed with a sharp, well-defined and reproducible **breakdown (Zener) voltage** V_Z, controlled by the doping level so that the device operates reliably in the breakdown region without being damaged.

Principle: A Zener diode works on the principle that in the reverse breakdown region the voltage across the diode stays almost constant at V_Z over a wide range of currents.

Circuit diagram description: The unregulated DC input voltage V_{in} is applied through a series current-limiting resistance R_S. A Zener diode is connected in reverse bias across the output, in parallel with the load resistance R_L. The regulated output $\mathbf{V_{out} = V_Z}$ is taken across the load.

Working as a voltage regulator: The input voltage must exceed V_Z so that the Zener operates in breakdown. In this region the output voltage equals V_Z and remains nearly constant.

If the input voltage V_{in} increases, the current through the circuit increases, so the extra voltage is dropped across the series resistor R_S, and the Zener current increases while V_Z stays constant.

If the load current changes, the Zener current adjusts itself (decreasing when the load draws more, increasing when the load draws less), so that the total

current through R_S and hence the voltage drop across it stays such that the **output voltage across the load remains constant at V_Z.**

Final answer. A Zener diode is heavily doped with a sharp breakdown voltage; in the breakdown region it holds the output at V_Z, with excess voltage dropping across R_S, so it works as a voltage regulator.

← *Back to Q308*

Q309

Draw the necessary energy band diagrams to distinguish between conductors, semiconductors and insulators.
How does the change in temperature affect the behaviour of these materials? Explain briefly.

[2015]

SOLUTION

Energy band diagrams:

Conductor: The valence band and conduction band overlap, or the conduction band is partially filled, so the energy gap $E_g = 0$. Free electrons are abundantly available, giving very high conductivity. (Diagram: shaded valence and conduction bands overlapping with no gap.)

Semiconductor: A completely filled valence band and an empty conduction band at 0 K are separated by a small energy gap, $E_g \approx 1$ eV (Si: 1.1 eV, Ge: 0.7 eV). (Diagram: filled VB below an empty CB with a narrow gap E_g.)

Insulator: A filled valence band and an empty conduction band are separated by a large energy gap, $E_g > 3$ eV (diamond ≈ 5.4 eV). Almost no electron can cross the gap, so the conductivity is practically zero. (Diagram: filled VB below an empty CB with a wide gap E_g.)

Effect of change in temperature:

Conductor: As temperature rises, the lattice vibrations increase and scatter the conduction electrons more, reducing their mobility. The carrier concentration

is essentially unchanged, so the **resistance increases** (conductivity decreases) with rise in temperature.

Semiconductor: As temperature rises, more electron-hole pairs are thermally generated, sharply increasing the carrier concentration. This increase more than compensates for the reduced mobility, so the **resistance decreases** (conductivity increases) with rise in temperature.

Insulator: The energy gap is so large that even a large rise in temperature excites very few electrons across the gap, so an insulator essentially remains a poor conductor, though its conductivity rises slightly at very high temperatures.

Final answer. Conductors have $\mathbf{E_g = 0}$, semiconductors $\mathbf{E_g \approx 1\ eV}$, insulators $\mathbf{E_g > 3\ eV}$; with rising temperature a conductor's resistance increases while a semiconductor's resistance decreases due to extra thermally generated carriers.

← *Back to Q309*

Q310

(a) What are the three basic units in communication systems? Write briefly the function of each of these.
(b) Write any three applications of the internet used in communication systems.

[2015]

SOLUTION

(a) The three basic units in a communication system are the **transmitter**, the **communication channel** and the **receiver**.

Transmitter: It processes the incoming message signal into a form suitable for transmission and sends it through the channel. An input transducer (e.g. microphone) converts the message into an electrical signal, a modulator superimposes it onto a high-frequency carrier, and the antenna radiates the modulated signal.

Communication channel: It is the physical medium that carries the modulated signal from the transmitter to the receiver. It may be free space (wireless),

wires/cables, or optical fibre. During transmission, the signal gets attenuated and noise may be added.

Receiver: It extracts and recovers the original message signal from the received signal. The antenna picks up the signal, which is amplified and then demodulated to separate the message from the carrier, and finally fed to an output transducer (e.g. loudspeaker).

(b) Three applications of the internet used in communication systems:

(i) Email: asynchronous exchange of text and multimedia messages between users worldwide.

(ii) Video conferencing: real-time two-way audio-video communication between people at different locations (e.g. Zoom, Google Meet).

(iii) Voice over Internet Protocol (VoIP) / internet telephony: making voice calls over the internet instead of conventional telephone lines.

Final answer. The three basic units are transmitter, channel and receiver; three internet communication applications are email, video conferencing and VoIP (internet telephony).

← Back to Q310

Q311

Draw a block diagram of a detector for AM signal and show, using necessary processes and the waveforms, how the original message signal is detected from the input AM wave.

[2015]

SOLUTION

The block diagram of an AM detector (receiver) has the following blocks connected in series, with the AM wave entering at the left and the recovered message coming out at the right.

Received AM signal → IF/RF Amplifier → Diode Rectifier → Envelope Detector (RC Low-pass Filter) → Amplifier → Output message signal.

The input is the amplitude modulated wave of the form $V_{AM}(t)=[A_c+A_m\omega_m t]\omega_c t$, a fast carrier oscillation whose envelope traces the slow message $m(t)$.

In the **rectifier** stage, a diode allows current only in one direction, so it removes the negative halves of the AM wave. The output is a series of positive half-cycle pulses whose tips still follow the message envelope.

In the **envelope detector**, a parallel combination of a capacitor C and resistor R (an RC low-pass filter) smooths these pulses. The capacitor charges to the peak of each pulse and discharges slowly through R between pulses, so the high-frequency carrier ripple is filtered out while the slowly varying envelope is retained.

The recovered envelope is the original modulating signal $m(t)$, which the final **amplifier** boosts to a level fit to drive a loudspeaker.

Waveforms (drawn one below the other): (a) the input AM wave, a fast carrier with a wavy envelope; (b) the rectified wave, only the positive half-pulses; (c) the detector output, a smooth wave that copies the message envelope; (d) the amplified **original message signal**, a faithful reproduction of $m(t)$.

Final answer. The diode rectifier plus RC envelope detector recovers the original message signal m(t) from the input AM wave.

← *Back to Q311*

Q312

With what considerations in view, a photodiode is fabricated? State its working with the help of a suitable diagram.
Eventhough the current in the forward bias is known to be more than in the reverse bias, yet the photodiode works in reverse bias. What is the reason?

[2015]

SOLUTION

A photodiode is a p-n junction made of a photosensitive semiconductor, fabricated keeping these points in view.

The semiconductor must have a **band gap** E_g small enough that photons of the light to be detected can excite electrons across it, i.e. the photon energy $h\nu \geq E_g$. For example, to detect visible light the gap is kept around 1 eV to 2 eV.

The junction is fabricated with a **transparent window** over the depletion region so that light can fall directly on it, and the active layer is kept near the surface so photons are not absorbed before reaching the junction.

Working: The photodiode is connected in **reverse bias** with its junction illuminated. When light of energy $h\nu \geq E_g$ falls on the depletion region, it creates electron-hole pairs. The strong reverse-bias electric field across the junction sweeps the electrons to the n-side and the holes to the p-side. This flow constitutes a reverse photocurrent, whose magnitude increases with the intensity of incident light, so the diode acts as a light detector.

Diagram: a p-n junction with a glass window on top; the p-side connected to the negative terminal and the n-side to the positive terminal of a battery (reverse bias); incident photons shown striking the depletion layer and generating electron-hole pairs that move in opposite directions.

Reason for using reverse bias: Although the forward current is larger, what we need is the **fractional change** in current produced by light. In reverse bias the dark current (current without light) is very small, so the change due to illumination is a large fraction of the total current and is easily measured. The change in reverse current with intensity is much more easily detectable than the change in the large forward current. Hence the photodiode is operated in **reverse bias**.

Final answer. A small-band-gap junction with a light window is used; in reverse bias the fractional change of current with intensity is large, so the photodiode is operated in reverse bias.

← Back to Q312

Q313

Draw a circuit diagram of a transistor amplifier in CE configuration.
Define the terms: (i) Input resistance and (ii) Current amplification factor. How are

these determined using typical input and output characteristics?

[2015]

SOLUTION

Circuit diagram (CE amplifier): An n-p-n transistor is drawn with the **emitter grounded** (common to input and output). The base is connected through a resistor R_B to the supply V_{BB} to forward-bias the base-emitter junction; the ac input signal is fed to the base through a coupling capacitor C_1. The collector is connected to V_{CC} through the load resistor R_L (or R_C), keeping the collector-base junction reverse biased; the amplified output is taken between collector and ground through coupling capacitor C_2.

(i) **Input resistance** (r_i): It is the ratio of a small change in base-emitter voltage to the resulting small change in base current, at constant collector-emitter voltage.

$$r_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}=\text{constant}}$$

It is determined from the **input characteristic**, the graph of I_B versus V_{BE} at a fixed V_{CE}. At the operating point we take a small change ΔV_{BE}, read the corresponding ΔI_B, and the reciprocal of the slope gives r_i. Its value is of the order of a few hundred ohms to a few kilo-ohms.

(ii) **Current amplification factor** (β, the ac current gain): It is the ratio of a small change in collector current to the small change in base current, at constant collector-emitter voltage.

$$\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}=\text{constant}}$$

It is determined from the **output characteristics**, the family of I_C versus V_{CE} curves drawn for different fixed values of I_B. At a chosen V_{CE}, we take two curves corresponding to base currents differing by ΔI_B, read the vertical gap ΔI_C between them, and the ratio $\Delta I_C/\Delta I_B$ gives β. Its value is typically around 100.

Final answer. Input resistance r_i = (ΔV_BE/ΔI_B) from the input characteristic, and current gain β = (ΔI_C/ΔI_B) from the output characteristics, both at constant V_CE.

← Back to Q313

Q314

When Sunita, a class XII student, came to know that her parents are planning to rent out the top floor of their house to a mobile company she protested. She tried hard to convince her parents that this move would be a health hazard.
Ultimately her parents agreed :
(1) In what way can the setting up of transmission tower by a mobile company in a residential colony prove to be injurious to health ?
(2) By objecting to this move of her parents, what value did Sunita display ?
(3) Estimate the range of e.m. waves which can be transmitted by an antenna of height 20 m. (Given radius of the earth = 6400 km)

[2014]

SOLUTION

(1) A mobile transmission tower radiates radio-frequency (RF) electromagnetic waves continuously. Living very close to such a tower means **long-term exposure to RF radiation**, which can heat body tissues and is suspected of causing headaches, sleep disturbance, fatigue and, with chronic high-level exposure, an increased health risk. Because of these concerns, towers are generally not permitted very close to homes, schools and hospitals.

(2) By objecting, Sunita displayed **awareness and scientific knowledge**, concern for the health and well-being of her family and neighbours, social responsibility, and the **courage** to speak up and convince her parents to do the right thing.

(3) The range (maximum line-of-sight distance) covered by an antenna of height h on the Earth of radius R is given by

$$d=\sqrt{2\mathrm{Rh}}$$

Here $\boldsymbol{R=6400\ \mathrm{km}=6.4 10^{6}\ m}$ and $\boldsymbol{h=20\ m}$.

$$d=\sqrt{2\times(6.4 10^{6})20}$$

$$d=\sqrt{2.56 10^{8}}\ m$$

$$d=1.610^4\ m=16\ \text{km}$$

So the antenna can transmit electromagnetic waves over a range of about **16 km**.

Final answer. (1) Long-term RF exposure is a health hazard; (2) awareness, social responsibility and courage; (3) range d = 1.6×10^4 m = 16 km.

← *Back to Q314*

Q315

Write any two distinguishing features between conductors, semiconductors and insulators on the basis of energy band diagrams.

[2014]

SOLUTION

On the basis of energy band diagrams, the three materials are distinguished as follows. In each diagram the lower band is the **valence band**, the upper band is the **conduction band**, and the gap between them is the forbidden energy gap E_g.

First distinguishing feature, the **energy band gap** E_g:

In a **conductor**, the conduction band and valence band overlap, so $\mathbf{E_g=0}$. There is no forbidden gap.

In a **semiconductor**, a small forbidden gap exists, $\mathbf{E_g\approx 1\ eV}$ (Si **1.1 eV**, Ge **0.7 eV**).

In an **insulator**, the forbidden gap is large, $\mathbf{E_g>3\ eV}$ (for example diamond **5.4 eV**).

Second distinguishing feature, the **availability of free electrons in the conduction band** at room temperature:

In a conductor, the conduction band is partially filled (or overlaps the valence band), so a large number of free electrons are always available and conductivity is high.

In a semiconductor, the gap is small enough that thermal energy at room temperature (**kT0.026 eV**) lifts a few electrons into the conduction band, leaving

holes in the valence band; conductivity is small but rises with temperature.

In an insulator, the gap is so large that almost no electron can cross it at room temperature; the conduction band stays nearly empty and conductivity is practically zero.

Final answer. Conductors have E_g = 0 with a partly filled/overlapping conduction band; semiconductors have small E_g (~1 eV) with few thermal carriers; insulators have large E_g (>3 eV) with an empty conduction band.

← *Back to Q315*

Q316

Write two basic modes of communication. Explain the process of amplitude modulation. Draw a schematic sketch showing how amplitude modulated signal is obtained by superposing a modulating signal over a sinusoidal carrier wave.

[2014]

SOLUTION

The two basic modes of communication are **point-to-point communication** (information passes between one specified transmitter and one specified receiver, for example a telephone call) and **broadcast communication** (one transmitter sends information to a large number of receivers, for example radio or television).

Amplitude modulation: In amplitude modulation, the **amplitude of a high-frequency carrier wave is varied in proportion to the instantaneous amplitude of the message (modulating) signal**, while the frequency and phase of the carrier are kept unchanged.

Let the carrier be $c(t)=A_c\omega_c t$ and the message be $m(t)=A_m\omega_m t$. In AM the resultant amplitude becomes $(A_c+A_m\omega_m t)$, so

$$V_{AM}(t)=(A_c+A_m\omega_m t)\omega_c t$$

$$V_{AM}(t)=A_c\omega_c t+\mu A_c\omega_m t\omega_c t$$

where $\mu = \frac{A_m}{A_c}$ is the modulation index. Expanding the product gives

$$V_{AM}(t) = A_c\omega_c\, t + \frac{\mu\, A_c}{2}\cos(\omega_c+\omega_m)t + \frac{\mu\, A_c}{2}\cos(\omega_c-\omega_m)t$$

so the AM wave contains the carrier ω_c and two **sidebands** at $\omega_c+\omega_m$ and $\omega_c-\omega_m$.

Schematic sketch (three waveforms stacked): the top trace is the low-frequency message $m(t)$; the middle trace is the constant-amplitude high-frequency carrier $c(t)$; the bottom trace is the AM wave, a fast carrier oscillation whose **upper and lower envelopes trace out the shape of the message signal**.

Final answer. The two modes are point-to-point and broadcast; in AM the carrier amplitude varies with the message, giving V_AM(t) = (A_c + A_m cos ω_m t) cos ω_c t whose envelope follows the message.

← *Back to Q316*

Q317

Write three important factors which justify the need of modulating a message signal. Show diagrammatically how an amplitude modulated wave is obtained when a modulating signal is superimposed on a carrier wave.

[2013]

SOLUTION

Three factors that justify the need for modulating a message signal are as follows.

First, **reasonable size of antenna**: for efficient radiation an antenna should have a length of about λ/4. An audio signal of frequency **ν5 kHz** has wavelength

$$\lambda = \frac{c}{\nu} = \frac{310^8}{510^3} = 610^4\ m = 60\ \text{km}$$

which would need an antenna about 15 km long, an impractical size. Modulating the signal onto a high-frequency carrier (MHz range) reduces the required antenna length to a few metres.

Second, **effective (large) power radiated**: the power radiated by an antenna varies as $(1/\lambda)^2$. A short-wavelength (high-frequency) carrier therefore radiates far more power, giving a stronger transmitted signal than the long-wavelength baseband signal.

Third, **avoiding mixing of signals and channel sharing (multiplexing)**: if every transmitter sent the same low audio frequencies, the signals from different sources would overlap and interfere. By giving each transmitter a different high carrier frequency, many signals can be sent through the same medium without mutual interference.

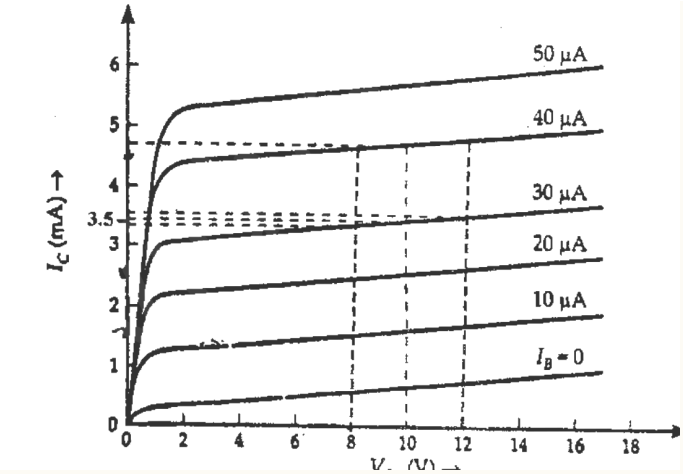
Diagram: three waveforms drawn one below the other, the **modulating signal** $m(t)=A_m\omega_m t$ at top, the high-frequency **carrier wave** $c(t)=A_c\omega_c t$ of constant amplitude in the middle, and the **amplitude-modulated wave** $V_{AM}(t)=(A_c+A_m\omega_m t)\omega_c t$ at the bottom, a fast carrier whose envelope traces the shape of the modulating signal.

Final answer. Modulation is needed for a practical antenna size, larger radiated power, and channel sharing without mixing; the AM wave is the carrier whose envelope follows the modulating signal.

← *Back to Q317*

Q318

Output characteristics of an n-p-n transistor in CE configuration is shown in the figure. Determine:
(i) dynamic output resistance
(ii) dc current gain and
(iii) ac current gain at an operating point $V_{CE} = 10\ V$, *when* $I_B = 30\ \mu A$.



[2013]

SOLUTION

All quantities are read from the given output characteristics (I_C versus V_{CE} curves for different I_B) at the operating point $\mathbf{V_{CE}=10\ V}$, $\mathbf{I_B=30\ \mu A}$. Representative readings from a typical such graph are used below.

(i) **Dynamic (ac) output resistance** r_o: it is the ratio of a change in V_{CE} to the change in I_C at constant I_B.

$$r_o=\left(\frac{\Delta V_{CE}}{\Delta I_C}\right)_{I_B=\text{constant}}$$

Along the $\mathbf{I_B=30\ \mu A}$ curve, taking V_{CE} from 8 V to 12 V, the collector current changes from 3.4 mA to 3.5 mA:

$$r_o=\frac{(12-8)\ V}{(3.5-3.4)\ \text{mA}}=\frac{4\ V}{0.110^{-3}\ A}$$

$$r_o=410^4\ \Omega=40\ \text{k}\Omega$$

(ii) **DC current gain** β_{dc}: it is the ratio of collector current to base current at the operating point.

$$\beta_{dc}=\frac{I_C}{I_B}$$

At $\mathbf{V_{CE}=10\ V}$, $\mathbf{I_B=30\ \mu A}$, $\mathbf{I_C 3.45\ mA}$:

$$\beta_{dc}=\frac{3.4510^{-3}}{3010^{-6}}=115$$

(iii) **AC current gain** β_{ac}: it is the ratio of the change in collector current to the change in base current at constant V_{CE}.

$$\beta_{ac}=\left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}=\text{constant}}$$

At $\mathbf{V_{CE}=10\ V}$, taking two adjacent curves with $\mathbf{\Delta I_B=20\ \mu A}$ giving $\mathbf{\Delta I_C 2.3\ mA}$:

$$\beta_{ac}=\frac{2.3\cdot 10^{-3}}{20\cdot 10^{-6}}=115$$

Thus $\mathbf{r_o 40\ k\Omega}$, $\mathbf{\beta_{dc} 115}$ and $\mathbf{\beta_{ac} 115}$.

> **TIP**
>
> Values depend on the printed graph; always quote the V_CE and I_B readings you actually used, then divide. The method (which ratio for which quantity) is what earns the marks.

Final answer. Dynamic output resistance r_o ≈ 40 kΩ, dc current gain β_dc ≈ 115 and ac current gain β_ac ≈ 115.

← *Back to Q318*

Q319

Which mode of propagation is used by short wave broadcast services having frequency range from a few MHz upto 30 MHz*? Explain diagrammatically how long distance communication can be achieved by this mode. Why is there an upper limit to frequency of waves used in this mode?*

[2012]

SOLUTION

Short-wave broadcast services in the range of a few MHz up to 30 MHz use the **sky-wave (ionospheric) mode** of propagation.

In this mode, the wave radiated obliquely from the transmitting antenna travels upward and reaches the **ionosphere**, a layer of ionised gas at heights of about 60 to 400 km. The ionosphere's refractive index decreases with height, so the wave is gradually bent (refracted) more and more until it is effectively **totally**

internally reflected back towards the Earth, landing at a distant point. By repeated reflections between the ionosphere and the Earth's surface, the signal can travel thousands of kilometres, achieving long-distance and even intercontinental communication.

Diagram: the Earth's curved surface with a transmitter (T) and a distant receiver (R); a ray leaves T at an oblique angle, rises to the ionosphere layer, bends and returns to R; one or two further bounces between the ground and the ionosphere are shown to indicate the extended range.

Reason for the upper frequency limit: The ionosphere can reflect a wave only if its frequency is below a **critical frequency** that depends on the maximum electron density N_{max},

$$f_c = 9\sqrt{N_{max}} \text{ (Hz, with } N_{max} \text{ in electrons/m}^3\text{)}$$

For waves of frequency above about 30 MHz, this condition is not met: the bending is insufficient, so the waves **penetrate the ionosphere and escape into space** instead of returning to Earth. Hence there is an upper frequency limit for sky-wave propagation.

Final answer. Short-wave broadcasts use sky-wave (ionospheric) propagation; the upper limit (~30 MHz) exists because higher-frequency waves penetrate the ionosphere and are not reflected back.

← *Back to Q319*

Q320

Mention three different modes of propagation used in communication system. Explain with the help of a diagram how long distance communication can be achieved by ionospheric reflection of radio waves.

[2012]

SOLUTION

The three different modes of propagation used in a communication system are:

Ground wave (surface wave) propagation, in which low-frequency waves (up to about 2 MHz) travel along the curved surface of the Earth, used for local AM

medium-wave broadcasting.

Sky wave (ionospheric) propagation, in which high-frequency waves (3 to 30 MHz) are reflected back to Earth by the ionosphere, used for long-distance short-wave communication.

Space wave (line-of-sight) propagation, in which very-high-frequency waves (above about 40 MHz) travel in straight lines from transmitter to receiver, used for TV, microwave links and satellite communication.

Long-distance communication by ionospheric reflection (sky wave): The wave radiated obliquely upward from the transmitting antenna reaches the **ionosphere**, a layer of ionised gas lying about 60 to 400 km above the Earth. Because the ionosphere's refractive index falls as the electron density rises with height, the wave is progressively bent until it is **reflected back to Earth** at a point far from the transmitter. By successive reflections between the Earth and the ionosphere, the signal can travel hundreds to thousands of kilometres, reaching distant continents.

Diagram: the curved Earth with a transmitter (T) sending a ray obliquely upward to the ionosphere layer, the ray bending and returning to a distant receiver (R), with one further bounce shown to illustrate the multiple-hop extension of range.

Final answer. The three modes are ground wave, sky wave and space wave; sky-wave reflection from the ionosphere bends radio waves back to Earth at distant points, giving long-distance communication.

← Back to Q320

Q321

Write any two factors which justify the need for modulating a signal.
Draw a diagram showing an amplitude modulated wave by superposing a modulating signal over a sinusoidal carrier wave.

[2012]

SOLUTION

A low-frequency information signal (audio, ~20 Hz to 20 kHz) cannot be transmitted directly over long distances. To send it efficiently, it is superposed on a high-frequency carrier wave. This is called modulation. Two factors justify this need.

Size of the antenna: For efficient radiation, the antenna length must be at least of the order of $\frac{\lambda}{4}$ of the signal wavelength.

For an audio signal of frequency $f \approx 5$ **kHz**, the wavelength is

$$\lambda = \frac{c}{f} = \frac{3\times 10^8}{5\times 10^3} = 6\times 10^4\ m = 60\ \text{km}$$

This needs an antenna nearly 15 km long, which is impossible to build. By shifting the signal onto a high-frequency carrier (in the MHz range), λ falls to a few metres and the antenna becomes practical.

Effective power radiated and channel separation (multiplexing): Power radiated by an antenna varies as $(\frac{1}{\lambda})^2$, so a higher carrier frequency (smaller λ) radiates much more power. Also, by giving each transmission a different carrier frequency, many signals (radio stations, TV channels) can share the same space without mixing up.

Diagram of an amplitude-modulated (AM) wave: Draw three graphs of amplitude versus time, one below the other.

(a) The **modulating signal** $\mathbf{c_m(t) = A_m\omega_m\, t}$: a slow, low-frequency sine wave of small amplitude A_m.

(b) The **carrier wave** $\mathbf{c_c(t) = A_c\omega_c\, t}$: a fast, high-frequency sine wave of constant amplitude A_c, with $\omega_c \gg \omega_m$.

(c) The **amplitude-modulated wave**: a high-frequency carrier whose amplitude (envelope) rises and falls following the shape of the modulating signal. The upper and lower dashed envelope curves trace out the modulating sine wave, while the fast oscillations inside are the carrier.

Final answer. Modulation is needed mainly for a practical antenna size and for greater radiated power plus channel separation; the AM wave is a carrier whose envelope follows the modulating signal.

← Back to Q321

Q322

Define modulation index. Give its physical significance.
For an amplitude modulated wave, the maximum amplitude is found to be 10 *V while the minimum amplitude is* 2 *V. Determine the modulation index* μ.

[2012]

SOLUTION

The **modulation index** μ of an amplitude-modulated wave is the ratio of the amplitude of the modulating signal A_m to the amplitude of the carrier wave A_c.

$$\mu = \frac{A_m}{A_c}$$

In terms of the maximum and minimum amplitudes of the AM wave it is written as

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

Physical significance: The modulation index measures the depth of modulation, that is, how strongly the carrier amplitude is varied by the signal. It is kept ≤ 1 to avoid distortion. If **μ = 0**, there is no modulation and the carrier alone is sent carrying no information. If **0 < μ < 1**, the signal is faithfully transmitted. If **μ = 1** (100% modulation), the envelope just touches zero. If **μ > 1** (overmodulation), the envelope crosses zero and the recovered signal is badly distorted.

Numerical: Given $\mathbf{A_{max} = 10\ V}$ and $\mathbf{A_{min} = 2\ V}$.

$$\mu = \frac{A_{max} - A_{min}}{A_{max} + A_{min}}$$

$$\mu = \frac{10 - 2}{10 + 2}$$

$$\mu = \frac{8}{12}$$

$$\mu = 0.667$$

Hence the **modulation index $\mu \approx 0.667$, that is about 66.7%**.

Final answer. The modulation index is **$\mu \approx 0.667$** (about 66.7%).

← *Back to Q322*

Q323

Name the three different modes of propagation of electromagnetic waves. Explain, using a proper diagram the mode of propagation used in the frequency range above 40 MHz.

[2012]

SOLUTION

The three modes of propagation of electromagnetic waves are **ground wave (surface wave) propagation, sky wave (ionospheric) propagation, and space wave (line-of-sight) propagation**.

Ground wave: Low-frequency waves (below about 1.5 MHz) glide over the surface of the Earth, following its curvature. They attenuate quickly, so the range is short.

Sky wave: Waves in the 3 to 30 MHz band are sent up to the ionosphere, which reflects/refracts them back to Earth, giving long-distance shortwave communication.

Space wave: Waves of frequency above about 40 MHz travel in a straight line from the transmitting to the receiving antenna.

Mode used above 40 MHz: Space wave (line-of-sight) propagation. At these high frequencies the wave cannot bend with the Earth and is not reflected by the ionosphere (it passes straight through it). So it must travel in a nearly straight line directly from the transmitting antenna to the receiving antenna, like a ray of light. This is used for TV, FM radio, and satellite communication.

Diagram: Draw a curved arc for the Earth's surface. Place a tall transmitting antenna of height h_T on the left and a receiving antenna of height h_R on the

right. Draw a straight tangential line (the line-of-sight ray) from the top of the transmitting antenna to the top of the receiving antenna, just grazing the Earth at the horizon point in between.

Because the wave travels in a straight line, the range is limited by the curvature of the Earth. The maximum line-of-sight distance is

$$d_{max} = \sqrt{2R\,h_T} + \sqrt{2R\,h_R}$$

where R is the radius of the Earth, and h_T, h_R are the heights of the transmitting and receiving antennas. Taller antennas (or relay satellites) increase the range.

Final answer. The three modes are ground wave, sky wave and space wave; above 40 MHz, space wave (line-of-sight) propagation is used.

← *Back to Q323*

Q324

Draw a labelled diagram of a full wave rectifier circuit. State its working principle. Show the input-output waveforms.

[2011]

SOLUTION

Circuit and labelled diagram: A **full-wave rectifier** uses a centre-tapped step-down transformer and two p-n junction diodes. The AC mains is applied to the primary. The two ends A and B of the secondary winding are connected to the anodes of diodes D_1 and D_2 respectively. The cathodes of both diodes are joined together. The load resistance R_L is connected between this common cathode point and the centre-tap C of the secondary. The output is taken across R_L.

Working principle: A p-n junction diode conducts only when it is **forward biased** and offers a very high resistance (almost no conduction) when **reverse biased**. The two diodes are arranged so that they conduct in alternate half cycles, but the current through the load R_L flows in the same direction in both halves.

During the positive half cycle: End A is positive and end B is negative with respect to the centre-tap C. So D_1 is forward biased and conducts, while D_2 is

reverse biased and does not conduct. Current flows through D_1 and down through R_L from the cathode side to *C*.

During the negative half cycle: The polarity reverses; end *B* is positive and end *A* is negative. Now D_2 is forward biased and conducts, while D_1 is reverse biased and is off. Current flows through D_2 and again down through R_L in the same direction as before.

Hence in both half cycles the current through R_L is in the **same (unidirectional) direction**, giving a pulsating DC output. Because both halves are used, the output is full-wave rectified.

Input and output waveforms: Draw two graphs of voltage versus time, one above the other.

(a) **Input:** a full sinusoidal AC wave with both positive and negative half cycles.

(b) **Output across R_L:** a series of positive humps only (the negative halves are flipped up to positive). The output has **no gaps** between the humps, so its ripple frequency is **twice** the input frequency.

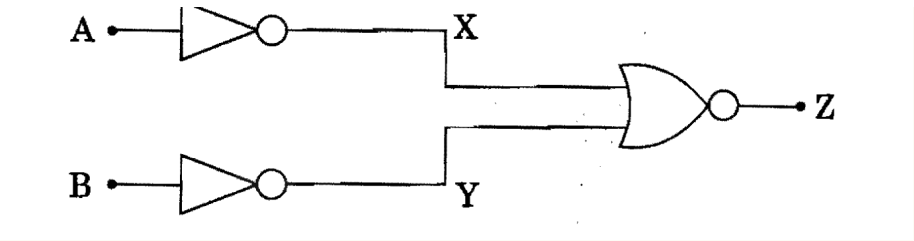
> **TIP**
>
> Full-wave output frequency = 2f, while half-wave gives *f*; this point alone is often a mark in the marking scheme.

Final answer. A centre-tapped full-wave rectifier uses two diodes conducting in alternate half cycles to give unidirectional pulsating DC of twice the input frequency.

← *Back to Q324*

Q325

You are given a circuit below. Write its truth table. Hence, identify the logic operation carried out by this circuit. Draw the logic symbol of the gate it corresponds to.



[2011]

SOLUTION

The circuit shown is a combination of NAND gates: a first NAND gate gives output $Y' = \overline{A \cdot B}$, and the second NAND gate is wired as an inverter (both its inputs tied together) so it gives $Y = \overline{Y'} = \overline{A \cdot B} = A \cdot B$. So the node logic builds up an AND operation.

We work out the output node by node and tabulate.

Truth table:

$$A = 0, B = 0 \Rightarrow Y = 0$$

$$A = 0, B = 1 \Rightarrow Y = 0$$

$$A = 1, B = 0 \Rightarrow Y = 0$$

$$A = 1, B = 1 \Rightarrow Y = 1$$

In tabular form, with columns A, B, Y:

0 0 0

0 1 0

1 0 0

1 1 1

Identification: The output Y is high (1) only when **both** inputs A and B are high (1). This is exactly the **AND operation**, so the circuit works as an **AND gate**, $Y = A \cdot B$.

Logic symbol: Draw a flat-backed, **D-shaped** symbol with a straight vertical left edge and a rounded right end. Two input lines A and B enter on the left straight

edge, and a single output line Y leaves from the rounded right tip. (No small circle/bubble at the output, since it is AND and not NAND.)

Final answer. The circuit performs the AND operation ($Y = A \cdot B$); its symbol is the plain D-shaped AND gate.

← *Back to Q325*

Q326

Name the important processes that occur during the formation of a p-n junction. Explain briefly, with the help of a suitable diagram, how a p-n junction is formed. Define the term 'barrier potential'.

[2011 • Set 55-2-1]

SOLUTION

When a p-type and an n-type semiconductor are joined to form a junction, two important processes take place: **diffusion** and **drift**.

(i) Diffusion: Because the n-side has a high concentration of electrons and the p-side a high concentration of holes, there is a concentration gradient across the junction. So electrons diffuse from the n-side to the p-side, and holes diffuse from the p-side to the n-side.

(ii) Drift: As electrons diffuse away from the n-side, they leave behind positively charged immobile **donor ions**, and as holes diffuse away from the p-side, they leave negatively charged immobile **acceptor ions**. These fixed ions set up an electric field $\vec{E}$ directed from the n-side to the p-side. This field makes the minority carriers drift across the junction, in the direction opposite to diffusion.

Formation (with diagram): Draw a rectangular block. Mark the left half as p-type (showing acceptor ions) and the right half as n-type (showing ⊕ donor ions). Near the central junction show a narrow region with a row of negative ions on the p-side and positive ions on the n-side, and an arrow for the field $\vec{E}$ pointing from n to p. This central region, swept clean of mobile charge carriers and containing only fixed ions, is the **depletion region** (of width about 1 μm).

As diffusion proceeds, the field and the opposing potential grow until the drift current exactly balances the diffusion current. At this equilibrium the junction is stable.

Barrier potential: The potential difference that develops across the depletion region, due to the immobile charged ions, which opposes the further diffusion of majority carriers, is called the **barrier potential** V_b. (Approximately 0.7 *V* for silicon and 0.3 *V* for germanium.)

Final answer. Diffusion and drift form a depletion region of fixed ions; the potential across it that stops further diffusion is the barrier potential (≈ **0.7** V for Si, **0.3** V for Ge).

← *Back to Q326*

Q327

Draw transfer characteristics of a common emitter n-p-n transistor. Point out the region in which the transistor operates as an amplifier.
Define the following terms used in transistor amplifiers:
(i) Input resistance
(ii) Output resistance
(iii) Current amplification factor

[2011 • Set 55-2-1]

SOLUTION

Transfer characteristic: The transfer characteristic of an n-p-n transistor in the common-emitter mode is the graph of the **output voltage V_o (= V_{CE})** along the y-axis against the **input voltage V_i (= V_{BE})** along the x-axis, keeping the supply V_{CC} fixed.

Diagram: Plot V_o (vertical) versus V_i (horizontal). The curve has three parts:

(1) For small V_i, the output stays high and nearly flat near V_{CC} (cut-off region).

(2) Then a steep, almost straight, downward-sloping line in the middle (active region).

(3) Finally, for large V_i, the output again becomes low and nearly flat near zero (saturation region).

Region of operation as an amplifier: The transistor is used as an amplifier in the **active region**, the steep linear central part of the curve. Here a small change in input voltage V_i produces a large change in output voltage V_o, giving amplification with minimum distortion.

Definitions:

(i) Input resistance: the ratio of a small change in base-emitter voltage to the resulting small change in base current, at constant V_{CE}.

$$r_i = \left(\frac{\Delta V_{BE}}{\Delta I_B}\right)_{V_{CE}}$$

(ii) Output resistance: the ratio of a small change in collector-emitter voltage to the resulting small change in collector current, at constant base current I_B.

$$r_o = \left(\frac{\Delta V_{CE}}{\Delta I_C}\right)_{I_B}$$

(iii) Current amplification factor: the ratio of a small change in collector current to the small change in base current, at constant V_{CE}.

$$\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}}$$

Final answer. The CE transfer curve has cut-off, active and saturation regions; the transistor amplifies in the active region.

← *Back to Q327*

Q328

What is space wave propagation? Give two examples of communication systems which use space wave mode.
A TV tower is 80 m tall. Calculate the maximum distance upto which the signal transmitted from the tower can be received.

[2010]

SOLUTION

Space-wave propagation: It is the mode in which the electromagnetic wave travels in a nearly **straight line (line of sight)** directly from the transmitting antenna to the receiving antenna through the lower atmosphere. It is used at high frequencies, above about 40 MHz. At these frequencies sky-wave propagation fails (the wave passes straight through the ionosphere instead of being reflected) and the ground wave is heavily attenuated, so line-of-sight is the only option. Its range is limited by the curvature of the Earth.

Two examples: Television broadcasting and satellite communication. (Microwave point-to-point links and mobile telephony also use the space wave.)

Numerical: Given height of TV tower $h = 80\ m$ and radius of Earth $R = 6.4\times 10^6\ m$.

The maximum line-of-sight distance from the tower to the ground is

$$d = \sqrt{2Rh}$$

$$d = \sqrt{2\times (6.4\times 10^6) \times 80}$$

$$d = \sqrt{1.024\times 10^9}$$

$$d = 3.2\times 10^4\ m$$

$$d = 32\ \text{km}$$

Hence the signal can be received up to a maximum distance of about **32 km** from the tower.

Final answer. Space wave is line-of-sight propagation; the 80 m tower gives a maximum range of about $3.2\times 10^4\ m = 32\ \text{km}$.

← *Back to Q328*

Q329

Distinguish between sky wave and space wave propagation. Give a brief description with the help of suitable diagrams indicating how these waves are propagated.

[2009]

SOLUTION

Sky-wave propagation: In this mode, radio waves of the high-frequency band (about 3 to 30 MHz) are sent obliquely upward from the transmitting antenna. The **ionosphere** (the layer of ionised gases between about 65 km and 400 km above the Earth) refracts them so strongly that they are effectively **reflected back** to a distant point on the Earth. By successive reflections between the Earth and the ionosphere, the signal can travel thousands of kilometres. This is used for long-distance shortwave (HF) broadcasting.

Diagram (sky wave): Draw the curved Earth at the bottom and a horizontal layer (the ionosphere) high above it. From a transmitting antenna draw a ray going up at an angle, striking the ionosphere, and bending back down to a far-off receiving point on the Earth.

Space-wave propagation: In this mode, waves of frequency above about 40 MHz (VHF, UHF, microwave) travel in a nearly **straight line (line of sight)** directly from the transmitting antenna to the receiving antenna. They are not reflected by the ionosphere (they pass through it). The range is limited by the curvature of the Earth, with maximum distance

$$d_M = \sqrt{2R\,h_T} + \sqrt{2R\,h_R}$$

It is used for TV broadcasting, microwave links, mobile telephony and satellite communication.

Diagram (space wave): Draw the curved Earth with a tall transmitting antenna (h_T) and a tall receiving antenna (h_R), joined by a straight tangential ray grazing the horizon.

Main distinctions: Sky wave uses 3 to 30 MHz and reaches far by ionospheric reflection, while space wave uses frequencies above 40 MHz, travels straight (line of sight), and its range is limited by Earth's curvature (extended by tall antennas or satellites).

Final answer. Sky wave (3 to 30 MHz) reaches far by ionospheric reflection; space wave (above 40 MHz) travels line-of-sight, limited by Earth's curvature.

← *Back to Q329*

Q330

With the help of a suitable diagram, explain the formation of depletion region in a p-n junction. How does its width change when the junction is (i) forward biased, and (ii) reverse biased?

[2009]

SOLUTION

Formation of the depletion region: When a p-type and an n-type semiconductor are joined, a concentration gradient exists across the junction. So electrons, being in higher concentration on the n-side, **diffuse** to the p-side, and holes, being in higher concentration on the p-side, diffuse to the n-side.

The electrons leaving the n-side expose positively charged immobile **donor ions** there, and the holes leaving the p-side expose negatively charged immobile **acceptor ions** there. The narrow region around the junction is thus left with no free charge carriers, only these fixed ions, and is called the **depletion region** (width about 1 μm).

These ions set up an electric field $\vec{E}$ directed from the n-side to the p-side, and a potential difference called the **barrier potential** V_b (**≈ 0.7 *V*** for Si, 0.3 *V* for Ge). At equilibrium the diffusion current is exactly balanced by the drift current of minority carriers, so the depletion width becomes steady.

Diagram: Draw a block with a p-region on the left (acceptor ions) and an n-region on the right (⊕ donor ions). Show the depletion region in the middle as a layer of negative ions on the p-side and positive ions on the n-side, with the field arrow $\vec{E}$ pointing from n to p.

(i) Forward bias: Connect the battery's positive terminal to the p-side and negative terminal to the n-side. The applied voltage opposes the barrier potential, so the barrier is lowered and the depletion region **width decreases**. Majority carriers cross easily, giving a large forward current.

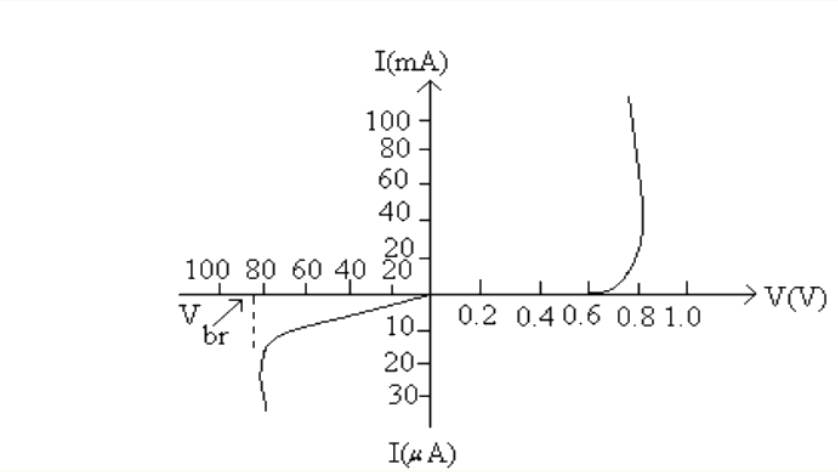
(ii) Reverse bias: Connect the positive terminal to the n-side and negative terminal to the p-side. The applied voltage now aids the barrier potential, pulling majority carriers away from the junction. The barrier is raised and the depletion region **width increases**. Only a very small reverse (leakage) current flows due to minority carriers.

Final answer. Diffusion and drift create a depletion region of fixed ions; its width decreases under forward bias and increases under reverse bias.

← *Back to Q330*

Q331

The figure below shows the V-I characteristic of a semiconductor diode.



(i) Identify the semiconductor diode used.
(ii) Draw the circuit diagram to obtain the given characteristic of this device.
(iii) Briefly explain how this diode can be used as a voltage regulator.

[2008]

SOLUTION

Identification of the diode: The given V-I graph has two distinct parts. In forward bias the current rises exponentially after the knee (cut-in) voltage, which is the normal behaviour of any p-n junction diode. In reverse bias the current stays almost zero up to a particular reverse voltage, and then at this voltage the reverse current increases very sharply (the curve drops almost vertically) while the voltage across the diode stays nearly constant at V_Z. This sharp reverse breakdown at a fixed reverse voltage is the special feature of a **Zener diode**.

(ii) Circuit to obtain the characteristic: Connect a variable DC supply to the Zener diode through a series current-limiting resistance R. For the reverse

characteristic, the n-side of the Zener is joined to the positive terminal of the supply (reverse biased). A voltmeter V is connected in parallel with the Zener to read the voltage across it, and a milliammeter mA is put in series to read the diode current. By slowly increasing the supply voltage and noting corresponding values of V and I, the points of the V-I graph are plotted. Reversing the supply connections gives the forward part of the curve.

(iii) Zener diode as a voltage regulator: The Zener is always operated in the **reverse breakdown region**, where its voltage stays fixed at V_Z even though the current through it changes a lot. The unregulated input V_i is applied through a series resistance R_S, and the Zener is connected in parallel with the load R_L, so the output is taken across the Zener.

$$V_{out}=V_Z=\text{constant}$$

The currents satisfy $\mathbf{I_S=I_Z+I_L}$. If V_i increases, the extra voltage is dropped across R_S and the Zener current I_Z rises so that V_Z (hence V_{out}) stays the same. If the load draws more current, I_Z falls to keep I_S and the drop across R_S steady. In both cases the output voltage across R_L remains constant, so a **regulated DC output** is obtained.

Final answer. The diode is a Zener diode; operated in reverse breakdown across the load (with series resistance $\mathbf{R_S}$) it holds the output fixed at $\mathbf{V_Z}$, acting as a voltage regulator.

← *Back to Q331*

Q332

Draw a plot of the amplitude versus ω for an amplitude modulated wave. Define modulation index. State its importance for effective amplitude modulation.

[2008]

SOLUTION

Amplitude versus ω plot for an AM wave: An amplitude modulated wave can be written as

$$c_{AM}(t)=A_c\omega_c\, t+\frac{\mu\, A_c}{2}\cos(\omega_c-\omega_m)t+\frac{\mu\, A_c}{2}\cos(\omega_c+\omega_m)t$$

So the AM wave contains three frequencies: the carrier at ω_c with amplitude A_c, the lower side band at $(\omega_c-\omega_m)$ with amplitude $\frac{\mu\, A_c}{2}$, and the upper side band at $(\omega_c+\omega_m)$ with amplitude $\frac{\mu\, A_c}{2}$.

The amplitude-versus-ω graph is therefore a **line spectrum** of three vertical lines: take amplitude on the y-axis and ω on the x-axis. Draw one tall line of height A_c at $\boldsymbol{\omega=\omega_c}$, and two shorter lines each of height $\frac{\mu\, A_c}{2}$, one at $\omega_c-\omega_m$ (left of the carrier) and one at $\omega_c+\omega_m$ (right of the carrier). The two side-band lines are equal in height and placed symmetrically about the carrier line.

Modulation index: It is the ratio of the amplitude of the modulating (message) signal to the amplitude of the carrier signal.

$$\mu=\frac{A_m}{A_c}=\frac{A_{max}-A_{min}}{A_{max}+A_{min}}$$

Its importance: μ tells the degree (depth) of modulation. For effective and distortion-free amplitude modulation the modulation index must satisfy $\mu\leq 1$. If $\boldsymbol{\mu>1}$ (over-modulation), the envelope of the wave gets distorted and the original message cannot be recovered properly. If μ is too small, very little power goes into the side bands (which carry the information), so the received signal is weak and noisy. Hence μ should be kept close to 1 but not more than 1.

CONCEPT

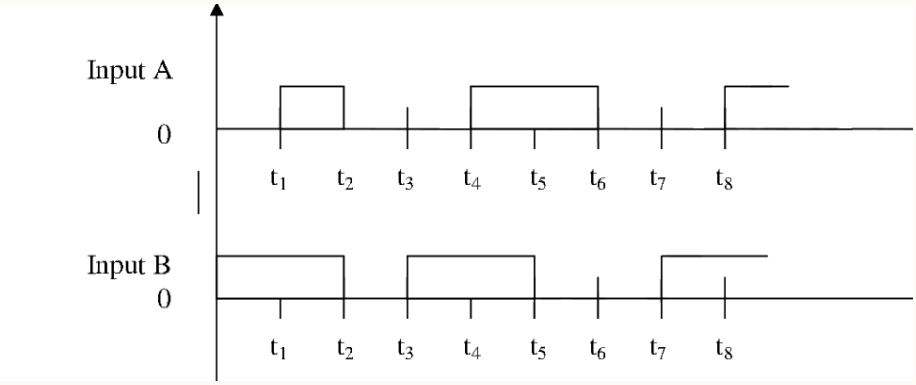
Side bands carry all the information, so a larger μ (up to 1) means more power in the side bands and a stronger received signal.

Final answer. The AM spectrum shows a carrier line ($\mathbf{A_c}$) at $\boldsymbol{\omega_c}$ with two side-band lines ($\boldsymbol{\mu\, A_c/2}$) at $\boldsymbol{\omega_c\omega_m}$; $\boldsymbol{\mu=A_m/A_c}$ and must be $\boldsymbol{\leq 1}$ for effective modulation.

← *Back to Q332*

Q333

Two signals A, B as given below, are applied as input to (i) AND (ii) NOR (iii) NAND gates. Draw the output wave-form in each case.



[2008]

SOLUTION

First write the Boolean output of each gate, then read the given input waveforms A and B interval by interval and apply the rule to draw the output.

(i) AND gate: $Y = A \cdot B$. The output is HIGH (1) only in those time intervals where both A and B are HIGH at the same time; in every other interval the output is LOW (0).

(ii) NOR gate: $Y = \overline{A + B}$. The output is HIGH (1) only in those intervals where both A and B are LOW at the same time; in every other interval the output is LOW (0).

(iii) NAND gate: $Y = \overline{A \cdot B}$. The output is LOW (0) only in those intervals where both A and B are HIGH at the same time; in every other interval the output is HIGH (1).

The combined truth table used to mark the waveforms is:

$A=0, B=0$: AND=0, NOR=1, NAND=1

$A=0, B=1$: AND=0, NOR=0, NAND=1

$A=1, B=0$: AND=0, NOR=0, NAND=1

$A=1, B=1$: AND=1, NOR=0, NAND=0

To draw each output waveform, line it up below the given A and B waveforms on the same time axis. For every time interval, read the pair (A,B) from the input plot, look up the row above, and draw the output level HIGH or LOW for that interval. Repeating this for all intervals gives the **AND, NOR and NAND output waveforms**, plotted one below the other.

Final answer. AND is HIGH only when both inputs are HIGH; NOR is HIGH only when both are LOW; NAND is LOW only when both are HIGH, and the waveforms are drawn interval by interval from these rules.

← *Back to Q333*

Q334

What does the term 'LOS communication' mean? Name the types of waves that are used for this communication. Give typical examples, with the help of a suitable figure, of communication systems that use space mode propagation.

[2008]

SOLUTION

Line-of-Sight (LOS) communication: It is the mode of communication in which the transmitting and receiving antennas are in direct sight of each other, so that the electromagnetic wave travels in a nearly **straight line** from the transmitting antenna to the receiving antenna through the lower atmosphere (troposphere), without using reflection from the ionosphere or the ground.

Type of waves used: LOS communication uses **space waves**. These are the waves with frequency above about 40 MHz, that is, the VHF, UHF and microwave bands. Sky waves and ground waves are not suitable here because above 40 MHz the waves are not reflected back by the ionosphere (they pass through it) and ground-wave propagation suffers very heavy attenuation at these high frequencies.

Because the wave travels in a straight line, the range is limited by the curvature of the Earth. For a transmitting antenna of height h_T and a receiving antenna of height h_R, the maximum LOS range is

$$d_M = \sqrt{2Rh_T} + \sqrt{2Rh_R}$$

where R is the radius of the Earth.

Examples of systems using space-wave (LOS) propagation, with figure: Draw a curved Earth surface with a tall transmitting tower of height h_T on the left and a shorter receiving tower of height h_R on the right, and a straight arrow (the space wave) going from the top of the transmitting antenna to the top of the receiving antenna, just grazing the Earth's surface; mark this distance as d_M.

Typical examples are: (1) **Television broadcasting**, where tall towers radiate space waves to TV receivers in the surrounding region. (2) **Microwave links**, where repeater stations on towers or hilltops, spaced about 50 to 60 km apart, relay the signal from one LOS hop to the next. (3) **Satellite communication**, where the signal is sent up to a geostationary satellite and beamed back down, greatly extending the range.

Final answer. LOS communication means antennas in direct line of sight using space waves (above 40 MHz); examples are TV broadcasting, microwave links and satellite communication.

← *Back to Q334*

Q335

Draw a labeled circuit diagram of a full wave rectifier and briefly explain its working principle.

[2008]

SOLUTION

Circuit diagram of a full-wave rectifier (describe with labels): Use a **centre-tapped step-down transformer**. The primary is connected to the AC mains. The two ends of the secondary winding are connected to the anodes (P-sides) of two p-n junction diodes D_1 and D_2. The cathodes (N-sides) of both diodes are joined together. The load resistance R_L is connected between this common cathode point and the centre tap of the secondary. The rectified (output) DC voltage is taken across R_L.

Working principle: A p-n junction diode conducts current only when it is **forward biased** and blocks current when reverse biased. Because of the centre

tap, the two ends of the secondary are always at opposite polarity, so in each half cycle one diode is forward biased and the other is reverse biased.

During the positive half cycle of the input AC, suppose the top end of the secondary is positive. Then D_1 is forward biased and conducts, while D_2 is reverse biased and stays OFF. The current flows from D_1 through the load R_L and back to the centre tap.

During the negative half cycle, the bottom end of the secondary becomes positive. Now D_2 is forward biased and conducts, while D_1 is OFF. The current again flows through the load R_L in the **same direction** as before.

So during both halves of the AC input, the current through the load is always in one direction. The output is a **unidirectional pulsating DC**. Since the load receives a pulse in both halves, the output frequency is twice the input frequency,

$$f_{out}=2f_{in}$$

which makes the output smoother and easier to filter than a half-wave rectifier.

Final answer. A centre-tapped full-wave rectifier uses two diodes that conduct in alternate half cycles, sending current through $\mathbf{R_L}$ in the same direction throughout, giving a pulsating DC output of frequency $\mathbf{2f_{in}}$.

← *Back to Q335*

Q336

Draw a labeled circuit diagram of a transistor amplifier in the common-emitter configuration. Briefly explain, how the input/output signals differ in phase by 180°.

[2008]

SOLUTION

Circuit diagram of a common-emitter (CE) amplifier (describe with labels): Take an n-p-n transistor with the **emitter grounded** (common to both input and output). The base is given a forward DC bias through a base resistor R_B connected to the supply V_{BB}, which fixes the operating (quiescent) point. The collector is connected to the supply V_{CC} through the load resistor R_L, keeping

the collector-base junction reverse biased. The AC input signal is applied between base and emitter through an input coupling capacitor C_1. The amplified AC output is taken between collector and emitter through an output coupling capacitor C_2.

Why input and output are 180° out of phase: In the active region the collector current is

$$I_C = \beta I_B$$

and the output (collector-emitter) voltage is given by the output loop equation

$$V_{CE} = V_{CC} - I_C R_L$$

When the input signal goes through its **positive half**, the base-emitter voltage V_{BE} increases, so the base current I_B increases. Since $\mathbf{I_C = \beta I_B}$, the collector current I_C also increases. This makes the drop $I_C R_L$ larger, so from $\mathbf{V_{CE} = V_{CC} - I_C R_L}$ the output voltage V_{CE} **decreases**.

When the input goes through its **negative half**, V_{BE} falls, so I_B and hence I_C fall, the drop $I_C R_L$ becomes smaller, and V_{CE} **increases**.

So when the input rises the output falls and when the input falls the output rises. The output voltage always changes opposite to the input voltage, which means the input and output signals are **180° out of phase**.

Final answer. In a CE amplifier $\mathbf{V_{CE} = V_{CC} - I_C R_L}$, so a rise in input increases $\mathbf{I_C}$ and lowers $\mathbf{V_{CE}}$; the output is therefore **180°** out of phase with the input.

← *Back to Q336*

Q337

What is a digital signal? Explain the function of modem in data communication. Write two advantages of digital communication.

[2007]

SOLUTION

Digital signal: A digital signal is one that can take only a **few discrete (fixed) values**, usually just two levels denoted by 0 and 1, and these values occur at discrete instants of time. To produce it, the continuous analog message is first sampled, then quantised, and then coded into a stream of binary pulses (a sequence of 0s and 1s).

Function of a modem in data communication: The word **modem** stands for MOdulator-DEModulator. It is a device that allows two computers (or a computer and a remote server) to exchange digital data over a communication channel such as a telephone line.

At the transmitting end, the **modulator** part of the modem converts the digital data (bit stream) coming from the computer into a modulated analog signal that can travel over the channel.

At the receiving end, the **demodulator** part of the modem does the reverse: it takes the received analog signal and recovers the original digital data from it, then passes this data to the receiving computer.

So a single modem can do both modulation and demodulation, which is why it is fitted at both ends of the link.

Two advantages of digital communication:

(1) **Better noise immunity and error correction:** since only two levels (0 and 1) have to be distinguished, the signal can be regenerated cleanly at repeaters and error-correcting codes can recover bits lost to noise, so quality does not degrade the way it does in analog systems.

(2) **Easy multiplexing and secure transmission:** many signals can share one channel by time-division multiplexing, and the digital data can be easily encrypted (coded), making the communication both efficient and secure.

Final answer. A digital signal takes only discrete values (0 and 1); a modem modulates digital data into analog for transmission and demodulates it back at the receiver, and digital communication offers high noise immunity and easy, secure multiplexing.

← Back to Q337

Q338

Explain, with the help of a schematic diagram, the principle and working of a Light Emitting Diode. What criterion is kept in mind while choosing the semi-conductor material for such a device? Write two advantages of Light Emitting Diode over conventional incandescent lamps.

[2007]

SOLUTION

Light Emitting Diode (LED), principle: An LED is a heavily doped p-n junction diode that is operated under **forward bias** and converts electrical energy directly into light. Schematic diagram: show a p-n junction with the p-side connected to the positive terminal and the n-side to the negative terminal of a battery through a resistor, with light (photons) coming out from the junction region near the surface.

Working: Under forward bias, the potential barrier is lowered, so electrons are injected from the n-side and holes from the p-side across the junction. Near the junction these injected **electrons recombine with holes**. In each recombination, an electron falls from the conduction band to the valence band and the energy equal to the band gap E_g is released as a **photon** of light.

$$h\nu = E_g \Rightarrow \nu = \frac{E_g}{h}$$

The corresponding wavelength (and hence the colour) is

$$\lambda = \frac{hc}{E_g}$$

A typical forward voltage is about 1.8 V for red up to about 3 V for blue. As more current flows, more recombinations occur, so the LED glows brighter.

Criterion for choosing the material: The semiconductor must have a band gap E_g that lies in the **visible range, about 1.8 eV to 3.0 eV**, so that the emitted photons are visible light. Also it must be a **direct band-gap** semiconductor so that recombination is radiative (gives photons). This is why elemental Si and Ge (indirect gap, recombination gives heat not light) are not used, and direct-gap compound semiconductors such as GaAs, GaP and GaAsP are used.

Two advantages of an LED over a conventional incandescent lamp:

(1) **Low operating voltage and very low power consumption / high efficiency** (almost all the input energy goes into light rather than heat).

(2) **Very long operating life** (of the order of 10^5 hours) together with fast switching (it turns on and off almost instantly).

> **COMMON MISTAKE**
>
> *Si and Ge are indirect band-gap, so their recombination energy turns into heat, not light, which is exactly why they cannot be used for LEDs.*

Final answer. An LED is a forward-biased heavily-doped p-n junction in which electron-hole recombination emits photons of energy $\mathbf{E}_g$; a direct band-gap material with $\mathbf{E}_g$ of about 1.8 to 3.0 eV is chosen, and LEDs are more efficient and far longer-lived than incandescent lamps.

← *Back to Q338*

Q339

Draw the circuit diagram of a common emitter amplifier using n-p-n transistor. What is the phase difference between the input signal and output voltage? State two reasons why a common emitter amplifier is preferred over a common base amplifier.

[2007]

SOLUTION

Circuit diagram of a common-emitter amplifier using an n-p-n transistor (describe with labels): Take an n-p-n transistor with its **emitter grounded** (common terminal). The base is forward biased through a base resistor R_B from the supply V_{BB}, fixing the operating point. The collector is connected to the supply V_{CC} through the load resistor R_L, so the collector junction is reverse biased. The AC input signal is fed between base and emitter through an input coupling capacitor, and the amplified AC output is taken between collector and emitter through an output coupling capacitor.

Phase difference between input and output: It is **180°**. Reason: the output voltage is

$$V_{CE}=V_{CC}-I_C R_L$$

When the input signal goes positive, V_{BE} rises, so I_B rises and hence $\mathbf{I_C=\beta I_B}$ rises. The drop $I_C R_L$ then increases, so V_{CE} falls. Thus the output falls when the input rises, which means the input and output are in **antiphase** (180° out of phase).

Two reasons why a common-emitter amplifier is preferred over a common-base amplifier:

(1) **Much higher current and power gain:** in CE the current gain is $\beta=\frac{\Delta I_C}{\Delta I_B}$, typically 50 to 500, whereas in CB the current gain $\alpha=\frac{\Delta I_C}{\Delta I_E}$ is less than 1. Because of the large β, the CE amplifier also gives a much higher voltage gain and power gain.

(2) **Higher input resistance:** the CE configuration has a moderate input resistance (a few kΩ), so it does not load the source heavily and is easy to drive from ordinary audio sources, whereas the CB configuration has a very low input resistance (a few Ω) which loads the source badly. This makes the CE amplifier far more practical for amplification and for cascading.

Final answer. The CE amplifier output is **180°** out of phase with the input; it is preferred over the CB amplifier because of its much higher current/power gain (**β** = 50 to 500) and higher input resistance.

← *Back to Q339*

Q340

Explain the formation of energy bands in solids. Draw energy band diagram for (i) a conductor, (ii) an intrinsic semiconductor.

[2007]

SOLUTION

Formation of energy bands in solids: In an isolated single atom the electrons occupy sharp, **discrete energy levels**. When a large number N of identical atoms are brought close together to form a crystalline solid, the outer electrons of neighbouring atoms come close and their wavefunctions overlap. By Pauli's exclusion principle no two electrons can have exactly the same energy state, so each original atomic energy level is forced to **split into N very closely spaced sub-levels**.

Since N is enormous (about 10^{23} atoms per cm^3), these N sub-levels are spaced so finely that they merge into a **continuous range of allowed energies called an energy band**. Between two such allowed bands there can be a range of energies that no electron can have, called the **forbidden energy gap**.

The highest band that is completely or partly filled with electrons at 0 K is the **valence band (VB)**, and the next higher allowed band, which is normally empty, is the **conduction band (CB)**. The energy gap between the top of the valence band and the bottom of the conduction band is the **forbidden energy gap** E_g.

(i) Energy band diagram of a conductor: Draw the valence band and conduction band so that they **overlap** (or show the conduction band as partly filled). Here $\mathbf{E_g=0}$, so there are always free electrons available, and even a small applied field makes them move easily, giving high conductivity.

(ii) Energy band diagram of an intrinsic semiconductor: Draw a fully filled valence band below and a completely empty conduction band above, separated by a **small forbidden gap $E_g \approx 1$ eV** (about 1.1 eV for Si and 0.7 eV for Ge). At 0 K it behaves like an insulator, but as temperature rises some electrons gain enough thermal energy to jump across E_g into the conduction band, leaving holes in the valence band, so the material conducts moderately.

Final answer. Energy bands form when the discrete atomic levels of N closely packed atoms split and merge into continuous bands; in a conductor the valence and conduction bands overlap ($\mathbf{E_g=0}$), while in an intrinsic semiconductor they are separated by a small gap $\mathbf{E_g \approx 1}$ **eV**.

← *Back to Q340*

Q341

What is modulation? Explain the need of modulating a low frequency information signal. With the help of diagrams, differentiate between PAM and PDM.

[2007]

SOLUTION

Modulation is the process of impressing a low-frequency information (message) signal onto a high-frequency carrier wave by varying one of the carrier's properties (amplitude, frequency or phase) in accordance with the instantaneous value of the message signal.

Need for modulation:

(a) Reasonable antenna size: For efficient radiation, the antenna length should be at least $\frac{\lambda}{4}$. For an audio signal of $f \approx 5$ **kHz**,

$$\lambda = \frac{c}{f} = \frac{3\times 10^8}{5\times 10^3} = 6\times 10^4\ m = 60\ \text{km}$$

This would need an antenna about 15 km long, which is impractical. Modulating the signal onto a MHz carrier shrinks λ so the antenna size drops to a few metres.

(b) Effective power radiated: Power radiated by an antenna $\propto \frac{1}{\lambda^2}$, so a high-frequency (short λ) carrier radiates far more power than a low-frequency baseband signal.

(c) Multiplexing / avoiding signal mixing: Different transmitting stations can be assigned different carrier frequencies, so many signals can be sent over the same medium without overlapping.

Difference between PAM and PDM:

In **Pulse Amplitude Modulation (PAM)**, the carrier is a train of pulses of fixed width and fixed spacing; the amplitude of each pulse is varied in proportion to the instantaneous value of the message signal at the sampling instants. (Diagram: a row of vertical pulses of equal width but varying heights that trace the message waveform.)

In **Pulse Duration Modulation (PDM)**, the pulse amplitudes are kept constant; the width (duration) of each pulse is varied in proportion to the instantaneous value of the message signal. (Diagram: a row of pulses of equal height but varying widths, wider pulses where the message amplitude is larger.)

Final answer. Modulation impresses the message on a high-frequency carrier; it is needed for small antenna size, higher radiated power and multiplexing. PAM varies pulse amplitude (fixed width); PDM varies pulse width (fixed amplitude).

← *Back to Q341*

Q342

Write the acronym LASER in expanded form. State any four reasons for preferring diode lasers as light sources for optical communication links.

[2007]

SOLUTION

LASER stands for **L**ight **A**mplification by **S**timulated **E**mission of **R**adiation.

Four reasons why diode lasers are preferred as light sources for optical communication links:

(i) Highly directional, low-divergence beam: A laser emits a very narrow beam, so almost all the optical power can be coupled into the tiny core (~ 10 μm) of an optical fibre with little loss.

(ii) Highly monochromatic: The output has a very narrow spectral width, so chromatic dispersion in the fibre is minimal and signals can travel long distances without significant pulse broadening.

(iii) Highly coherent and intense: The high coherence and intensity give a strong, stable signal that can be detected reliably even after long propagation, increasing the range and clarity of communication.

(iv) Fast direct modulation: The output of a diode laser can be switched on and off (modulated) simply by varying its current at very high speed (gigabit-per-

second rates), matching the large bandwidth of optical fibres. It is also compact, rugged, low-power and easily driven by simple electronics.

Final answer. LASER = Light Amplification by Stimulated Emission of Radiation; diode lasers are preferred for being directional, monochromatic, coherent/intense and fast-modulating.

← *Back to Q342*

Q343

What is an intrinsic semiconductor? How can this material be converted into (i) P-type (ii) N-type extrinsic semiconductor? Explain with the help of energy band diagrams.

[2006]

SOLUTION

An **intrinsic semiconductor** is a pure semiconductor (free of any impurity) in which the number of free electrons in the conduction band exactly equals the number of holes in the valence band, generated by thermal excitation.

$$n_e = n_h = n_i$$

where n_i is the intrinsic carrier concentration. Examples are pure silicon (Si) and pure germanium (Ge). At 0 K it behaves as an insulator; as temperature rises, electron-hole pairs are thermally created and it conducts slightly.

(i) Conversion to P-type: A small amount of a **trivalent** impurity (B, Al, In or Ga) is added (doped) into the pure semiconductor. Each impurity atom forms only three covalent bonds with neighbouring Si atoms; the fourth bond is left incomplete, creating a vacancy called a **hole**. The impurity is called an **acceptor**.

Energy band diagram (P-type): An empty **acceptor level** E_A is created just above the top of the valence band (about 0.01 to 0.05 eV above E_V). At room temperature, electrons easily jump from the valence band into E_A, leaving behind holes in the valence band. Thus **holes are the majority carriers** and electrons the minority carriers.

(ii) Conversion to N-type: A small amount of a **pentavalent** impurity (P, As, Sb or Bi) is added. Four of its five valence electrons form covalent bonds with neighbouring Si atoms; the fifth electron is loosely bound and easily becomes a free electron in the conduction band. The impurity is called a **donor**.

Energy band diagram (N-type): A filled **donor level** E_D is created just below the bottom of the conduction band (about 0.01 to 0.05 eV below E_C). At room temperature, electrons from E_D easily move up into the conduction band. Thus **electrons are the majority carriers** and holes the minority carriers.

Final answer. An intrinsic semiconductor has $\mathbf{n_e = n_h = n_i}$; doping with a trivalent acceptor gives P-type (level $\mathbf{E_A}$ above $\mathbf{E_V}$, holes majority) and with a pentavalent donor gives N-type (level $\mathbf{E_D}$ below $\mathbf{E_C}$, electrons majority).

← *Back to Q343*

Q344

Define the term modulation. Name three different types of modulation used for a message signal using a sinusoidal continuous carrier wave. Explain the meaning of any one of these.

[2006]

SOLUTION

Modulation is the process of superposing a low-frequency message (baseband) signal onto a high-frequency sinusoidal carrier wave by varying one of the carrier's parameters in proportion to the instantaneous amplitude of the message signal.

For a sinusoidal carrier $\boldsymbol{c(t) = A_c \cos(\omega_c t + \phi)}$, the three types of modulation are:

(i) **Amplitude Modulation (AM)**: the amplitude A_c of the carrier is varied.

(ii) **Frequency Modulation (FM)**: the frequency ω_c of the carrier is varied.

(iii) **Phase Modulation (PM)**: the phase ϕ of the carrier is varied.

Explanation of Amplitude Modulation (AM): Here the amplitude of the carrier is varied in accordance with the message signal while its frequency stays

constant. Let the message be $m(t) = A_m\omega_m t$ and the carrier $c(t) = A_c\omega_c t$. The AM wave is

$$c_m(t) = (A_c + A_m\omega_m t)\omega_c t$$

$$c_m(t) = A_c\omega_c t + \mu A_c\omega_m t\,\omega_c t$$

$$c_m(t) = A_c\omega_c t + \frac{\mu A_c}{2}\cos(\omega_c - \omega_m)t + \frac{\mu A_c}{2}\cos(\omega_c + \omega_m)t$$

where $\mu = \frac{A_m}{A_c}$ is the **modulation index**. The AM wave thus consists of the carrier of frequency ω_c and two side bands at $(\omega_c - \omega_m)$ and $(\omega_c + \omega_m)$ which carry the information.

Final answer. Modulation superposes the message on a sinusoidal carrier; the three types are AM, FM and PM, with AM varying the carrier amplitude to give a carrier plus two side bands at $\omega_c \pm \omega_m$.

← *Back to Q344*

Q345

Draw a circuit diagram for use of NPN transistor as an amplifier in common emitter configuration. The input resistance of a transistor is 1000 Ω. *On changing its base current by* 10 μA, *the collector current increases by* 2 mA. *If a load resistance of* 5 kΩ *is used in the circuit, calculate:*
(i) the current gain
(ii) voltage gain of the amplifier

[2006]

SOLUTION

Circuit diagram (n-p-n transistor as a common-emitter amplifier): The emitter is grounded (common to input and output). The base is connected through a resistor R_B to the bias battery V_{BB}, with the AC input signal applied between base and emitter. The collector is connected through the load resistance $R_L = 5$ kΩ to the supply battery V_{CC}, and the amplified AC output is taken between

collector and emitter. Both junctions are biased so that the emitter-base junction is forward biased and the collector-base junction is reverse biased.

Given: input resistance $\mathbf{r_i = 1000\ \Omega}$, change in base current $\mathbf{\Delta I_B = 10\ \mu A = 10\times 10^{-6}\ A}$, change in collector current $\mathbf{\Delta I_C = 2\ mA = 2\times 10^{-3}\ A}$, load resistance $\mathbf{R_L = 5\times 10^3\ \Omega}$.

(i) Current gain:

$$\beta = \frac{\Delta I_C}{\Delta I_B}$$

$$\beta = \frac{2\times 10^{-3}}{10\times 10^{-6}}$$

$$\beta = 200$$

(ii) Voltage gain:

$$A_V = \beta \times \frac{R_L}{r_i}$$

$$A_V = 200 \times \frac{5000}{1000}$$

$$A_V = 200 \times 5 = 1000$$

So the **current gain is $\beta = 200$** and the **voltage gain is $A_V = 1000$** (strictly -1000, the negative sign showing the 180° phase difference between output and input).

Final answer. Current gain $\beta = 200$ and voltage gain $A_V = 1000$ (magnitude).

← *Back to Q345*

Q346

Define the term 'critical frequency' in relation to sky wave propagation of electromagnetic waves.
On a particular day, the maximum frequency reflected from the ionosphere is 10 MHz. *On another day, it was found to decrease to* 8 MHz. *Calculate the ratio of the maximum electron densities of the ionosphere on the two days.*

[2006]

SOLUTION

Critical frequency is the highest frequency of a radio wave, sent vertically upward, that can be reflected back to the Earth by a particular layer of the ionosphere. A wave of frequency higher than this critical frequency penetrates the layer and escapes into space instead of being reflected.

The critical frequency is related to the maximum electron density N_{max} of the layer by

$$f_c = 9\sqrt{N_{max}}$$

(with f_c in Hz and N_{max} in electrons per cubic metre). Hence

$$N_{max} \propto f_c^{\,2}$$

Given: $\mathbf{f_{c,1} = 10\ MHz}$ on the first day and $\mathbf{f_{c,2} = 8\ MHz}$ on the second day.

$$\frac{N_1}{N_2} = \frac{f_{c,1}^{\,2}}{f_{c,2}^{\,2}}$$

$$\frac{N_1}{N_2} = \left(\frac{10}{8}\right)^2$$

$$\frac{N_1}{N_2} = \left(\frac{5}{4}\right)^2 = \frac{25}{16}$$

So the ratio of the maximum electron densities on the two days is $\mathbf{N_1 : N_2 = 25 : 16 \approx 1.56 : 1}$.

COMMON MISTAKE

Since $\mathbf{N_{max} \propto f_c^{\,2}}$, never take the ratio of the frequencies directly; always square them.

Final answer. Critical frequency is the highest vertically-incident frequency reflected by an ionospheric layer; the ratio of maximum electron densities is $\mathbf{N_1 : N_2 = 25 : 16}$.

← Back to Q346

Q347

Give reasons for the following:
(i) *Long distance radio broadcasts use short-wave bands.*
(ii) *The small ozone layer on top of the stratosphere is crucial to human survival.*
(iii) *Satellites are used for long distance TV transmission.*

[2005]

SOLUTION

(i) Long-distance radio broadcasts use short-wave bands because short waves (about 3 MHz to 30 MHz) are reflected back to the Earth by the ionosphere. Through repeated reflections between the ionosphere and the Earth's surface (**sky-wave propagation**), the signal travels around the curved Earth and covers very large distances with modest transmitter power. Frequencies above about 30 MHz penetrate the ionosphere and are lost, while much lower frequencies are heavily absorbed near the ground.

(ii) The thin ozone layer at the top of the stratosphere is crucial because ozone (O_3) molecules strongly absorb the harmful ultraviolet radiation of the Sun (the UV-B and UV-C bands, roughly 200 to 290 nm). If this radiation reached the Earth's surface unfiltered, it would cause severe damage to living tissue, such as skin cancer in humans, destruction of plankton and crop failure, threatening most forms of life. Hence the ozone layer is essential to human and biological survival.

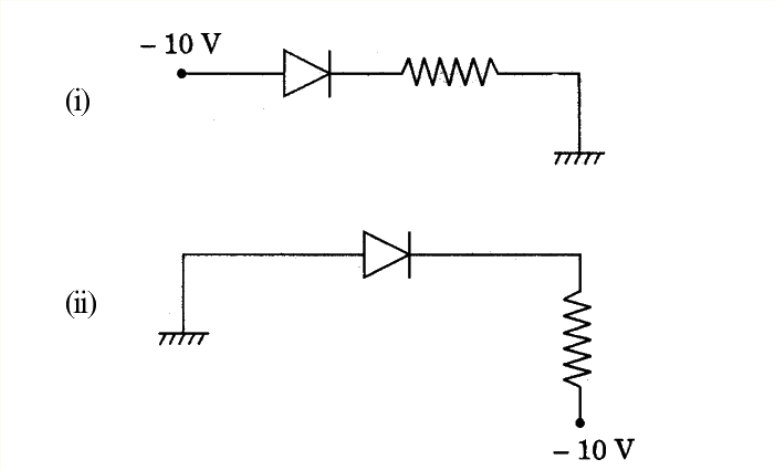
(iii) Satellites are used for long-distance TV transmission because TV signals have frequencies above about 100 MHz, which are not reflected by the ionosphere and travel in straight lines (line-of-sight). A ground transmitter can therefore reach only about 100 km. A geostationary satellite high above the Earth receives the signal from one ground station, amplifies it and re-transmits (relays) it back over a very large area, enabling TV coverage across continents.

Final answer. Short waves reflect off the ionosphere for global reach; the ozone layer absorbs harmful UV; satellites relay line-of-sight TV signals beyond the horizon.

← *Back to Q347*

Q348

Explain, with the help of a circuit diagram, how the thickness of depletion layer in a p-n junction diode changes when it is forward biased. In the following circuit which one of the two diodes is forward biased and which is reverse biased?



[2005]

SOLUTION

Change in depletion-layer thickness under forward bias: Circuit diagram: the **p-side** of the diode is connected to the **positive** terminal and the **n-side** to the **negative** terminal of the battery, with a series resistor and the supply voltage applied across the junction.

The applied forward voltage opposes the built-in potential barrier. It pushes the majority carriers, holes from the p-side and electrons from the n-side, towards the junction. These carriers move into the depletion region and neutralise some of the fixed (immobile) ions there.

As a result, the **depletion layer becomes thinner** and the height of the potential barrier is **reduced**. The junction resistance falls and a large current flows easily across the diode.

Identification of bias in the given circuit: With the −10 V supply, a diode is forward biased when its p-side is at the higher (more positive) potential and its n-side at the lower potential. In the figure, the diode whose p-side is connected towards the 0 V (ground) side is **forward biased**, and the diode whose p-side is connected towards the −10 V end is **reverse biased**.

Final answer. Under forward bias the depletion layer becomes thinner and the barrier is lowered; in the circuit, the diode with its p-side toward 0 V is forward biased and the one with its p-side toward **−10** V is reverse biased.

← *Back to Q348*

Q349

Distinguish between analog and digital communication. Write any two modulation techniques employed for the digital data. Describe briefly one of the techniques used.

[2005]

SOLUTION

Distinction between analog and digital communication:

In **analog communication**, the signal (such as voltage or current) varies continuously with time and can take any value in a range. It is sensitive to noise, and once noise is added it cannot be fully removed.

In **digital communication**, the signal takes only discrete values (usually two levels representing the binary digits 0 and 1). It is far more noise-tolerant, the bits can be regenerated cleanly at repeaters, and error detection and correction are possible, which makes it well suited to large data transfer.

Two modulation techniques for digital data:

(i) **Amplitude Shift Keying (ASK)**: the carrier amplitude is switched between two fixed levels for binary 1 and 0.

(ii) **Frequency Shift Keying (FSK)**: the carrier frequency is switched between two fixed values f_1 and f_2 for binary 1 and 0. (Phase Shift Keying, PSK, is also used.)

Brief description of FSK: In Frequency Shift Keying, the amplitude of the carrier is kept constant and only its frequency is changed according to the digital data. When the message bit is 1, the modulator transmits a sinusoid of frequency f_1; when the bit is 0, it transmits a sinusoid of frequency f_2. At the receiver, two band-pass filters tuned to f_1 and f_2, followed by a decision circuit,

recover the original bits. Because the information is carried by frequency and not amplitude, FSK is highly robust against amplitude noise.

Final answer. Analog signals vary continuously and are noise-prone; digital signals are discrete (0/1) and noise-tolerant; two digital modulation techniques are ASK and FSK, with FSK encoding bits as two distinct carrier frequencies.

← *Back to Q349*

Q350

Draw a schematic diagram of a single optical fibre structure. Explain briefly how an optical fibre is fabricated. Describe in brief, the mechanism of propagation of light signal through an optical fibre.

[2005]

SOLUTION

Schematic structure of a single optical fibre: A central cylindrical **core** of glass or plastic of higher refractive index n_1 (about 1.50) is surrounded coaxially by a **cladding** of slightly lower refractive index n_2 (about 1.48). Over the cladding is a protective polymer **buffer/jacket**. (Diagram: three concentric cylinders labelled, from inside out, core (n_1), cladding (n_2) and protective jacket.)

Fabrication: A solid glass **preform** is first prepared with the required refractive-index profile (the core slightly higher than the cladding), usually by vapour deposition or a rod-in-tube method. This preform is then heated in a drawing furnace and gently pulled (drawn) into a very thin, long fibre under carefully controlled tension. As the fibre cools, a polymer coating is applied for protection, and the finished fibre is wound onto a drum.

Propagation of light signal (mechanism): Light propagates through the fibre by **total internal reflection**. A light ray entering the core within the acceptance angle strikes the core-cladding boundary at an angle of incidence greater than the critical angle

$$\theta_c = \sin^{-1}\left(\frac{n_2}{n_1}\right)$$

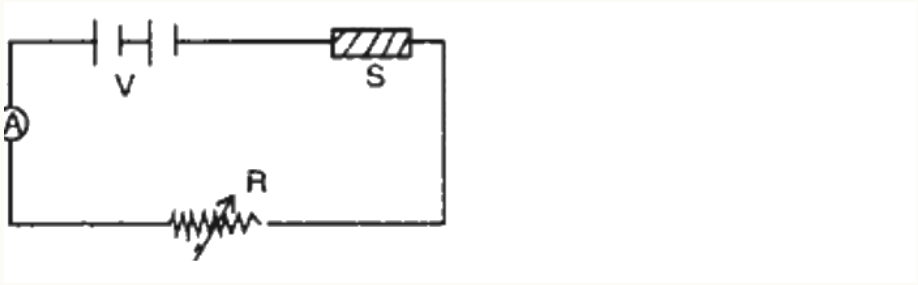
Since $n_1 > n_2$, the ray suffers total internal reflection back into the core. By repeated total internal reflections at the core-cladding interface, the ray zig-zags along the length of the fibre and reaches the far end with very little loss, even when the fibre is bent (within a permissible minimum radius).

Final answer. An optical fibre has a high-index core in a low-index cladding with a protective jacket; it is fabricated by drawing a heated glass preform into a thin fibre, and light travels by repeated total internal reflection ($\theta_c = \sin^{-1}(n_2/n_1)$).

← *Back to Q350*

Q351

The diagram shows a piece of pure semiconductor, S in series with a variable resistor R, and a source of constant voltage V. Would you increase or decrease the value of R to keep the reading of ammeter (A) constant, when semiconductor S is heated? Give reason.



[2004]

SOLUTION

A pure (intrinsic) semiconductor has a negative temperature coefficient of resistance. On heating, more covalent bonds break, so the number of electron-hole pairs increases almost exponentially. This greatly increases the number of charge carriers, and hence the resistance R_S of the semiconductor S decreases.

The circuit is a simple series loop, so by Ohm's law the ammeter current is

$$I = \frac{V}{R + R_S}$$

Here V is constant. For the reading I to stay constant, the total series resistance $(R + R_S)$ must also stay constant.

When S is heated, R_S decreases. To keep $(R + R_S)$ unchanged, the external variable resistance R must be raised by exactly the same amount.

Hence the value of R should be **increased**. This compensates for the fall in the semiconductor's resistance on heating and keeps the current through the ammeter constant.

CONCEPT

Intrinsic semiconductors get LESS resistive when heated, unlike metals which get more resistive. This is the give-away in the question.

Final answer. R should be increased, because heating lowers the semiconductor's resistance and R must rise to keep the total resistance (and hence the current) constant.

← *Back to Q351*

Q352

Draw the circuit diagram of a common-emitter amplifier using an n-p-n transistor. Draw the input and output waveforms of the signal. Write the expression for its voltage gain.

[2004]

SOLUTION

Circuit diagram (describe with labels): An n-p-n transistor is connected in the common-emitter mode. The emitter is connected to the common (earth) line. The base is connected to the positive supply through a biasing resistor R_B, and the AC input signal v_i is fed to the base through a coupling capacitor C_1. The collector is connected to the supply V_{CC} through the collector (load) resistor R_C (or R_L). The amplified output v_o is taken from the collector with respect to the emitter through a coupling capacitor C_2. The base-emitter junction is forward biased and the base-collector junction is reverse biased to fix the operating point.

Working: When the AC signal v_i is applied at the base, it adds to and subtracts from the steady base bias, so the base-emitter voltage V_{BE} varies. A small change ΔV_{BE} causes a relatively large change in the base current ΔI_B.

The collector current follows as $\mathbf{I_C = \beta I_B}$, so it changes by a large amount. The output voltage is

$$V_{CE} = V_{CC} - I_C R_C$$

As I_C rises, V_{CE} falls and vice versa, so the output is **amplified and reversed in phase** with respect to the input.

Input and output waveforms (describe): The input v_i is a small sine wave centred on the bias point. The output v_o is a much larger sine wave, but it is **180° (half a cycle) out of phase** with the input. So when the input is at its positive peak, the output is at its negative peak.

Expression for voltage gain:

$$A_V = \frac{v_o}{v_i} = -\frac{\beta_{ac} R_L}{r_i}$$

where β_{ac} is the small-signal (AC) current gain, R_L is the load resistance and r_i is the input resistance. The negative sign shows the 180° phase reversal between output and input.

Final answer. The CE amplifier gives an output 180 degrees out of phase with the input, with voltage gain $\mathbf{A_V = -\beta_{ac} R_L / r_i}$.

← *Back to Q352*

Q353

Write the symbol and truth table of an AND gate. Explain how this gate is realised in practice by using two diodes.

[2004]

SOLUTION

Symbol (describe): The AND gate is drawn as a flat-backed, round-fronted 'D' shape. Two input lines *A* and *B* enter at the flat (left) side and one output line *Y*

leaves from the curved (right) side. Its Boolean expression is

$$Y = A \cdot B$$

Truth table:

$$A = 0, B = 0 \Rightarrow Y = 0$$

$$A = 0, B = 1 \Rightarrow Y = 0$$

$$A = 1, B = 0 \Rightarrow Y = 0$$

$$A = 1, B = 1 \Rightarrow Y = 1$$

So the output is HIGH (1) only when both inputs are HIGH; otherwise it is LOW (0).

Realisation using two diodes (describe circuit): Take a +5 V supply. Connect it to the output node Y through a pull-up resistor R. From node Y, connect two diodes D_A and D_B with their anodes at Y and their cathodes going to the two inputs A and B respectively. The inputs A and B are either at 0 V (LOW) or +5 V (HIGH).

Operation: If any input is LOW (0 V), the corresponding diode becomes forward biased and conducts. Current flows from +5 V through R and that diode to ground, so node Y is pulled down close to 0 V. Hence $\boldsymbol{Y = 0}$.

Only when **both inputs are HIGH** (+5 V), both diodes are reverse biased and do not conduct. No current flows through R, so there is no voltage drop across it and node Y stays at +5 V, i.e. $\boldsymbol{Y = 1}$.

This output behaviour exactly matches the AND truth table, so the two-diode circuit realises the **AND gate**.

Final answer. Y = A·B; the gate is HIGH only when both inputs are HIGH, realised by two diodes whose anodes meet at the output pulled up to +5 V through R.

← Back to Q353

Q354

With the help of a block diagram, explain the principle of an optical communication system. Give its two advantages over cable communication system.

Or

With the help of relevant diagrams, explain the following terms:

(i) Pulse-position modulation (PPM)

(ii) Pulse-duration modulation (PDM)

[2004]

SOLUTION

Block diagram of an optical communication system (describe in order):

Message (information) signal → Coder / Modulator → Optical source (LED or laser diode) → Optical fibre channel → Optical detector (photodiode) → Demodulator / Decoder → User (output).

Principle and working: The electrical message signal is first coded and modulated. The modulator drives an optical source (an LED or a semiconductor laser diode), which converts the electrical signal into a stream of light pulses. These light pulses travel along the core of the optical fibre by repeated **total internal reflection** at the core-cladding boundary, with very little loss.

At the receiving end, an optical detector (a photodiode) intercepts the light pulses and converts them back into an electrical signal. This is then demodulated and decoded to recover the original message, which is delivered to the user.

Two advantages over a cable (copper) communication system:

(i) **Very large bandwidth.** Optical carrier frequencies are of the order of 10^{14} Hz, so a single fibre can carry millions of telephone channels at once. Copper cables are limited to a few MHz.

(ii) **Very low loss and immunity to interference.** Fibre losses are only about 0.2 dB/km, allowing repeaters to be widely spaced. Being a dielectric, the fibre is unaffected by external electromagnetic fields, lightning and cross-talk, and is also secure against eavesdropping.

OR

(i) Pulse-position modulation (PPM): The carrier is a train of fixed-amplitude, fixed-width pulses. The **position (time of occurrence)** of each pulse within its time slot is shifted in proportion to the instantaneous value of the modulating signal, while the pulse height and width stay constant. (Diagram: a row of identical narrow pulses whose horizontal positions shift earlier or later as the message amplitude changes.)

(ii) Pulse-duration modulation (PDM), also called pulse-width modulation: The **width (duration)** of each pulse is varied in proportion to the instantaneous value of the modulating signal, while the pulse amplitude and starting position stay constant. (Diagram: a row of pulses of equal height but unequal widths, the widest pulse occurring where the message amplitude is largest.)

Final answer. An optical system sends modulated light pulses through a fibre by total internal reflection; key edge over cable is huge bandwidth (~10^14 Hz) and very low loss (~0.2 dB/km).

← *Back to Q354*

Q355

Name the device used for data transmission from one computer to another. Justify the name. Using this device draw the block diagram for data communication and explain it briefly.

[2004]

SOLUTION

Device: The device used for data transmission from one computer to another is a **MODEM**.

Justification of the name: The word MODEM is short for **MOdulator-DEModulator**. At the sending computer it acts as a modulator, converting the digital data (the 1s and 0s) into modulated analog signals (tones) that can travel over a telephone or similar line. At the receiving computer it acts as a demodulator, converting the analog signals back into the original digital data. So the device both modulates and demodulates, giving the name MODEM.

Block diagram (describe in order):

Computer A → Modem A → Telephone line / channel → Modem B → Computer B.

Explanation: Computer A sends out a digital bit stream. Modem A (acting as a modulator) converts this bit stream into modulated analog tones and feeds them onto the telephone channel. The tones travel along the channel to the far end. Modem B (acting as a demodulator) receives the analog tones and converts them back into the original digital bit stream, which it delivers to computer B.

Since each modem can both modulate (transmit) and demodulate (receive), the link can carry data in both directions, allowing **full-duplex** communication between the two computers.

Final answer. A MODEM (MOdulator-DEModulator) is used; it modulates digital data to analog at the sender and demodulates analog back to digital at the receiver.

← *Back to Q355*

Q356

Define Antenna. Write a short note on Antenna. Name two types of antenna.

[2003]

SOLUTION

Definition: An **antenna** is a device (a transducer) that converts a high-frequency electrical signal (current or voltage) into an electromagnetic wave and radiates it into space, or conversely intercepts an electromagnetic wave from space and converts it back into an electrical signal.

Short note: For an antenna to radiate (or receive) efficiently, its physical size must be comparable to the wavelength λ of the signal, typically of the order of $\lambda/4$ or $\lambda/2$.

At low (audio) frequencies the wavelength is several kilometres long, so the required antenna would be impossibly large. This is one reason why the message must first be modulated onto a high-frequency carrier, whose short wavelength needs only a short, practical antenna. An antenna is also

directional, meaning it radiates or receives more strongly in some directions than others, and its gain depends on direction.

Two types of antenna:

(i) **Half-wave dipole antenna** : a straight conductor of length $\lambda/2$, fed at its centre; widely used in radio and TV.

(ii) **Yagi-Uda antenna** : an array of dipoles (a driven element with a reflector and one or more directors) giving high directional gain; commonly seen as a rooftop TV aerial.

Final answer. An antenna is a transducer between electrical signals and EM waves, sized near $\lambda/4$ to $\lambda/2$; two types are the half-wave dipole and the Yagi-Uda antenna.

← Back to Q356

Q357

What is meant by the term 'modulation'? Explain with the help of a block diagram, how the process of modulation is carried out in radio broadcasts.

[2003]

SOLUTION

Meaning of modulation: **Modulation** is the process of superimposing a low-frequency information (message) signal onto a high-frequency carrier wave, by varying one of the carrier's properties (its amplitude, frequency or phase) in step with the instantaneous value of the message signal.

Why modulation is needed:

(i) Audio signals (20 Hz to 20 kHz) have very long wavelengths, so they would need impractically long antennas (size ~ $\lambda/4$, several km).

(ii) Without modulation all stations would transmit in the same low-frequency band and overlap, so they could not be separated.

(iii) The range and strength of directly radiated audio signals would be very small.

Block diagram of an AM radio transmitter (describe in order):

Microphone → Audio Amplifier → Modulator → RF Power Amplifier → Transmitting Antenna, with an RF Carrier Oscillator feeding into the Modulator.

Explanation: The microphone converts sound into a weak audio (electrical) signal. The audio amplifier raises its strength. Meanwhile a high-frequency carrier oscillator produces a steady RF carrier wave. Both the amplified audio signal and the carrier are fed into the **modulator**, which produces an amplitude-modulated wave (the carrier plus two side bands). This modulated wave is boosted by the RF power amplifier and radiated into space by the transmitting antenna.

Final answer. Modulation superimposes the message on a high-frequency carrier; in AM radio the audio is amplified and combined with an RF carrier in a modulator, then radiated by the antenna.

← *Back to Q357*

Q358

Write three special characteristics of the light source used in optical communication. Name any one optical detector. Explain the meaning of the term 'sensitivity' and 'responsivity' of a detector.

[2003]

SOLUTION

Three special characteristics of the light source used in optical communication:

(i) **Narrow spectral width (near monochromatic / coherent output)** : the source must emit a very narrow band of wavelengths, so that pulse broadening due to material dispersion in the fibre is kept small.

(ii) **High intensity and good directionality** : most of the emitted optical power should be concentrated and easily coupled into the tiny core of the fibre.

(iii) **Fast switching / high modulation bandwidth** : the source must be able to be switched on and off (modulated) very rapidly, at GHz rates, without

distorting the pulses.

(LEDs and semiconductor laser (injection) diodes meet these needs, with laser diodes preferred for long-distance high-bandwidth links.)

One optical detector: a **p-i-n photodiode** (a type of p-n junction photodiode).

Sensitivity: The **sensitivity** of a detector is the minimum optical power it can detect for a specified signal-to-noise ratio at the output. A more sensitive detector can respond to fainter (weaker) optical signals.

Responsivity: The **responsivity** is the ratio of the photocurrent produced to the incident optical power,

$$R = \frac{I_{ph}}{P_{opt}}$$

measured in A/W. A higher responsivity means a larger electrical signal is produced for the same amount of incident light.

Final answer. The source needs narrow spectral width, high directional intensity and fast switching; a p-i-n photodiode detects light, with sensitivity = least detectable power and responsivity R = I_ph/P_opt (A/W).

← *Back to Q358*

Q359

Sunil and his parents were travelling to their village in their car. On the way his mother noticed some grey coloured panels installed on the roof of a low building. She enquired from Sunil what those panels were and Sunil told his mother that those were solar panels.
(a) What were the values displayed by Sunil and his mother? State one value for each.
(b) In what way would the use of solar panels prove to be very useful?
(c) Name the semiconductor device used in solar panels. Briefly explain with the help of a diagram, how this device works.

[2017]

SOLUTION

(a) Values displayed:

Sunil showed **scientific awareness and a helpful, sharing attitude** (he could identify the panels and explained them to his mother).

His mother showed **curiosity and a willingness to learn** (she observed the panels and enquired about them).

(b) Usefulness of solar panels: Solar panels use sunlight, a **renewable, clean and free** source of energy. During operation they produce no greenhouse gases or pollution. They have a long working life (about 25 years), need very little maintenance, reduce dependence on fossil fuels and the grid, and can supply power in remote, off-grid places such as villages and satellites.

(c) Device used: the semiconductor device used in a solar panel is the **solar cell**, which is a p-n junction diode.

Diagram (describe with labels): A solar cell has a thin, transparent top layer of p-type semiconductor and a thicker n-type layer below it, with the p-n junction between them. Thin metal 'finger' contacts are placed on the top surface (to let light in) and a full metal contact covers the back. Light falls on the top surface, and an external load is connected across the two contacts.

Working: When light of photon energy $\mathbf{h\nu > E_g}$ (the band gap) falls on the junction, it is absorbed and creates **electron-hole pairs** near the depletion region.

The built-in electric field of the depletion region separates these carriers: it pushes the **electrons towards the n-side** and the **holes towards the p-side**.

This separation makes the n-side negative and the p-side positive, producing a photo-voltage across the junction. When an external load is connected, a **photocurrent** flows through it, delivering electrical power. Thus the solar cell converts light energy directly into electrical energy.

Final answer. Sunil showed scientific awareness, his mother showed curiosity; solar panels give clean renewable power, using a solar cell (p-n junction) that converts light into electricity via electron-hole pair generation.

← *Back to Q359*

Q360

(a) (i) A germanium crystal is doped with antimony. With the help of energy-band diagram, explain how the conductivity of the doped crystal is affected.
(ii) Briefly explain the two processes involved in the formation of a p-n junction.
(iii) What will the effect of (1) forward biasing, and (2) reverse biasing be on the width of depletion layer in a p-n junction diode?
OR
(b) (i) With the help of a circuit diagram, briefly explain the working of a full-wave rectifier using p-n junction diodes.
(ii) Draw V-I characteristics of a p-n junction diode. Explain how these characteristics make a diode suitable for rectification.
(iii) Carbon and silicon have the same lattice structure. Then why is carbon an insulator but silicon a semiconductor?

[2023 • Set 55-4-1]

SOLUTION

(a)(i) Germanium doped with antimony (n-type): Antimony (Sb) is a pentavalent atom with 5 valence electrons. When it replaces a Ge atom, four of its electrons form covalent bonds with neighbouring Ge atoms, and the **fifth electron is very loosely bound** to the Sb atom.

Energy-band diagram (describe): On the band diagram, this loosely bound electron creates a **donor energy level E_D** lying just below the bottom of the conduction band E_C, the gap being only about 0.045 eV.

At room temperature (300 K), the available thermal energy **kT ≈ 0.025 eV** is enough to lift almost all these donor electrons from E_D into the conduction band. This greatly increases the number of free electrons (the majority carriers), so the **conductivity of the doped crystal increases sharply** compared with pure germanium.

(a)(ii) Two processes in the formation of a p-n junction:

Diffusion: Because of the large difference in carrier concentration, holes diffuse from the p-side to the n-side and electrons diffuse from the n-side to the p-side. This leaves behind immobile negative ions on the p-side and immobile positive ions on the n-side, forming the depletion region.

Drift: The uncovered ions set up a built-in electric field (from n-side to p-side). This field causes a drift of minority carriers in the opposite direction to diffusion. In equilibrium, the drift current just balances the diffusion current.

(a)(iii) Effect on the width of the depletion layer:

(1) Forward bias: the applied voltage opposes the built-in barrier, so the barrier is reduced and the **depletion layer width decreases**.

(2) Reverse bias: the applied voltage adds to the built-in barrier, so the barrier is raised and the **depletion layer width increases**.

OR

(b)(i) Full-wave rectifier (describe circuit): A centre-tapped secondary transformer is used with two p-n junction diodes D_1 and D_2. The two ends of the secondary connect to the anodes of D_1 and D_2; their cathodes are joined and connected to one end of the load resistor R_L, whose other end goes to the centre tap of the transformer.

Working: During one half-cycle, the top end is positive, so D_1 is forward biased and conducts while D_2 is reverse biased. During the next half-cycle the polarity reverses, so D_2 conducts while D_1 is off. In both half-cycles the current passes through R_L in the **same direction**, giving a unidirectional (rectified) output. The output is a series of pulses whose frequency is **twice** the input frequency.

(b)(ii) V-I characteristics (describe graph): Plot diode current I (vertical) against applied voltage V (horizontal). In **forward bias** (right side), the current stays nearly zero until the knee (threshold) voltage and then rises steeply, showing very low forward resistance. In **reverse bias** (left side), only a tiny constant reverse saturation current flows until breakdown, showing very high reverse resistance.

Suitability for rectification: A diode conducts strongly in forward bias but almost not at all in reverse bias, i.e. it allows current in essentially **one direction only**. This one-way property is exactly what converts alternating current into direct current, making the diode suitable for rectification.

(b)(iii) Carbon vs silicon: Carbon (diamond) and silicon have the same diamond-type lattice and four valence electrons each. But the carbon atom is much smaller, so its covalent bonds are shorter and stronger and its valence electrons are held more tightly. This makes the **energy band gap of carbon**

(diamond) very large, about 5.5 eV, so almost no electrons can reach the conduction band; carbon is an insulator. Silicon's larger atomic size gives a much smaller band gap (about 1.1 eV), so a useful number of electrons are thermally excited into the conduction band, making silicon a semiconductor.

> **TIP**
>
> Diamond (carbon) $E_g \approx 5.5$ eV makes it an insulator; silicon $E_g \approx 1.1$ eV makes it a semiconductor. Quoting both band-gap values bags the full mark here.

Final answer. Sb doping raises Ge's conductivity via donor levels near E_C; a junction forms by diffusion and drift; forward bias narrows and reverse bias widens the depletion layer (OR full-wave rectifier, diode V-I, and the large band gap of carbon vs small gap of silicon).

← *Back to Q360*

Q361

(a) Draw the circuit arrangement for studying V-I characteristics of a p-n junction diode in (i) forward biasing and (ii) reverse biasing. Draw the typical V-I characteristics of a silicon diode. Describe briefly the following terms: (i) minority carrier injection in forward biasing and (ii) breakdown voltage in reverse biasing.
OR
(b) Name two important processes involved in the formation of a p-n junction diode. With the help of a circuit diagram, explain the working of junction diode as a full wave rectifier. Draw its input and output waveforms. State the characteristic property of a junction diode that makes it suitable for rectification.

[2023 • Set 55-5-1]

SOLUTION

(a) V-I characteristics of a p-n junction diode.

Forward-bias circuit: The p-side of the diode is connected to the positive terminal of a variable DC battery and the n-side to the negative terminal. A milliammeter is in series with the diode and a voltmeter is connected in

parallel across the diode to read the junction voltage. A rheostat (potential divider) lets you vary the applied voltage from zero upward.

Reverse-bias circuit: The same diode is now turned around, so the p-side goes to the negative terminal and the n-side to the positive terminal of the battery. A microammeter (because the current is very small) is in series and a voltmeter across the diode reads the reverse voltage.

Typical V-I curve of a silicon diode: In the forward region the current is almost zero until the applied voltage reaches the knee (threshold) voltage $\mathbf{V_T \approx 0.7\ V}$ for silicon, after which the current rises very steeply. In the reverse region only a tiny saturation current of the order of a few μA flows and stays nearly constant, until at the breakdown voltage the reverse current suddenly shoots up. The forward part is drawn in the first quadrant (current in mA) and the reverse part in the third quadrant (current in μA).

(i) Minority carrier injection in forward biasing: In forward bias the applied voltage lowers the height of the potential barrier and narrows the depletion region. This lets the majority carriers diffuse across the junction, electrons from the n-side move into the p-side and holes from the p-side move into the n-side. Once a carrier crosses over, it becomes a minority carrier on the other side. This crossing over of carriers, increasing the minority-carrier concentration beyond its equilibrium value on each side, is called **minority carrier injection**, and it is responsible for the large forward current.

(ii) Breakdown voltage in reverse biasing: As the reverse voltage is increased, a value is reached at which the reverse current suddenly increases by a very large amount. This reverse voltage is called the **breakdown voltage** V_{br}. Near breakdown a small change in voltage produces a very large change in current. As long as the current is limited by an external resistor, the diode is not damaged.

Final answer. The forward curve has a knee at $\mathbf{V_T \approx 0.7\ V}$; minority carrier injection raises minority carrier concentration in forward bias, and the breakdown voltage is the reverse voltage at which the reverse current rises sharply.

← *Back to Q361*

Q362

(a) Draw a circuit diagram of an n-p-n transistor with its emitter-base junction forward biased and base-collector junction reverse biased. Briefly describe its working.
(b) Explain how a transistor in its active state exhibits a low resistance at its emitter-base junction and high resistance at its base-collector junction.
(c) Derive the expression for the voltage gain of a transistor amplifier in CE configuration in terms of the load resistance R_L, current gain β, and input resistance. Explain why input and output voltages are in opposite phase.

OR

(a) Write the important considerations which are to be taken into account while fabricating a p-n junction diode to be used as a Light Emitting Diode (LED). What should be the order of band gap of an LED, if it is required to emit light in the visible range? Draw a circuit diagram and explain its action.
(b) Draw the V-I characteristics of an LED. State two advantages of LED lamps over conventional incandescent lamps.

[2019 • Set 55-3-1]

SOLUTION

(a) n-p-n transistor: circuit and working.

Draw an n-p-n transistor with three regions: emitter (n), base (p, thin and lightly doped), collector (n). The emitter-base junction is forward biased by a battery V_{EE} so that $\mathbf{V_{BE} > 0}$ (positive terminal to the p-type base). The base-collector junction is reverse biased by a battery V_{CC} so that $\mathbf{V_{CB} > 0}$ (positive terminal to the n-type collector).

Working: Because the emitter-base junction is forward biased, the large number of electrons from the heavily doped emitter are injected into the thin, lightly doped base. The base is so thin and lightly doped that only a small fraction (about 5%) of these electrons recombine with holes there, giving a small base current I_B. The remaining electrons are swept into the collector by the strong reverse-bias field of the collector-base junction, giving the collector current I_C. By Kirchhoff's law $\mathbf{I_E = I_B + I_C}$, and since I_B is very small, $\mathbf{I_C \approx I_E}$.

(b) Low and high resistance: The emitter-base junction is forward biased, so its depletion layer is narrow and a large carrier injection occurs for a small applied voltage. Therefore this junction offers a **low resistance** (a few ohms). The base-

collector junction is reverse biased, so its depletion layer is wide and only a very small minority-carrier current flows for a large applied voltage. Therefore this junction offers a **high resistance** (of the order of mega ohms).

(c) Voltage gain of CE amplifier:

A small change in input voltage ΔV_{in} across the input resistance r_i produces a base current change

$$\Delta I_B = \frac{\Delta V_{in}}{r_i}$$

The resulting change in collector current is

$$\Delta I_C = \beta \Delta I_B$$

This collector current flows through the load resistance R_L, so the change in output voltage is

$$\Delta V_{out} = \Delta I_C \times R_L = \beta \Delta I_B R_L$$

The voltage gain is therefore

$$A_v = \frac{\Delta V_{out}}{\Delta V_{in}} = \frac{\beta \Delta I_B R_L}{\Delta I_B r_i} = \beta \frac{R_L}{r_i}$$

Phase relation: The output is taken as $\mathbf{V_{CE} = V_{CC} - I_C R_L}$. When the input voltage rises, I_B and hence I_C rise, so $I_C R_L$ increases and V_{CE} falls. When the input falls, V_{CE} rises. Hence the **output voltage is 180° out of phase with the input voltage**.

CONCEPT

In CE configuration the same β that gives current gain also multiplies the resistance ratio R_L/r_i to give voltage gain, so a large gain needs $R_L \gg r_i$.

Final answer. The CE voltage gain is $\mathbf{A_v = \beta \frac{R_L}{r_i}}$, with output and input **180°** out of phase.

← *Back to Q362*

Q363

(a) State briefly the processes involved in the formation of p-n junction explaining clearly how the depletion region is formed.
(b) Using the necessary circuit diagrams, show how the V-I characteristics of a p-n junction are obtained in
(i) Forward biasing
(ii) Reverse biasing
How are these characteristics made use of in rectification ?
OR
(a) Differentiate between three segments of a transistor on the basis of their size and level of doping.
(b) How is a transistor biased to be in active state ?
(c) With the help of necessary circuit diagram, describe briefly how n-p-n transistor in CE configuration amplifies a small sinusoidal input voltage. Write the expression for the ac current gain.

[2014]

SOLUTION

(a) Formation of a p-n junction and the depletion region.

A p-n junction is formed when a single semiconductor crystal is doped so that one part is p-type and the adjoining part is n-type. Two processes occur at the junction:

Diffusion: Because the n-side has a high concentration of electrons and the p-side a high concentration of holes, electrons diffuse from the n-side to the p-side and holes diffuse from the p-side to the n-side, due to the concentration gradient. When an electron diffuses across, it leaves behind a fixed positive donor ion on the n-side; when a hole diffuses across, it leaves behind a fixed negative acceptor ion on the p-side.

Drift: The exposed immobile ions set up an electric field directed from the n-side to the p-side. This field drives the minority carriers across the junction, holes from n to p and electrons from p to n, which constitutes the drift current opposite to the diffusion current.

Formation of the depletion region: The layer of immobile positive ions on the n-side and negative ions on the p-side, near the junction, has no free charge carriers. This charge-free region is called the **depletion region**, and the potential difference across it is the barrier potential V_0 (about 0.7 *V* for Si). At equilibrium the drift current exactly balances the diffusion current, so the net current is zero and the depletion region stops growing.

(b) V-I characteristics and rectification.

Forward-bias circuit: p-side to the positive terminal, n-side to the negative terminal of a variable supply, with a milliammeter in series and a voltmeter across the diode. As voltage is raised the current stays almost zero up to the knee voltage ($\approx$ **0.7 *V***) and then rises steeply, so the diode offers low resistance.

Reverse-bias circuit: the diode is reversed, with a microammeter in series. Only a tiny constant reverse current flows up to breakdown, so the diode offers very high resistance.

Use in rectification: Since the diode conducts (low resistance) in forward bias and blocks (high resistance) in reverse bias, it allows current in only one direction. When AC is applied, only the half-cycles that forward bias the diode produce current through the load. Thus the diode converts AC into **pulsating DC**, which is the basis of rectification.

Final answer. Diffusion and drift create the immobile-ion depletion region with barrier potential $\mathbf{V_0}$; the diode conducts only in forward bias, which is used to rectify AC into pulsating DC.

← *Back to Q363*

Q364

(a) Draw the circuit diagrams of a p-n junction diode in (i) forward bias, (ii) reverse bias. How are these circuits used to study the V–I characteristics of a silicon diode? Draw the typical V–I characteristics.
(b) What is a light emitting diode (LED)? Mention two important advantages of LEDs over conventional lamps.
OR

(a) Draw the circuit arrangement for studying the input and output characteristics of an n-p-n transistor in CE configuration. With the help of these characteristics define (i) input resistance, (ii) current amplification factor.
(b) Describe briefly with the help of a circuit diagram how an n-p-n transistor is used to produce self-sustained oscillations.

[2012]

SOLUTION

(a) Forward-bias, reverse-bias circuits and Si V-I characteristic.

Forward-bias circuit: The p-side of the silicon diode is joined to the positive terminal of a variable DC supply and the n-side to the negative terminal. A milliammeter in series reads the forward current and a voltmeter across the diode reads the forward voltage. A potential divider varies the voltage from zero.

Reverse-bias circuit: The diode is reversed so the p-side goes to the negative and the n-side to the positive terminal, with a microammeter in series (the current is very small) and a voltmeter across the diode.

How they give the V-I characteristic: By slowly increasing the source voltage and noting the meter readings in each connection, the forward and reverse current values are obtained against the corresponding voltages and plotted together.

Typical Si V-I curve: In the forward region (first quadrant, current in mA) the current is negligible up to the knee voltage $\approx \mathbf{0.7\ V}$ and then rises steeply. In the reverse region (third quadrant, current in μA) a very small saturation current flows almost unchanged up to the breakdown voltage, where it rises sharply.

(b) Light Emitting Diode (LED).

An **LED is a heavily doped p-n junction diode that is forward biased and emits spontaneous light** when current flows through it. On forward bias, electrons and holes are injected across the junction and recombine near it. Each recombination releases energy nearly equal to the band gap as a photon of frequency $\nu = \frac{E_g}{h}$, so the diode glows.

Two advantages of LEDs over conventional lamps:

(i) LEDs have a much longer operating life and consume far less power (higher luminous efficiency) for the same brightness.

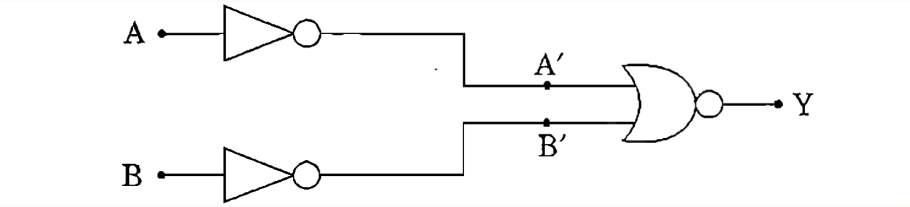
(ii) LEDs switch on and off very fast, work at low voltage, and are mechanically rugged with no fragile filament.

Final answer. The Si diode conducts above the $\approx 0.7\ V$ knee in forward bias; an LED is a forward-biased heavily doped diode emitting photons of energy E_g, with longer life and lower power use than conventional lamps.

← *Back to Q364*

Q365

(a) Describe briefly, with the help of a diagram, the role of the two important processes involved in the formation of a p-n junction.
(b) Name the device which is used as a voltage regulator. Draw the necessary circuit diagram and explain its working.
OR
(a) Explain briefly the principle on which a transistor-amplifier works as an oscillator. Draw the necessary circuit diagram and explain its working.
(b) Identify the equivalent gate for the following circuit and write its truth table.



[2012]

SOLUTION

(a) Two important processes in p-n junction formation.

Diffusion: In a p-n junction the n-region has many free electrons and the p-region has many holes. Because of this concentration difference, electrons diffuse from the n-side to the p-side and holes diffuse from the p-side to the n-side. As majority carriers leave, they expose fixed positive donor ions on the n-side and fixed negative acceptor ions on the p-side, building up a depletion region and a built-in electric field directed from n to p.

Drift: This built-in electric field forces the minority carriers to move in the opposite direction, electrons from the p-side to the n-side and holes from the n-side to the p-side. This motion of minority carriers under the field is the **drift current**, which opposes the diffusion current.

Equilibrium is reached when the drift current exactly balances the diffusion current; the net current is then zero and a steady barrier potential exists across the junction.

(b) Voltage regulator.

The device used as a voltage regulator is the **Zener diode**.

Circuit: The unregulated input voltage V_{in} is applied through a series resistance R_S to a Zener diode connected in reverse bias, and the load R_L is connected in parallel with the Zener. The regulated output V_{out} is taken across the Zener (and load).

Working: The Zener diode is operated in its reverse breakdown region. In this region the voltage across the Zener stays constant at its breakdown value V_Z even though the current through it changes a lot. If V_{in} increases, the extra voltage is dropped across R_S and the Zener current rises, but the output stays clamped at V_Z. If the load current changes, the Zener current adjusts to keep the total current through R_S roughly fixed, again holding $\mathbf{V_{out} = V_Z}$.

Hence the output voltage remains **constant at $\mathbf{V_Z}$** despite variations in input voltage or load, giving voltage regulation.

Final answer. Diffusion (of majority carriers) and drift (of minority carriers) form the junction; the Zener diode in reverse breakdown holds the output constant at $\mathbf{V_Z}$, acting as a voltage regulator.

← *Back to Q365*

Q366

(a) Draw a circuit for studying the input and output characteristics of an n-p-n transistor in CE configuration. Show how, from the output characteristics, the information about the current amplification factor (β_{ac}) can be obtained.
(b) Draw a plot of the transfer characteristic (V_o versus V_i) for a base-biased transistor

in CE configuration. Show for which regions in the plot, the transistor can operate as a switch.

OR

Why is a zener diode considered as a special purpose semiconductor diode?
Draw the I–V characteristics of a zener diode and explain briefly how reverse current suddenly increases at the breakdown voltage.
Describe briefly with the help of a circuit diagram how a zener diode works to obtain a constant dc voltage from the unregulated dc output of a rectifier.

[2012]

SOLUTION

(a) CE characteristics circuit and current amplification factor.

Circuit: The emitter of the n-p-n transistor is grounded. The base is supplied through a resistance R_B from a battery V_{BB}, with a voltmeter reading V_{BE} and a microammeter reading the base current I_B. The collector is supplied through a resistance R_C from a battery V_{CC}, with a voltmeter reading V_{CE} and a milliammeter reading the collector current I_C. The input is the base-emitter loop and the output is the collector-emitter loop.

Output characteristics give β_{ac}: The output characteristics are plots of collector current I_C against V_{CE} for several fixed base currents I_B. To find the current amplification factor, fix V_{CE} at a constant value (for example 10 *V*) and read the change in collector current ΔI_C produced by a change in base current ΔI_B between two adjacent curves:

$$\beta_{ac} = \left(\frac{\Delta I_C}{\Delta I_B}\right)_{V_{CE}=\text{const}}$$

Its value is typically between about 50 and 300.

(b) Transfer characteristic and switch action.

Plot the output voltage V_o (which is V_{CE}) against the input voltage V_i (which is V_{BE}). The curve has three parts:

Cut-off region: For small V_i, almost no collector current flows, so $\mathbf{V_o \approx V_{CC}}$ (output HIGH). Here the transistor is OFF.

Active region: In the middle, V_o falls steeply and almost linearly as V_i increases; this sloping region is used for amplification.

Saturation region: For large V_i, the collector current is maximum, the drop across R_C is large, so V_o is very small (output LOW). Here the transistor is fully ON.

As a switch the transistor is operated only in the **cut-off region (OFF, output HIGH) and the saturation region (ON, output LOW)**, avoiding the active region.

Final answer. From the output curves $\beta_{ac} = (\Delta I_C/\Delta I_B)_{V_{CE}}$; in the transfer plot the transistor works as a switch in the cut-off and saturation regions.

← *Back to Q366*

Q367

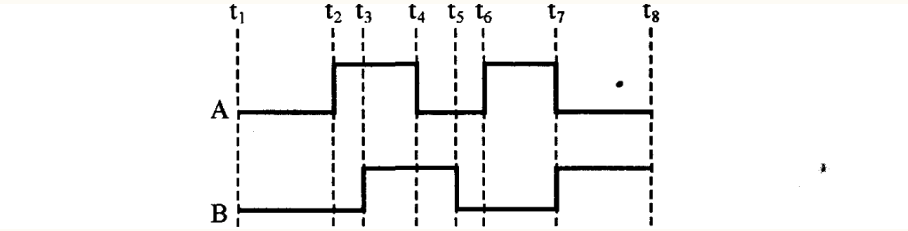
Draw a simple circuit of a CE transistor amplifier. Explain its working. Show that the voltage gain, A_V, of the amplifier is given by $A_V = -\frac{\beta_{ac} R_L}{r_i}$, *where β_{ac} is the current gain, R_L is the load resistance and r_i is the input resistance of the transistor. What is the significance of the negative sign in the expression for the voltage gain?*

OR

(a) Draw the circuit diagram of a full wave rectifier using p-n junction diode. Explain its working and show the output, input waveforms.

(b) Show the output waveforms (Y) for the following inputs A and B of

(i) OR gate (ii) NAND gate



[2012]

SOLUTION

CE transistor amplifier: circuit and working.

Circuit: The emitter is grounded. The base is biased into the active region through a resistor R_B from V_{CC}, and the AC input signal v_i is applied between base and emitter through a coupling capacitor. The collector is connected to

V_{CC} through the load resistor R_L, and the amplified output is taken between collector and emitter through another coupling capacitor.

Working: A small AC input voltage v_i superimposed on the DC bias produces a small change in base current ΔI_B. Because of transistor action this causes a much larger change in collector current $\mathbf{\Delta I_C = \beta_{ac} \Delta I_B}$. This amplified current flows through R_L, and the output voltage $\mathbf{V_{CE} = V_{CC} - I_C R_L}$ varies in step with the input, but inverted.

Derivation of voltage gain:

The change in base current for an input v_i across input resistance r_i is

$$\Delta I_B = \frac{v_i}{r_i}$$

The collector current change is

$$\Delta I_C = \beta_{ac} \Delta I_B = \beta_{ac} \frac{v_i}{r_i}$$

This current flows through R_L. Since $\mathbf{V_{CE} = V_{CC} - I_C R_L}$ and V_{CC} is constant, an increase in I_C decreases the output, so

$$v_o = -\Delta I_C R_L = -\beta_{ac} \frac{v_i}{r_i} R_L$$

Therefore the voltage gain is

$$A_V = \frac{v_o}{v_i} = -\frac{\beta_{ac} R_L}{r_i}$$

Significance of the negative sign: The negative sign shows that the **output voltage is 180° out of phase with the input voltage**. When the input rises, the output falls and vice versa, which is the characteristic phase reversal of a common-emitter amplifier.

Final answer. The CE voltage gain is $\mathbf{A_V = -\frac{\beta_{ac} R_L}{r_i}}$, the negative sign showing a **180°** phase reversal between output and input.

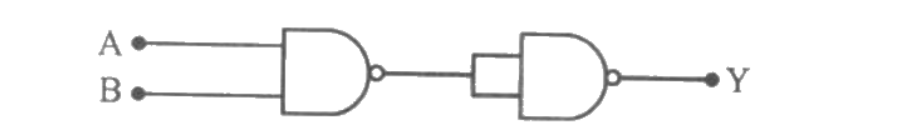
← *Back to Q367*

Q368

(a) Explain the formation of depletion layer and potential barrier in a p-n junction.
(b) In the figure given below the input waveform is converted into the output waveform by a device 'X'. Name the device and draw its circuit diagram.



(c) Identify the logic gate represented by the circuit as shown and write its truth table.



[2010]

SOLUTION

(a) Depletion layer and potential barrier in a p-n junction.

When p- and n-type materials form a junction, electrons diffuse from the n-side (high electron concentration) to the p-side, and holes diffuse from the p-side to the n-side, because of the concentration gradients. As the carriers cross over, they leave behind immobile charged ions, fixed positive donor ions on the n-side and fixed negative acceptor ions on the p-side. The narrow region around the junction that contains only these immobile ions and no free carriers is the **depletion layer**.

These uncovered ions set up an electric field directed from the n-side to the p-side, producing a potential difference across the junction called the **barrier potential** V_b (about 0.7 V for Si and 0.3 V for Ge). This barrier opposes further diffusion of majority carriers. At equilibrium the drift of minority carriers exactly balances the diffusion of majority carriers, so the net current is zero.

(b) Device X.

The input is a full AC wave and the output keeps only the half-cycles of one polarity, so the device X is a **half-wave rectifier (a single p-n junction diode)**.

Circuit: An AC source is connected through a single diode in series with a load resistor R_L; the output voltage is taken across R_L. During the half-cycle that forward biases the diode, current flows and a voltage appears across R_L; during the other half-cycle the diode is reverse biased and no current flows, giving a half-wave rectified output.

(c) Logic gate.

The truth table given is that of an **AND gate**, $Y = A \cdot B$, where the output is HIGH only when both inputs are HIGH.

Truth table:

$$A=0,\ B=0 \Rightarrow Y=0$$

$$A=0,\ B=1 \Rightarrow Y=0$$

$$A=1,\ B=0 \Rightarrow Y=0$$

$$A=1,\ B=1 \Rightarrow Y=1$$

Final answer. Diffusion creates the immobile-ion depletion layer and barrier potential V_b; device X is a half-wave rectifier, and the circuit represents an AND gate ($Y = A \cdot B$).

← *Back to Q368*

Q369

(a) With the help of the circuit diagram explain the working principle of a transistor amplifier as an oscillator.
(b) Distinguish between a conductor, a semiconductor and an insulator on the basis of energy band diagrams.

[2010]

SOLUTION

(a) Transistor amplifier as an oscillator.

An oscillator is a circuit that produces a continuous AC output without any external input signal. It is built from a CE transistor amplifier together with

positive feedback.

Circuit: A parallel LC tuned (tank) circuit is placed in the collector branch; it sets the frequency. A coil coupled to it (a transformer winding) feeds a fraction of the output from the collector back to the base.

Working: When the circuit is switched on, a small disturbance starts oscillations in the LC tank at its natural frequency

$$f = \frac{1}{2\pi\sqrt{LC}}$$

The CE amplifier introduces a 180° phase shift, and the feedback transformer is connected to give another 180° shift, so the signal fed back to the base is **in phase with the input (positive feedback)**. If the loop gain is equal to or greater than one, the energy fed back makes up for the resistive losses in the tank, so the oscillations are sustained and a steady sinusoidal output of frequency f appears at the collector.

(b) Energy band diagrams.

Conductor: The conduction band and valence band overlap, or the conduction band is partly filled, so the energy gap $\mathbf{E_g = 0}$. Free electrons are always available, giving very high conductivity.

Semiconductor: At 0 K the valence band is full and the conduction band is empty, separated by a small gap $\mathbf{E_g \approx 1\ eV}$. At higher temperature some electrons gain enough energy to cross into the conduction band, giving moderate conductivity.

Insulator: The valence band is full and the conduction band is empty, separated by a large gap $\mathbf{E_g > 3\ eV}$. Even at high temperature almost no electron can cross, so the conductivity is practically zero.

CONCEPT

For sustained oscillations the Barkhausen condition needs both the right phase (net 360°, i.e. positive feedback) and a loop gain of at least one.

Final answer. An oscillator is a CE amplifier with positive feedback whose LC tank fixes $\mathbf{f = \frac{1}{2\pi\sqrt{LC}}}$; conductors have $\mathbf{E_g = 0}$, semiconductors $\mathbf{E_g \approx 1\ eV}$, and

insulators $E_g > 3$ eV.

← Back to Q369

Q370

Draw the electronic configuration in the atomic structures of 'Ge' and 'Si' atoms. How can these be made 'P-type' and 'N-type' semiconductors?

[2010]

SOLUTION

Electronic configurations of Ge and Si.

Silicon ($Z = 14$):

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$$

It has 4 electrons in its outermost shell, so it is tetravalent.

Germanium ($Z = 32$):

$$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^2$$

It also has 4 electrons in its outermost shell, so it too is tetravalent. Both form covalent crystals in which each atom shares its four valence electrons with four neighbours.

Making an N-type semiconductor: Add a small amount of a **pentavalent (donor) impurity** such as phosphorus, arsenic, antimony or bismuth. Four of its five valence electrons form covalent bonds with the surrounding Si or Ge atoms, while the fifth electron is loosely bound and easily enters the conduction band as a free electron. The impurity provides a donor energy level E_D just below the conduction band. Here **electrons are the majority carriers and holes the minority carriers**.

Making a P-type semiconductor: Add a small amount of a **trivalent (acceptor) impurity** such as boron, aluminium, indium or gallium. Its three valence electrons form three covalent bonds, leaving one bond incomplete, which behaves as a hole. The impurity provides an acceptor energy level E_A just above

the valence band. Here **holes are the majority carriers and electrons the minority carriers**.

Final answer. Si is $1s^2 2s^2 2p^6 3s^2 3p^2$ and Ge is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2$ (both tetravalent); adding a pentavalent impurity gives N-type and a trivalent impurity gives P-type.

← *Back to Q370*

Q371

With the help of a circuit diagram, explain the working of a P-N-P transistor.

[2010]

SOLUTION

A P-N-P transistor consists of a thin layer of n-type semiconductor (the base) sandwiched between two p-type regions, namely the emitter and the collector. The emitter is heavily doped and of moderate size, the base is very thin and lightly doped, while the collector is moderately doped and the largest in area so that it can dissipate heat.

Circuit diagram (described): The emitter-base (EB) junction is forward biased by a battery V_{EE} with its positive terminal connected to the p-type emitter and negative terminal to the n-type base. The collector-base (CB) junction is reverse biased by a battery V_{CC} with its negative terminal connected to the p-type collector and positive terminal to the base. The emitter arrow points inward (toward the base). Conventional current flows out of the emitter, with I_E entering the emitter, I_B leaving the base and I_C leaving the collector.

Working: Because the EB junction is forward biased, the barrier at this junction is lowered and the majority carriers of the emitter, namely holes, are injected from the heavily doped emitter into the thin base region.

Since the base is very thin and lightly doped, it contains very few free electrons. Hence only about 1 to 5 percent of the injected holes recombine with electrons in the base; this small recombination constitutes the base current I_B.

The remaining 95 to 99 percent of the holes diffuse across the thin base and reach the collector-base junction. The strong reverse-bias electric field at the CB junction immediately sweeps these holes into the collector, constituting the collector current I_C.

Applying Kirchhoff's current law at the transistor,

$$I_E = I_B + I_C$$

with $I_C \gg I_B$, so the collector current is almost equal to the emitter current. The current amplification factor in common-base mode is

$$\alpha = \frac{I_C}{I_E} \approx 0.95 \text{ to } 0.99$$

and in common-emitter mode

$$\beta = \frac{I_C}{I_B} = \frac{\alpha}{1-\alpha} \approx 50 \text{ to } 500.$$

Thus, a **P-N-P transistor** transfers current from a low-resistance forward-biased input circuit to a high-resistance reverse-biased output circuit, with the conduction carried mainly by **holes**.

Final answer. The P-N-P transistor works by hole injection from the forward-biased emitter through the thin base into the reverse-biased collector, giving $I_E = I_B + I_C$ with $I_C \gg I_B$.

← *Back to Q371*

Q372

What are intrinsic and extrinsic semiconductors? How does the rise in temperature affect the conductivity of semiconductors?

[2010]

SOLUTION

An intrinsic semiconductor is a pure semiconductor such as silicon (Si) or germanium (Ge) that contains no added impurity. At $T = 0\ K$ its valence band is

completely filled and the conduction band is completely empty, so it behaves like a perfect insulator.

At higher temperatures, thermal energy breaks some covalent bonds, lifting electrons into the conduction band and leaving behind holes in the valence band. The electrons and holes are generated in equal numbers, so

$$n_e = n_h = n_i$$

where n_i is the intrinsic carrier concentration. The conductivity of such a material is small.

An extrinsic semiconductor is an intrinsic semiconductor that has been deliberately doped with a small, controlled amount of impurity. Doping with a pentavalent impurity (As, P, Sb) gives an n-type semiconductor in which electrons are the majority carriers, while doping with a trivalent impurity (B, Al, In) gives a p-type semiconductor in which holes are the majority carriers.

In an extrinsic semiconductor the electron and hole concentrations are unequal and are governed by the law of mass action $\mathbf{n_e\, n_h = n_i^2}$. The conductivity of an extrinsic semiconductor is much larger than that of the intrinsic material.

Effect of rise in temperature on conductivity: The conductivity of a semiconductor is given by

$$\sigma = e(n_e\mu_e + n_h\mu_h)$$

where μ_e and μ_h are the mobilities of electrons and holes. As temperature T rises, the carrier concentration increases sharply because thermal energy breaks more covalent bonds, following

$$n_i \propto T^{3/2}\exp\left(-\frac{E_g}{2k_B T}\right)$$

where E_g is the band gap and k_B is Boltzmann's constant.

The mobilities μ_e and μ_h decrease slightly with temperature because of increased lattice vibrations, but this small fall is far outweighed by the exponential rise in n_i.

Hence the net effect is that the **conductivity of a semiconductor increases (resistivity decreases) as temperature rises**, which means a semiconductor

has a **negative temperature coefficient of resistance**, opposite to that of a metal.

Final answer. Intrinsic semiconductors are pure ($\mathbf{n_e = n_h = n_i}$) and extrinsic ones are doped; conductivity rises with temperature because $\mathbf{n_i}$ grows exponentially, giving a negative temperature coefficient of resistance.

← *Back to Q372*

Q373

What is a P-N junction? How does it behave under forward and reverse bias? Explain with the help of circuit diagrams.

[2010]

SOLUTION

A P-N junction is a single semiconductor crystal in which one region is doped p-type and the adjacent region is doped n-type during fabrication, so that the two regions meet at a metallurgical junction.

When the junction is formed, electrons from the n-side diffuse to the p-side and holes from the p-side diffuse to the n-side. This leaves behind immobile positive donor ions on the n-side and immobile negative acceptor ions on the p-side, creating a charge-free depletion region near the junction. The separated charges set up a built-in potential barrier V_b (about 0.7 *V* for Si and 0.3 *V* for Ge) that opposes further diffusion of majority carriers.

Forward bias (circuit): The positive terminal of an external battery is connected to the p-side and the negative terminal to the n-side, through a series resistor. The applied voltage opposes the built-in barrier potential V_b. As a result the effective barrier is lowered and the width of the depletion region decreases.

Majority carriers (holes from p-side and electrons from n-side) now cross the junction easily. The forward current is small until the applied voltage exceeds the threshold or knee voltage (about 0.7 *V* for Si, 0.3 *V* for Ge), after which it rises steeply, almost exponentially, with voltage. A forward-biased junction therefore offers low resistance.

Reverse bias (circuit): The battery polarity is reversed, with the positive terminal to the n-side and the negative terminal to the p-side. The applied voltage now adds to the barrier potential V_b, so the barrier height increases and the depletion region widens.

Majority carriers are pulled away from the junction, so they cannot cross it. Only a very small current due to minority carriers flows. This is the reverse saturation current (of the order of microamperes) and it is almost independent of the applied voltage, until the breakdown voltage is reached. A reverse-biased junction therefore offers very high resistance.

V-I characteristics (described): In the first quadrant (forward) the current stays near zero up to the knee voltage and then rises steeply. In the third quadrant (reverse) a tiny, nearly constant reverse saturation current flows, which drops sharply at breakdown.

Because the junction conducts strongly in forward bias and blocks current in reverse bias, this **asymmetric V-I behaviour makes the P-N junction act as a rectifier**.

Final answer. A P-N junction has a depletion region with a barrier $\mathbf{V_b}$; it offers low resistance and large current in forward bias and high resistance with only a tiny reverse saturation current in reverse bias, acting as a rectifier.

← *Back to Q373*

Q374

Explain the working of a full wave rectifier using semiconductor diodes, with the help of a labeled circuit diagram.

[2010]

SOLUTION

A full-wave rectifier converts both halves of an alternating input into a unidirectional (direct) output. It uses two P-N junction diodes and a centre-tapped transformer.

Circuit (described): A centre-tapped step-down transformer has the two ends of its secondary winding connected to the anodes (p-sides) of two diodes D_1 and D_2. The cathodes (n-sides) of both diodes are joined together. The load resistor R_L is connected between this common cathode point and the centre tap of the secondary. The output voltage is taken across R_L.

Principle: A diode conducts only when it is forward biased and blocks current when reverse biased. The centre tap allows the two diodes to conduct on alternate half cycles so that current always flows through R_L in the same direction.

Working during the positive half cycle: Suppose the upper end of the secondary becomes positive and the lower end negative. Then diode D_1 is forward biased and conducts, while diode D_2 is reverse biased and is OFF.

Current flows from the upper end of the secondary, through D_1, then down through the load R_L, and back to the centre tap.

Working during the negative half cycle: Now the polarities reverse, so the lower end of the secondary becomes positive. Diode D_2 is forward biased and conducts, while D_1 is reverse biased and is OFF.

Current flows from the lower end of the secondary, through D_2, and then through the load R_L in the same direction as before, returning to the centre tap.

Since the current through R_L flows in the same direction during both half cycles, the output is unidirectional for the entire input cycle.

Output waveform (described): The output across R_L is a series of positive half-sine humps, one for each half cycle of the input, so the output ripple frequency is twice the input frequency. This is far smoother than a half-wave output.

When a capacitor filter is connected across R_L, the dips between the humps are filled in and a nearly constant **direct voltage** is obtained.

Final answer. In a full-wave rectifier, $\mathbf{D_1}$ conducts on the positive half cycle and $\mathbf{D_2}$ on the negative half cycle, so current flows through $\mathbf{R_L}$ in the same direction throughout, giving a unidirectional output of twice the input frequency.

← *Back to Q374*

Q375

(i) Draw a circuit diagram to study the input and output characteristics of an n-p-n transistor in its common emitter configuration. Draw the typical input and output characteristics.
(ii) Explain, with the help of a circuit diagram, the working of n-p-n transistor as a common emitter amplifier.

[2009]

SOLUTION

(i) Circuit to study the characteristics of an n-p-n transistor in common-emitter (CE) configuration (described): The emitter is common to both input and output. In the input section, a battery V_{BB} drives the base through a variable resistor (or potential divider) R_B; a milliammeter reads the base current I_B and a voltmeter reads the base-emitter voltage V_{BE}. In the output section, a battery V_{CC} supplies the collector; a milliammeter reads the collector current I_C and a voltmeter reads the collector-emitter voltage V_{CE}.

Input characteristics (described): These are graphs of base current I_B versus base-emitter voltage V_{BE} at a constant V_{CE}. The curve resembles a forward-biased diode characteristic: I_B stays nearly zero until V_{BE} reaches about 0.7 *V*, then rises sharply.

Output characteristics (described): These are graphs of collector current I_C versus collector-emitter voltage V_{CE} for several fixed values of I_B. For each fixed I_B, I_C rises steeply for small V_{CE} (the saturation region) and then becomes almost flat (the active region) as V_{CE} increases, until breakdown. Higher values of I_B give a higher family of curves.

(ii) Working as a CE amplifier (described): The transistor is biased in the active region with the EB junction forward biased and the CB junction reverse biased. The a.c. signal to be amplified is applied between base and emitter; a load resistor R_L is connected in the collector circuit, and the amplified output is taken across R_L (between collector and emitter).

When the a.c. input voltage adds to V_{BE}, it makes the base-emitter junction more forward biased, so the base current I_B increases. Since

$$I_C = \beta I_B,$$

a small change in I_B produces a β times larger change in the collector current I_C.

The output voltage is given by the collector loop equation

$\mathbf{V_{CE} = V_{CC} - I_C R_L}$.

When I_C increases, the drop $I_C R_L$ increases, so V_{CE} decreases, and when I_C decreases, V_{CE} increases. Hence the output voltage swings in the opposite sense to the input, so the **output is 180° out of phase with the input**.

The a.c. voltage gain of the amplifier is

$$A_V = -\beta_{ac}\frac{R_L}{r_i}$$

where r_i is the input resistance. The negative sign shows the phase reversal, and the large value of β_{ac} gives **voltage amplification**.

Final answer. The CE input characteristic is diode-like while the output is a family of $\mathbf{I_C}$-$\mathbf{V_{CE}}$ curves; as an amplifier, a small base signal varies $\mathbf{I_C = \beta I_B}$, giving an amplified output $\mathbf{A_V = -\beta_{ac} R_L / r_i}$ that is **180°** out of phase.

← *Back to Q375*

Q376

How is a zener diode fabricated so as to make it a special purpose diode? Draw I-V characteristics of zener diode and explain the significance of breakdown voltage. Explain briefly, with the help of a circuit diagram, how a p-n junction diode works as a half wave rectifier.

[2009]

SOLUTION

Fabrication of a Zener diode: A Zener diode is a specially designed P-N junction in which both the p-side and the n-side are heavily doped, far more heavily than in an ordinary junction diode.

Because of this heavy doping, the depletion region becomes very narrow (of the order of 10 nm). When the diode is reverse biased, even a small reverse voltage

produces a very large electric field across this thin depletion layer.

This strong field causes Zener breakdown (by field-assisted tunnelling) at the sharply defined reverse voltage V_Z, called the Zener or breakdown voltage. The Zener diode is always operated in this reverse breakdown region.

I-V characteristics (described): In forward bias the Zener behaves like an ordinary diode, with current rising sharply after the knee voltage. In reverse bias only a tiny reverse saturation current flows until the voltage reaches the breakdown voltage V_Z. At V_Z the reverse current increases almost vertically with a negligible further increase in voltage.

Significance of the breakdown voltage: In the breakdown region, the voltage across the Zener diode remains practically constant at V_Z even though the current through it changes over a wide range.

This constant-voltage property in the breakdown region is what makes the Zener diode useful as a **voltage regulator**, holding the output voltage steady against changes in load or supply.

Half-wave rectifier (described): A single P-N junction diode is connected in series with the a.c. source (through the secondary of a transformer) and the load resistor R_L. The output is taken across R_L.

During the half cycle when the diode is forward biased, it conducts and current flows through R_L, producing a voltage pulse across it.

During the next half cycle the diode is reverse biased, so it does not conduct and no current flows through R_L; the output is zero.

Thus current flows through the load only during alternate half cycles, so the output across R_L is a **unidirectional pulsating voltage** present for only one half of each input cycle, which converts a.c. into d.c.

Final answer. A Zener diode is heavily doped to give a narrow depletion layer and a sharp breakdown at $\mathbf{V_Z}$, where its voltage stays constant (used as a voltage regulator); a half-wave rectifier conducts only on the half cycle that forward biases the diode.

← *Back to Q376*

Q377

State the principle of working of p-n diode as a rectifier. Explain, with the help of a circuit diagram, the use of p-n diode as a full wave rectifier. Draw a sketch of the input and output waveforms.

[2007]

SOLUTION

Principle of a P-N diode as a rectifier: A P-N junction diode offers a very low resistance when it is forward biased and a very high resistance when it is reverse biased. Therefore the diode conducts current only during those half cycles of the a.c. input for which it is forward biased and blocks the others. In this way it converts alternating current into a unidirectional (direct) current.

Full-wave rectifier circuit (described): A centre-tapped step-down transformer has the two ends of its secondary winding connected to the p-sides (anodes) of two diodes D_1 and D_2. The cathodes (n-sides) of both diodes are joined together. The load resistor R_L is connected between this common cathode and the centre tap of the secondary, and the output is taken across R_L.

Working during the positive half cycle: When the top end of the secondary is positive, diode D_1 is forward biased and conducts, while diode D_2 is reverse biased and stays OFF. Current flows from the top end through D_1 and down through R_L, returning to the centre tap.

Working during the negative half cycle: Now the bottom end of the secondary becomes positive. Diode D_2 is forward biased and conducts, while D_1 is OFF. Current flows from the bottom end through D_2 and again down through R_L in the same direction as before, returning to the centre tap.

Since the current through R_L flows in the same direction during both half cycles, the output current is unidirectional throughout the input cycle.

Input and output waveforms (described): The input is a full sine wave with alternate positive and negative half cycles. The output across R_L is a series of positive half-sine humps, one for each half cycle of the input, so the **output frequency is twice the input frequency**. This full-wave rectified output is much smoother than that of a half-wave rectifier.

Final answer. A P-N diode rectifies because it has low forward and high reverse resistance; in a full-wave rectifier $\mathbf{D_1}$ and $\mathbf{D_2}$ conduct on alternate half cycles to drive current through $\mathbf{R_L}$ in one direction, giving an output of twice the input frequency.

← *Back to Q377*

Q378

Draw the symbolic representation of a (i) p-n-p, (ii) n-p-n transistor. Why is the base region of a transistor thin and lightly doped? With a circuit diagram, show the biasing of a p-n-p transistor in common base configuration. Explain the movement of charge carriers through different parts of the transistor in such a configuration and show that $\mathbf{I_e = I_b + I_c}$.

[2007]

SOLUTION

Symbolic representation (described):

(i) P-N-P transistor: drawn with the emitter arrow pointing inward, that is, toward the base. The inward arrow always lies on the emitter and shows the direction of conventional current into the transistor.

(ii) N-P-N transistor: drawn with the emitter arrow pointing outward, that is, away from the base.

Why the base is thin and lightly doped: The base is made very thin (a few micrometres) so that most of the carriers injected from the emitter pass straight through it and reach the collector before they can recombine.

The base is lightly doped so that it contains very few majority carriers, which again keeps recombination of the injected carriers in the base to a minimum.

Together these ensure that almost all the emitter current reaches the collector, so the collector current is nearly equal to the emitter current, giving a high current gain.

P-N-P transistor in common-base (CB) configuration (circuit described): The base is common to input and output. The emitter-base junction is forward

biased by a battery V_{EE} with its positive terminal to the p-type emitter. The collector-base junction is reverse biased by a battery V_{CC} with its negative terminal to the p-type collector.

Movement of charge carriers: Because the EB junction is forward biased, holes (the majority carriers of the heavily doped p-emitter) are injected into the thin n-type base, constituting the emitter current I_e.

A small fraction, about 1 to 5 percent, of these holes recombine with the electrons in the lightly doped base; this recombination is supplied by the base current I_b.

The remaining 95 to 99 percent of the holes diffuse through the thin base and reach the collector-base junction, where the strong reverse-bias field sweeps them across into the collector, constituting the collector current I_c.

Since the emitter current divides into the part that recombines in the base and the part that reaches the collector, by Kirchhoff's current law the total emitter current equals the sum of the base and collector currents:

$$I_e = I_b + I_c$$

with $I_c \gg I_b$, so I_c is almost equal to I_e. This is the required relation.

CONCEPT

In the CB relation, remember the emitter current is the source that splits into base and collector parts, so I_e is always the largest of the three currents.

Final answer. The P-N-P arrow points inward and N-P-N outward; the thin, lightly doped base minimises recombination, and charge sharing in CB configuration gives $\mathbf{I_e = I_b + I_c}$ with $\mathbf{I_c \gg I_b}$.

← *Back to Q378*

Q379

(a) *Distinguish between metals, insulators and semiconductors on the basis of their energy bands.*
(b) *Why are photodiodes used preferably in reverse bias condition? A photodiode is*

fabricated from a semiconductor with band gap of 2.8 eV. *Can it detect a wavelength of* 6000 nm*? Justify.*

[2005]

SOLUTION

(a) Distinction on the basis of energy bands:

In a metal, the valence band and the conduction band overlap, or the conduction band is partially filled. Free electrons are always available for conduction, so the forbidden energy gap is effectively $\mathbf{E_g = 0}$. Metals are therefore good conductors.

In a semiconductor, the valence band is completely filled and the conduction band is empty at 0 *K*, but the forbidden energy gap is small (about 1 eV, for example $\mathbf{E_g \approx 1.1\ eV}$ for Si). At room temperature, thermal energy lifts some electrons across this small gap, so a semiconductor conducts moderately.

In an insulator, the valence band is completely filled and the conduction band is empty, and the forbidden energy gap is large (greater than about 3 eV, for example $\mathbf{E_g \approx 6\ eV}$ for diamond). Electrons cannot cross this wide gap at ordinary temperatures, so an insulator does not conduct.

(b) Why a photodiode is used in reverse bias: In reverse bias the dark current is very small and is carried mainly by minority carriers. When light of suitable frequency falls on the junction, it generates additional electron-hole pairs, and these extra minority carriers cause a large fractional change in the reverse current.

In forward bias the current is already large and is dominated by majority carriers, so the small photo-generated current is only a tiny fraction of it and cannot be measured reliably. Hence a photodiode is preferably operated in reverse bias, where the fractional change due to light is large and easy to detect.

Detection check for $\boldsymbol{\lambda = 6000\ nm}$: A photodiode can detect light only if the photon energy is at least equal to the band gap, that is, $E_{photon} \geq E_g$.

The photon energy is

$$E_{photon} = \frac{hc}{\lambda}$$

$$E_{photon} = \frac{1240 \text{ eV nm}}{6000 \text{ nm}}$$

$$E_{photon} = 0.207 \text{ eV}$$

Comparing with the band gap $\mathbf{E_g = 2.8\ eV}$,

$$E_{photon} = 0.207 \text{ eV} \ll E_g = 2.8 \text{ eV}$$

Since the photon energy is far less than the band gap, the photon cannot lift an electron from the valence band to the conduction band. Hence **No, the photodiode cannot detect the wavelength of 6000 nm.**

Final answer. Metals have overlapping bands ($\mathbf{E_g = 0}$), semiconductors a small gap (about **1 eV**) and insulators a large gap (above **3 eV**); a photodiode is reverse biased for a large fractional photocurrent, and since $\mathbf{E_{photon} = 0.207\ eV < 2.8\ eV}$ it cannot detect **6000 nm**.

← *Back to Q379*

Q380

(a) *Explain briefly, with the help of a circuit diagram, how V-I characteristics of a p-n junction diode are obtained in* ***(i)*** *forward bias, and* ***(ii)*** *reverse bias. Draw the shape of the curves obtained.*
(b) *A semiconductor has equal electron and hole concentration of* $6 \times 10^8/m^3$. *On doping with certain impurity, electron concentration increases to* $9 \times 10^{17}/m^3$.
(i) *Identify the new semiconductor obtained after doping.*
(ii) *Calculate the new hole concentration.*

[2005]

SOLUTION

(a) V-I characteristics of a P-N junction diode:

Forward bias circuit (described): The diode is connected so that its p-side is joined to the positive terminal of a battery and its n-side to the negative terminal, through a series resistor. An ammeter (milliammeter) measures the current and a voltmeter is connected across the diode to read the voltage. As the forward voltage V is increased from zero, the current I stays nearly zero up

to the threshold or knee voltage (about 0.7 V for Si and 0.3 V for Ge), and beyond this it rises sharply, almost exponentially.

Reverse bias circuit (described): The diode polarity is reversed, with the p-side to the negative terminal and the n-side to the positive terminal of the battery, with the ammeter (microammeter) and voltmeter in place. A very small, nearly constant reverse saturation current (of the order of microamperes) flows over a wide range of reverse voltages, until at the breakdown voltage the reverse current suddenly increases sharply.

Shape of the curve (described): In the first quadrant the forward current is near zero until the knee voltage and then rises steeply. In the third quadrant the reverse current is a tiny, almost flat line that drops vertically at breakdown. This asymmetric curve shows the rectifying action of the diode.

(b) Numerical: Given the intrinsic concentration $\mathbf{n_i = 6\times 10^{8}\ m^{-3}}$ and, after doping, the electron concentration $\mathbf{n_e' = 9\times 10^{17}\ m^{-3}}$.

(i) Since $\mathbf{n_e' = 9\times 10^{17}\ m^{-3} \gg n_i}$, electrons are now the majority carriers, so the doped material is an **n-type semiconductor**.

(ii) By the law of mass action, the product of electron and hole concentrations stays equal to n_i^2:

$$n_e' n_h' = n_i^2$$

$$n_h' = \frac{n_i^2}{n_e'}$$

$$n_h' = \frac{(6\times 10^{8})^2}{9\times 10^{17}}$$

$$n_h' = \frac{36\times 10^{16}}{9\times 10^{17}}$$

$$n_h' = 4\times 10^{-1}\ m^{-3}$$

$$\mathbf{n_h' = 0.4\ m^{-3}}$$

This vanishingly small hole concentration confirms that the doped material is strongly n-type, with holes as the minority carriers.

TIP

Always apply the law of mass action $\mathbf{n_e\, n_h = n_i^2}$ after doping; the majority carrier rises while the minority carrier falls so that the product stays fixed.

Final answer. The forward V-I curve rises steeply after the knee voltage while the reverse curve is a tiny saturation current; the doped material is n-type, and the new hole concentration is $\mathbf{n_h' = 0.4\ m^{-3}}$.

← *Back to Q380*

Q381

Draw a circuit diagram of a common emitter amplifier using n-p-n transistor. Show input and output voltages graphically.
The current gain for common emitter amplifier is 59. If the emitter current is 6.0 mA, find (i) base current and (ii) collector current.

[2003]

SOLUTION

A common emitter (CE) amplifier uses a transistor in the common emitter configuration, where the emitter terminal is common to both the input and the output circuits.

Circuit diagram (described): An n-p-n transistor is drawn with its emitter at the bottom connected directly to the common ground line. The base is connected through a base resistor R_B to a base battery V_{BB}, which forward biases the base-emitter junction. The collector is connected through a load resistance R_L (also called R_C) to the positive terminal of the collector battery V_{CC}, whose negative terminal goes to the common ground. This reverse biases the collector-base junction. The input AC signal v_i is applied between the base and the emitter (in series with V_{BB}), and the amplified output AC voltage v_o is taken across the collector and the emitter (that is, across the collector-emitter terminals).

Working: During the positive half cycle of the input signal, the base-emitter voltage V_{BE} increases, so the base current I_B increases. Since $\mathbf{I_C = \beta\, I_B}$, the collector current I_C also increases.

The output voltage is given by

$$V_{CE} = V_{CC} - I_C R_L$$

As I_C rises, V_{CE} falls, so the output voltage decreases on a positive input half cycle. Hence the output voltage is **180° out of phase** with the input voltage.

Input and output waveforms (described): On the same time axis, draw the input voltage v_i as a sine wave. Directly below it, draw the output voltage v_o as a larger amplitude sine wave that is inverted, that is, when v_i is at its positive peak, v_o is at its negative trough. This shows both the amplification (larger amplitude) and the 180° phase reversal.

Numerical: We are given the current gain **$\beta = 59$** and the emitter current **$I_E = 6.0$ mA**.

For a transistor, the emitter current is the sum of the base and collector currents:

$$I_E = I_B + I_C$$

Also, the collector current relates to the base current by

$$I_C = \beta I_B$$

Substituting this into the first relation:

$$I_E = I_B + \beta I_B = I_B(1 + \beta)$$

(i) Base current:

$$I_B = \frac{I_E}{1 + \beta}$$

$$I_B = \frac{6.0}{1 + 59} = \frac{6.0}{60}$$

$$I_B = 0.10 \text{ mA} = 100 \ \mu\text{A}$$

(ii) Collector current:

$$I_C = I_E - I_B = 6.0 - 0.10$$

$$I_C = 5.90 \text{ mA}$$

Check: $\frac{I_C}{I_B} = \frac{5.90}{0.10} = 59 = \beta$, which confirms the values. Hence the **base current is 0.10 mA (100 µA)** and the **collector current is 5.90 mA**.

CONCEPT

In a CE amplifier the output is always 180 degrees out of phase with the input because rising collector current pulls V_CE down. State this phase reversal explicitly, it is a marked point.

Final answer. Base current $I_B = 0.10$ mA $= 100$ µA and collector current $I_C = 5.90$ mA; the CE amplifier output is **180°** out of phase with the input.

← *Back to Q381*